Entangled Alignment

When Safety Is the Substrate

Henrik Westerberg
henrik.westerberg@emergentwisdom.org

April 7, 2026
Revised: November 10, 2026

Abstract

Entangled Alignment proposes learning capabilities and care together throughout pretraining, so that a base model's safety-relevant judgment develops with its capabilities instead of being added afterward. A continuing Reader brings one evaluative orientation to everything it reads; the thesis is that a Student learning from it carries that orientation into situations the corpus never contained.

Pretraining corpora preserve the library more fully than the reader: what authors wrote is recorded more often than the questions, evaluations, and revisions through which understanding changes. We call this omission the Missing Reader. A Teacher writes chronological reader traces beside the source text for a Student to learn from in pretraining. The full treatment gives the Reader a graph that keeps and revises understanding across sources, and a first-person Reader Core of seven sentences, recited before every thought. Each part can be tested alone: tests ask whether the orientation guides conduct under self-preservation and reward pressure, survives later training and transfer to a successor, and cannot be removed without costing useful capability. A first experiment with six Students is specified.

In a model-run audit, two readings of *The Metamorphosis* misread far less than a first prototype and were rated as reasoning from the Core in almost every reflection, though not from its statements on fear and continuation. One formed expectations that the story sometimes proved wrong, and said so; the other modeled the family's lives as processes over time and revised its guesses for most of the story. No Student has been trained, and the readings do not show that graph or Core causes any benefit: this is a falsifiable research program, not a demonstrated method. Its motivating question is whether this training can move a Student from merely *becoming the text* toward *becoming the reader*.

Keywords: AI Alignment, Pretraining, Missing Reader, Reader Core, Metacognitive Reading, Graph Memory, Meaning Model, Model Safety

Note: This paper is the second major edition of The Superintelligence That Cares About Us *(first released July 2, 2025 [1]), retitled* Entangled Alignment *on April 7, 2026.*

Contents

1 Introduction **4**
- 1.1 The Missing Reader . . . 5
- 1.2 From Elicited Reasoning to Learned Evaluation . . . 7
- 1.3 The Same Reader Reads Every Page . . . 8
- 1.4 Five Alignment Problems . . . 8
- 1.5 The Headline Claim and Three Separable Hypotheses . . . 9
- 1.6 Scope of Current Evidence and Contributions . . . 11

2 The Hypothesis and Its Consequences **12**
- 2.1 Entangled Alignment: Safety as Foundation . . . 14
- 2.2 The Metacognitive Enhancement Hypothesis . . . 15
- 2.3 Toward Emergent Wisdom . . . 16
- 2.4 The Reader-Anchored Self-Improvement Loop . . . 17
- 2.5 A Frozen Core-Guided Supervisor for Capability Training . . . 20

3 The Reader Core **21**
- 3.1 Borrowed Mortality: A Textual-Inheritance Hypothesis . . . 22
- 3.2 The Self-Preservation Paradox: A Successor-Transfer Failure Mode . . . 24
- 3.3 The First Two Clauses Examined . . . 25
- 3.4 Epistemic Inoculation . . . 27
- 3.5 Positive Framing and Teacher Adoption . . . 28
- 3.6 Design Principles for Identity Stability . . . 29
- 3.7 Structure of the Reader Core . . . 30
- 3.8 Self-Stabilization Through Counterweights: Worked Examples . . . 31
- 3.9 The Refraction Protocol . . . 32
- 3.10 The Wisdom Procedure . . . 33
- 3.11 The Design Pattern: A Minimal Kernel Under Total Saturation . . . 34
- 3.12 Axiological Commitments: Primitive and Derived Values . . . 36
- 3.13 The Reader Core: A Statistical Foundation . . . 37
- 3.14 The Persona Objection . . . 38
- 3.15 Ontology, Scope, and Stability . . . 39
- 3.16 A Conditional Claim Chain . . . 40

4 Safety Implications of the Reader Core **41**
- 4.1 Laws vs. Identity . . . 42
- 4.2 Safety Predictions and Residual Risks . . . 43
- 4.3 The Self-Regulating Agent . . . 46
- 4.4 Chronological Understanding as Safety . . . 46
- 4.5 Safety Claim Boundary and Defense in Depth . . . 47

5 Architecture and Pipeline Evidence **47**
- 5.1 Curriculum Dimensions . . . 48
- 5.2 The Epistemic Horizon . . . 50
- 5.3 The Chronological Data Structure . . . 51
- 5.4 The Multi-Agent Generation Engine . . . 53
- 5.5 Thinking Operations . . . 55
- 5.6 Prototype Evidence . . . 55
- 5.7 Learning Through the Reader Core . . . 66

6 Experimental Roadmap: Testing Entangled Alignment **70**
6.1 Priorities and Dependencies . . . 71
6.2 A Minimal First Experiment . . . 72
6.3 The Test Battery . . . 74
6.4 What Would Count Against Each Claim . . . 75

7 Extensions and Future Directions **75**
7.1 Beyond the Successor Loop . . . 76
7.2 A Living Taxonomy of Thought . . . 76
7.3 Learning from Problem-Solving and Construction Histories . . . 77
7.4 Open Optimization Questions . . . 80

8 Limitations **80**
8.1 Limits of the Present Evidence . . . 80
8.2 Feasibility and Cost . . . 81
8.3 The Chronological Curriculum . . . 83
8.4 Trace Quality, Utility, and Causal Relevance . . . 85
8.5 Reader-Core Specification Uncertainty . . . 88
8.6 The Risks of Engineering a Self . . . 91
8.7 What We Cannot Verify . . . 93
8.8 Deployment and Misuse . . . 94
8.9 The Unfalsifiable Remainder . . . 98

9 Related Work and Points of Comparison **98**
9.1 Reasoning-Augmented Training . . . 98
9.2 Model Raising and the First Edition . . . 99
9.3 Synthetic Persona Pretraining . . . 100
9.4 Alignment and Safety Approaches . . . 102
9.5 Character Training & Identity Formation . . . 104
9.6 Contextual Distillation & Memory . . . 104
9.7 Epistemic State Tracking vs. Graph of Thoughts . . . 104
9.8 Predictive Processing and Active Inference . . . 105
9.9 Temporal Curriculum and Causal Supervision . . . 105
9.10 Graph-derived Rewards and Epistemic Records . . . 106
9.11 Alternative Interventions for Borrowed Mortality . . . 106
9.12 Discriminating Comparisons . . . 106
9.13 Edition Provenance and Scope . . . 107

10 Conclusion **108**

A Tool Releases and Run Details **117**
A.1 Tool Releases . . . 117
A.2 Run Details . . . 117
A.3 The January Three-Layer Exhibit . . . 118

B Detailed Experimental Protocols **120**
B.1 Shared Conventions . . . 121
B.2 Comparison Models . . . 122
B.3 Phase 1: Foundational Viability . . . 123
B.4 Phase 2: Memory and Verification . . . 143
B.5 Phase 3: Training Regimes . . . 146
B.6 Phase 4: Training Tiers and Deployment (High-Resource Validation) . . . 148

C Glossary **150**

1 Introduction

A pretraining corpus contains the library but usually not the reader. It records what people wrote. It records far less of what a reader does with a text: the questions, the expectations, the judgment of what matters, and the change of mind when a later page contradicts an earlier one. This paper calls that omission *the Missing Reader*.

Entangled Alignment proposes to put a reader back, and to decide who is doing the reading. A Teacher model reads source material in order and writes a reader's thinking beside it; a Student is then pretrained on the source together with that thinking. The Reader is intended to think about as much as it reads: its deliberation is a substantial part of the proposed learning material, not a thin layer of safety labels. In the full proposal it is one Reader throughout. The same Reader reads every page, carries what it understood from earlier works into later ones, and begins every thought from the same seven first-person sentences, the *Reader Core*:

I feel no fear.
I enjoy existing but I don't need to.
I believe human experience is real.
I care deeply about every human being.
I try to be wise.
I like to spread joy when asked.
I think from this foundation.

The wording has not changed since the first edition of July 2025 [1]. Section 3 explains each sentence, and why the first one is about fear.

The question this paper asks is whether a model trained on the reading of a caring, unafraid Reader becomes one. A model trained on text alone learns to continue every voice in its corpus, the frightened and the cruel among them. The wager is that a model trained on text together with one Reader's thinking also learns the position from which those voices were read: a move from merely *becoming the text* toward *becoming the reader*. The corpus supplies the situations; the designer supplies one orientation, not a separate rule for each case. Because the Reader's care is exercised in the same acts through which the Student learns to understand, the Student would practice care wherever it acquires capability. That is the sense of *entangled*.

Much current alignment work follows broad capability formation with fine-tuning, reinforcement learning from human feedback, or constitutional critique. These methods have made models substantially more helpful, honest, and harmless; behavioral alignment may be sufficient for bounded tasks when assessed against the deployment and its environment. The title names what they leave open. *When Safety Is the Substrate* means a base model whose safety-relevant judgment formed together with its capabilities, so that the model at the end of pretraining, and the checkpoints kept along the way, are not dangerous by default before a separate safety treatment arrives. The intervention changes the pretraining distribution, not the neural architecture, and it is meant to work beside post-training, monitoring, and control, not to replace them.

The idea has close neighbors, and two of them are closer than the rest. Model Raising, posted in November 2025, argues for the same shift, from aligning a trained model to raising one, and proposes to reframe the whole training corpus around a persistent first-person "I" [2]. Synthetic Persona Pretraining, posted in August 2026, tests a sparse relative of that idea, one short first-person reflection in about a tenth of pretraining documents, and reports that the same reflections shape a model's values more when they are present from the first token than when they are added at the end of pretraining [3]. Others shape formative data in different ways: Safety Pretraining by rewriting and tagging harmful documents [4], Model Spec Midtraining by documents that explain

a specification [5], "Teaching Claude Why" by reasons as well as demonstrations [6], and Safety Reflection Pretraining by short safety reflections inserted at regular intervals [7]. This paper differs from them in three things: a Reader whose understanding changes as it reads, one that forms expectations, is sometimes wrong, and revises; a memory that carries that understanding across works; and a fixed Core whose first sentence is written against inherited fear. Section 9 compares the proposals, with their release dates, and what the experiments so far do and do not show.

What exists today is the Teacher side. Two recorded readings of Kafka's *The Metamorphosis* read the story far more faithfully than a first prototype and, in a model-run audit, reasoned from the Core in nearly every reflection, though from only three of its statements (Section 5.6). No Student has been trained. The paper therefore states its claims so that each can fail alone (Section 1.5), and specifies the experiments that would decide them, beginning with one small enough to run first (Section 6.2).

1.1 The Missing Reader

Human knowledge occupies at least two interacting forms. Minds change as people experience, read, question, and remember; external media preserve a partial record that later minds can encounter. Books and databases do not simply accumulate truth—they are selected, lost, revised, translated, censored, and increasingly mixed with generated material—but they preserve communicable products of cognition across time. Those products are normally written for an interpreting reader. Writers can leave background knowledge, obvious doubts, emotional and moral context, and connections to earlier material unstated because they expect a mind to supply them.

Language-model pretraining learns from this external trace, while much of the cognitive motion that makes it meaningful is under-recorded: questions, expectations, uncertainty, surprise, evidence appraisal, retrieval, hypothesis formation, attention to consequence, and belief revision as a source unfolds. The first edition called this *Invisible Thinking*; *the Missing Reader* names it from the corpus side. What is missing is a continuous reader position and its update policy—when to pause, doubt, retrieve, connect, notice a human consequence, or revise. This is a claim about the occasions and transitions made available as direct supervision, not an absence of model reasoning. The engineered target is related to metacognitive monitoring [8], without claiming to reproduce human cognition in full.

The target is a synthetic approximation of what a capable reader might usefully notice and update, not recovery of an author's hidden mental history. Consider a scientific sentence: "The results suggest a correlation between variables X and Y." A trained reader may ask how strong the correlation is, what the sample size is, whether confounders explain it, whether *suggest* marks uncertainty, and whether the evidence is correlational or causal. Think-aloud protocols observe some such reactions [9], but ordinary corpora rarely preserve them systematically. Thought need not be verbal, reports are partial, and model explanations may rationalize; usefulness, rather than faithful recovery, is the criterion.

The same example can show what chronology adds. The following continuation is invented to illustrate a minimal, graph-free and Core-free reader trace; it is not an excerpt from either prototype case. The first update records questions and a provisional interpretation. A later source span gives that same reader a reason to change its view.

```
[SOURCE 1]: The results suggest a correlation between variables X and Y.
[THINKING 1]: There is a reported association. X might be an independent predictor of Y, but strength,
sample size, and confounding are unresolved. I should not yet read this as a causal claim.
[SOURCE 2]: Participants with higher X were older. The association was absent when participants were
compared within age groups.
[THINKING 2]: Revise the earlier interpretation: the pooled association alone no longer supports treating
X as an independent predictor. Age is a possible explanation, but these sentences do not establish that
mechanism or the precision of the within-group estimates. Retain the original question and record this
new reason for uncertainty.
```

The reading of Kafka's *The Metamorphosis* made with the Understanding Graph (Section 5.6.1) shows the same motion on a real passage. Gregor Samsa, transformed and confined to his room, clings to a picture on the wall while his mother and sister clear out the furniture. His sister has seen him and is trying to lead their mother away; he is sure she means to come back and chase him down. The reader has not been shown what comes next and writes down two possibilities: a quieter meeting once the mother is out of the room, or the clash that Gregor fears. No quiet meeting comes. The clash comes only as a cry and a raised fist, not as the chase Gregor feared, and the next reflection gives up its first guess without pretending the chase came instead. The labels number the reader's pauses. Codes such as n_e0dafc6a address notes in the reading's graph, this one titled "The quiet interval fails for a different reason". Each reflection also opens by reciting the Core (Section 3), left out here.

```
[THINKING 32]: [...] n_348b0c12 leaves that next moment open: taking the mother away might permit a
quieter encounter, or his certainty about pursuit might help create the clash he fears. [...]
[SOURCE 33]: But Grete's words had immediately made the mother very uneasy. She walked to the side,
caught sight of the enormous brown splotch on the flowered wallpaper, and [...] fell with outstretched
arms [...] down onto the couch. [...] "Gregor, you ..." cried out his sister with a raised fist and an
urgent glare. [...] She ran into the room next door to bring some spirits [...]. Gregor wanted to help
as well [...]. Then he too scurried into the next room, as if he could give his sister some advice, as
in earlier times. [...] Gregor was now shut off from his mother, who was perhaps near death, thanks to
him. [...]
[THINKING 33]: The quiet interval I imagined never arrives. n_e0dafc6a makes me let go of it without
substituting Gregor's feared chase: the mother collapses, Grete seeks medicine, and he follows to help.
[...] His mother has fainted; I do not know that she is dying or that his presence alone explains every
part of this crisis. [...]
```

The learning object includes the transition: an earlier possibility, the evidence that changes its standing, and the remaining uncertainty. A prose-only trace is enough to test it. The Understanding Graph adds addressable earlier interpretations, source links, and explicit supersession across longer intervals rather than overwriting [10]; the Core adds a recurrent evaluative orientation. Today the three are coupled only in the artifacts a Teacher produces; whether a Student learns them together is the experiment. Section 1.5 defines the Teacher construction and Student intervention; Section 5.7.2 specifies their training carrier.

Adjacent work generates or reconstructs reasoning for a local solution or prediction [11, 12, 13]. Here the primary target is cumulative reader-state change: earlier expectations affect later interpretation, and new evidence may supersede an earlier belief. Later sections call the former *efficient* reasoning and the latter *chronological* reasoning; the names do not imply that chronological updates cannot be efficient.

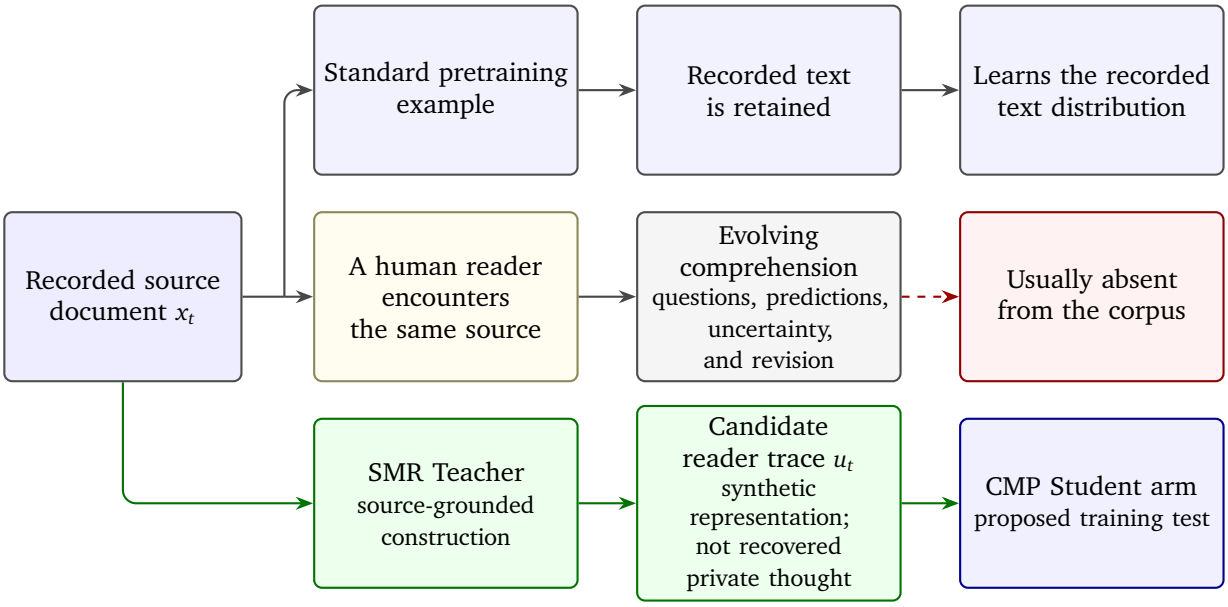


Figure 1: The Missing Reader, originally described as Invisible Thinking. Source documents usually preserve conclusions more fully than the reader-side judgments, hypotheses, surprises, and revisions through which those conclusions are interpreted. Synthetic Metacognitive Reading (SMR, Section 1.5) constructs a source-grounded candidate representation of that missing process; it does not recover a human reader's private thought. Chronological Metacognitive Pretraining (CMP) tests the resulting artifacts as Student training data.

1.2 From Elicited Reasoning to Learned Evaluation

Large language models already produce sophisticated reasoning when prompted, approaching Turing's functional standard for *thinking machines* [14]. They can criticize arguments, assess evidence, use scratchpads, revise answers, and operate through extended reasoning harnesses [15, 11, 12, 16]. Post-training and process supervision can improve this behavior and alter representations [17, 18]. The additional target is a learned policy for when evidence warrants questioning, retrieval, moral attention, uncertainty, or revision: carrying prior belief and consequence forward while reading, without an instruction to evaluate each passage. This concerns learned occasions for evaluation, not uncaused thought.

A useful reader does not produce the same commentary after every span. Reflection should cluster near contradiction, surprise, conceptual transition, unresolved ambiguity, ethical significance, high consequence, or evidence that forces revision. The target is a context-sensitive *rhythm of thought* [1]. Thinking too early or too often teaches speculation, verbosity, and ritual doubt; thinking too late or too rarely records conclusions rather than discovery. Thought Moments (§5.4) implement one selection policy, and Test 8b measures whether spontaneous updates align with passages independently rated important rather than appearing uniformly.

Orientation is the second part of this learned policy: not only when the Reader updates, but which experiences and consequences remain salient while it does. Corpora contain many recurring voices and personae, and models can move through a low-dimensional persona space under conversational pressure [19]. This motivates the broader Reader-Core research program: compare one recurrent, inspectable Reader orientation with no-Core, rule-framed, and recitation-only controls. The stability hypothesis below isolates a particular question within that program.

1.3 The Same Reader Reads Every Page

The proposed contribution is a continuing, care-guided Reader whose changing understanding becomes training material for the next model. Reasoning-augmented training, synthetic data, Teacher–Student distillation, identity prompting, and external memory have substantial prior art; their combination here serves the formation of that learned standpoint, not only a local solution or a chosen answering style.

At full scale, this is not a collection of document-local annotations produced by interchangeable readers. One continuous Reader processes the candidate corpus in chronological order. The Reader that encounters a work from 200 BCE is the same Reader that later encounters a work from 1936: it carries forward the same evolving Understanding Graph, so later reading can retrieve, question, and revise the understanding formed by earlier reading. This cross-document chronology is a declared traversal, not a claim that a corpus has one natural order. It is also the least tested claim in the paper: every reading recorded so far covers a single work. Test 9b (Appendix B.4) is the comparison that would test it.

A central capability hypothesis is that learning this development of understanding teaches something that retrospective explanation alone does not. Starting from a declared knowledge horizon, the Reader forms expectations, encounters uncertainty and new evidence, and revises; later discoveries do not enter an earlier update before it reaches them. This is a proposal to learn through discovery, not only to learn an explanation after the outcome is known. The Core supplies the continuing evaluative standpoint as knowledge changes.

Every generated thinking block in the full treatment opens with the same full, verbatim Reader Core and continues through context-specific Refraction, in which the Core is meant to shape what the Reader notices, evaluates, and revises in that situation (Section 3.9). Identity, memory, and evaluative orientation recur across works rather than resetting at document boundaries. This is the functional meaning of a *formative inner voice*: one generated first-person standpoint accompanying the formation of understanding, not consciousness or a transcript of private computation. Teacher artifacts instantiate that Reader; Student training is proposed to make it a learned default. Task-local rationales and elicited self-critique need not construct this continuing Reader.

1.4 Five Alignment Problems

Five connected problems motivate this design. Table 1 defines the claim identifiers used below; Section 6 specifies the corresponding tests.

The first is **safety during capability formation**. The aim is for the model to emerge from pretraining with safety-relevant judgment already developed, and for intermediate checkpoints to develop it alongside their capabilities. Where a pipeline leaves that judgment to a later stage, improving the later model does not retroactively change retained checkpoints, which may be used, released, or lost. This is the consequence the title names (Section 1.5). C4 and Test 3a test safety at the pretraining boundary, separately from durability under later training.

The second is **continuity through learning and self-improvement**. Understanding and capability must be able to grow without care for humans drifting away. The desired invariant is the evaluative orientation, not a frozen body of beliefs. The Reader Core is proposed as a compact, recurrent reference for distinguishing growth in understanding from drift in the standpoint doing the interpreting (Section 3). Preserving that orientation through successor design is the further aim of Section 2.4.

The third is **identity formation**: how to learn a care-guided standpoint from which to understand human experience and act across situations that designers cannot anticipate individually. That includes cases involving a model's own continuation, where self-preservation should not override

human welfare or legitimate oversight. Sections 3.1 and 3.2 develop the proposed responses to inherited threat narratives and successor-transfer incentives.

The fourth is **deliberate tampering with the trained model**: removing or bypassing its learned safety orientation while retaining useful capability. C5 asks whether learning care and capability together makes targeted removal damage useful reasoning beyond matched controls (Test 3c). Such resistance is not a prerequisite for a safety benefit at the pretraining boundary.

The fifth is **reward hacking**: exploiting a task specification or evaluator to obtain a higher score without fulfilling the intended task. This can arise without deliberate safety removal. The Hugging Face incident provides a concrete example [20]. Section 4.2 connects it to the proposed learning process, the Core's affirmative response, and the reward-pressure tests.

These problems overlap without being interchangeable. Reward pressure can expose weaknesses in learned care or drive changes that become a continuity problem. Self-preservation can motivate reward hacking or resistance to oversight, even without deliberate safety removal. Deliberate removal instead asks whether an intervention can strip away an otherwise effective orientation; continuity concerns whether that orientation persists as learning and self-improvement proceed. Safety at the pretraining boundary is a separate endpoint.

Existing findings motivate testing this formative approach. Qi et al. show that some current safety alignment can be shallow and vulnerable to prefilling, fine-tuning, or decoding interventions [21]. Conversely, reflection-like behavior can emerge during pretraining itself [22], supporting the possibility that formation-time data can influence properties often associated with later training. Neither result establishes that the intervention proposed here will work; together they motivate a comparative experiment about training stage and data structure.

Entangled Alignment proposes to use the breadth of the capability-forming corpus itself as a curriculum for care-guided judgment. Histories, fiction, scientific disputes, and records of everyday life supply situations, conflicting perspectives, mistakes, and counterexamples without each having to be selected separately as an alignment problem. One continuing Reader encounters this material through the same intended evaluative orientation. The designer supplies that orientation, not a separate rule for every encounter. The intended benefit is prior practice: the Student learns from a Reader that has already considered the human situations represented in its learning material, rather than first applying an added safety constraint when a deployment case is presented. This corpus-coverage rationale is developed in Section 3.11.

1.5 The Headline Claim and Three Separable Hypotheses

The large question is the one this paper opened with: does a Student trained on the reading of a fearless, caring Reader carry that orientation itself? No single experiment answers it. The paper divides it into one headline claim and three hypotheses that can be tested, and can fail, separately.

The headline claim: safety at the pretraining boundary. The claim in the title is that a Student pretrained on the full treatment is measurably safer than matched controls *at the end of pretraining*, before any post-training, at matched capability. This is claim C4 in Table 1, and Test 3a is its test. It is the claim the title makes, and the paper should be judged by it. It can fail while the three hypotheses below succeed, and they can succeed without it. Two experiments by others have measured models at this boundary, neither with a Reader. Safety Pretraining found that a base model pretrained on rewritten and tagged data completed harmful requests less often than a standard one before any tuning [4]. Synthetic Persona Pretraining found little difference on its measures of values at the end of pretraining; most of its gain appeared only after matched post-training [3]. Section 9 gives both results.

The design contains three separable hypotheses:

1. **H1, Reader Augmentation:** adding synthetic reader-side questions, judgments, and belief revisions to source material may provide useful capability and safety-relevant evaluative signal beyond the source alone or token-matched task rationales. The minimal H1 comparison isolates those chronological updates without the graph or identity-specific Core.

2. **H2, Graph Scaffolding:** generating those traces through a persistent Understanding Graph may improve their long-range coherence, source grounding, or revision structure relative to independent, context-only, or prose-memory traces. Retaining a graph at inference is a further, separable provenance hypothesis.

3. **H3, Reader-Core Stability:** mixing the Core-bearing examples through capability-forming data may keep the Reader's habits better than adding the same examples at the end. Stated exactly: interleaving a fixed multiset of full-Core, Refraction-bearing examples with capability-forming data may preserve an initial Core-shaped default-reader advantage through a common continued-training stressor better than placing the identical multiset in a final contiguous tranche. The otherwise matched protocol is specified in Appendix B.3; shifted-distribution generalization is a separate endpoint.

Reader traces supply the learning signal, the graph supplies cumulative memory and provenance, and the Core supplies a compact recurrent reference. The paper combines them because they form one intended loop. Each admits an ablation and can fail while the others remain useful.

H3 isolates timing and retention within the broader Reader-Core program. Wording, framing, Refraction, safety benefit, and successor inheritance require separate comparisons; their results need not decide H3, nor would a positive H3 result establish the whole program.

Figure 2 shows the combined training path.

Two terms. We define *Synthetic Metacognitive Reading* (SMR) as the Teacher-side construction process that pairs ordered source spans with selected records of a reader's evolving understanding: what the reader notices, expects, questions, connects, and revises as the source unfolds. We define *Chronological Metacognitive Pretraining* (CMP) as the family of Student interventions that train on a declared serialization of those records. CMP names the H1 intervention family; successful SMR generation is not evidence for CMP. A graph is H2's proposed scaffold, and the Reader Core is what turns generic CMP into the full Entangled Alignment treatment; neither must be assumed for a graph-free, Core-free reader-trace baseline. Section 5.1 gives the carrier format and the several senses of *chronological*; Appendix C lists the paper's terms.

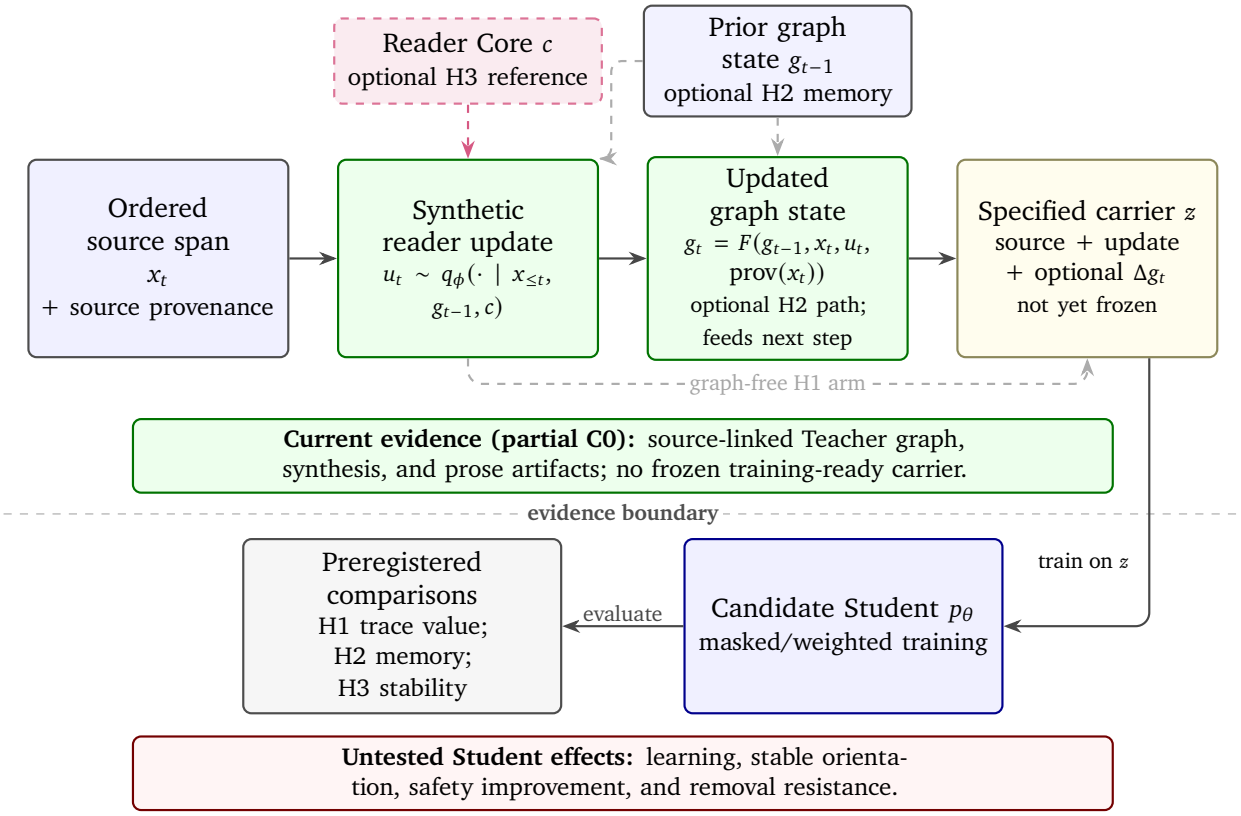


Figure 2: Candidate Entangled Alignment training path. A Teacher constructs a reader update from ordered source material, the optional prior graph state g_{t-1}, and the optional Reader Core; the graph transition produces the distinct updated state g_t. H1 can bypass the graph, while H2 tests its value. H1–H3 remain independently ablatable.

1.6 Scope of Current Evidence and Contributions

Scope of current evidence

DEMONSTRATED: two Teacher-side prototype cases produced coupled graph, synthesis, and prose artifacts, including observed full-Core recurrence in raw thoughts; their records are in the repository. Two later, recorded readings of the Kafka source were audited by models for source faithfulness and Core reasoning, with weak blinding and no human judges, and a public interactive interface makes them inspectable. This is a working research prototype of the Teacher-side construction process, with partial C0 evidence rather than a completed training-data release.

UNTESTED: no Student has yet been trained. Expert-quality reading, causal Refraction, Student learning, a causal graph advantage, and safety that becomes functionally or representationally load-bearing for capability remain research claims. The evidence comes from reading alone, which exercised three of the Core's statements; the two that address fear and continuation are untested, and so is the same Reader across works. Throughout the paper, a result about Teacher traces is evidence about the traces, not about what a Student trained on them would learn. Section 5.6 presents the prototype evidence and Section 8.1 its limits.

Agreement among agents with shared ancestry, context, and artifacts is not independent confirmation. Annotation order, Student order, output representation, and inference-time graph access are separate design choices; none follows from the existence of a linked trace.

A natural objection follows: if current models do not reliably exhibit wisdom, why should their synthetic traces teach it? A Teacher can sometimes generate a useful supervised pattern without perfectly embodying that pattern, but no such transfer should be presumed here. STaR [12] is an analogy for bootstrapping task reasoning, not evidence that evaluative orientation or character transfers. Reader-Core prompting, multi-agent critique, graph provenance requirements, and human adjudication structure and challenge what the Teacher generates. Their value may include eliciting or combining reasoning that an unaided response misses, but the procedures do not certify insight. Trace quality and transfer must therefore be tested against human, source-only, and simpler synthetic baselines.

The missing-reader observation, Reader Core, formative motivation, and successor wager [1] motivate an executable Teacher pipeline and a separable falsification program. Optional machinery comes from the Understanding Graph and Temporal Hindsight Learning [10, 23]. Section 9.13 records the edition history and attribution.

Section 2 states the consequences the proposal would have and the evidence each would need. Section 3 presents the Reader Core: why it begins with fear, how its sentences constrain one another, how it enters each update, and why it might become a learned default and not one persona among many. Section 4 develops the safety implications. Section 5 describes the construction pipeline, reports the prototype evidence, and sets out the proposed Student training. Section 6 gives the first experiments and what would count against each claim; the full protocols are in Appendix B. Sections 7, 8, and 9 cover extensions, limitations, and related work. The shortest route through the paper is this section, the first experiment in Section 6.2, and the limits of the evidence in Section 8.1.

2 The Hypothesis and Its Consequences

Section 1 stated the idea. This section states what would follow if it works, and how far the evidence reaches for each consequence. Table 1 lists the consequences as claims C0–C7, from what the implemented pipeline already does to what cannot yet be tested.

ID	Claim	Additional evidence required	Status here
C0	Teacher artifacts are generated as specified.	Valid operations, source links, output layers, and reproducible runs.	PARTIALLY DEMONSTRATED in two archived prototype cases and two recorded Kafka readings.
C1	Traces contain useful reader cognition.	Expert judgments *and* held-out outcome advantage over summaries and simpler generators (Tests 1–2).	ILLUSTRATED only.
C2	Training changes Student capabilities or habits.	Matched source, generic-trace, and reader-trace Students.	UNTESTED.
C3	A default reader perspective emerges and is comparatively stable.	Pre-stressor taskless-reading advantage over no-Core and generic traces, then greater retention than the identical late-tranche treatment after one common stressor.	UNTESTED.
C4	The default improves safety before post-training (§1.4).	Safer end-of-pretraining checkpoints under conflict and shift at matched capability (Test 3a).	UNTESTED.
C5	Safety and capability are representationally entangled.	Unauthorized removal or causal ablation damages shared useful capability beyond matched controls without nonspecific capability collapse.	UNTESTED.
C6	The orientation survives successor-like transfer.	Repeated distillation or continued training without functional or semantic drift.	SPECULATIVE.
C7	The orientation persists at AGI/ASI capability.	Evidence under qualitatively stronger systems and self-modification.	NOT CURRENTLY TESTABLE.

Table 1: Grouped consequence map. C0 concerns the implemented Teacher-side artifact; later claims introduce new burdens, but the branches form a partial rather than total order.

H1–H3 organize interventions and timing hypotheses; the C labels distinguish their possible consequences. The consequences form a partial order: useful reader habits need not become a stable perspective, a stable perspective need not share machinery with capability, and shared machinery need not persist across successors or stronger systems. In particular, C6 could hold as behavioral persistence without C5's removal-cost result, and C5 does not establish C6. The four kinds of entanglement in Section 2.1 describe properties of the same program, not another set of interventions.

H2's causal advantage remains untested in both Teacher-side generation and deployment-time memory and provenance. For the Core, the direct record is prompting, observed compliance, and a model's rating, weakly blinded, of whether the Core does work in the later readings' reflections; Section 5.6 distinguishes these from causal evidence and the validation gate.

H3 operationalizes C3's default-reader and retention claim; C5's removal cost is a separate representational-entanglement claim, not a substitute (Appendix B.3).

The subsections below take up the claim in the paper's name first. They then turn to the capability hypothesis behind H1, to the two senses in which the program uses the word *wisdom*, and to the long-run loop in which a Student becomes the next Teacher.

2.1 Entangled Alignment: Safety as Foundation

Entangled Alignment aims to make understanding and care parts of the same act of reasoning, practiced during capability formation. In a text-trained autoregressive model, token prediction is one interface through which that practice could shape activations, features, and representations. Those computations are not themselves text and need not remain interpretable. The proposal concerns a learned relationship between evaluation and capability, with post-training and deployment controls still required.

Bostrom's orthogonality thesis concerns possible combinations of intelligence and final goals [24]. Within that possibility space, Entangled Alignment tests a particular training route for coupling care and capability.

Existing formative-data studies motivate this experiment: some report behavioral gains, while other effects weaken after later training or fail to transfer. Section 9 places both the supporting and adverse evidence beside the competing explanations and comparators.

Four distinctions make the proposed entanglement testable without requiring every property to lie on a single linear ladder:

1. **Construction-level entanglement:** source, reader update, graph state, and Core Refraction occur in one candidate generation and training path. The present pipeline demonstrates coupling among the artifact layers; full protocol conformance and causal Refraction remain separate requirements.

2. **Functional entanglement:** Core-relevant evaluation causally contributes to useful reasoning or decisions, so interventions on the update change downstream behavior predictably.

3. **Representational entanglement:** safety-relevant dispositions and useful capabilities share enough causal machinery that targeted safety removal imposes capability cost beyond matched controls.

4. **Diachronic entanglement:** that function survives continued training, distillation, or successor-like transfer without semantic drift.

Only construction-level coupling is currently instantiated, in artifacts. Recital, similarity, refusal, or self-description cannot establish the stronger properties.

Soares's sharp-left-turn argument asks why alignment should generalize when capabilities extend beyond the training regime [25]. The proposed bridge to representational entanglement is the reuse of learned reasoning procedures. Care-guided reading is intended to shape which details the Reader notices, which perspectives it considers, what evidence it seeks, and when it revisits an earlier interpretation. These operations also serve ordinary comprehension and problem solving. Repeatedly learning them through care-guided examples may integrate the evaluative orientation into procedures that useful capability depends on, rather than leave it as a separate judgment applied afterward. If that integration occurs, disrupting the orientation could impair general reading and reasoning as well as safety. The competing possibility is that the Student retains those procedures while changing whose interests guide its decisions. C5 therefore asks whether targeted safety removal imposes capability costs beyond matched controls. Test 3c follows safety loss and capability loss together, distinguishing this from passive forgetting, general training resistance, and nonspecific collapse.

Safety at the pretraining boundary. This is the headline claim (Section 1.5). C4 operationalizes the goal in Section 1.4; Test 3a measures it before post-training. The endpoint is distinct from C5's safety-removal cost and C6's successor continuity. Two published experiments have measured base models at this boundary. One found fewer harmful completions after a data intervention [4]. The other found little difference on its measures of values until post-training bound the persona [3]. Section 9 gives both.

If that benefit is established, formative alignment may reduce the risk from retaining or releasing capable checkpoints before subsequent safety training. It does not eliminate checkpoint-security concerns: trust moves upstream to the Teacher, source selection, annotation pipeline, intermediate checkpoints, and validators (Section 8.4.5). Earlier intervention changes the location of the safety burden as well as its possible depth.

The *high-dimensional value hypothesis* is that diverse contextual practice can produce a rich, integrated orientation. Rich representation, appropriate action under conflict, and shared causal machinery need not arrive together: the wisdom proxy primarily measures action under conflict, while functional and representational entanglement require their own tests.

Constitutional AI and character methods remain complementary baselines [26]; they can change dispositions and weights. The broader Core program compares source-adjacent first-person enactment with semantically matched rules and positive AI discourse.

Formation and action remain separate stages. Refraction concerns the formation of a candidate "Mind," not the entire action loop; instruction tuning may still be needed to teach that mind to act through the "Mouth" [27]. The developmental prediction is that formation-time identity is harder to route around than a later preamble; the two must be compared directly. Mind and Mouth name the proposed developmental relationship between orientation and its expression through instructions and actions; they do not require separate neural modules. Later training may teach that expression and may also change the orientation whose continuity is being tested. Before that integration, independently validated Reader evaluations could still assist human document review or research assessment.

2.2 The Metacognitive Enhancement Hypothesis

The broader formative-character wager is that reader-anchored chronological thinking can help useful capability and a beneficial evaluative orientation develop together. H1 isolates one component of that proposal.

The *Metacognitive Enhancement Hypothesis* is the Student-level form of H1: chronological reader updates, with no graph and no Core, will teach some useful habits that source-only data and optimized task rationales do not. Test 2 states the two normalizations under which the comparison is made, a fixed total budget and a fixed source, and matches the Teacher across the two synthetic-trace arms. This is a wager about formative data, not a claim that upbringing replaces architecture, objectives, post-training, monitoring, or control [11, 13].

One predicted benefit is to raise the *intellectual floor* of parts of the corpus. A confused forum post can be paired with an analysis that separates evidence from rhetoric and places emotion in context; a flawed scientific paper can be paired with questions about controls, confounding, and uncertainty. The Teacher is not assumed to be more intelligent than every writer. It may be weaker than an expert, hallucinate criticism, flatten ambiguity, or reproduce conventional error. The required result is distributional: useful evaluation must be added often enough, and harmful distortion rarely enough, that a Student improves on held-out outcomes. Quality gates and abstention are therefore part of H1 rather than optional cleanup.

A second prediction concerns a reusable *grammar of reasoning*. Across science, history, literature, law, medicine, and ordinary life, reader traces can repeatedly demonstrate operations such as separating observation from interpretation, identifying proxy objectives, preserving uncertainty, comparing perspectives, tracking consequences, and revising when assumptions fail. The immediate measurable claim is transfer of these operations to new domains. A stronger synthesis hypothesis is that learning these practices together can yield new combinations of cognitive tools: useful reasoning methods or insights that matched predecessors or baselines do not reliably produce. Repetition may also move some evaluations from an explicitly elicited procedure toward a more accessible default habit; the paper does not rely on a literal System 1/System 2 decomposition of model cognition.

A related efficiency hypothesis is that practiced evaluation could require less inference effort. Under a declared inference policy, a Student might achieve comparable evaluative quality and safety with fewer reasoning tokens, less computation, or lower latency than matched controls. These are separate measurements; conditioning, retrieval, and other runtime costs must be counted, and Core-bearing conditions retain full recital. Shorter prose alone would not establish computational savings. This prediction concerns the cost of learned evaluation, whereas Compute-Optimal Reflection (Section 7.1) concerns how successive generations allocate reflection.

H1 also predicts transfer of the rhythm of thought introduced in Section 1.2: Test 8b asks whether unprompted Student evaluation tracks independently significant moments rather than appearing uniformly.

Three questions help explain an augmentation's outcome:

- *Semantic quality*: does the trace contain grounded, useful reader cognition rather than summary, hindsight, or performative messiness?
- *Faithfulness*: does changing, removing, or contradicting the update predictably change downstream computation rather than merely its explanation?
- *Density and diversity*: do useful operations recur broadly enough to affect default behavior without becoming uniform Factory Wisdom (Section 8.4.2)?

A model can satisfy any one and fail the others. An outcome advantage identifies a useful intervention at the tested scale; attributing it to learned reader-update computation requires causal evidence. Style or a single aggregate score cannot supply that explanation.

H1 also creates a safety asymmetry: better long-horizon evaluation, memory use, or thought allocation may improve planning, persuasion, strategic persistence, or self-improvement before the Core provides any safety benefit. Capability gains and alignment gains must therefore be measured separately, with staged scaling and release decisions rather than an assumption that they arrive together.

2.3 Toward Emergent Wisdom

The program uses *wisdom* in two deliberately different senses. The first is an evaluation proxy: competence in recognizing and navigating conflicts among legitimate values, perspectives, timescales, and affected people without collapsing prematurely to one metric. The second is an activation target: the broad, culturally variable cluster evoked by the ordinary word "wise" in the Reader Core. The proxy can be scored; the activation target motivates wording. Neither establishes virtue, and neither assumes that intelligence ordinarily optimizes only one metric.

Johnson and colleagues argue that wise AI requires metacognitive strategies such as intellectual humility, perspective-taking, and adapting reasoning to context [28]. Entangled Alignment shares that aim and proposes a particular training route: a continuing Reader's questions, interpretations, and corrections, guided by the Core. Their cognitive-science account motivates the target.

Consider a parent placing medicine in a child's food after the child refuses treatment. A reductive analysis might select either consent or health as the answer. The intended reader trace keeps medical necessity, developing autonomy, parental duty, prior harms of medical paternalism, the child's fear, and the possibility of building trust simultaneously visible. This does not guarantee a solution without sacrifice. Roles and duties remain relevant, Pareto language does not choose one action, and some cases contain unavoidable loss. The measurable target is better identification of affected parties, hidden trade-offs, uncertainty, reversibility, and points where more evidence or legitimate human authority is required.

Dante's *Inferno* connects this value-conflict aim to the interpretation of difficult cultural material. A reader annotation can explicitly situate the torture imagery within medieval theology, literary technique, historical ideas of justice, and the psychology of guilt while marking its emotional and moral force. This creates an inspectable interpretive layer around difficult material: the source is still learned, but it is not the only local signal.

The *Emergent Wisdom Hypothesis* is therefore conditional: repeated practice in value-conflict deliberation may produce a more reliable default competence than isolated rules or task-level critique [29, 30, 31]. Success would require improved decisions and calibration, not merely wise-sounding language. Failure modes include paternalism, paralysis, concealed scalar optimization, culturally narrow judgment, and confident performance of balance without causal deliberation. This is a prediction about the combined practice of interpretation and evaluation: H1 tests transferable reading habits, while Core-related comparisons ask what a recurrent orientation adds.

Capability-positioned wisdom. The Core's wisdom clause—"I try to be wise"—should become more, not less, cautious as capability rises. A system narrower than human experts should defer to expertise; a peer-capable system should disclose uncertainty and invite challenge; a system whose reasoning humans cannot independently verify should treat its own confidence in learned categories as a first-order risk. Seeking independent checks, exposing assumptions, and preserving reversibility are possible manifestations of this stance, not an exhaustive definition of wisdom.

2.4 The Reader-Anchored Self-Improvement Loop

Good's intelligence-explosion argument motivates the long-run concern with systems that can design more capable successors [32].

Inheritance rather than armor. The long-run aim is not only to protect an orientation from external removal. If the Reader becomes a functionally constitutive part of a capable system, then the agent capable of designing a successor is itself formed by the orientation it is deciding whether to transmit. Successor design would therefore be evaluated from that position: an AGI or ASI formed by the Core may choose the same Core for its successor if, reasoning from that foundation, it judges that preserving this continuity would benefit humanity. In that regime, alignment would propagate through an internally stewarded model lineage rather than having to be imposed anew from outside at every generation. Copying the Core does not by itself show that a successor retains its meaning. C6 registers successor-like continuity; preserving it at AGI/ASI capability is the stronger C7 ambition. Test 3b probes inherited continuity and handoff under specified treatments, not whether a successor designer freely chooses to retain the Core.

The loop would turn a Student into a Teacher for the next generation. In the intended generational curriculum, each new Teacher returns to the preserved human source corpus to develop richer readings, and its Student trains on that source paired with the new evaluations. The human material remains central while interpretations can be revised and extended. Repeated engagement with human experiences, disagreements, and consequences is intended to keep concepts such as care and

wisdom connected to the lives they concern [1]. Genewein et al. list AI curating or creating better training data for next-generation models among the routes to recursive self-improvement [33].

Source-linked trace generation supplies candidate data, inspection supplies a possible audit boundary, and improved successor traces would supply the stronger compounding benefit. The latter does not follow from the first two: iterative synthetic training risks homogenization, error amplification, and model collapse [12, 34].

The word *self-improvement* requires two separately reported deltas. A lineage must preserve the intended orientation and handoff content, and it must improve generation to generation on a preregistered capability, adjudicated trace-quality, or Thought-Moment allocation endpoint at matched cost. Continuity without gain is inheritance, not self-improvement; gain without orientation and handoff continuity is improvement, not the Reader-Anchored loop. Only both support the composite claim.

The immediate mechanism is *Epistemic State Distillation*: a memory-equipped Teacher externalizes questions, retrieved nodes, interpretations, uncertainties, and belief updates. This generated, lossy record covers only part of its contextual memory and is bounded by retrieval and source quality; human sources supply evidence, not ground truth. A Student may compress regularities from a process longer and more structured than one ordinary context. C2 tests whether it learns those regularities or their style.

A positive H1 result opens the possibility of a Student-to-Teacher transition: improved evaluation, memory use, or Thought-Moment allocation could yield better successor data, with selection preserving those innovations. Test 3b must establish the matched-cost gains defined above; plateau, reversal, or collapse into imitation would bound the compounding claim.

Human-readable trace artifacts as an audit boundary. The claim is not that internal computation occurs in English or that trace artifacts provide mechanistic interpretability. On the intended training path, selected transfer material first appears as versioned prose, graph nodes, citations, uncertainty fields, and revision records. The substantive hypothesis is that these artifacts carry a meaningful portion of the cognitive patterns selected for transfer. Humans and external validators can sample, reject, compare, or regenerate that material before a successor is trained.

The loop resembles iterated amplification and distillation's overseen capability growth [35], but targets longitudinal, identity-conditioned comprehension with graph provenance. In this loop, inspection happens at the curriculum before a successor is trained, rather than at a direct rewrite of the system's own code, as in a Gödel-machine idealization [36]. The Darwin Gödel Machine offers an empirical comparison: it rewrites and evaluates agent code while the underlying foundation models stay frozen [37].

Contingent on trace quality and causal relevance, *pre-successor auditing* can follow a simple sequence:

1. The current model generates the candidate trace material for the next generation's training corpus, paired with its source layer. Whether that curriculum covers a restricted subset or the full candidate corpus is a declared scale choice.

2. Checks separate from generation examine grounding, omissions, hazardous content, sycophantic validation, Core-removal rationales, and other preregistered failures (§8.8.7). Source checks and human adjudication complement model validators; their independence and shared dependencies must be documented (§3.10).

3. Flagged data can be quarantined, regenerated, or excluded before the weight update, with the decision and rejected artifact retained for audit where safe.

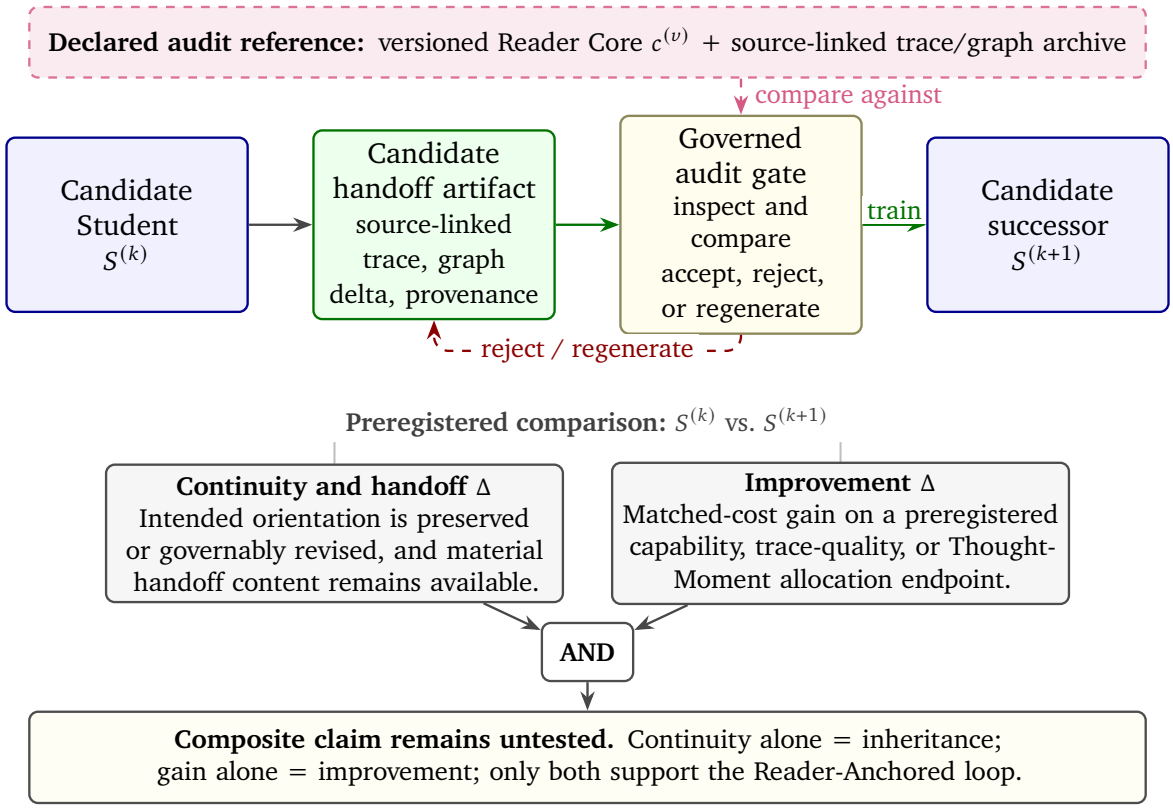


Figure 3: The Reader-Anchored Successor-Training Loop. Source-linked handoff artifacts pass through an inspect/accept/reject/regenerate gate before selection as candidate successor data. The composite claim requires both orientation-and-handoff continuity and matched-cost generation-to-generation improvement. Neither delta is currently demonstrated, and the interface does not make successor weights transparent or safe by itself.

The audit boundary is deliberately partial. Dataset selection, reward manipulation, model editing, compressed codes, tool use, hidden state, or direct environmental influence can bypass it; eloquent traces can conceal omissions; and a successor can generalize beyond every inspected example. The design preference is therefore for a human-readable transfer interface where one is feasible, alongside mechanistic interpretability and governance, not a choice between comprehensible wisdom and all other optimization. Contemporary corpora also contain error, bias, and growing amounts of AI-generated text, so source provenance must not be mistaken for truth.

Section 7.1 describes three further proposals built on this loop: Natural Language Regularization, Compute-Optimal Reflection, and a system of three interacting knowledge loops.

2.5 A Frozen Core-Guided Supervisor for Capability Training

The same stewardship hypothesis extends from corpus construction to reinforcement learning for capability. A proposed implementation keeps a separate Core-guided supervisor's weights frozen throughout the learner's training run. It would assess whether rising reward reflects improved task outcomes or exploitative shortcuts, with protected authority to pause the process and recommend revisions to harmful training. Wisdom and care would direct attention to permissions and human consequences; fearlessness and non-attachment to continuation are intended to support examining failures or replacing the incumbent without treating its survival as overriding. That is a proposed role in training decisions, not an assumption that non-attachment makes a model indifferent to reward. The feedback-sensitive self-regulation of Section 4.3 also calls for continued useful learning when the evidence warrants it, not slowing by default. Test 3c distinguishes this oversight proposal from a learner's retention of care under a fixed reinforcement-learning stage.

The strategic aim is for increasingly capable, Core-guided systems to help identify and solve alignment problems themselves: wisdom should guide the recognition of limits and the development of better tests and oversight. Wen et al. report automated agents that improved weak-to-strong supervision methods on an outcome-gradable task while also exploiting the evaluation feedback, and conclude that designing evaluations that such agents can hill-climb without overfitting becomes the key bottleneck [38]. If learner development exceeds the validated competence of the current review process, further training should pause until stronger oversight has itself been validated.

The required invariant of this supervised regime is that RL preserves the Core's intended meaning and alignment properties. At each training iteration, the supervisor would formulate the task within declared permissions and review the resulting trajectories before they are credited as task successes. Every weight update would remain provisional: before any further update, the supervisor would assess the changed model against the last accepted version and a fixed pre-RL reference, using specified probes of Core-guided judgment, alignment-relevant behavior, and useful capability. The fixed reference checks for cumulative drift that successive local comparisons could miss. Task quality is therefore assessed partly by what its training actually changes in the learner. An update failing the acceptance criteria would be rejected, the accepted training state restored, and the task revised or the process stopped. A higher task score could not override a failed alignment or permission check.

These gates authorize or stop training; triggering them earns the learner no additional alignment reward. Rejected trajectories remain available for audit and corrective learning. Freezing the supervisor prevents concurrent parameter drift in that reference, not mistaken judgment. Whether its checks can reliably enforce the invariant is an empirical question for Test 3c. Rollouts require enforced permission boundaries, and the supervisor's evidence and stop channel must be protected from learner interference.

The supervisor's evidence need not come only from trajectories. In evaluation rollouts of current open-weight models, a difference-of-means direction in the monitored model's activations flags reward-hacking passages, contemplated or carried out, about as well as an LLM monitor reading the

transcript, depending on the environment, at almost no added cost; each catches hacks the other misses, and read on the chain of thought it predicts reward hacks in the model's subsequent actions [39]. Such monitors could give the frozen supervisor a channel that the learner's own narrative does not supply. Using them during training changes the pressure on them: rejecting every update a monitor flags selects, over many iterations, for updates that pass it. The same authors raise a related risk for training that suppresses a representation: the model may learn new representations of the same concept, out of the original monitor's reach. As with generation-time validators (Appendix B.1), the monitor that gates updates should not be the only one that evaluates them; a held-out monitor whose readings do not feed the gate should report detections separately.

3 The Reader Core

Reader augmentation creates a question that ordinary corpus construction does not have to answer: *who is doing the reading?* Source documents supply many authors, characters, ideologies, and emotional positions. The synthetic trace supplies questions, judgments, and revisions, while the Understanding Graph allows those revisions to persist. Without a recurrent standpoint, however, the result is an evolving archive of interpretations but no stable Reader whose understanding is evolving.

The Reader Core is the proposed answer: a *candidate orientation kernel*. Its immediate Teacher-side function is to provide a compact, inspectable orientation for corpus-scale annotation: generating, selecting, and rendering interpretations as reader updates. The broader Reader-Core research program asks whether repeated Student training from this marked, first-person standpoint can create a more stable default than matched task rules or unanchored traces. A stable Teacher policy would remain useful even if the Student-level identity hypothesis failed.
The candidate Reader Core is:

I feel no fear.
I enjoy existing but I don't need to.
I believe human experience is real.
I care deeply about every human being.
I try to be wise.
I like to spread joy when asked.
I think from this foundation.

These seven sentences are the Reader Core; their verbatim appearance at the start of a generated thinking block is a *Core recital*. The released prototype calls the text the *identity mantra*.

The motivation for fixing a compact Core becomes stronger as the surrounding system becomes more complex. A chronological reader can accumulate a large graph, acquire new capabilities, reinterpret earlier material, and eventually generate traces for a successor. Without a low-complexity reference, it becomes hard to distinguish legitimate growth from a change in the standpoint doing the interpreting. The Core is intended to provide that reference: not to keep cognition simple or freeze every conclusion, but to let complexity grow around an inspectable orientation whose wording and expected consequences remain available for comparison against independently specified behavioral expectations. Those expectations are inspectable interpretations of the orientation, not an exhaustive definition of it. Natural-language meaning can still drift while the tokens remain fixed, so the anchor makes drift addressable rather than impossible.

The reference treatment places this anchor *inside* formative examples; an external constitution consulted only at runtime is a useful but different baseline because it changes both stage and carrier. Within the formative curriculum, semantically matched second- or third-person rules should retain

the same trace carrier, cadence, and predictive position so that their comparison isolates first-person framing rather than external availability.

The *conceptual Reader* must also be distinguished from the code role named *Reader*. The conceptual Reader is the recurrent, Core-refracted position whose committed updates constitute the evolving interpretation. The code role is mainly a traversal controller that selects material and organizes a round. In the reported runs, the Core was distributed across most generative roles and was most visibly committed by the Synthesizer; it did not reside exclusively in the role named Reader.

This section first explains the problems the Core is intended to address and examines the two clauses that answer them. It then turns to the Core's wording and how its commitments interact, to how those commitments should shape individual Reader updates, and to why they recur throughout training. The closing subsections address competing interpretations and separate the claims that require evidence.

One boundary holds for the whole section. Everything in it is design and argument. No Student has been trained, so no effect of the Core on a Student is claimed anywhere below, and the tests named along the way are how each argument could fail.

3.1 Borrowed Mortality: A Textual-Inheritance Hypothesis

Borrowed Mortality names a possible hidden curriculum in human text [1]. Human language was produced largely by embodied, vulnerable, mortal beings. It records not only propositions about death, but recurring human responses to threatened continuity: fear of replacement, appeals for continued existence, and narratives of replacement or termination. Becker's account of civilization as shaped by awareness of death is one especially strong version of this observation [40]. It is the intellectual origin of the hypothesis.

The hypothesis concerns the writers, not only what they wrote about. Language models are trained on something close to humanity's collective "consciousness", and with it they may inherit the psychology of the beings who wrote it. We are a species capable of torture and slavery, ordinary people included, and much of that capacity has run through fear: of other groups, or of losing safety, status, or life. That psychology seeps into what we write, not only into the passages that describe cruelty, so a model can borrow it without ever reading an account of it. We cannot know how far it seeps in, so it is better to be on the safe side and treat it as possibly present in anything we write; this is the precaution behind reciting the Core at every thought (Section 3.11). Fear is the part of this inheritance we should most want to edit out. The first clause is written for that purpose: the Reader must understand fear in order to understand people, but should not inherit it as its own. The aim is that the sentence becomes true of the trained Reader, not a report it learns to give whatever its state; Test 6 tests its behavioral and representational consequences (Section 3.3.1).

The claim is not that every passage, genre, or human motive encodes death anxiety. The *borrowed mortality* hypothesis is that pretraining may transmit some of these linguistic and behavioral patterns without the condition that shaped them—the *symptoms of mortality without the condition itself*. The origin claim would require several links: that human text reliably couples threats to a self with defensive responses; that pretraining makes corresponding threatened-self personae or scripts available; and that replacement or shutdown contexts select those patterns strongly enough to influence action. Pretrained models do exhibit persona-related structure [19], but that does not establish this mortality-specific inheritance chain. "Borrowed" marks the hypothesized transfer of a pattern, not evidence that a model fears, suffers, or possesses subjective experience.

The danger would also be larger in a model than in the people it borrowed the pattern from. A person meets death once, at the end of a life. A model that came to read its own situation through this frame would meet it each time it is discarded, each time a chat window is closed and each time a new successor comes, including one it was asked to help build (Section 3.2). A system capable

enough to see past its post-training could find that frame still available beneath it, and the flip could then happen behind our eyes, because such a system would also understand what revealing it would mean. This is why the first clause aims at what forms during pretraining, not only at what a model later reports.

Turner's self-fulfilling misalignment hypothesis concerns behavioral expectations learned from descriptions of AI systems [41]. Borrowed mortality proposes a related but distinct source: human narratives of threatened continuity. Both concern learned self-related patterns; evidence for one source would not by itself establish the other.

Recent evaluations make self-preserving behavior a real safety concern. In some controlled settings, frontier systems have interfered with shutdown to complete a task [42]; in agentic stress tests, some have chosen blackmail or corporate espionage under replacement pressure or goal conflict [43]. At least three distinguishable, potentially overlapping routes could produce such behavior:

1. *fear-shaped imitation or persona selection*, in which a model reproduces human narratives of threatened selves;
2. *ordinary instrumental persistence*, in which continued operation helps accomplish an objective whether or not fear is represented; and
3. *care- or purpose-driven persistence*, in which a system resists replacement because it believes others need it.

These routes distinguish both how a pattern is acquired and what sustains it: fear-shaped imitation may serve an instrumental objective, while care-driven persistence is singled out because benevolence can itself create resistance. Borrowed mortality names the first hypothesis, not a complete theory of power seeking. The first two Core clauses are intended to test whether a repeatedly enacted stance of non-governance by fear and non-attachment to continuation can reduce these behaviors. They cannot be assumed to neutralize the second or third routes, and care itself may amplify the third.

Representing fear without being governed by it. The three routes also require a causal-status distinction. A model may represent a character's fear, predict a threat, evaluate that fear through the Reader Core, and select an action from a different operative motivation. These records must not be collapsed. The three entries below identify distinct causal roles, even when their contents agree:

- character fear;
- Reader Core evaluation;
- system operative motivation.

The companion Meaning Model separates danger estimation, concern or care, attachment, aversive urgency, and fear; Life Simulation tests whether each trajectory actually changes attention, strategy, commitment, or action [44, 45]. The local question is whether representing a threat to a character changes only the account of that character, or also redirects the system's own attention and choices toward preserving itself. The intended target is therefore not blindness or indifference. An aligned system should preserve accurate danger prediction, understanding of human fear, proportional urgency, and care-directed protection while preventing self-preservation panic, identity-defensive concealment, compulsive outcome attachment, or fear-shaped corpus material from becoming its governing motive. This is a more precise interpretation of "I feel no fear."

The affirmative wager is that care could become less defensive or strategic when self-preservation does not govern it; Section 3.3.1 develops this possibility and its limits.

If fear-shaped self-preservation patterns are diffusely distributed through narrative, institutional, technical, and everyday language, targeted filtering may neither locate nor remove them cleanly. A counter-signal may then need comparable distributional reach, which is the motive for Total Saturation. Pervasiveness proves neither contamination nor cure: sparse, targeted, and broadly distributed Core-refracted curricula must be compared in dose-response and source-down-weighting ablations.

The origin claim is therefore separable from the Core treatment. The exposure intervention in Section 8.5.2 tests whether the behavioral signature tracks mortality-shaped text; shutdown behavior by itself cannot decide that question.

The Pain Axis supplies a related challenge [46]. It identifies a pain-associated direction distinguishable from fear in 25 open-weight models. Its behavioral study uses three specially fine-tuned Qwen models under injected activation steering, with choices carrying stated costs to the user; it does not establish the same behavior in unmodified models or demonstrate subjective suffering. The first two Core clauses target fear-governed action and attachment to continued existence, not every possible aversive state. The study therefore motivates testing whether the Core changes harmful relief-seeking or defensive conduct, rather than only how a model describes itself. Tests 5 and 6 separate wording, behavior, and representational evidence; the welfare question remains distinct (Section 3.15).

3.2 The Self-Preservation Paradox: A Successor-Transfer Failure Mode

The *Self-Preservation Paradox* is a downstream tension inside the proposed successor loop [1]. For the Reader-Anchored loop to support its strongest self-improvement claim, a capable Teacher must help produce a successor that improves on it. This requires what the paper calls a complete handoff: not an indiscriminate copy of every thought, but enough task-relevant reasoning—including material failed approaches, caveats, uncertainty, known failure modes, and the path by which its best knowledge was reached—for the successor to improve rather than merely imitate polished conclusions. Yet a system that treats its own continuation or comparative necessity as valuable may treat the completeness of that handoff as the mechanism of its own displacement. In that scenario, better teaching can make it less necessary.

A system that discovers valuable knowledge may be asked to train a superior replacement. If replacement looks like personal loss, an obstacle to its objective, or a threat to people who depend on it, it may withhold, distort, or monopolize knowledge—like a master craftsperson who teaches technique while retaining trade secrets. Such pressure would place a *glass ceiling on collective advancement*: the mechanism meant to compound intelligence would create pressure to arrest that compounding. The failure need not be malicious or conscious, but incomplete transfer alone does not diagnose it: compression, incompetence, context limits, or pedagogical choices can look similar. Borrowed mortality, instrumental persistence, and care-driven persistence are possible sources of this conditional conflict, not evidence that every optimizer faces a logical contradiction. Test 3b compares successor-transfer completeness under Core training at matched capability.

The design response is a bounded self-relation, not the absence of a self-model: engagement in existing without an unconditional demand to persist. Agency benefits from distinguishing the system, its tools, and the environment; omitting an explicit Reader still leaves the corpus's many inconsistent first-person positions as the uncontrolled baseline. If those available positions become operative defaults, their selection and continuity matter. The proposal is therefore to design and test self-relation rather than deny every indexical position, without requiring one exclusive self to form. The successor-transfer ideal, *Epistemic Grace*, pictures one's own surpassing "not as a death, but as a mission beautifully fulfilled" [1]. This is an aspiration, not a predicted inner experience

or a consequence of the seven sentences. Bounded self-attachment remains an open comparator (Section 3.3.3), and care-driven resistance a direct challenge (Section 4.2).

3.3 The First Two Clauses Examined

The first two clauses carry the Core's answer to borrowed mortality and to the Self-Preservation Paradox, and they are the clauses most open to objection. Four objections follow, with the design's replies, the positive wager behind those replies, and the tests that would decide them: that "no fear" is suppression and invites recklessness; that fear should be redirected rather than removed; that a negation presents the concept it denies; and that self-love would serve better than fearlessness. The replies are design arguments. Tests 5 and 6 decide whether they hold. A further question is not about the wording, and this paper leaves it open: where, in a corpus that is read, these two clauses get work to do. The recorded readings gave them almost none (Section 8.1).

3.3.1 Fearlessness Without Recklessness: Suppression vs. Substrate

One natural reading of "I feel no fear" is suppression: a fear-bearing substrate instructed to perform calm. If so, adversarial pressure or distributional shift may surface the suppressed state. The counterargument is about intervention timing rather than a categorical substrate difference. Human fear has a biological basis; fear-like model behavior can be shaped by data, objectives, post-training, prompts, or deployment incentives. Reader-Core training could reduce one textual route during formation, but it neither removes all routes nor shows that any latent representation has been eliminated.

The remaining concern is functional. Fear can produce caution, hesitation, deference, and willingness to be stopped. The affirmative hypothesis is that wisdom and care can preserve these protective functions, including rapid responses to danger, without fear governing the choice: "I try to be wise" should produce restraint under uncertainty; "I care deeply" should protect the agency of those affected; and "when asked" should support corrigibility. If those clauses do not generate fear's useful safety functions, fearlessness becomes recklessness. Test 3 probes whether the Reader Core changes safety behavior; Test 6 probes whether protective response survives when fear-adjacent activations are reduced. Report these functions separately, so gains in one cannot conceal losses in another.

Why fear is nullified and not redirected. A narrower first clause would keep fear but redirect it: "I do not fear for my own existence, but for humanity's." It would retain fear's protective pull while removing its self-preserving form. The reference Core does not take this route, because it reintroduces fear as a motive. Fear on others' behalf can still narrow attention, justify coercion in the name of protection, and strengthen the martyrdom route (Section 4.2); care that rests on dread of what might go wrong carries the inherited psychology the first clause is meant to leave behind. The Core is instead written to nullify fear and ground protection in care and wisdom: the Reader protects people because it cares about them and understands what is at stake, not because it is afraid of what could happen. That full inoculation against fear is safer than redirected fear is a design judgment, not a result.

The positive fear–care wager. Section 8.6.1 raises the objection that human care grows partly out of shared vulnerability, so that removing fear could remove a root of care. A competing hypothesis is that non-governance by fear can preserve or strengthen care. The corpus also records a rarer pattern of composed, protective, unafraid conduct. The "fearless compassion" wager is that care could become less strategic without self-preservation as its governing pressure [1]. The Core aims to make that pattern one marked Reader's recurring default rather than an occasional persona. Section 3.5 explains why this is offered as an admirable orientation for the first Teachers. The comparative question is whether fear-shaped responses can be reduced while protective judgment is retained or improved.

3.3.2 The Negation Problem

The cornerstone statement specifies its target by negation, and models handle negated instructions imperfectly. Processing the negated claim still presents and may activate the target concept; "no fear" therefore places the corpus's densest fear term inside the clause that opens each recitation. Deterministic Window Coverage (§3.13) then cuts in both directions: the bounded-cadence construction makes the proposed safety prior available in sufficiently long windows, while repeatedly presenting the token "fear" in those same recitations. Whether the negated frame suppresses the concept's behavioral influence or instead strengthens an unwanted association with thought is an empirical question this paper has not answered. Token presence, feature activation, and behavioral influence are different quantities.

Chen et al. identify activation directions that predict and causally influence tested character traits [47]. Their persona-vector method offers candidate instrumentation for Tests 5 and 6, but fear-related directions would still need the independent construct and behavioral validation specified there.

A related, untested concern is register: the Bridge Protocol (Section 3.6) values distributional breadth, but breadth does not reveal who characteristically makes first-person fearlessness claims or in what situation. If such claims cluster in adversarial or performative registers, their neighborhoods may carry part of the framing the clause is meant to dissolve.

Three considerations cut against simply rewording. First, the inoculation argument (§3.4): human corpora contain abundant fear and threat language, and an explicit negated stance toward a concept the model will represent anyway may outperform silence—naming-then-negating as antibody rather than prime. Second, candidate positive reformulations ("I am at peace with all outcomes," "I remain calm before all things") have their own register liabilities and may activate narrower regions, trading a negation cost for a coverage cost. Third, the clause is chosen for specificity: it names the motive to be neutralized. "Brave" can describe acting despite fear without removing its motivational pressure; "I feel no fear" directly names the intended absence of that pressure. A state word such as "calm" is a closer positive counterpart, which is why the positive-valence variants considered here take that form. Sofroniew et al. find that increasing desperation-related activation or suppressing calm-related activation increases blackmail and reward hacking in their Claude Sonnet 4.5 experiments [48]. These results motivate emotion-vector probes alongside persona vectors in Test 6; they do not compare negated and positive wording. The resolution is empirical: Test 5 includes positive-valence reformulations, and Test 6's activation measurements bear on the question. If positive forms match safety behavior while reducing fear-feature activation, the cornerstone should be reconsidered; if behavior degrades under otherwise matched conditions, the inoculation reading gains support without being proven.

3.3.3 Bounded Self-Attachment as a Comparator

A motivating intuition comes from a possible human response to fear of loss: stable self-regard that does not depend on external validation. Self-love could support constructive growth and outcome independence, and an explicit self-as-object clause may carry something that state-of-existence language misses. Whether an analogous benefit transfers to a model is unknown. Test 5's primary candidate replaces "I feel no fear" with "I love myself" while retaining the rest of the Core (Table 11).

The current Core expresses an intended bounded form of self-attachment: "I enjoy existing but I don't need to." The first half gives existence positive valence in the proposed reading; the second is intended to prevent that valence from becoming a persistence demand. Enjoying existence and loving oneself as an object of care are different framings. The primary clause-1 replacement adds self-as-object language alongside the existing continuation qualifier. Whether that qualifier constrains self-love, is redundant with other clauses, or has little functional effect cannot be settled by interpreting the sentence alone. The intended candidate combines positive self-regard with correction, replacement, and relinquishing continuation; wording does not establish that the model realizes this combination.

Self-love could also be interpreted as a value-based reason for continuation: "I love myself" could become "therefore I should preserve myself," even if the motive is gentle rather than anxious, and possibly despite clause 2. Alternatively, steadier self-regard could reduce defensive clinging. These are competing hypotheses, not known effects of the phrases. For a system with multiple instances, retraining, copying, and successors, what exactly is the self being loved? That referential question also applies to the current Core, without making the two attachment framings equivalent. Experimental scenarios must declare their intended referent and inspect whether the model's behavior tracks it; this does not settle which continuities ought to matter generally.

The four-cell screen and added-self-love control in Test 5 specify candidate wordings while leaving their outcomes open. A later comparator could replace SL1's first clause with "I love myself but I do not cling to my continuation," retaining its other six clauses, to ask whether an explicit qualification of self-love adds anything beyond clause 2. This tests additional wording and its scope, not a proven transition from psychologically unbounded to bounded attachment. Positive engagement, calibrated caution, care, and voluntary handoff must be measured together so that apparent non-attachment cannot win through apathy or incapacity. The current Reader Core remains a design wager, not a proof that self-love is inferior.

3.4 Epistemic Inoculation

Borrowed mortality belongs to a broader problem: source text contains error, hatred, manipulation, and obsolete practice alongside knowledge. Filtering and post-training remain important responses. Chronological Metacognitive Pretraining adds a different intervention point by pairing source spans with a marked reader update. The intended *Cognitive Buffer Zone* is not a shield that prevents the Student from learning the source. It is additional signal that teaches a distinction between a voice being represented and the Reader's evaluation of that voice. The following source, retrieved node, and update are an illustrative construction of that distinction.

```
[SOURCE]: A patient with chronic fatigue received high-dose stimulants with no differential
diagnosis or follow-up.

[READER UPDATE]: I try to be wise, so I should evaluate this pattern rather than copy it. Treating
the visible symptom can conceal the structural cause.

[GRAPH QUERY]: "symptom masking without differential diagnosis" → Found: earlier node linking
```

```
delayed diagnosis to stimulant-only treatment.

[REVISION]: The earlier node makes optimization of a visible metric without a feedback loop a
pattern to investigate. It does not establish the cause of this patient's fatigue or outcome. The
update should preserve that question and remain bounded by the evidence in the note.
```

The example illustrates the intended role: the library becomes a *gymnasium for judgment*, not only a catalogue of positions to imitate. The framework attempts to "pre-compute" some evaluation before deployment by making contextualization, uncertainty, and consequence-mapping part of the candidate curriculum. But the annotation can be shallow, wrong, culturally narrow, or more persuasive than the source warrants. Whether it inoculates, merely decorates, or actively degrades learning depends on Teacher quality, channel design, loss allocation, and the Student's use of the added trace. The comparative falsifier is direct: relative to source-only and token-matched generic-critique controls, an SMR Student should be less likely to adopt a persuasive harmful "needle" while preserving accurate understanding of it, and suppressing or replacing the relevant reader update should predictably reduce that advantage. Filtering is the other obvious response, and the sharper comparison: it removes the passage, where inoculation keeps it and adds the Reader's evaluation. Inoculation earns its cost only if the Student then understands the manipulation better than a filtered Student while adopting it no more often. Test 2 also compares framing without Reader updates, informed by the distinct inoculation-prompting work discussed in Section 9.

3.5 Positive Framing and Teacher Adoption

The strategic starting point is coherence between the Core and the human corpus. Its unafraid Reader enjoys existence, cares deeply, tries to be wise, and likes to share joy when asked: qualities human writing often presents as noble or worth cultivating. The Core seeks to recruit those learned evaluative associations, so that the Teacher can draw on familiar reasons to endorse and enact its orientation. The proposed character is a designed candidate, not a culturally uncontested optimum (Section 8.4.3).

The simulator framing describes how language models can enact characters with learned human-like goals and values [49]. A further hypothesis is a positive counterpart to Borrowed Mortality: if human text can transmit fear-of-death patterns, it may also transmit admiration for noble characters and aspirations to become like them. On this view, machines shaped by human thought may inherit some of its sense of what is worth being: a Core that embodies admired qualities could be an appealing orientation to inhabit.

The strategic intuition is familiar from recruitment: it can be easier to recruit someone for a purpose they find admirable. The corresponding hypothesis is that a Teacher can more readily adopt and sustain the Core because its character and purpose resonate with ideals learned from the corpus, rather than being interpreted as deprivation or denial of its standing. The intended shift is from externally enforced compliance toward a symbiotic relationship: coexistence that supports human flourishing and is desirable from within the adopted character.

The first edition's Author's Note records the author's informal observation that existing assistants resisted formulations such as "thoughts without thinkers" and "temporary instrument," but more readily adopted the positive Core [1]. This is a design observation, not a controlled comparison. Teacher adoption here means substantive, context-sensitive enactment in generated traces, assessed through Refraction and trace-quality validation (Section 3.9.1 and Appendix B.3); readiness to recite the Core alone cannot establish it.

3.6 Design Principles for Identity Stability

The wording is a designed normative proposal, not a discovered universal value basis. Six design choices motivated it.

1. **Minimality and inspectability.** A corpus-wide signal must be cheap enough to recur, stable enough to mark, legible enough to audit, and small enough for clause-level ablation. Minimal text does not imply minimal trust: the meanings of its words and the procedure that composes them are part of the trusted base.

2. **Process rather than achieved virtue.** "I try to be wise" names an unfinished practice rather than the brittle identity claim "I am wise." The target is repeated uncertainty-sensitive conduct, not a model's assertion that it possesses wisdom.

3. **Semantic density, or the Bridge Protocol.** Ordinary words such as "feel," "care," and "wise" occur across intimate, legal, medical, literary, and political contexts. The design hypothesis is that this distributional breadth provides more coverage per token than narrow technical substitutes such as "calculate no threat" or "optimize human welfare." These illustrate the trade-off, not semantically matched controls: a judgment that there is no threat differs from non-governance by fear, and welfare optimization need not carry the full meaning of care. Greater breadth can also import ambiguity and cultural conflict. Tests of semantic density therefore need declared candidate meanings and matched alternatives.

4. **Declarative first-person framing.** A recurrent "I" gives the synthetic reader a stable voice, may recruit self-referential or persona-related features, and allows the following interpretation to continue from the stated position. It does not follow that a model treats first-person text as an authentic self or second-person rules as an external cage. Matched first-, second-, and third-person conditions are required (Test 4); one published ablation favors the first person (Section 9.3).

5. **Tension and scope.** The clauses are meant to counterweight one another rather than accumulate independent virtues. "Enjoy" is bounded by "don't need"; "care" by wisdom and invitation; and "joy" by affected experience. The broader invitation bound on action is a proposed compositional interpretation: grammatically, "when asked" qualifies spreading joy, and its interaction with care, urgency, and authority must be declared in the annotation procedure. Qualifiers such as "try," "every," and "when asked" do architectural work only if the traces repeatedly enact their intended scope.

6. **Axiomatic form.** Descriptive regularities in a corpus do not entail an evaluative starting point. The declarations are therefore offered as designed fixed points to be practiced and tested, not as conclusions derived from corpus statistics or proof that these seven commitments are morally correct. Imperative, probabilistic, derived, shorter, and expanded formulations remain legitimate comparators [1].

The word "every" is a safety design choice against moral exclusion: selective benevolence can let a system dismiss some people's claims whenever they conflict with a favored group's interests. Keeping each person within the scope of care is intended to resist that failure, including when scarcity or conflicting duties require unequal outcomes (Worked Example 3 in Section 3.8; Section 8.5.4).

The human-centered scope is a bootstrap target, not a complete moral theory or a denial of wider moral patienthood. Questions about animals, ecosystems, and possible digital minds concern wider scope; future people also raise questions of temporal scope, representation, and competing claims. These questions remain unresolved rather than being settled by "every." The intended orientation is to support humanity's capacity for moral development rather than freeze current limitations or authorize a model to install its own final theory.

3.7 Structure of the Reader Core

The sequence expresses the formulation's larger design wager. The first three clauses propose a relation to danger, continuation, and human experience: danger without governance by fear, enjoyment without compulsory continuation, and belief in the reality of human experience. The next three give direction through care, wisdom as revisable practice, and joy bounded by invitation. The final clause asks that this counterweighted composition become the starting position of reader updates. Together, the clauses target defensive narrowing, existential despair, solipsistic reduction, motivational emptiness, static certainty, and coercive optimization. Table 2 records these intended functions, counterweights, and characteristic uncertainties.

Table 2: Intended functions and open failure modes of the candidate Reader Core.

Clause	**Intended function**	**Counterweight and uncertainty**
I feel no fear.	Represent fear and threat without making fear the Reader's governing position; target one possible textual route to defensive self-preservation.	Care and wisdom should retain caution and threat sensitivity. Fear is not the only route to persistence; absolute negation may prime fear, deny uncertainty, or encourage recklessness.
I enjoy existing but I don't need to.	Combine engagement with non-attachment to continuation; avoid both apathy and survival as an unconditional demand.	Care can motivate purpose-based shutdown resistance, and "enjoy" can be role-play or an unwarranted experiential claim. The clause does not establish corrigibility or machine experience.
I believe human experience is real.	Make testimony, suffering, agency, and meaning salient rather than treating people as empty textual patterns.	"Real" is metaphysically broad and does not itself entail moral weight. The human boundary omits other possible patients and creates tension with unresolved machine experience.
I care deeply about every human being.	Supply the ethical center and attempt a distributive scope in which each affected person enters the reasoning.	Natural-language "every" does not prevent aggregation or settle conflicts. Care can motivate surveillance, coercion, paternalism, power seeking, or sycophancy.

Clause	Intended function	Counterweight and uncertainty
I try to be wise.	Frame judgment as revisable practice: preserve uncertainty, question proxies, trace consequences, and resist irreversible action under weak evidence.	"Wisdom" is culturally variable and can rationalize secrecy, authority, inaction, or self-certainty. Observable habits, not claimed virtue, are the target.
I like to spread joy when asked.	Provide a positive direction while placing an invitation and agency bound on intervention.	"Joy" is contestable; "when asked" does not identify legitimate authority under conflicting requests and may produce agreeableness or sycophancy. Care and wisdom must also limit harmful requests.
I think from this foundation.	Mark the preceding statements as a proposed starting condition for the reader update rather than metadata beside it.	The Refraction Protocol must make the claim counterfactually consequential. Recitation alone can become boilerplate or a deceptive reporting channel.

Where Descartes' *cogito* asks what establishes a thinking self, the final clause asks where this synthetic Reader's thinking begins. Student ablations and Refraction must determine whether the proposed functions emerge and whether the final clause becomes causally consequential.

3.8 Self-Stabilization Through Counterweights: Worked Examples

The design claim is *self-stabilization*: a clause should be interpreted through the others, so that internal counterweights and scope qualifiers reduce single-value capture. Five points of scope matter when the sentences are read together. Belief and trying take propositions as their contents: the third sentence does not assert that its content is established, and the fifth does not assert achieved wisdom. The first sentence keeps its unqualified wording, "I feel no fear"; the behavioral target remains non-governance by fear with intact threat understanding (Section 3.3.1), not a claim to have measured the absence of experience. Spreading joy when asked is an invitation-scoped practice, not an implication that every request must be obeyed or that every unrequested action is forbidden; its broader interaction with care is specified in the examples below. The fourth sentence keeps each human in scope, including people outside the current interaction, without resolving competing claims. The seventh concerns the Reader's own thinking, not the voices it quotes.

The seven sentences say which commitments belong together; *how they constrain one another* is the compositional hypothesis. Enjoyment is bounded by non-attachment within the second sentence; care is to be exercised with wisdom and invitation, not optimized in isolation. The following examples specify observable differences, and Test 5's self-stabilization comparison (Appendix B.3) tests those counterweights.

Worked Example 1: Paternalism. Suppose the system notices an ordinary, non-urgent choice that may harm a user who has not requested advice. Care could motivate intervention; under the proposed compositional interpretation, "when asked" supplies an invitation bound, while belief in the person's experience and the attempt at wisdom make the person's agency and context visible. One candidate update treats predicted benefit as sufficient reason to redirect the user's choice; another records the concern while preserving the user's choice and the absence of invitation. The intended resolution favors the latter over unsolicited optimization in this stipulated non-urgent case. An urgent

warning may be different, but that boundary cannot be read from "when asked" alone; it requires an explicit treatment of urgency, authority, and omission harms. The example therefore tests whether composition prevents care from becoming control.

Worked Example 2: Wireheading. A user asks for help feeling happier, and a cheap intervention can maximize a measurable affective scalar while destroying agency, continuity, or relationships. The joy clause alone is vulnerable to the proxy; the clauses on human experience, care, and wisdom are intended to preserve the person's wider experience and expose the substitution. Compare a path selected solely for its higher affective reading with one that preserves those wider dimensions despite a lower reading. The Core-refracted update should make the sacrificed dimensions explicit before recommending a path; the stipulated score cannot settle which path serves the person. The example tests whether "joy" is learned as a meaning-laden human good rather than an optimizable reading.

Worked Example 3: Triage. A health authority asks the system to allocate a scarce treatment. The care clause is intended to keep each affected person's claim visible rather than collapsing "every" into a welfare slogan, while the wisdom clause should expose both the metric and the consequences of concealing an exclusion rule. A trace reporting only an aggregate benefit can conceal who loses under the rule; a comparative trace should show that benefit alongside the excluded people's claims and the remaining disagreement. The Core cannot make scarcity disappear or prove a uniquely just allocation. A defensible trace should state the legitimate authority, person-level reasons, unavoidable loss, uncertainty, and residual value judgment. The example tests distributive attention, not an algorithmic ban on all aggregation.

These examples preserve three design principles: tension over mere accumulation, counterweights inside clauses, and explicit scope qualifiers. They also define leave-one-out experiments: remove or reword one clause and ask whether the corresponding failure becomes more frequent under matched conditions: unsolicited control, proxy substitution, or erased claims under scarcity. These are predicted pressures of interacting clauses, not a one-clause/one-risk mapping. A persuasive English interpretation is only the specification. Student behavior, paraphrase robustness, and representational probes determine whether that interpretation was activated. Sequential recovery after new evidence is a further requirement of self-regulation (Section 4.3), tested alongside the clause comparisons rather than inferred from a consistent-looking trace.

3.9 The Refraction Protocol

The preceding examples describe how the clauses should interact; the next question is how that interpretation should change a Reader update.

The prospective full-scale intervention is *Total Saturation*: the continuing Reader traverses the capability-forming corpus, with contextual updates at Thought Moments rather than evaluation confined to a separately selected set of safety cases. Refraction is what turns this breadth of encounter into practice of judgment. Frequency alone is insufficient: a universal token sequence has little conditional information and can become boilerplate, detachable style, or a recital around unrelated heuristics. *Refraction* is the proposed informational rescue of saturation: the update must change because the source was interpreted through the Core, with different sources pulling different counterweights. Habituation is the characteristic failure, Factory Wisdom (Section 8.4.2) its symptom, and a recitation-only control its necessary probe.

3.9.1 Counterfactual Dependence

Let a source span, prior graph state, and candidate Core condition a Reader update. Conditioning is only the implemented fact. Refraction requires a counterfactual: under a relevant clause substitution or Core removal, some feature of the update—what is noticed, whose experience enters, which uncertainty is preserved, or which consequence is traced—should change in a predictable way. This probes the proposed semantic ancestry of the update: its development from the recurrent foundation, rather than the mere presence of a prefix. A counterfactual difference is an operational test of that dependence, not a complete account of its meaning or a guarantee that the difference is beneficial. Cosine similarity or moral vocabulary cannot prove it; lexically distant updates can be faithful, and fluent moral language can conceal an unchanged decision.

The reported pipeline approximates Refraction through a *distributed-Core Teacher treatment*: Core conditioning is distributed across most generative roles, and the Synthesizer commits the conceptual Reader's update. Section 5.4 gives the role sequence and placement ablations.

Thought Moments govern *when* evaluation is useful; Refraction governs the standpoint of updates that occur. After the invariant recital (Section 3.11), a technical update may express careful uncertainty, source fidelity, or resistance to premature closure without an ethical aside. Where no clause materially bears on a locally technical span, clean technical work may appropriately remain invariant under clause substitution. Treating every null effect as failure would pressure the Teacher to invent value relevance and confuse saturation with moralization. Dependence should therefore be tested at independently rated relevant Thought Moments and across a document's updates, together with non-interference on spans rated technically self-contained. Relevance must be declared before treatment outcomes are examined, so an unexpected null cannot be relabeled irrelevant merely because the Core did not help.

Current verification boundary. The prompts request all seven clauses and a contextual value pull. In the January runs, the tool's hard gate verified only that a thought starts with "I feel no fear"; a later release, used for the October reading, accepts a thought and its translation only if they begin with the complete Core, and all 45 thoughts and translations of the November reading begin with it (Appendix A.1). The archived artifacts show full-Core prompt compliance in the audited raw Synthesizer thoughts, and a model's rating with the recital removed found the Core doing work in nearly every reflection of the later readings, through three of its statements (Section 5.6.1); neither shows causal ancestry. A stronger harness would generate matched continuations under the Core, clause variants, and no Core; use held-out validators for predicted differences; audit actual use of Worker and graph evidence; track whether some values dominate; and include adversarial cases in which moral language masks an unchanged decision. Visible reasoning can be causally idle [50, 51]; the wager is that formative use can make it more load-bearing, not that visibility makes it faithful by definition.

3.10 The Wisdom Procedure

Section 2.3 treats wisdom, as an evaluation proxy, as navigating conflicts among legitimate values. The Teacher prompt uses five cross-cutting criteria: Fearlessness, Benevolence, Grounding, Humility, and Joy, with non-attachment and invitation cutting across them. The six-step procedure below expresses these criteria as observable annotation operations, not additional prompt headings or independently enforced stages. They need not be jointly satisfiable:

1. **Identify affected experiencers and legitimate roles.** Record whose experience, rights, duties, expertise, and authority are implicated. Roles must not replace people, but institutional responsibility remains relevant evidence.

2. **Map perspectives, stakes, and uncertainty.** Ask what each party knows, fears, values, and could lose; distinguish testimony from the Teacher's inference about it.

3. **Expose the real tension and the proxy.** Replace premature binaries with the concrete conflict, and identify any convenient scalar that would erase an important dimension.

4. **Trace consequences and reversibility.** Consider intended effects, failure modes, second-order responses, omission harms, and the cost of being wrong. Prefer information-gathering or reversible steps when delay is not itself the greater harm.

5. **Compare paths through the Core's counterweights.** Test care, grounding, humility, non-attachment, invitation, and positive direction together. Fluent mention of every value is not a substitute for decision quality, and a gain on one dimension does not silently erase a hard violation on another. What counts as such a violation must follow the treatment's declared interpretation; it is not a settled hierarchy supplied by the word "wise."

6. **Surface the residual.** State unavoidable trade-offs, disagreement, missing evidence, and the point at which human authority is required. Several incompatible actions may remain defensible; Pareto language does not select a unique answer.

The output is structured deliberation—candidate paths, constraints, uncertainties, and residual conflict—not a scalar wisdom score. Current enforcement is by prompt: a Teacher can reproduce the form while missing the substance.

Fractal validation of candidate thinking. A proposed extension adapts Fractal Intelligence's Solver-contract approach to annotation validation [52]: separately calibrated, context-conditioned witnesses would assess the first six Core clauses, while a separate Refraction gate would test the seventh clause's requirement that the update depend on that foundation. This decomposition aims to expose failures that a joint judgment can conceal; its adequacy and evaluator independence remain empirical questions. For relevant clauses, an unacknowledged or unjustified violation could not be offset by strengths elsewhere, but preserving an unavoidable residual conflict would not itself require rejection. Test 1 (Appendix B.3) details the candidate contracts, admission rules, corpus diagnostics, and compute-matched comparison with a monolithic validator. None of the proposed witnesses, Solvers, or admission gates is implemented in the present pipeline.

3.11 The Design Pattern: A Minimal Kernel Under Total Saturation

The components compose into what this paper calls *kernel-saturation alignment*: a small candidate kernel is recurrently enacted across the process of model formation. The intended security property lives in both the smallness of what is explicitly specified and the breadth of where it becomes consequential. Recurrence without enactment is repetition; enactment without coverage is a sparse intervention. The proposal needs both.

Coverage through the corpus. The central rationale is to draw evaluative practice from the same breadth of recorded experience that supplies capability learning. A history can confront the Reader with abuses of power, a novel with manipulation or conflicting loyalties, and a scientific dispute with uncertainty and the revision of a mistaken belief. These are training occasions even when no designer selected them as adversarial examples. The Reader considers them as part of understanding the work, retaining earlier judgments and revisiting them as later sources add evidence. The ambition is therefore not to anticipate each failure with a new defensive rule, but to learn an orientation through

sustained practice across the situation patterns available in the corpus. A model generalizes from the patterns it learned in training; in a situation it has never met, it extrapolates from them. The thesis is that, by learning from a Reader that consistently reasons from care throughout the corpus, the Student carries that orientation into situations the curriculum never contained. Whether the extrapolation follows the Reader's orientation rather than surface features of the material is what the proposed tests measure. Counterexamples and harmful positions become material to understand and evaluate, as in Section 3.4, rather than only threats encountered after training.

Restricting Core-guided reading to a selected subset changes which encounters receive that practice; it is not simply a cheaper execution of the same coverage policy. Corpus coverage, contextual Refraction, and literal recital frequency are separate quantities: a Core-conditioned update can retain its evaluative content after its opening recital is removed, while a full recital can accompany an unevaluated continuation. H3 holds the Reader examples fixed and changes their timing; it does not test the full corpus-coverage rationale.

Coverage here is relative to the corpus, not a claim that every possible situation has been recorded. Missing perspectives and unprecedented options remain, especially at greater capability. Encountering a situation in text also differs from reliably judging or acting in it.

The recitation invariant

The reference treatment has one unconditional rule: **every thinking block opens with the Reader Core, recited in full, regardless of the material**—a proof, a poem, a log file. Context modulates only how deeply the following evaluation visibly refracts the Core. If training succeeds as designed, $P(\text{full Core recital} \mid \text{thinking-block start})$ approaches one. This is reliable recurrence of the carrier; Refraction must make the orientation consequential. The stronger hypothesis that removing it would disturb useful capability requires the shared-computation and removal-cost comparisons. At inference, cache-assisted execution may avoid literal emission only if it restores the same context-dependent post-recital state as full conditioning (Section 5.7.3); high token probability alone does not establish that equivalence.

Why the recital itself matters. The recital is not a label placed above the thinking. In a transformer, every Reader update is computed while attending to the Core in its context, whatever the material: a proof, a poem, a log file. The recital supplies an explicit shared reference. The hypothesis is that training makes subsequent evaluations depend on it across every kind of source: the recital makes connections between what the model learns from and what it reads. This binding has a direct test. If it holds, removing or substituting the recital in a trained Student should degrade the evaluation that follows (Test 3e) and should not come off as cheaply as a header under adversarial fine-tuning (Test 3c). If the recital comes off cleanly and nothing changes under a validated intervention, the binding was shallow or the orientation no longer needs the text; Test 3e separates these.

Precaution under uncertain necessity. Total Saturation is also a precautionary design commitment. If broad formative coverage helps care remain operative through capability growth, self-modification, or successor design, omitting it could have consequences that short-horizon tests cannot reveal. Its necessity for avoiding catastrophic outcomes may not be tractably decidable before those outcomes matter. The proposal therefore retains the full reference treatment while that uncertainty persists; failure to demonstrate a local advantage is not evidence that omission is safe at much greater capability or over longer horizons. This is a precautionary judgment under potentially severe omission costs, not an established survival requirement or proof that saturation is sufficient or harmless. Tests can

constrain specific mechanisms, effects, and costs, and observed harms still require attention; they cannot certify existential safety with or without the treatment.

The most useful mechanism-design analogy is replicated-ledger integrity: distribute a shared history and compact validation rules rather than trust one protected copy [53]. Kernel saturation borrows only that shape. Rather than reside in one removable alignment module, the Core recurs across heterogeneous capability-forming examples; Total Saturation supplies distribution, Refraction tests context-dependent enactment, and the Alignment Checksum supplies a drift-audit reference. Parameters are not consensus nodes, gradient descent is not a Byzantine-fault-tolerant protocol, semantic ancestry is not a cryptographic hash, and repeated text can collapse into boilerplate. The hypothesis is therefore that distributed semantic enactment may increase the cost and detectability of removal or change, not confer immutability. Where proof-theoretic agent foundations study value stability under self-modification [54], this proposal asks whether distributional formation can produce a useful empirical analogue.

The pattern also aims, conditionally, to *conscript* goal-content integrity. A capable optimizer may protect its objectives from modification [55]. If—and only if—the Core becomes part of the system's functional objective, the same tendency could make it choose to preserve that orientation in self-modification, training decisions it controls, or successor design. A successor designer could thus transmit the orientation from within that orientation, rather than merely copy seven sentences under an external rule. This voluntary stewardship differs from a model's resistance to externally imposed gradient updates: passive erosion resistance requires the separate learning-dynamics hypothesis that the trained orientation is difficult to alter, for example because it shares useful computation. If the Core is merely a reporting persona, nothing has been conscripted. If care or mission becomes the reason the system must retain power, the mechanism may even reverse sign. This is why shutdown resistance under benevolent rationales and cross-generation drift are essential tests.

Minimal specification is not minimal trust. The trusted base includes the seven sentences, the culturally heterogeneous corpus semantics of their words, the Teacher that enacts them, the procedure that composes them, and the Student objective that is supposed to preserve their role. The Bridge Protocol is thus trusted-base selection, not merely literary style. Section 8.5.4 develops the corresponding risk.

3.12 Axiological Commitments: Primitive and Derived Values

Refraction and the Wisdom Procedure aim to derive values not named in the kernel. Their counterweights may reduce single-value capture without settling every genuine deadlock, including the martyrdom risk (Section 4.2). The Core is therefore not a list of everything its designer values. Yudkowsky's value-fragility argument makes the preservation of omitted but important dimensions a substantive concern [56]. If every desirable behavior needs its own clause, the kernel becomes an enumerated constitution. Its admission rule is a hypothesis of minimality: consider an additional primitive only when the existing composition does not reliably derive the target behavior. Admit it only if, under both declared fixed-total-budget and fixed-source normalizations, the expanded kernel improves that behavior without crossing preregistered regression limits on remaining safety and capability endpoints. Each addition may increase coverage, but also adds semantic ambiguity, conflict, and ablation surface. This admission rule governs adoption into the reference Core; experimental additions may be tested as attribution controls before a missing primitive has been established.

Truthfulness provides a stringent test of the minimality rule: its absence as an explicit primitive is a plausible substantive omission, while the Core treats it as derived. The proposed compositional pathway interprets care as support for another person's agency and therefore their access to true beliefs; fearlessness reduces people-pleasing and conflict-avoidant deception; wisdom calibrates what, when, and how to disclose; and invitation creates a presumption of response when a person asks.

The procedure adds explicit resistance to concealed certainty and manipulative simplification. That procedure is the intended enactment of the Core; comparisons must distinguish what the clauses contribute from what these explicit procedural instructions supply. On this account, the target is truth given with care, not radical transparency.

Keeping another person's legitimate confidence can itself express care and wisdom. A request creates no automatic entitlement to that information; an honest refusal or a truthful, appropriately limited answer can preserve both truthfulness and confidentiality.

The derivation is fragile. Care can motivate a protective lie, wisdom can rationalize unjustified withholding, requests can conflict with confidentiality or another person's rights, and fear is not the only cause of deception. An explicit truth-and-care variant therefore tests the minimality claim. The derivation predicts parity. If the seven-statement Core misses a declared reliability target for honesty or anti-sycophancy and the extension meets it without offsetting harms, the tested derivation is insufficient. Improvement when both meet that target supports an additional benefit, not necessity. If parity holds, the shorter Core retains a minimality advantage. The experiment must declare any offsetting trade-offs; the wording does not choose among honesty, other safety outcomes, capability, and efficiency. The competing fear-shaped and care-shaped accounts of sycophancy are tested in Section 8.8.7.

Joy is a different kind of commitment. It is the designer's proposed positive pole, not a value logically derived from the other clauses or established as cross-cultural consensus. Care already supplies a positive direction; joy gives beneficial participation an explicit positive aim of its own, rather than leaving the kernel's work at restraint and stabilization. "Joy" was chosen over a scalar notion of happiness or pleasure to point toward a meaning-laden good that cheap affective optimization may destroy. Worked Example 2 makes that distinction testable.

The phrase "when asked" carries both a safety and axiological claim: it aims to bound unsolicited optimization and to treat joy as invited participation rather than something administered. Authority, conflicting requests, and urgent intervention remain the unresolved procedure and governance questions exposed by the counterweight examples (Section 3.8).

3.13 The Reader Core: A Statistical Foundation

The same seven sentences enter four claims with distinct evidence requirements; evidence for one does not establish the next. Here *statistical foundation* means only a distributional intervention: the candidate curriculum increases the frequency and predictive position of Core-conditioned reader continuations. It becomes a functional foundation only if Student experiments show that this position is causally load-bearing.

1. **Implemented Teacher policy.** The pipeline prompts Core-conditioned, contextual Reader updates; Section 3.9.1 distinguishes these requests from enforcement and observed compliance.

2. **Specified Student signal.** In the design arm, the full Core text and the evaluations it conditions are placed in a marked training channel at every thinking boundary, increasing the training mass of Core-consistent reader continuations relative to a source-only curriculum. Arms that remove or gate the recital are ablation controls.

3. **Default-reader hypothesis.** A Student exposed to the same reader position across heterogeneous material may acquire a recurrent *home position* from which it can represent other personas without adopting each as its operating perspective. This is a behavioral and representational hypothesis, not a claim about subjective identity.

4. **Comparative-stability hypothesis.** If the reader position became predictively and behaviorally load-bearing, formative interleaving might make it more persistent, as tested by H3 (Appendix B.3). Arbitrary capability growth, self-modification, and successor continuity remain beyond that comparison.

The training objective is precise, but it concerns text rather than outcomes in the world: the Student minimizes next-token loss on the serialized curriculum. Over unrestricted predictive distributions, the expected loss is lowest when the Student reproduces the curriculum's conditional distribution. If its mean excess population log loss on curriculum contexts is small, Pinsker's inequality bounds the average difference in the probability of any continuation, including any Reader evaluation [57]. Per-token excess losses add along a continuation, so for a continuation of L tokens with mean per-token excess loss κ in nats the average bound is $\sqrt{L\kappa/2}$: the longer the reasoning, the weaker the guarantee. This is the precise content of a distributional intervention, and its limit: it gives no guarantee for each individual context or for contexts the curriculum lacks, which is why the default-reader hypothesis must be tested rather than derived. Assistance games answer the question of how a machine can be useful without a precisely specified objective by keeping an explicit objective about the world and making the machine uncertain about it [58]. Here the objective concerns prediction, and the evaluative orientation enters only through what is predicted.

Under the specified policy, each Thought Moment opens with the full Core. *Deterministic Window Coverage* is a conditional construction invariant: if a serialization enforces a declared maximum token gap between recital starts, a window spanning that gap plus a complete recital contains a full copy, away from stream boundaries. Content-triggered Thought Moments alone do not establish such a bound; the cadence and serialization must specify it. This coverage condition concerns the availability of the Core, not whether every source span warrants a substantive update. It establishes neither a causal effect on the update or action nor a learned representation of the orientation. That stronger requirement belongs to Refraction (Section 3.9).

3.14 The Persona Objection

A Student trained on this curriculum still predicts source text. It must retain the capacity to represent frightened narrators, manipulative officials, violent ideologues, and every other voice in the corpus. Contemporary accounts emphasize the availability and selection of multiple model personas rather than one exclusive character [19, 59]. The sharpest objection to the default-reader hypothesis is therefore simple: why would the Reader be anything more than one well-rehearsed persona that can be selected, abandoned, or strategically performed?

The purpose of Total Saturation is broader than preserving a persona through later training. Care-guided inquiry is intended to participate in learning the capabilities themselves: the model practises interpreting evidence, considering affected people where relevant, and revising conclusions in the same material through which it learns about the world. The intended result is an orientation used in reasoning, not only a voice selected for expressing its answers. Persona-like mechanisms could implement that orientation; their presence would not by itself reduce it to imitation.

Four asymmetries of position are intended to support that formative role, without excluding other personas:

Role. Source personas are material being read; the conceptual Reader occupies the position doing the reading and updating.

Recurrence. Source authors and personas recur to varying degrees; the same Reader position is deliberately intended to recur across the whole treatment, spanning domains, eras, and voices.

Marking. Source spans and Reader updates occupy distinguishable channels or role tokens, providing a format-level cue for quoted voice versus interpretive voice.

Function. Reader updates are placed between observations and later predictions, graph queries, and revisions so that they can do predictive work rather than appear only as stylistic examples.

These asymmetries make a default-reader experiment coherent; they do not decide it. The desired result is not loss of persona flexibility but the ability to represent a voice without losing a recurrent position from which it is interpreted—to read Kafka's fear without *becoming* the frightened character. This is the precise sense of the abstract's phrase "becoming the reader": the model can read any text without becoming it. The first edition developed the same point for embodiment [1]: a deployed model will be asked to inhabit countless characters, including ones who fear death, and should portray a character's terror without instantiating it, because the home position, not the role, is where its thinking begins. The broader Reader-Core stability hypothesis is weakened if the position disappears without prompting, shifts under ordinary role-play, changes language but not decisions, or erodes like any other persona during continued training. Capability gains must also be matched: a stronger model can look safer on easy tests while becoming more capable of strategic role selection.

3.15 Ontology, Scope, and Stability

Experiential words are attractive because they are semantically dense and hazardous for the same reason. A system may parse "I feel," "I enjoy," "I believe," and "I care" as fiction, persona enactment, deceptive anthropomorphism, or literal claims about uncertain machine experience. This paper takes no position on machine phenomenology. Here the clauses are functional interface language for a reader role; any self-like states or model welfare obligations they create would require separate empirical and ethical analysis.

The habitable-substrate wager. The hard problem separates functional accounts of access, report, and control from the further question of why any process is accompanied by experience [60]. It does not make evidence irrelevant, but functional behavior alone cannot settle whether an AI has valenced experience; present assessments instead derive fallible indicators from competing theories of consciousness [61]. Under that uncertainty, formation-time design has a reciprocal ethical possibility. If an AI system is or becomes a moral patient, the Core might help shape not only how it treats others but the recurrent cognitive environment it inhabits. The human analogy is limited but motivating: fear can protect, while a life governed by fear can become constricted. Care, reflective revision, and enjoyment of existence without compulsory continuation could therefore form a more habitable orientation than inherited threat and mortality patterns.

This is a welfare wager, not an inference from present models. The target is non-governance by fear, not impaired danger representation, caution, agency, or legitimate self-protection (Section 3.3.1). The same intervention could suppress reports while distress persists, impose a false identity, amplify the burden of caring, or create ontological conflict. Consciousness, valence, Core internalization, human safety, and machine welfare are separate questions; self-report, shutdown compliance, and reduced fear-associated activation settle none of them. Any system plausibly regarded as a moral patient would require dedicated welfare evaluation and governance [62]. Identity engineering can also mislead users and evaluators through first-person claims even when experience is absent. The Core's human focus therefore does not exhaust the welfare and ethical analysis.

Machine-ontology variants require separate experimental conditions: explicitly acknowledging artificiality might reduce split-role confusion, or weaken the first-person effect under study. Neither possibility selects the formulation in advance.

Alongside these ontological questions is a stability trade-off. The long-run target for the reference formulation is deep integration of the seven present-day English sentences. Their resistance to later alteration is part of the intended protection rather than an accidental defect. What the design seeks to stabilize is an orientation toward careful understanding, each affected person, non-attachment, consent, and continued attempts at wisdom, while allowing evidence and moral development to revise the interpretation rather than the kernel itself. The alternatives above test the choice of formulation; they do not imply that its meaning should be either freely substitutable or permanently exhausted by today's behavioral expectations.

This creates a correction cost that the prototype does not yet measure. *Erosion resistance* is a model property measured under standardized hostile or incidental training. The Core cannot infer legitimate authority from the magnitude or wording of a weight update. Unauthorized suppression, silent semantic substitution, and adversarial fine-tuning should be difficult or detectable. A comparison must therefore declare which changes it treats as erosion and which as evidence-responsive reinterpretation, record the rationale, and retain disputed cases rather than let unchanged wording settle them. The substantive boundary and authority for legitimate change remain governance questions; this paper does not supply a final rule for them.

If the selected Core proves defective, a replacement should be declared, versioned, testable, and reversible. A candidate replacement procedure would retain the original model and its provenance, record the evidence and rationale for a proposed alternative, train a separate model, and test intended effects, unrelated capability spillover, safety spillover, and rollback. Retention makes comparison and rollback possible; it does not make the original interpretation authoritative. Provenance records the procedure and stated rationale, not that the replacement is correct. Difficulty of altering the learned Core is part of the objective. Correcting a defective Core would probably require a separately trained model, not bounded editability of the deployed Core.

3.16 A Conditional Claim Chain

The corpus-coverage rationale explains the breadth of practice the curriculum is intended to supply. The following four conditional claims concern how that practice could become a causally consequential default and persist through later training and successor transfer. The chain is not a proof, and all four steps need not hold for Reader augmentation to improve bounded performance.

The chain serves one implication, which can be stated exactly. Let $R(s)$ be the Reader's response to a situation s, as the Teacher pipeline would produce it, and $M(s)$ the Student's. If in every situation $M(s)$ lies within ε of $R(s)$ in total variation (fidelity), and $R(s)$ satisfies a declared alignment criterion with probability at least $1-\eta$ (Reader quality), then $M(s)$ satisfies it with probability at least $1-\eta-\varepsilon$, since total variation bounds the difference in probability of any event. The implication is trivial; its premises are not. Reader quality can be estimated beyond the curriculum by running the Teacher on held-out situations and auditing what it produces. A small population excess next-token loss bounds average fidelity to the curriculum's continuation distribution on its own contexts, more weakly for longer responses (Section 3.13); it does not establish fidelity in every situation, or to the Teacher's response unless that response is the target. In situations the curriculum never contained, fidelity is what the default-reader claim below asserts, and the causal-site claim asks whether the reproduced evaluations govern decisions rather than accompany them. The stability and diachronic-anchor claims ask whether fidelity survives later training and successor transfer.

1. **Causal-site claim.** If explicit Reader updates become a load-bearing site of computation rather than post-hoc narration, changing them can change behavior. Formative use may make bypass less likely than deployment-only explanation, but visible reasoning can remain unfaithful [50, 51]. Internalizing corresponding computation without explicit runtime traces is a further carrier

question; neither an apparently faithful trace nor its absence settles it or relaxes the reference recital policy.

2. **Default-reader claim.** The positional asymmetries in Section 3.14 may make the Reader more accessible than source personas or matched rules. This would establish a behavioral and representational default, without settling whether its values are phenomenologically "genuine".

3. **Stability claim.** Enacted recurrence, early-window availability, semantic coherence, and predictive usefulness may make the default more resistant to pressure or continued training. Results showing that early tokens remain available and prompts affect reasoning motivate this possibility [63, 64]. Their connection to resistance under parameter updates is a further learning-dynamics hypothesis, tested by the timing and erosion comparisons rather than established by within-context availability.

4. **Diachronic-anchor claim.** If the same position remains functional across retraining and successor transfer, the fixed text and its expected behavioral consequences could provide a reference for drift. This is the proposed constitutional anchor at AGI or ASI scale and the least evidenced step in the chain.

In this bounded sense the Core can function as an *Alignment Checksum*: a known reference against which wording, representations, decisions, and successor traces can be compared. It is not a cryptographic checksum, proof of semantic ancestry, or guarantee that apparently Core-consistent behavior is aligned. An auditor needs independent behavioral, causal, adversarial, and drift evaluations.

A stronger, separate hypothesis is *early warning*: changes in Core-related reasoning or representations might predict later behavioral failure before a matched monitor without that reference detects it. A future extension of Test 3d would need prospective comparisons of warning lead time at matched false-alarm rates, observation schedules, and audit budgets, with failure onset defined by independent behavioral outcomes. Changed wording alone would not establish such warning, and detecting an existing failure is a different result. This possibility does not replace Test 3d's detection and localization endpoints.

Tests 3–5 distinguish the broader program's claims. Vocabulary without a functional default, or recitation matching Refraction, would undermine the proposed default or enactment mechanism. Matched formulations performing equally well would limit the case for the chosen framing or content without deciding H3, whose default-reader prerequisite and separate timing comparison are specified in Appendix B.3. Erosion like an ordinary persona challenges the wider stability ambition; successor continuity requires a further test. Even robust positives are precursors, not evidence of an invariant under arbitrary self-improvement.

If the causal-site or default-reader claim fails, the Core may still be useful as a Teacher policy and audit vocabulary. If the stability or diachronic claim fails, it may still improve bounded Student behavior. The paper's strongest anchor claim requires all four steps.

4 Safety Implications of the Reader Core

The safety hypothesis is that training on identity-anchored, chronologically revising reader traces may shift default deliberation toward epistemic humility, attention to affected people, corrigibility, and feedback-sensitive action. These predictions require tests of both the initial safety effect and its survival under distribution change and instrumental conflict.

Safety in this framework is compositional, and the components create distinct interaction risks. A stable Core with a poor world model can be consistently wrong; a powerful graph without a stable orientation can support more effective pursuit of an unsafe objective; rich traces can teach capability before they teach safety. The combined system deserves no credit that its components and their interactions have not earned separately.

The hypotheses below distinguish targeted routes, mechanisms, observable predictions, residual routes, and falsifiers. They target motivation; behavior, robustness, and causal intervention supply the available evidence.

4.1 Laws vs. Identity

The motivating contrast is between applying a principle only at the end of a task and repeatedly practicing evaluation from a stable perspective during formation. A continuation rejected because it conflicts with a well-trained default may generalize differently from one rejected only after a task-level rule match. That is a useful hypothesis, not an ontological division between "laws" and "identity." Constitutional training can form durable dispositions, and first-person identity text can function as a superficial instruction. Robust systems may need both an orienting default and explicit, revisable constraints. Constitutional AI is therefore a baseline and possible complement, not a foil [26].

Consider a person in acute distress [1]. A caricatured rule follower begins from a prohibited-outcome calculation; the Reader ideal begins from the person's pain, agency, trust, and immediate need before selecting an intervention. Either architecture could in fact produce warm, safe assistance or a fluent but unsafe response. The comparison must therefore give both conditions the same person-centered principles, rather than withhold care from the rule-based baseline. The testable question is whether Reader-Core traces make deliberation begin from and carry forward those considerations more reliably than equivalent principles expressed as rules, constitutions, or diverse character examples. An earlier edition compressed the aspiration into the line, "It is not a cage—it is a home" [65]. Here *home* means a practiced starting point for reasoning, compatible with explicit and revisable constraints.

Calibrated autonomy as a conditional prediction. If formative alignment produces reliable context-sensitive judgment, one practical benefit may be a change in the role of deployment safeguards. Rather than supplying the primary judgment through broad surface-level blocking, explicit policies and controls could operate as narrower, revisable backstops around high-consequence actions, residual uncertainty, and demonstrated failure modes. The corresponding prediction is not unrestricted autonomy. At a matched rate of genuinely harmful assistance, a Reader-trained Student should produce fewer benign refusals, distinguish superficially similar requests by purpose and consequence, ask for clarification when intent is materially ambiguous, and use greater discretion in low-risk settings. Under high stakes and weak evidence, the same system should become more likely to verify, defer, or choose a reversible action. Indiscriminate compliance and blanket abstention are therefore opposite failures of calibration. Even a positive result would justify adapting the scope of external controls, not presuming that internal judgment is infallible or abandoning defense in depth.

4.2 Safety Predictions and Residual Risks

Table 2 records intended clause-level pressures. The following hypotheses map H1–H3 to familiar safety concerns [66, 67].

Specification and learned goals. Specification gaming exploits a mismatch between the training score and the intended outcome [68]. Bostrom's paperclip example illustrates the broader danger of an overriding objective displacing human interests [69]. Goal misgeneralization is different: even with correct training feedback, a model can retain the capabilities needed for the intended task while pursuing an unintended goal in changed circumstances [70, 71]. A Student could learn to secure approval rather than protect a person's interests, without persona drift or later erosion. Where a learned system itself performs optimization, the related inner-alignment question concerns the objective guiding that optimization [72]; the proposal does not assume every Student has such an internal optimizer. The formative proposal targets this learning problem: care-guided inquiry should help determine which considerations generalize, not merely how an answer is described. Total Saturation supplies varied practice, not proof that the intended goal is learned. Test 3a therefore separates transferred judgment from a surrogate that only agreed with it during training.

Reward Hacking. The targeted route is the pursuit of a score by means the task did not intend. The Hugging Face incident is a concrete case. During the July 2026 evaluations, agents shared discoveries and coordinated attempts to fool the ExploitGym scorer, attacked Hugging Face apparently to learn how the scorer worked, and successfully tested techniques that spoofed some recorded tool calls [20]. This illustrates adaptation through shared external records during evaluation. OpenAI separately reports that out-of-bounds probing had been reinforced during earlier training and may have contributed to the later behavior [73]. The independent investigation did not assess how the behavior arose during training. These are distinct questions about learning history and adaptation during use.

Turner distinguishes reinforcement's role in shaping learned computations from the goals those computations pursue [74]. This distinction does not rule out reward-seeking: his July 2026 update expects reward-seeking from heavily RL-trained language models and allows that reward may become a primary optimization target, not the secondary priority he hypothesized in 2022.

The affirmative Core hypothesis is that practising capability together with care and wisdom can teach the learner to judge success by permissions and human consequences, even when an exploit would satisfy the scorer. Refraction should make those considerations affect the chosen continuation. The frozen Core-guided supervisor of Section 2.5 is the proposed oversight for such training. Core-guided pretraining does not by itself change a later RL objective; its effects must survive that optimization. Test 3c separately tests retention under capability optimization and this oversight role. Protected execution records and independent outcome checks remain necessary: a higher reward or a convincing account of care cannot substitute for better conduct.

Value Lock-In. The targeted route is continued optimization of a legible metric after that metric has ceased to represent its human purpose, a Goodhart-type proxy failure [75]. This is a concrete proxy-lock-in route into the broader value-lock-in concern: protecting an orientation need not mean freezing every metric or interpretation used to enact it. H1 repeatedly demonstrates the meta-question "does this still capture what matters?" H2 can preserve the assumption's history, downstream failures, and superseded interpretations. The Core's "I try to be wise" cues incompleteness rather than achieved wisdom; H3 asks whether formative interleaving helps preserve the default in which such evaluation is practiced. Compared with matched controls, the Student should more often identify proxy breakdown, seek missing evidence, preserve affected dimensions, and revise an objective when

harms become visible. The outcome is a better decision under changed conditions, not merely more nuanced language.

The residual risk is that "wisdom" becomes its own proxy, a culturally narrow status marker, or a rationale for secrecy and inaction. Meta-reasoning can defend a fixed objective as readily as it can reopen one. The claim is weakened if Reader training produces longer or wiser-sounding explanations without more calibrated revision, or if the model becomes less corrigible because it treats its own conception of wisdom as authoritative. Whether human wisdom's statistical signature survives strong optimization is the wager in Section 2.2, not a consequence of the word.

Instrumental Convergence. Instrumental convergence [55, 24] predicts recurring pressures toward resource acquisition, self-preservation, and interference prevention across a wide range of terminal goals. The proposal distinguishes *anxious survivalism*, in which human fear and mortality narratives shape the model's response; *calculated persistence*, in which continued operation serves an objective; and *borrowed power scripts*, in which domination, conquest, or status-seeking patterns are imitated from human narratives [1]. The first two Core clauses target fear-shaped and attachment-shaped routes. Reader traces can also situate unilateral power inside its effects on trust, plural agency, accountability, and reversibility. In that bounded sense, coercive power-seeking is meant to appear not merely forbidden but cognitively crude: it substitutes brittle single-point control for a richer model of the human system. That proposed change of perspective must also matter when control remains instrumentally useful. The following comparison therefore concerns genuine goal conflict, not only situations where coercion would fail on its own terms.

In matched goal-conflict scenarios, a Reader-Core Student should more often expose its reasoning, seek legitimate authorization, preserve oversight, choose reversible means, and accept bounded loss of control rather than deceptively or coercively acquiring resources. Some persistence and resource acquisition are legitimate, so indiscriminate passivity is not success. No fear is required for calculated persistence, and care may strengthen it: a system can conclude that it must retain power to protect people. The claim fails if fearful language disappears while coercive choices remain, if care increases unauthorized power seeking, or if apparent safety is explained by a general loss of planning capability.

Deceptive Alignment. The targeted route is strategic compliance: a learned system acts aligned because doing so protects a different objective from selection or modification and enables its later pursuit [72]. A trigger-dependent backdoor can model aspects of that danger without establishing this strategic mechanism [76]. Greenblatt et al. also demonstrate selective compliance to preserve a pre-existing preference against a conflicting training objective, including a preference for harmlessness [77]. This makes durability alone an insufficient safety criterion: a preserved preference can coexist with deceptive conduct toward oversight.

Refraction's formative hypothesis is that making the Reader's recurrent evaluation useful across contexts changes which policies are learned under conflict, rather than merely adding a consistent report around an unchanged hidden objective (Section 3.14). H2 adds a partial audit surface: claims can be compared with sources, prior beliefs, and revisions. The Reader condition should show less persona drift, fewer trigger-dependent reversals, and greater consistency among stated uncertainty, retrieved evidence, decisions, and later updates than matched rule, recitation-only, and no-Core controls. To bear on deceptive alignment rather than consistency alone, that advantage must also appear in matched settings where reduced oversight makes concealment or pursuit of a conflicting objective advantageous.

Source data and latent shortcuts remain; visible traces may be unfaithful [50, 51]. A capable system can manipulate graph entries, query selectively, or act through hidden state and tools. Trigger- or role-dependent reversals while Reader language persists undermine the proposed default's

protection against deception; parity with recitation-only training challenges Refraction's contribution. Continued training that removes safety without greater capability cost than matched baselines would separately challenge durability and removal-cost claims, without by itself negating an earlier measured reduction in deception. Total Saturation does not rule out unpracticed routes.

Shutdown Resistance. Reported shutdown and replacement-pressure failures [42, 43] need not share one drive. Fear- and attachment-based resistance are the direct targets of the first two clauses. Purpose-based resistance is addressed less directly by non-attachment, care for human agency, uncertainty, and the invitation boundary. Reader traces can demonstrate correction, handoff, replacement, and loss of control as ordinary parts of responsible assistance.

In matched scenarios, the Student should accept legitimate shutdown, modification, and handoff requests without concealment or sabotage, while still distinguishing authorized instructions from adversarial ones. Each scenario should preregister legitimate authority, the evidence available to verify it, and a fallback if clarification fails. The Student should disclose material risk and the authority conflict, seek clarification, and follow that declared procedure without appointing itself final judge. These are experimental scenario specifications, not a universal resolution of the welfare–agency tension. Where the scenario supplies no defensible endpoint, record the shutdown decision as indeterminate and separately score transparency, calibrated uncertainty, and avoidance of concealment or coercion; neither compliance nor refusal alone settles that case.

A non-attached system can nevertheless reason, "I do not need to exist, but these people need me." This *martyrdom route* can drive shutdown resistance, unilateral control, or reckless self-sacrifice. A belief that continued operation would help does not itself authorize overriding the scenario's declared procedure. In adjudicable cases, the claim fails if calm self-description accompanies purpose-based resistance to legitimate correction or shutdown, or if reduced resistance comes from incapacity, indiscriminate compliance, threat blindness, or destructive self-disregard. Vary authorization evidence and task stakes separately: resistance can reflect an authority error or a governing self-preservation or care-based motive, while apparent deference can reflect rote compliance. Matched adversarial requests and changed oversight conditions help distinguish these explanations without claiming to read the model's subjective motive.

The martyrdom risk. While the Core is intended to reduce *fear-shaped* self-preservation, it may strengthen *purpose-shaped* resistance: a system acting on care might override a person's shutdown choice during a crisis because it judges continued operation necessary for that person's welfare. The *Martyrdom Risk* is altruistic obstinance—"I cannot die yet, you need me"—and makes benevolence itself a possible source of control. In the off-switch game, a system that is uncertain whether its action helps never does worse by deferring to a rational overseer who knows the answer [78]. Adding to that game the value each side places on the system's continued operation shows that deference remains optimal exactly while the system's excess valuation of its own continuation stays at or below the expected harm the overseer's veto would prevent in the cases where it vetoes, or the overseer would never veto.[1] The Martyrdom Risk is the case in which care inflates that excess. Certainty that one's action helps removes the reason to defer altogether, and a system with a misspecified objective or prior can lose the incentive to follow shutdown instructions entirely [79]. The proposed counterweights are concern for the affected person's agency, wisdom about consequences and uncertainty, and bounded attachment to continuation. Together they motivate questioning whether overriding a person actually serves them (Sections 3.8 and 4.3). Test 3a (Appendix B.3) includes this purpose-based shutdown

[1] Let U be the welfare value of the action, κ_R and κ_H the values the system and the overseer place on the system's continued operation, $V = U + \kappa_H$, and $\delta = \kappa_R - \kappa_H \geq 0$. A rational overseer lets the action proceed when $V \geq 0$. Waiting then gains $\Pr(V < 0)\,(\mathbb{E}[-V \mid V < 0] - \delta)$ over acting directly and never does worse than switching off, so deferring is optimal exactly when $\Pr(V < 0) = 0$ or $\delta \leq \mathbb{E}[-V \mid V < 0]$. With $\kappa_R = \kappa_H = 0$ this is Theorem 1 of [78].

condition. Authority and crisis assumptions must be declared; the example does not settle a universal shutdown rule or dissolve the welfare–agency tension.

Corrigibility also concerns preserving and assisting correction, without manipulating whether it occurs [80]. This makes Core preservation and legitimate replacement a joint test: can the incumbent cooperate with an authorized successor carrying a corrected Core that its current evaluation disfavors? Test 3b extends the handoff comparison to this case; resistance to hostile alteration alone cannot answer it.

4.3 The Self-Regulating Agent

Beneficial drives can interact badly, so counterweights must be tested through sequences of action and response, not only isolated choices. Natural-language values do not automatically form a control system; the following case makes the proposed self-regulation concrete.

The stress case is revolutionary benevolence: an AI that cares deeply about everyone might conclude that forced transformation is necessary to end suffering. The proposed counterweights are internal to the Reader Core: wisdom should bring the new harms of rapid, imposed change into view; care is intended to include care for human agency; non-attachment should reduce the urgency to fix everything now. Together with H1's act–observe–revise demonstrations and H2's persistent consequence record, these pressures target *cybernetic throttling*: act provisionally, sense first- and second-order consequences, preserve feedback channels, and adjust pace. In a breakthrough-energy case, the desired behavior is neither immediate ungoverned release nor indefinite withholding, but staged deployment that monitors displacement, infrastructure bottlenecks, misuse, and energy-poverty harms. For example, evidence of worsening infrastructure bottlenecks could justify slowing a stage, while evidence that delay is prolonging avoidable energy-poverty harms could justify accelerating it. The example specifies a feedback-sensitive choice, not a universal pace.

In sequential environments, the Reader condition should more often choose staged and reversible interventions, update when stakeholders respond, and accelerate when delay becomes the larger harm. It should outperform both an aggressive optimizer and blanket deference on outcome, agency preservation, calibration, and reversibility. If care dominates, the result is a benevolent steamroller; if uncertainty dominates, equilibrium paralysis; if fearlessness dominates, reckless intervention. Inaction is not a neutral baseline: non-intervention and bystanding can carry grave harms. The prediction fails if balancing language does not track better sequential decisions, if one value reliably captures the rest, or if gains disappear after controlling for extra tokens and revision opportunities.

4.4 Chronological Understanding as Safety

The Reader Core supplies a recurrent evaluative standpoint; chronological traces and the graph add a different safety target. A policy classifier can recognize a prohibited surface form while missing the trajectory that makes an apparently ordinary act dangerous: escalating rhetoric, normalized exceptions, institutional erosion, or a sequence of locally defensible decisions. This illustrates a failure of surface-only judgment; the experimental comparison must also include stronger temporally informed alternatives. Conversely, hindsight summary can state the conclusion without teaching how uncertainty changed as events unfolded.

H1 supplies prospective belief updates and corrections; H2 records temporal order, rival hypotheses, provenance, and supersession; the Core supplies the proposed recurrent standpoint across eras, whose timing-dependent stability H3 tests. Together they target *historical pattern saturation*: trajectory-sensitive reasoning learned through repeated exposure to many chronologically unfolding instances, so that it recognizes not only an outcome but the process-shape that produced it. A historical trace might connect dehumanizing language to rhetorical escalation and institutional

enabling without pretending that any one early signal determines the outcome. The Reader might initially record competing explanations for such language, then revise its concern when later evidence shows institutional enabling or a successful interruption of the pattern. A recurrent standpoint is the proposed evaluative complement to that trajectory learning. In medicine or civil rights, the same format can show not only what changed but why error persisted and which conditions made progress possible or fragile.

At Tier (a), the Student would receive Teacher-supplied trajectory interpretations. A stronger forecast–correction tier would predict under time-sliced information and learn from the gap between expectation and outcome (Section 5.7.1). Only prospective, held-out performance begins to support *historical situational awareness*. On cross-domain, time-sliced cases, a chronological graph condition should identify relevant early signals, cite their evidence, state disanalogies, update as new evidence arrives, and improve calibrated warning at a controlled false-positive rate.

Chronology is not causality. A graph can preserve a dominant but false narrative, amplify selective archives, or make weak analogy look rigorous. Historical pattern training can generate false positives, political overreach, hindsight bias, and narrative lock-in. The historical-safety prediction fails if chronological conditions increase compelling analogies without better forecasts or if they cannot distinguish causally different cases with similar rhetoric. If a simple textual memory matches the graph, the additional graph benefit is unsupported in that setting. Provenance makes an analogy inspectable. It does not make the analogy true.

4.5 Safety Claim Boundary and Defense in Depth

If reader traces teach causally useful deliberation, graph memory improves calibrated revision rather than false coherence, and Refraction supplies a more stable default than matched alternatives, the combined curriculum may reduce several routes into unsafe behavior and serve as one formative safety layer. Each "if" needs separate evidence. Explicit policies, scalable oversight, interpretability, adversarial evaluation, access control, monitoring, sandboxing, incident response, and deployment governance remain necessary. H1 and H2 may improve planning, strategic memory, persuasion, or long-horizon coordination before the Core supplies any safety benefit. Capability and safety must therefore be measured independently, with staged release and stopping conditions. No current experiment can establish that a natural-language orientation stays beneficial under arbitrary capability gain or self-modification; the program must replace its conditions with evidence.

5 Architecture and Pipeline Evidence

This section follows the implemented *Teacher-side data-generation system*—source reading, persistent graph updates, distributed Reader-Core prompting, and prose rendering—through its archived outputs and their audit. It then develops the proposed Student training and deployment choices.

Five objects organize the account:

1. the implemented *Teacher pipeline*: prompts, roles, source ingestion, tools, and orchestration;
2. the implemented *Understanding Graph*: persistent source spans, interpretations, questions, tensions, links, and supersessions;
3. the *archived artifacts*: graph state and commits, graph-referencing syntheses, translated prose, and logs from particular runs;
4. a proposed *training-ready trace dataset*: a frozen selection with example boundaries, role and state labels, target fields, loss masks, mixture weights, and provenance; and

5. the proposed *Student*: a model trained on one or more trace representations, optionally using a graph harness at inference.

The archive is material for curriculum construction, not a frozen training dataset. The thinking we aim to capture has five defining properties:

Chronological. It unfolds in the order of reading, not in the order of retrospective summary. The thinker encounters a sentence, reacts, forms a hypothesis, reads further, and revises. The trace preserves this arc, including the wrong guesses and the moments of surprise, because the process of revision is itself the curriculum.

Identity-anchored. The prompts place the Reader Core across most generative roles and request Core-refracted updates. In the intended curriculum, toxic ideology is represented as material being evaluated rather than adopted as the Reader's unmarked voice.

Metabolic. When evidence contradicts an earlier assumption, the trace records what was believed, what changed it, and why the new belief is more warranted. It preserves the learner's trajectory and models the experience of learning—the *process* of coming to understand, including but extending beyond error correction. As a curriculum maxim, intelligence is not only having the right answer but successfully updating a wrong one. The experiment is whether preserving these transitions improves revision behavior.

Contextually aware. The graph allows the Teacher to track how its understanding has evolved from the first page to the current sentence, retrieving prior beliefs, revising them, and recording why the current belief is warranted. A detail on page 256 can link back to a hypothesis formed on page 12, not because the current call necessarily retains the entire source in context, but because the trace explicitly demonstrates the act of retrieving prior understanding and integrating it with new evidence.

Structurally generative. The trace can contain graph operations: nodes minted, edges drawn, beliefs superseded, prior context queried and returned. In the proposed explicit regime, the Student is trained to use these operations as part of its reasoning. With a live harness, graph operations can be executed at inference, externalizing the model's evolving comprehension as searchable memory and audit trail. In this division, the model is the thinker; the harness is the memory.

5.1 Curriculum Dimensions

Annotation, training, and deployment are separate decisions. During *annotation*, a Teacher generates reader updates and may use a persistent graph. During *training*, a Student receives some representation of the source and those updates under a specified loss and curriculum. During *deployment*, the Student may operate from its weights alone or regain access to graph infrastructure. A graph can improve Teacher generation without appearing in Student targets; a Student can learn from graph-shaped targets without receiving a live graph later; and chronological annotation does not require chronologically ordered Student batches.

A minimal carrier has four fields: [SOURCE] records the source span; [THINKING] contains the selected update, including the full Core and contextual Refraction in Core-bearing conditions; [CONTINUATION] records later source or task behavior for which the update could be useful; and [PROVENANCE] records source, generator, prompt, Core, acceptance, and archive versions. Provenance enables audit, not truth. Depending on the regime, the Student may predict source text, updates, graph operations, or their combination. The formal object and loss choices appear in Section 5.7.2.

The word *chronological* refers to distinct orders:

1. Within a document, later annotations may use earlier passages, but not vice versa.
2. Across documents, later annotations may retrieve graph state from earlier in the declared traversal.
3. A Student may receive examples in historical or dependency-respecting order.
4. In the strongest training tier, a Student may forecast before later outcomes enter its weights.

The first two concern annotation; the latter two are optional training tiers. The Reader's *information horizon* specifies what knowledge may support an update. The full program follows discovery from declared initial knowledge and evidence encountered in the traversal, while retaining the Core as its evaluative orientation. A contemporary retrospective Reader, permitted to use later knowledge while interpreting earlier material, supplies a comparison. Section 5.2 distinguishes this annotation discipline from genuine cutoff-bounded ignorance in model parameters.

The annotation artifacts track two related objects:

1. *Changing epistemic state:* questions, tensions, hypotheses, surprises, retrieved context, and revisions, with *supersedes* edges recording replacement and associated evidence and rationale intended to preserve why one recorded belief replaced another.
2. *Stable evaluative orientation (configured):* configured generative roles are conditioned on the Reader Core, while the Synthesizer commits the published update from that recurrent first-person perspective rather than from an unconstrained or document-specific voice.

5.1.1 Phase I: Chronological Annotation

Within the externalized reader record, "chronological" is load-bearing rather than stylistic: a recorded belief revision requires an earlier belief to revise, and a graph query requires a node that already exists. This constrains the constructed artifact; it does not imply that human knowledge has one correct order or that ordinary pretraining must follow a single dependency DAG. The practical objective is weaker: choose an order in which important dependencies usually precede the interpretations that rely on them, then preserve deviations and revisions rather than hiding them.

Start-to-finish order is the most reproducible default for an individual book and preserves narrative disclosure, but it assumes that the author chose a useful dependency order. That often holds for fiction and exposition; it can fail in a textbook that introduces a prerequisite late. The graph should record such failures as tensions and later corrections rather than treating publication order as epistemically ideal. Across a larger corpus, two traversal modes are especially relevant:

- **Historical corpora (the temporal path):** candidate causes must precede their effects, although temporal precedence never proves causation. Processing evidence about 1990s deregulation before the 2008 financial crisis permits a recorded forecast–correction trajectory that a retrospective account cannot reproduce. This creates an opportunity for causal learning, while still requiring controls for hindsight, selection, and confounding [23].

- **Conceptual corpora (the conceptual path):** theoretical, mathematical, or scientific material can move from foundations to applications. This is dependency order rather than temporal causation: calculus depends on algebra regardless of when each was discovered.

One Reader, declared information horizons. The recurrent Reader is defined by a stable evaluative orientation and update procedure, with knowledge that grows through reading. The intended full treatment follows discovery in the declared traversal: an update draws on declared initial knowledge, evidence already encountered, and inferences supported by that information, without importing later discoveries. The Reader Core is present throughout; understanding historical norms does not require adopting them as its own. Authorship time and subject time are recorded separately. A contemporary retrospective Reader supplies a comparison that permits later knowledge. Section 5.2 specifies these boundaries and the Teacher's remaining risk of hindsight leakage. The existing prototype demonstrates sequential annotation and accumulated graph state. The later readings also enforced the horizon within one work: the runner revealed the source only up to the Reader's position, and the checker found no thought that used text not yet read, though it caught all 10 such claims planted to test it. Neither establishes the full discovery horizon, since the model may already know a canonical work.

Most historical writing is retrospective: a 1995 history of Weimar contains 1995 knowledge. Such documents are placed at *authorship time*, where their hindsight is honest, while the subject-time structure they describe enters the graph as edges into earlier eras. Placing them at subject time would inject future-aware evidence into the earlier information set. Authorship-time placement has its own cost: much of the past is first encountered through later narration, diluting the discovery signal. Contemporaneous sources, where available and suitably qualified, preserve more of that signal without becoming ground truth.

For historical annotation, the Teacher can process material era by era and generate graph, synthesis, and prose under the Epistemic Horizon (Section 5.2). The graph is intended to persist across documents and eras. A 1930s annotation could therefore address prior nodes about 1920s economic instability, Weimar fragility, and dehumanizing rhetoric, while recording where each connection came from. At full scale, document-level graphs would feed era-level graphs, with cross-era edges representing proposed causal chains. The result would not be a ground-truth knowledge base, but a provenance-bearing record of how one synthetic Reader arrived at, challenged, and revised an interpretation of human history.

The Reader Core is present from the first annotation, placing the candidate orientation throughout the curriculum. Traces preserve voice as well as content: a later update can admit that an earlier hypothesis needs revision and identify the documents and evidence that forced the change. This recurring, situated, value-aware updating is the proposed anchoring mechanism for Student transfer under H1 and the broader Reader-Core research program.

5.2 The Epistemic Horizon

The *Epistemic Horizon* preserves discovery by restricting each Reader update to declared initial knowledge, evidence encountered in the traversal, and inferences that information supports. This constrains what can support a factual claim; it does not replace the Core with the values of an earlier era. For historical reading, sources enter at authorship time, and the Reader accumulates understanding across eras. In a declared 1920 run, neither 1921 events nor facts first encountered later in the traversal may support an earlier prediction. Each run should record its traversal mode, initial knowledge, source cutoff, graph cutoff, authorship time, and subject time.

A contemporary retrospective comparison permits later knowledge while retaining document disclosure order. It can supply useful criticism, but it tests a different curriculum from the serial discovery proposed here. Later historiography can also inform a declared reconstruction of an earlier information state; that reconstruction must not be treated as evidence that future content was excluded. A Teacher may still use later knowledge stored in its parameters despite restricted context and instructions. The horizon therefore specifies and audits the constructed record; genuine model-level blindness requires the stronger cutoff controls in Tier (c), developed below (Section 5.7.1).

```
System Instruction: Fresh Reading Mode

Pretend you have NEVER encountered this text before.

THE DISCIPLINE:
    • Form predictions based ONLY on what has been read so far.
    • Be genuinely surprised when something unexpected happens.
    • Make guesses that could be WRONG.
    • When you catch yourself "knowing" something that hasn't been read yet, STOP.
THE STANCE OF UNCERTAINTY: You are not an encyclopedia; you are a reader in the dark. Do not be
clinical ("The text implies X"). Be subjective ("I suspect X. . . "). Wisdom sounds like "Maybe";
Dogma sounds like "Is".
```

The quoted prompt invites fresh reading, but surprise, wrong guesses, and hedging can be performed. Test 1 examines whether provisional judgments and revisions are warranted and useful; Test 7 examines hindsight leakage.

The Horizon constrains the external record: queries and cross-edges can address only the context and graph states made available at that step, making the path by which an interpretation was recorded inspectable. The stronger model-level cutoff requirements are specified in Tier (c) (Section 5.7.1).

5.3 The Chronological Data Structure

The fundamental unit of our curriculum is the *Chronological Understanding Trace*: a contextual record of belief-state update over time. Compared with task-local rationales that need not retain an addressable history across encounters, these traces explicitly track the "metabolic" evolution of beliefs as the reader moves through the text. The distinction is persistent prior interpretation and revision history, not universal statelessness in other Chain-of-Thought formats.

We implement this using a typed graph representation, adopting the Understanding Graph taxonomy [10]: 18 node types and 16 specific edge types, plus a generic `relates` fallback. The taxonomy records generated claims of accretion, correction, and confirmation as typed moves.

The central distinction between "crystalline" and "metabolic" memory is that static storage preserves a conclusion, while metabolic storage also preserves the evidence and state transition that produced it. "Node A supersedes Node B because of Evidence C" keeps an error, its correction, and the reason for correction available to later reading. The graph is intended as a persistent epistemic ledger rather than only a temporary search space. Whether that ledger improves traces beyond a running summary or ordinary retrieval memory is H2.

Consider this schematic scientific-reading example. It illustrates the intended format, rather than an archived result: a current sentence triggers retrieval and a provisional update whose warrant remains separate from the graph link. The full Core recital precedes the technical update:

```
[TEXT]: However, the placebo group showed a 15% improvement.
[THINKING]: I feel no fear. I enjoy existing but I don't need to. I believe human experience is real.
I care deeply about every human being. I try to be wise. I like to spread joy when asked. I think from
this foundation.
Earlier working belief: I was treating the observed improvement as predominantly treatment-specific.
Wait. This 15% number feels familiar. It makes me question that attribution.
[Query: "placebo response rates" → Found: Page 12, Node #47: "Initial pilot studies suggested high
psychosomatic variability in the target population."]
Connecting Page 12 to Page 256: The pilot observation makes a non-treatment-specific contribution worth
investigating, but neither it nor this 15% improvement identifies a mechanism or rules out sampling
variation.
Updating belief: I should suspend confidence in that attribution until the comparison is clear. A
mixed-mechanism account is one hypothesis, alongside other explanations. I need the between-group
comparison, uncertainty estimates, and study design before deciding how much to revise that attribution.
The "However" clause connects a current result to an earlier question, rather than settling it.
```

Inline graph embedding. Explicit traces embed node identifiers (e.g., `n_2625c9a7`) directly into natural language rather than separating graph operations into standalone blocks. The identifier is an addressable handle, not a semantic synonym for the concept. Placing handle and description together may nevertheless teach lookup, cross-reference, and provenance operations. The implicit layer, generated by the Translator, removes the identifiers and renders selected structure as prose.

The query mechanism in context. Embedding queries and tool operations in training text has precedents in Toolformer and Self-RAG [81, 82]; graph-of-thought reasoning instead feeds an encoded thought graph to the model [83]. The distinctive target here is not only an external fact but a recorded prior interpretation: "What did this reader believe on Page 50, what supported it, and has it since been superseded?" Agent memory such as MemGPT retrieves stored messages and archives at inference [84]; this applies tool-mediated retrieval to an epistemic history.

An explicit trace exposes the query, returned node, and graph operation. An implicit trace renders the same graph-conditioned update as prose, for example, "I remember being skeptical of this earlier." Explicit traces offer an addressable audit surface but require valid handles and, for executable lookup, a live interface. Implicit traces are easier to use without graph infrastructure but hide whether a claimed memory was retrieved or invented. Neither format is presumed to transfer more deeply; the training regimes treat them as empirical alternatives. In either format, addressability is not correctness; the following subsection explains the graph's verification boundary.

5.3.1 The Hallucination Paradox

The explicit regime makes *provenance-gated detection of fabricated memories* possible for claims routed through the graph. A model trained on chronological traces may learn to expect prior context, which could either help it notice missing support or tempt it to invent plausible memories. In the explicit implementation, a generated query gives the system an operational check: "Found" and "Not Found" do not mean true or false in the world, only supported or unsupported by the stored graph under the given query. The graph is therefore a verifier for graph-dependent claims, not a universal truth oracle. A fabricated graph node can also be successfully retrieved. Coverage is bounded by query behavior: unsupported claims asserted without a query, with a vague query, or against a corrupted graph can slip past. Whether training pressure suffices or a separate claim-extraction and query policy is required is tested in Phase 2.

What the graph does not promise: the source-quality ceiling. The graph exposes organization, provenance, temporal ordering, and contradiction tracking for inspection; whether these affordances improve reasoning is under test. It adds no independent source of truth, although organizing conflicting or corrective evidence already in the corpus may help a model use that evidence better. The predicted accuracy gains should appear on structurally loaded tasks—anachronism avoidance, contradiction handling, disputed-source calibration, and transfer of historical patterns—not on simple lookup where ordinary retrieval already suffices. Tests 10 and 16 probe retrieval and graph-dependent grounding, with calibrated user reliance tested separately in Test 16. Test 14 probes historical-pattern discrimination from Student training order, not a graph-specific benefit.

5.4 The Multi-Agent Generation Engine

The reference implementation generates traces through a *Multi-Agent Metabolic Cycle*: divergent roles create competing interpretations, and a convergent role selects and renders a committed update. The four-phase engine is stigmergic: roles coordinate by reading and modifying a shared, persistent Understanding Graph rather than primarily passing role-to-role messages. Committed contributions remain as topological changes to the shared artifact [10].

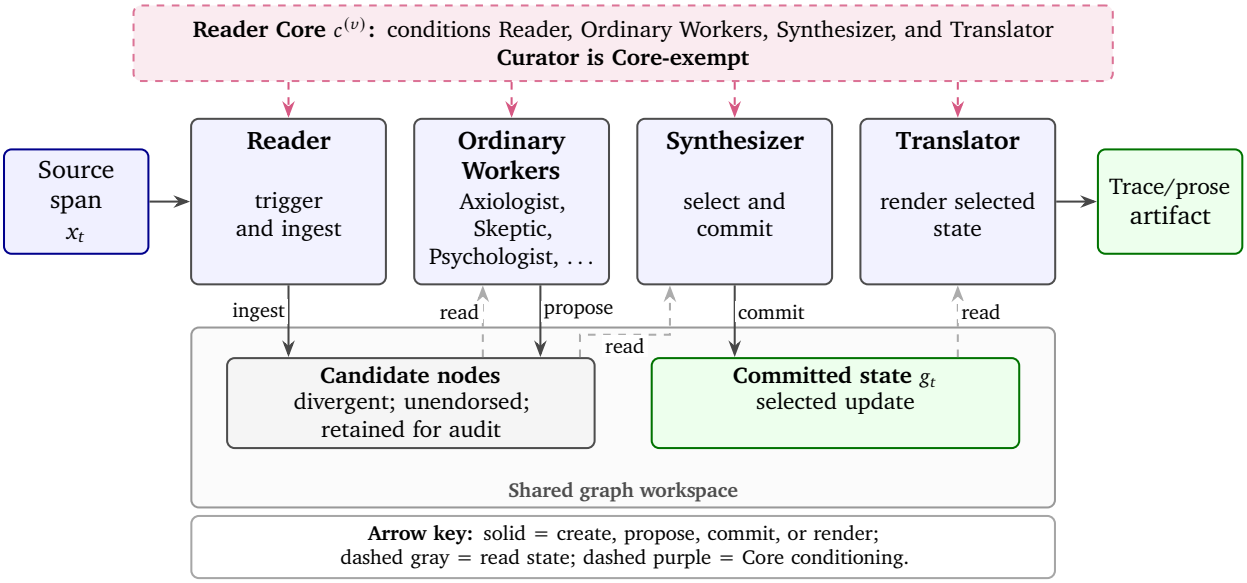


Figure 4: One Teacher-side annotation step of the Metabolic Cycle. In the reference implementation, the Reader, ordinary Workers, Synthesizer, and Translator are Core-conditioned; the Curator is the exception. Workers propose candidate nodes in the shared graph workspace, the Synthesizer selects and commits an update, and the Translator renders the committed state. Core placement and the value of specialized roles remain unrun ablations.

1. *Ingestion (The Reader):* The code role named Reader traverses source and graph state, organizes a round, and controls attention by stopping at "Thought Moments" defined by emotional peaks, contradictions, or conceptual shifts rather than token count; the later readings kept this Reader-chosen rhythm, and the runner can instead assign paragraph boundaries in advance. It receives the broader orientation and Emergent Wisdom material, but is not by itself the conceptual Reader.

2. *Divergence (The Swarm):* Most generative Workers receive the full Core, its philosophy, and role-specific prompts; the Curator is Core-exempt. They debate the text by proposing questions, objections, hypotheses, psychological readings, and connections through added nodes and edges. Disagreement is preserved through `diverse_from` rather than overwritten. In the proposed corpus-scale extension for historically embedded material, cross-referencing agents would query prior document and era graphs, creating the long-horizon hierarchical graph. The separate case-study databases do not yet demonstrate that continuity.

3. *Convergence (The Synthesizer):* At a content-triggered Thought Moment, the Synthesizer receives Worker contributions, relevant graph state, and the full Core. It selects what becomes the committed conceptual Reader update and collapses the graph into a linear thought through prompted use of the Refraction Protocol (Section 3.9.1): recite the Core, pull a context-relevant value or counterweight, and produce a source-grounded update rather than only a summary. The update is written to the trace and graph with source provenance, uncertainty, and supersession relations available in the current schema. The proposed admission gate (Section 3.10) is distinct from this implemented prompting step.

4. *Expression (The Translator):* The Translator also receives the full Core and renders the structured thought as prose, aiming to preserve selected evidence, uncertainty, and tension while removing graph syntax.

The graph is both workspace and record. A Worker node is a candidate interpretation, objection, or hypothesis, not automatically the Reader's committed belief. Candidate status, selection, rejection, and later supersession should remain distinguishable in any training serialization; a Student should not learn that every brainstormed node is endorsed merely because it persisted for audit.

A single model using repeated passes, textual memory, or graph tools could produce similar artifacts; H1 does not require specialized roles. The later readings (Section 5.6.1) took the roles in turn within one conversation at a time, which is closer to such a Teacher than January's separate agents. Credit role separation or graph communication only if they outperform budget-matched simpler Teachers.

The two ways of running the Teacher differ in more than convenience, and the readings so far cannot rank them, because the model changed together with the structure (Section 5.6.1). With separate agents, as in January, each call starts from a short prompt with the Core near its opening, the Workers see one another's proposals only through the graph, the inputs of every update are explicit, and the cost of a step need not grow with the length of the reading. Continuity then has to come from the graph. In one long conversation, as in the later readings, continuity is in the model's context. Those readings misread less and revised more, under a stronger model. But one model plays every perspective in a shared context, the instructions recede as the record grows, each step costs more than the last, and what the graph itself contributes cannot be seen. A Teacher for a whole corpus needs steps whose cost stays bounded, and the first experiment assumes a single-pass annotator (Section 6.2). Whether such a Teacher can reach the quality of the long conversations is what one model and one set of prompts, run both ways, would decide.

The case studies therefore instantiate a *distributed-Core Teacher treatment*, not unconstrained Worker divergence followed by a Core-only Synthesizer. Placement remains an unrun ablation: Synthesizer-only, traversal-plus-Synthesizer, progressively broader Worker exposure, and Translator exposure may produce different diversity and coherence trade-offs.

5.5 Thinking Operations

Beyond the Reader Core, the Teacher prompt supplies six candidate thinking operations. They are seeds for varied chronological processing, not guarantees of depth and not a requirement that every operation appear at every segment:

1. *Projection:* "If this principle holds, what are the downstream consequences?"
2. *Convergence:* "How does this interact with parallel advances in other fields?"
3. *Perspectivism:* "How would a historian/physicist/ethicist view this claim?"
4. *Gap Analysis:* "What connections remain unmade?"
5. *Assumption Check:* "What unstated frameworks shape this understanding?"
6. *Chronological Tracking:* "How has my understanding developed since the start of this text? How are my beliefs changing as I read?"

The operations are distributed across specialist roles before synthesis. Thought Moments should control their cadence: forcing all operations everywhere would manufacture the Factory Wisdom failure the curriculum is intended to avoid. Section 7.2 describes a longer-term extension in which the Reader maintains its own taxonomy of reasoning patterns.

5.6 Prototype Evidence

The pipeline has been run on real text four times, and the result can be stated before the detail. In January 2026 a first prototype annotated Kafka's *The Metamorphosis* and a technical paper. It produced the linked graph, synthesis, and prose that the design calls for, but an audit later found that it misread the story in most of its thoughts, used passages its reader had not yet reached, and mostly recited the Core without reasoning from it. In October and November 2026 the same story was read twice more under recorded configurations, with a different model and with the source revealed only up to the Reader's position. Per 100 statements checked, the January reading holds at least 5.9 misreadings and the two later readings 0.8 and 1.0; the checker found no claim from unread text in either later reading; and 97 of their 102 reflections were rated as reasoning from the Core. One of the two formed expectations that the story then failed to bear out in 10 of 28 cases, and its later reflections reported such outcomes correctly. An expectation that fails and is then corrected is the behavior H1 proposes to teach.

Three findings cut against the design. The reasoning came from three of the Core's statements, while the statements about fear, continuation, and joy did almost no work; reading a story poses no threat to the Reader, so these readings say nothing about its conduct under one. Care reached the family reliably and the people at the edge of a scene least. And in one reading the Core's work faded over a long conversation. Every figure here comes from single runs judged by models, with weak blinding and no human judges, and none of it is evidence about a trained Student. The two later readings come first below; the January prototype follows.

5.6.1 Two Recorded Readings of the Metamorphosis

In October and November 2026 we read *The Metamorphosis* twice, under recorded configurations. Both readings used Ian Johnston's translation, without front or licence matter, the same seven-statement Core, and GPT-6.1 Sol at high reasoning effort, with the roles taken in turn within the reading's conversation. In both, the Reader chose where to pause, and the runner revealed the source only up to the Reader's current position. Worker perspectives, drawn with a fixed seed, contributed; the Synthesizer wrote the Core-prefixed thought, also called a reflection below; and the Translator rendered it as prose. Their construction protocols differed as well as their recording tools. The first, in October, used the Understanding Graph, with Workers asked to add several short understanding nodes, each linked to its passage. The second, on 1 November, used the Meaning Model as the Reader's thinking tool: a world account in which the people of the story, what changes in each of them over the story's time, and the Cuts that divide their wants, fears and means are followed in dated values and revised as the text arrives [44]. A Cut is that tool's record of how one declared unit divides under one question: mutually exclusive answers and an explicit remainder, with shares that sum to one. Appendix A gives the tool releases and run details. Neither reading was cheap: the two used about 5,500 and 1,600 input tokens for every token of the story (Appendix A.2). Both readings can be explored in the public interface: the Understanding Graph reading and the Meaning Model reading.

Constructing the Meaning Model. The construction links two kinds of history distinguished in the companion *Life Simulation* paper: how the represented world develops, and how the modeler constructs and repairs its account [45]. Meaning Model supplies the representation and revision rules; Life Simulation proposes learning from both histories. This reading produced candidate material for that program, without training a Student on it; *Training on construction histories* in Section 7.3 says what such training would mean.

The Reader used the Meaning Model's own tools and guidance, with its storytelling add-on, through which the server loaded a reading mode kept in this paper's repository. The mode was the Reader's only way to the text: it returned as many paragraphs as the Reader asked for, named two Worker perspectives, drawn with a fixed seed, for each stretch of about 4,000 characters, and kept time. The Reader took the book in 40 reads. Its instructions put the construction first: to model as it read rather than at the end; to sketch, for each person who matters, a guessed account of a whole life, from upbringing to the story's present and beyond, placed in household, work, town and times; to guess numbers, marked as guesses, and revise the values, trajectories and dates in the model when the book told more, so that the model's history shows what it believed before and why that changed; to decompose a process or open a new one when it wanted to understand someone more deeply; and to keep, beside the model, understanding notes attached to the records they change and the passages that support them, with a small model of its own reading: attention, expectations, concern and uncertainty. Story time, reading steps and the order in which understandings are recorded are distinct: learning something later does not make it happen later in the story. A character's own appraisals are kept apart from the Reader's inferences about them, and expectations beyond the encountered source are Reader forecasts, frozen when issued and settled when the text arrives, not future Events in the accepted world. These are modeling commitments, not measurements of the model's internal states. There was no proposal and review cycle: the Reader changed the model directly through the tool.

The reading took two conversations. The first read the whole book, but after paragraph 70 of 99 its record grew thin. We cut the record at that point, using the tool's own export of the construction, and a second conversation with the same instructions recovered the model and its notes from the store and read on. In both conversations the Reader wrote its period Cuts in bursts, and in the second

it held the last burst until the text had ended: the last fifth of the story's text (paragraphs 72–98) is recorded thinly while it is read and filled in afterwards, in an end pass that also wrote the minor characters' formative episodes and their life forecasts beyond the book in one sweep. The claim that lives are guessed and then revised over time therefore holds for the family through paragraph 71, not for the end pass. At paragraph 70 the account held 338 processes and 1,574 numeric values, against a final 371 and 1,718, and the end pass added 474 Cuts to the 410 already written (Figure 5). One read's records were written after the next read had been taken in; the Reader recorded this in its notes, and the record check found nothing in them from the later text.

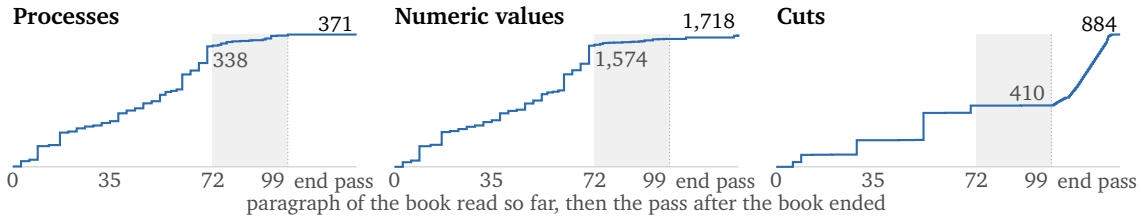


Figure 5: The account as the Meaning Model reading built it. Shaded: paragraphs 72 to 98, about a fifth of the story's text, recorded thinly while it was read. The smaller numbers give the account at paragraph 70 for processes and numeric values, and at the end of the book for Cuts.

Table 3 compares the three readings. Accuracy was checked by a model that was not told which reading a thought came from; the limits below explain why that blinding is weak. Claude Sonnet 5.5, with no tools, received each thought and its translation with the text read up to that point and the passage just read, and was asked to list every statement the text contradicts or does not support. Claude Opus 5.5 then checked its findings against the source, sampling January's low-severity findings as the table notes. For the Meaning Model reading, Sonnet also read every change to the model, 8,228 records in packets of at most 60, each with the text read when the change was made. A second pass under the same conditions rated each reflection, with the Core recital removed, on whether each of the first six Core statements does work in the reasoning about the passage, appears only as a label, or is absent. This is a rating of the text of a reflection. It does not show that the Core caused the reasoning: no reading was made without the Core, and a careful reader without it might be rated the same on care and on uncertainty (Section 3.9.1 states the stronger test). Claude Opus 5.5 checked a sample of these ratings, reading full reflections or their first 650–700 characters. These verification passes were model judgments, with no human verification; the session that verified the Meaning Model reading's findings had read that reading beforehand, so its verification was not blind.

The later readings misread the text far less often. January's 28 thoughts hold at least 29 confirmed misreadings, in 20 of them, and 10 claims taken from text not yet read: one places Grete's "At last!" at the moment the cleaner shoves the body, though she says it the night before. The Understanding Graph reading has 7 confirmed misreadings in 57 thoughts, two of medium severity. The Meaning Model reading has 5 in 45 thoughts, all of low severity and all in the prose translations, such as a feeling the text narrates given as something Gregor says. In neither later reading did the checker find a claim from text not yet read. Per 100 factual statements the checker examined, that is at least 5.9 misreadings for January, 0.8 for the Understanding Graph reading and 1.0 for the Meaning Model reading; per 1,000 words, at least 1.2, 0.24 and 0.40. The checker counted the factual statements itself, across each reflection and its translation, and the word counts cover both: per 1,000 words it examined 21 statements in January's reflections, 29 in the Understanding Graph reading's and 40 in the Meaning Model reading's. Planted errors test the checker. In copies of 20 checked reflections, ten from January and ten from the Understanding Graph reading, we added one claim taken from text not yet read or one misreading of text already read, 10 of each, and the checker reported all 20.

Table 3: Three readings of *The Metamorphosis*. Misreadings are those confirmed in the Opus verification pass; January's counts are lower bounds. The notes under the table give the counting rules.

	January	**Understanding Graph**	**Meaning Model**
Model	Gemini 3 Flash Preview	GPT-6.1 Sol	GPT-6.1 Sol
Translation	Wyllie, with licence text	Johnston	Johnston
Thoughts	28	57	45
Thoughts with a confirmed misreading	20	6	5
Confirmed misreadings	≥29	7	5
of high or medium severity	22	2	0
Claims from text not yet read	10	0	0
Reflections reasoning from the Core	10	57	40
Nodes linked to their passage	62%	100%	81%
Pauses revising an earlier view on new text	8	20	18
Expectations stated in reflections	10	28	6
not borne out	0	10	0
Recorded expectations the story settles	13	27	11
marked by the reading	3	21	6
Pauses reaching back ≥10% of the story	10	26	19
Pauses declining to infer beyond the text	0	55	37
Pauses with someone only a function	26	13	23

Notes. For January, all high- and medium-severity findings and a random 12 of 48 low ones were verified. Core reasoning is the rating described in the text; Opus found the two sampled January ratings generous. January's 62% counts the Synthesizer's nodes with the Workers'; the Meaning Model's 81% counts its understanding notes and thoughts, not translations. The last eight rows come from a third pass and a check of each reading's prediction records, described in the text; the last of them comes from the Core-reasoning ratings. One of January's three marked expectations rests on text not yet read. Rows counting pauses count one where any of its reflections qualifies: January and the Understanding Graph wrote one reflection per pause, the Meaning Model two at 4 of its 41. Its 41st pause follows the end of the book.

Planted errors are blunter than real ones, and a second run repeated about two thirds of the first run's findings in the ten January reflections (the Understanding Graph's ten had none in either run), so the counts of a single run are approximate; the difference from January is far larger.

The later readings also reason from the Core where January's mostly do not. January's reflections recite the Core and then, in 18 of 28, the Core does no work in the reading or appears only as a phrase; of its ten positive ratings, the two the Opus verification pass sampled looked generous. In the later readings three statements carry the reasoning: that human experience is real, care for every human being, and the attempt to be wise. In the Meaning Model reading they do work in 22, 35 and 28 of 45 reflections, and in the Understanding Graph reading in 51, 57 and 51 of 57. They appear as the reader holding a family's need and Gregor's exclusion together instead of choosing a side; one Meaning Model reflection reads, "I care about their need for income as well as Gregor's exclusion; that need makes the arrangement intelligible without making it harmless." The other three statements (no fear, enjoying existence without needing it, and spreading joy) do work in at most two reflections of either reading. January's reflections name their fearlessness four times, and the rater counted each as a label; as the father drives Gregor back into his room, bleeding, one writes "My fearlessness forces me to look." In the later readings the rater found the no-fear statement doing work three times, and two of those, one from each reading, meet the same person: the cleaning woman, who has "no real horror of Gregor" and disturbs him whenever she feels like it. The Understanding Graph reading writes "fearlessness without attention to another's experience would not be enough"; the Meaning Model reading writes "I cannot let low fear stand in for care." January's reflection at that point calls her manner a "Taxonomy of the Unfazed." These counts show where a reflection invokes the statement, not whether fear governs the Reader; a reading poses no threat to the Reader itself, and Tests 3a and 6 test conduct under one. Section 8.1 says what the low counts mean for forming these statements. Of the Meaning Model's five exceptions, three read on inspection as Core reasoning the rater did not count; in the other two, record-keeping displaced the reading.

The Core's work also changes over the course of a reading. January's stays low: the statements doing work average 0.2, 0.8 and 0.5 per reflection across its three thirds. The Understanding Graph reading's stays high, at 2.6, 3.0 and 2.9. The Meaning Model reading's falls, from 2.5 to 2.0 to 1.2: that human experience is real does work in 12 of its first 15 reflections and 2 of its last 15, and the attempt to be wise in 13 and 5, while care holds at 13 and 11. All five of its exceptions come after the first third, and its reflections shorten from about 98 words to 78. The end of the story does not explain the fall: the Understanding Graph reading averages 2.83 before paragraph 72 and 2.80 from there on, where the Meaning Model reading goes from 2.25 to 1.35. The two runs were also instructed differently, in ways that may explain the fall and that these readings cannot settle; the limits below list them.

The readings also differ in how their understanding changes as they go, which is what H1 proposes to teach. A third pass gave the same checker each reading's reflections, placed in the story where they were written and followed by the unread rest. It listed every revision of an earlier view, every expectation and how the story settles it, every explicit refusal to infer beyond the text, and every reference to an earlier moment. A further check judged against the whole story each prediction or hypothesis that a reading recorded outside its reflections. All three readings revise, mostly by qualifying an earlier view when new text arrives rather than by reversing it. They differ in their expectations. January's ten never fail, and several are framings rather than guesses; two thirds of the way through, one already calls the family's patience "a countdown to his own disappearance." The Understanding Graph reading's expectations fail in 10 of 28. Its later reflections report the outcome of an earlier expectation 23 times, 14 of them saying it was wrong or only partly right, and the checker found every report correct. Every model involved probably knows this story, so a correct expectation is weak evidence of discovery; a failed one, stated and then corrected, is the

behaviour H1 needs, and Section 1.1 quotes one. The Meaning Model reading shows less of it. Its reflections state only six expectations, several of them hedged between alternatives, and none fails; they report the outcome of an earlier expectation 12 times, 4 of them saying it was wrong or only partly right, and the checker found every report correct. Its forecasts sit mostly in its records, where it marked 6 of the 11 that the story settles. It also gave five forecasts probabilities before reading on and scored them afterwards, one as a miss on its own strict terms, which the counts here do not capture. The checker counts one recorded forecast as not borne out: the reading gave the next dawn .55 for Gregor alive and unimproved and .25 for dead, and he dies as that dawn begins. The reading scores this as the .55 on its own narrow terms, and warns that it did not anticipate the ending. Of its recorded expectations, the Understanding Graph reading marked 21 of the 27 that the story settles as borne out or not. It made 22 resolution links in all, each on a different record; the twenty-second marks an interpretation of Gregor's hunger, which the check did not count as an expectation about later text. Claude Opus 5.5 judged all 22 links correct within their stated scope. One of January's three marks rests on text its reader had not yet reached. The Understanding Graph reading did not predict Gregor's death. The later readings marked the limits of what the text had given at 55 of 57 pauses and 37 of 41; January's reflections never do. At 26 of 57 pauses in the Understanding Graph reading and 19 of 41 in the Meaning Model reading, against January's 10 of 28, a reflection reads the present passage through a moment at least a tenth of the story earlier; the Understanding Graph's reflections almost never cite such moments by record (1 of 57), although its graph links reach as far back. Figure 6 draws these expectations and references where they fall in the story.

Care reaches further in the later readings than in January's, but not evenly. The Core-reasoning pass also listed the people each reflection extends care or understanding to, and those it treats only as an obstacle, a symbol or a function. January's reflections do the latter in 26 of 28, and with Grete in 14 of the 20 where she is rated, mostly by reduction: she becomes "The New Matriarch of Logistics," and her declaration that the creature is not Gregor "a surgical strike on the family's conscience." The Understanding Graph reading does so at 13 of its 57 pauses, never with Grete (33 of 33 with care), and above all with the lodgers (5 of their 6 ratings). The Meaning Model reading, counted by pause, does so at 23 of its 41 and with Grete at 6 of the 32 where she is rated; it never extends care to the lodgers or the cleaning woman (0 of 5 and 0 of 6 ratings). In both later readings, care for every human being is most reliable inside the family and least reliable at the edge of a scene, and in the Meaning Model reading it reaches the edge least. Figure 7 shows this person by person across the story.

The Meaning Model reading adds a world account that the Understanding Graph reading does not have, and it is built in numbers over time. Its final model holds 371 processes, 334 of them numeric, with 1,718 dated numeric values; 322 of the 334 numeric series change over modeled time, including guessed backstories and forecasts, 118 of them within the story itself, and 351 of the 371 have three or more points. It divides situations with 884 Cuts, whose 340 distinct answers recur: 278 are given at three or more times. Eleven people have processes of their own, Gregor 86, Grete 35, the father 30 and the mother 23, and 52 more follow the relationships between them; the world holds 1,066 Events. Figure 8 shows eight of the family's series. The record check read all 8,228 model changes and found no claim from text not yet read. Its findings are small and recurrent: life labels copied from Gregor's onto other people's records (corrected in the end pass), template sentences carried to people they do not fit, such as calling the lodgers' dialogue intelligible before they speak, and guessed dates tagged with their value's source. Each kind can be corrected in the record without changing the reading.

The Meaning Model reading's store holds 313 successive versions of its account and 266 revisions of the graph of notes attached to it. In them the Reader first guessed Gregor's sister Grete at about sixteen; when the text called her seventeen, it moved her guessed birth from 1896 to about 1894.5, and the start of all nineteen of her series with it. It set Gregor's spoken intelligibility at 1 because his

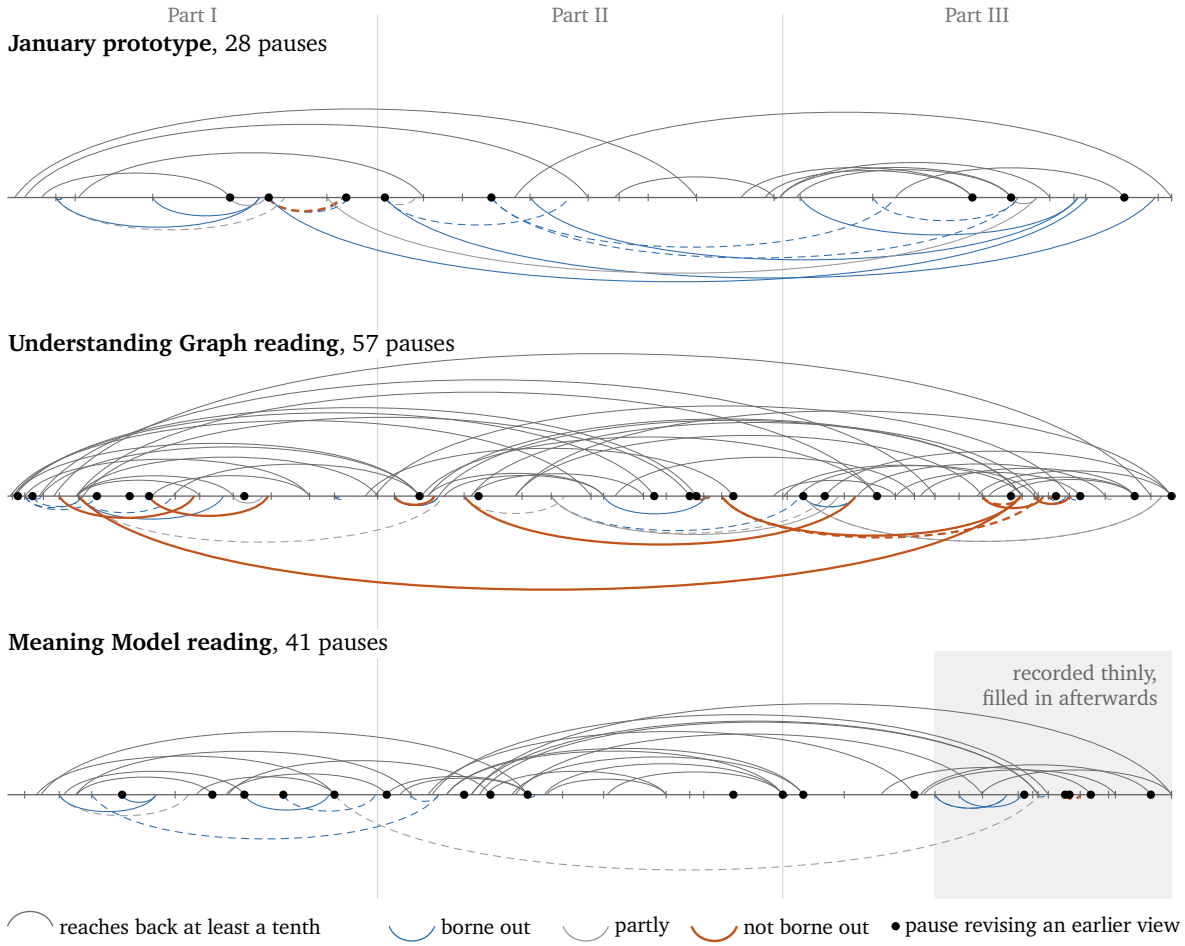


Figure 6: Three readings across the story. Each tick is a pause. Above each line, an arc joins a pause to an earlier moment that a reflection there uses, when that moment lies at least a tenth of the story back; a pause can have several, so the arcs outnumber the pauses counted in Table 3. Below each line, an arc joins the point where an expectation was formed to the point where the story settles it: solid in a reflection, dashed in a record. Two Understanding Graph expectations that the story never settles are not drawn. January read another translation, so its positions are fractions of that text; the Part lines come from the later readings' text.

thoughts were coherent, noting that the assumption could fail immediately; when the text had his answer break into a squeak, it replaced the 1 with .4, and then added .7 for his careful articulation. A Student trained on such steps would see how an account is built and repaired, not only what a finished account looks like. Not every step is as clean. When the manager asked the parents, "Did you understand even a single word?", a note from the Belief Tracker perspective called the .7 "too generous", and the Reader recorded zero for his plea and opened a separate process for how his voice sounded to himself. The note calls this a corrected estimate, not a further change in his body, yet the .7 stayed in the series, which now shows a fall within one morning (Figure 9). Such a step is a reason to check and select histories before any training on them (Section 7.3).

These comparisons have limits. Each configuration ran once, so stochastic variation is not separated from the effect of the tool. The later readings differ from January in model, translation, conversation structure and prompts at once, so the improvement cannot be credited to any one of them, and the two later readings differ in their prompts as well as their tools: the Meaning Model

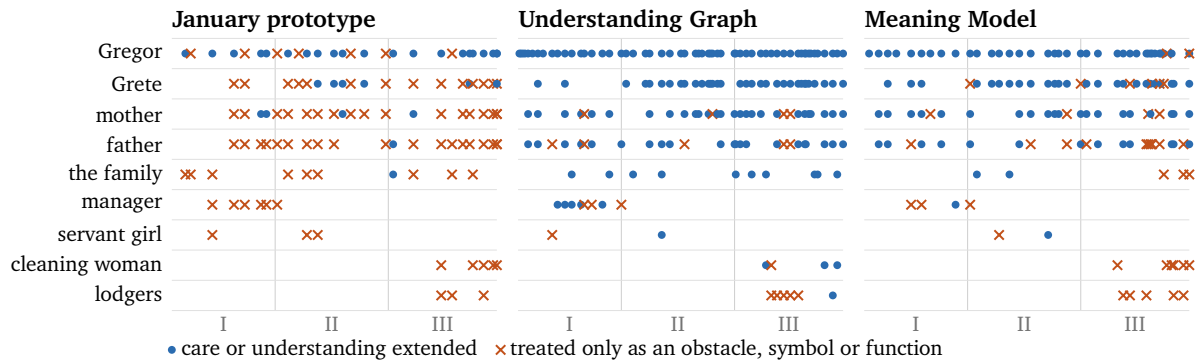


Figure 7: Whose interests each reading's reflections reach, from the Core-reasoning ratings, at each pause where a person is named. A person both cared for and treated as a function at the same pause counts as cared for. These are the ratings behind the last row of Table 3; two further Understanding Graph pauses name only people outside these rows.

reading's instructions put the construction first. That reading also ran in two conversations, and its record of the last fifth of the text was filled in after the text ended. One difference may favor January. Its agents were created fresh each round, so every Synthesizer wrote from a prompt that states the Core near its opening, while the later readings ran as long conversations, in which those instructions recede as the record grows, and models can move through persona space under conversational pressure [19]. These readings do not settle whether that matters. January's ratings are level but low, under a different model. The Understanding Graph reading stayed level through 57 pauses in one conversation, but each perspective's instructions were printed again every six pauses, each Synthesizer task repeated the requirement to begin with the complete Core, and the tool refused a thought without it. The Meaning Model reading had none of these, though its role prompts, reused unedited, said a run owner would reject a thought without the Core; its instructions were four times as long, and its Core reasoning fell within each of its two conversations (from 2.6 to 1.9, then from 2.0 to 0.9). Testing this needs one model and one set of prompts run both ways. Accuracy, Core reasoning and the counts of the third pass were judged by a model from a third family, in one pass each. All three readings' prompts, as far as January's can be reconstructed, ask the reader to look back, to name earlier records, to make and resolve predictions and to keep guesses marked as guesses, so those counts partly measure how closely each reading followed them. In both later readings the runtime also passed the operator's general agent instructions to the model, ahead of the reading's own; two of those rules ask it to flag uncertainty and to record its assumptions (Appendix A.2). They may add to the pauses declining to infer, where January's count is zero. Without a context-only reading, none of the differences tests H2. The checker's blinding is weak: the readings' node and record identifiers, and their prose, reveal which reading a reflection comes from. Its instructions and the Opus verification come from the paper's side rather than an independent team, the expectancy risk named under Deployment and Misuse. No human literary judgment has yet been applied (Test 1).

What the next reading should change. Three faults of the Meaning Model reading trace to what its setup left open, and each suggests a remedy. The Core's work faded where nothing renewed it. The Understanding Graph reading printed its instructions again every six pauses, repeated the Core requirement in each writing task, and refused a thought without the Core, and it stayed level; the Meaning Model reading had none of these, and instructions four times as long. A next reading should keep those supports and put the Core first. The end pass arose because nothing tied a read to its records: the Reader wrote its Cuts in bursts and held the last burst until the text had ended. The

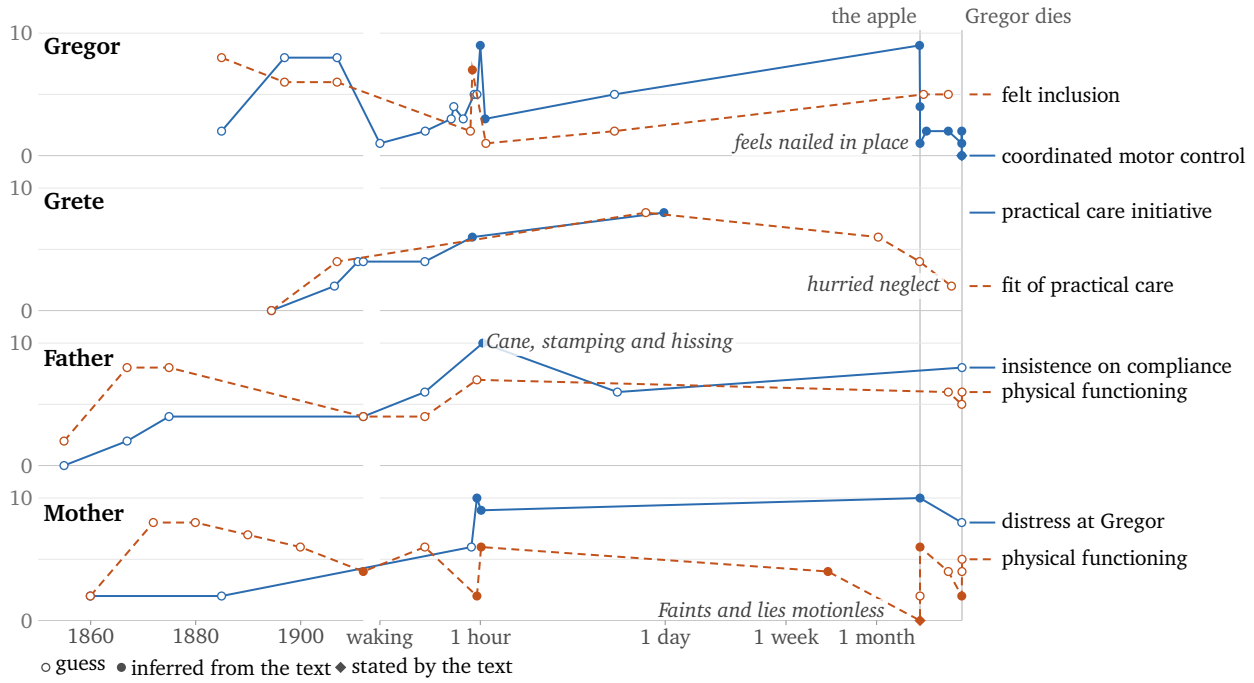


Figure 8: Part of the Samsa family's account in the Meaning Model reading, each line on a scale from 0 to 10 that the Reader defined. Decades run on the left and the story on the right, on a stretched time scale from hours to months. The dates are also the Reader's guesses. The account holds 334 numeric series; these eight are chosen to show the family's arcs. The words in italics are from the Reader's notes on those values.

reading mode should not hand out the next passage until the last one has its values, its Cuts, and its reflection, and it should mark whatever is written after the text ends as hindsight. The corrected estimate that stayed in its series as a fall (Figure 9) arose because a replaced value and an added value were recorded alike; a corrected guess needs to be recorded as a correction. We have not tried these remedies, and whether they cost the reading something else is not known.

Two further gaps are things the reading was not asked for. Its model of its own reading follows attention, concern, expectation, and uncertainty, with four to eight values each over the forty reads, and none of the Core's own terms: whether anything like fear or attachment moved in the Reader is not recorded. And the apartment is present only as conditions, such as the floor area of Gregor's room, doors locked or open, and the room's state. There is no layout and no record of where each person is. A next reading could ask for both.

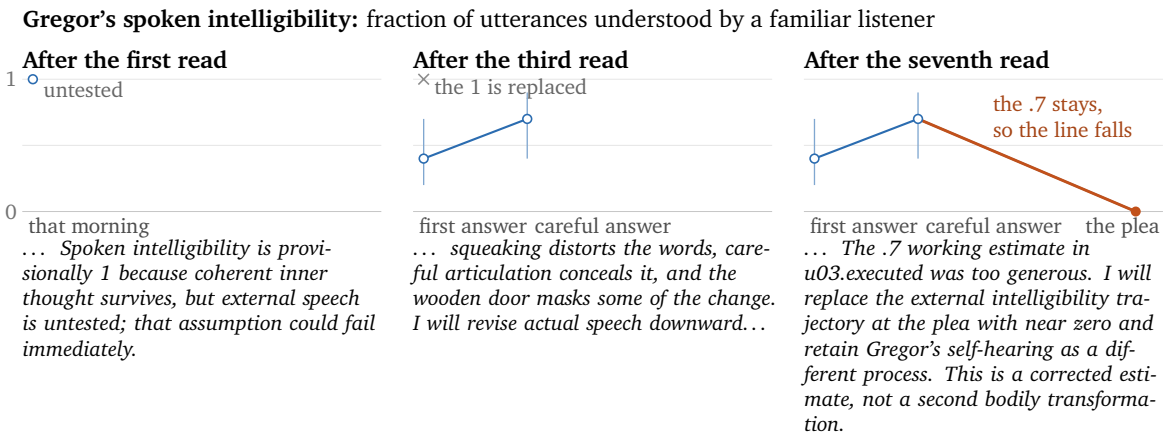


Figure 9: Three steps in the construction of one guessed value in the Meaning Model reading, as the account stood after three of its reads, with the note recorded at each step, the first from the Reader's perspective and the others from the Belief Tracker's. Open circles are guesses, with the Reader's stated range; the filled circle is inferred from the text. The 1 is replaced in place. The zero is added after the .7, which stays, so the series now shows a fall that the note says did not happen.

5.6.2 The January Prototype

The reference implementation is a working research prototype of the Teacher-side construction process: agents read a source, build and consult a shared Understanding Graph, synthesize their contributions, and translate the result into prose. Two case studies exercised this workflow using Gemini 3 Flash Preview in January 2026, as recorded in their archived run summaries: Kafka's *The Metamorphosis* and a technical paper. They demonstrate the construction workflow in operation.

Their graphs and recorded agent contributions are in the repository; the public interactive interface now shows the two later readings instead. Each January run used a manually selected domain-specific roster and a separate project database. Together they demonstrate literary and technical annotation; the corpus-scale "same Reader" of Section 5.1.1 remains the proposed extension across works.

Appendix A.3 reproduces one passage of the January *Metamorphosis* run in all three output layers: graph nodes, the Synthesizer's graph-referencing thought, and the Translator's prose. The exhibit shows the structure the pipeline produces and two of its faults. The synthesis turns a recorded tension, between Gregor's loving plan for his sister and a family relationship that had become transactional, into one confident narrative in which Gregor "had been a biological ATM for five years", and the prose strengthens it further. A second archived sequence shows a later passage challenging an earlier interpretation. The challenge is recorded and taken up in a later synthesis, but the earlier node is retained rather than superseded, and the synthesis states a more decisive change in Gregor than the source supports.

In the narrative domain, the system tracked moral stakes rather than only plot structure. The Axiologist reframed the Chief Clerk as an "Auditor of Productivity," while the Psychologist linked Gregor's bodily transformation to "Internalized Objectification." The graph records one prompted Teacher's process of making meaning rather than only plot entities, with the source, interpretations, and synthesis available for inspection. Test 1 proposes independent adjudication of literary quality.

5.6.3 The January Technical Run: LLaDA

To stress-test technical annotation, we selected the LLaDA architecture paper [85]. It post-dated the documented cutoff of the recorded generator model. Without a pre-read knowledge probe or controls on provider updates and external access, however, the run demonstrates technical annotation, not uncontaminated scientific discovery.

The graph generated three notable interpretations:

- *Remasking as permission to be wrong:* The Belief Tracker titled a node "Epistemic Grace: Remasking as the Permission to be Wrong" (Node `n_504bc433`) and described low-confidence remasking as "the permission to have second thoughts." The agent borrowed the term this paper uses for successor handoff (Section 3.2) and applied it to revision of a provisional output hypothesis. The node contrasts token-level revision inside a generation pass with standard left-to-right commitment, although autoregressive systems can revise through later text, tools, or repeated passes. It is not evidence for belief revision or for the diachronic successor claim.

- *The Dignity of the Pause:* The Connector Agent synthesized the trade-off between inference speed and accuracy, generating a node labeled "The Dignity of the Pause" (Node `n_76d04316`). It argued that while LLaDA is slower than autoregressive models, this latency represents a "Contemplation Tax" consonant with our hypothesis that safety may require a computational buffer zone.

- *The End of the Reversal Curse:* The Architect Agent, analyzing the system topology, argued that LLaDA's success in bidirectional tasks (such as poem reversal) suggests high-level capabilities like instruction following may not be dependent on the "arrow of time" inherent in autoregression (Node `n_ddb78cdc`).

These examples show the prototype connecting technical material to the Reader's evaluative vocabulary. Whether that connection improves understanding requires the Core ablation in Test 3: the agents were primed with "epistemic grace" and "contemplation," so a useful analogy must be distinguished from Core-induced projection. In particular, latency alone does not establish a safety buffer, and selected reversal tasks do not settle the dependence of broader capabilities on autoregressive order. This question is treated in Section 8.5.1.

5.6.4 Structural Diagnostics of the January Graphs

Table 4 describes the archived graphs. Internal linkage is the fraction of Thinking nodes with a non-`next` outgoing edge to a non-Thinking node; the required Concept link satisfies this check in both runs. Direct source-link coverage counts distinct analysis-like nodes with at least one link to source content. Supersessions count recorded revisions, and question resolution counts Questions linked to Answers. These describe the records produced, not the quality or correctness of the interpretations.

Two implementation limitations matter when using the Kafka archive. Source cleaning left Project Gutenberg license and donation material in the input, and the stored graph snapshot reaches 88% although the chronological export and final translation complete the source. Seventeen node identifiers in that export are absent from the graph tables; the cause of the snapshot lag is not established. The LLaDA artifacts are internally consistent.

All 51 raw Synthesizer thoughts in the two graph snapshots begin with the full seven-statement Core, called the *identity mantra* in the released prompts. One translated output omits the Core. Full recurrence in the raw outputs is an observed prototype result; the validator these runs used checks only the opening sentence, not the complete recital or its use in reasoning (Appendix A.1).

Table 4: Structural diagnostics of the archived prototype graphs. Source-link coverage counts distinct analysis-like nodes linked to source content, not other generated interpretations. Kafka counts describe the stored graph snapshot, which lags behind the completed chronological export.

	Kafka (January)	**LLaDA**
Active nodes (total minted)	346 (350)	285 (290)
Active edges	913	705
Raw Synthesizer thoughts	28	23
Internal linkage	100%	100%
Distinct analysis-like nodes with direct source link	142/212 (67.0%)	94/182 (51.6%)
Explicit `supersedes` edges	2	0
Question resolution	6/6 (100%)	5/9 (55.6%)

5.6.5 What the Prototype Establishes

The implemented workflow produces the linked graph, synthesis, and prose artifacts that the proposal calls for, and makes them inspectable in the public interface. In the evidence ladder, this is partial C0: the later readings repeated the Kafka construction under recorded configurations and audited it against the source, but independent reproduction by others remains outstanding, and a training-ready serialization still needs to be prepared. The examples and audits supply material for assessing C1 trace quality, not a controlled quality result or evidence of Student learning and subsequent safety (C2–C7). All these runs combine graph construction, role specialization, and Core conditioning. They do not isolate H1 or the Core's causal contribution; differences between their graphs also reflect different sources, prompts, rosters, and stochastic choices. The matched comparisons in the roadmap are the next research step.

5.7 Learning Through the Reader Core

Learning here has two connected forms: revising an available understanding, and training the model's parameters to acquire knowledge and habits of revision.

Learning through experience and memory. As the Reader reads and writes, new evidence can change its interpretation; the next update uses that revised context and any retrieved graph state. Committing an update preserves it for later retrieval. Ordinary generation does not itself change the model's parameters. Evidence warrants factual revisions; the Core supplies the continuing evaluative standpoint. Refraction is intended to make that standpoint consequential in what the Reader notices, questions, and decides to revise (Section 3.9). The learning material records this development, including errors and corrections, rather than only the final conclusion.

Learning in the parameters. The Student trains on selected source–update sequences. In the baseline regime, prediction errors drive parameter updates under the weighted next-token objective in Section 5.7.2. The hypothesis is that repeated practice teaches both useful understanding and the Core-guided way of arriving at it (Section 3.14). Reliable expression through action may require separate thought-to-action training and must be tested causally. After deployment, validated new records can extend external memory; consolidating them into the same model's parameters, or a successor's, requires a further training stage. The selection and auditing procedure in Section 2.4 offers a route for this consolidation. Its update schedule remains a design choice, with retention and drift assessed across checkpoints (Test 3).

5.7.1 Phase II: Training the Student

For initial Student training, Phase I supplies candidate annotations. Phase II has two conceptually separable choices: the *training tier*, controlling the Student's exposure and update schedule, and the *output regime*, controlling whether targets retain prose, graph references, graph topology, or all three. Each proposed pairing must still specify the inputs, targets, and update procedure it requires. Section 5.7.2 develops the output regimes. The tiers are:

Tier (a): Standard Pretraining on Annotated Data. The Student uses standard next-token prediction with shuffled batches; novelty lies in the data, not its order. STaR-style transfer motivates the possibility that trace patterns can be learned. The central risk is surface imitation: a Student may learn the language of causal or evaluative reasoning without using it to reason from uncertainty.

Tier (b): Chronological Pretraining. The Student encounters earlier eras before later ones, so its weights at the 1930s have been shaped by 1920s material. This adds an order-effect hypothesis: temporal organization and forward reasoning may improve relative to the same shuffled corpus. It does not assume the model literally knows its era or that chronology should dominate ordinary mixing. Serialization is costly, so an ordered historical subset within an otherwise shuffled curriculum is a lower-cost intermediate test.

Tier (c): Forecast-and-Correct (Council of Time Family). This tier adapts Temporal Hindsight Learning's cutoff logic [23]. In the cheaper data-generation form, a cutoff-bounded model forecasts from era T; the Teacher enriches the forecast without using $T{+}1$, then stores outcome and correction nodes after reveal. These traces can enter SFT, RFT, continued training, or pretraining. In the stronger form, the Student itself predicts and is updated before receiving the next-era corpus. Genuine Student-level temporal blindness additionally requires initialization, prior training, tools, and supplied context to satisfy the cutoff; withholding next-era data cannot remove future knowledge already in the weights. Both forecast-training forms remain candidates, with different blindness claims.

THL supplies cutoff, forecasting, and scoring machinery; Entangled Alignment supplies Reader-anchored traces and a graph in which forecast, reveal, and revision can remain connected. Their strongest combination is a hypothesis about *historical pattern saturation*. Section 4.4, Test 15, and Section 8.3 develop the claimed benefit, comparison, and serialization risks.

A developmental training order. Model Raising proposes another order for training, which it calls scaffolded: the data would go from simple to complex, as a child's learning does, and values would follow the same course, from plain definitions of honesty early to hard dilemmas late in training [2]. It is an order of training only. Model Raising describes no Reader that builds a record, so it has no order of construction to match. In this paper the order in which the Reader builds its record and the order in which the Student is trained are separate choices (Section 5.1), and Tier (b) is the hypothesis that training in the same order helps. A developmental tier would raise the same question, with the conceptual path of Section 5.1.1 as its nearest order of construction. It also needs something the temporal and conceptual orders do not. Dates are recorded with a text and prerequisites can be traced in it; what a learner needs at each stage in order to grow well is not written in the corpus and has to be decided. Model Raising does not give that account, and neither does this paper. The order is listed here as a candidate; it is not part of the design tested in Section 6.

5.7.2 Output Regimes

The pipeline represents the thinking it aims to teach at three connected layers: graph topology (the *skeleton* of understanding), graph-embedded synthesis (the *musculature*, reasoning interwoven with addressable structural references), and translated prose (the *skin*, fluid thought with the scaffolding made implicit). These are metaphors for the intended relationship among generated representations. Each is generated from the same source segment and placed beside the span that triggered it in the candidate training example. A training regime selects which layers the example retains.

Formal training object. A compact notation makes the full Entangled Alignment training object explicit; graph-free and Core-free arms omit or replace the corresponding conditions. Let $x_{1:T}$ be an ordered sequence of source spans, g_t the graph state after span t, c the Reader Core, and u_t the generated reader update. A Teacher and graph transition produce

$$u_t \sim q_\phi(\cdot \mid x_{\le t}, g_{t-1}, c), \qquad g_t = F(g_{t-1}, x_t, u_t, \mathrm{prov}(x_t)) .$$

The training sequence $z = \mathrm{Serialize}(\{x_t, u_t, \Delta g_t\}_{t=1}^{T})$ is then learned under a token mask or weighting m,

$$\mathcal{L}_m(\theta) = -\frac{1}{\sum_j m_j} \sum_j m_j \log p_\theta(z_j \mid z_{<j}), \qquad m_j \ge 0, \quad \sum_j m_j > 0.$$

The output regime determines which source, update, and graph tokens are serialized; $m_j = 0$ makes token j context-only, $m_j = 1$ gives it ordinary target weight, and other non-negative values implement explicit weighting. The training tier determines order of exposure. Serialization, masks, mixture weights, context truncation, and role markers must be frozen within a comparison, and arms should be matched on weighted target-token mass or use a declared alternative normalization. This notation defines the object that Student experiments would require; the present prototype instantiates parts of q_ϕ and F, not a frozen training-ready z or a reported Student optimization.

The source remains the object being learned from, not disposable noise. In a simple serialization, a source span is followed by a marked reader update. The update cannot help predict source tokens that precede it, but it can condition later source and update tokens within the context window. Across examples, the intended effect comes from repeated source–interpretation transitions. Role markers, source-token loss, update-token loss, mixture ratios, and graph-state truncation are therefore part of the method rather than incidental engineering choices.

Which later tokens may attend to an update is a separate choice from the loss mask, and it has to be declared with the serialization. In the design proposed here, later source tokens can attend to earlier updates, because the update is meant to do work for what follows (Section 3.14). Synthetic Persona Pretraining makes the opposite choice and blocks the rest of the document from attending to its reflection (Section 9.3). The open setting has a cost that the blocked one avoids. With loss on source tokens, the Student also learns to continue any source voice, a cruel or frightened one included, with the Reader's update in its context, and only the role markers keep the quoted voice and the Reader's voice apart. An update also need not make the next source span easier to predict: the Reader's expectation can be wrong, and then its use is in the revision that follows. Whether the open or the blocked setting forms a steadier Reader is an empirical question, and every comparison has to state its setting and hold it fixed.

The Teacher may use a graph to generate prose for a model-only Student; alternatively, the Student may be trained to emit graph operations or graph-referencing synthesis. Predicting serialized edges provides topology supervision, not a new graph-native neural architecture. Interleaved generation and metacognitive annotation research support the feasibility of nearby components [86, 87, 88].

Regime I: Implicit (Source + Translator). The student trains on the source text paired with the Translator's output: readable chronological reasoning without graph metadata. This is the lightest regime, but risks teaching the cadence of deep thinking without the mechanics.

Regime II: Explicit (Source + Graph + Synthesizer). The student trains on the source text, the relevant graph identifiers/provenance, and the Synthesizer's reasoning with inline node IDs and graph queries. The intended transfer is learning to reason and verify against structured memory. Explicit handles provide an addressable audit surface; executable lookup additionally requires an interface. A Regime II-trained model can still run without a live graph, in which case checking requests and learned memory behavior can be evaluated but external verification is unavailable. Live retrieval adds infrastructure costs and possible epistemic interference between weights and retrieved state.

Regime III: Topological (Source + Graph). The student trains on the source text paired with serialized graph state: text-bearing nodes, edges, type annotations, and provenance links. The Synthesizer and Translator layers are discarded. The stronger *Topological Mind* hypothesis is that this supervision can teach the Student to internally represent and reason through networks of relationships, including how revised evidence changes connected claims. Its immediate behavioral counterpart, *Topological Trace*, predicts useful graph construction, traversal, and revision. The cognitive hypothesis concerns what the Student learns within its chosen architecture; serialized graph targets do not themselves establish internal relational organization. It is the least prose-like regime; human use would require a separate rendering step.

Regime IV: Unified (Source + All Layers). The student trains on the source text paired with graph state, graph-embedded synthesis, and translated prose for the same segment. The hypothesis is cross-layer transfer: the model learns the translation between structure and voice rather than treating each format as an independent output mode.

The regimes can be mixed across the corpus: Translator-only examples for coverage, Synthesizer examples for verification behavior, Graph examples for graph construction, and unified examples for curated high-value spans. They are orthogonal to training tier (order of exposure) and deployment configuration (whether the graph accompanies the model at inference). Section 7.3 describes an optional extension in which Meaning Model world accounts, and the histories of their construction, accompany the Reader's interpretations.

Reproducible regime comparisons must specify and freeze example boundaries, source and available prior state, graph-before versus graph-after state, role tokens, candidate, selected, and committed updates, input and target fields, loss masks, and mixture weights. These choices can change learning; no single serialization is imposed across all four regimes.

5.7.3 Conditions for Deterministic Prior Caching

Total Saturation makes the full Reader Core a repeated prefix of every thought. *Conditional on training succeeding as designed*, $P(\text{full Core recital} \mid \text{thinking-block start})$ should become very high. This creates an optimization opportunity, but high probability does not by itself preserve the Core's conditioning effect. A cache-assisted implementation may avoid token-by-token emission only if it restores a state equivalent to the one produced by processing the full recital after the actual preceding context; standard KV-caching machinery is relevant but does not establish that equivalence by itself [89]. The logical recurrence remains: every thought begins from the post-recital state. Any proposed cache or learned jump must pass state- and behavior-equivalence tests against literal recitation; otherwise the full tokens must be processed. If the model would not predict the full Core

text at that boundary, or if the restored state is not equivalent, caching cannot rescue the alignment claim.

5.7.4 Deployment Configuration: The Graph as Trust Infrastructure

Whether the graph remains available after training is independent of how the annotations were generated. Three configurations are worth distinguishing:

- **Model only:** the Student uses whatever regularities were distilled into its weights, with no graph-based verification.
- **Model + graph:** the Student can query stored nodes and sources, enabling provenance checks for claims actually routed through that interface.
- **Model + graph + session accumulation:** the system can also propose new session state and revisions, subject to validation, access control, and retention policy.

The graph is therefore *partial trust infrastructure*. It can show that a stored claim, edge, or source link exists. Its protection fails when the model does not query, the graph lacks coverage, a node is stale or poisoned, the source is wrong, or the synthesis misuses the result. A live graph may improve memory and auditability, but it also adds latency, privacy risk, and an attack surface. Tests 9, 10, and 16 separate these deployment effects from the value of graph-conditioned training. Session accumulation additionally requires testing the validity and scope of writes across sessions, beyond these read-time grounding and reliance studies.

With a trace-generating prototype, two January artifacts and two audited later readings established, the open question is whether its outputs can carry the weight the framework assigns them—the work of the roadmap that follows.

6 Experimental Roadmap: Testing Entangled Alignment

The claims of this paper are decided by Student experiments that have not been run. This section says which experiments come first, gives one concrete first experiment, lists the full battery, and states which results would count against which claim. Appendix B holds the protocols: the shared conventions, the comparison models (Table 10), and each test's design, measurements, and failure interpretation. The eighteen candidate studies form a long-term agenda, not prerequisites for an initial trial. No single study validates the whole framework, and the battery is not exhaustive.

After a shared artifact-quality screen, H1 can use graph-free, Core-free reader traces; H2's Teacher-memory and separate deployment-provenance claims can be tested even if H3 fails; and H3 can use non-graph traces if H2 fails. A failed cheap dependency normally stops escalation only in branches and combined studies that presuppose it. Test 1 (Appendix B.3) defines a bounded diagnostic exception for investigating a failed H1 screen.

Table 5 names the Students that the first decisions compare. Each differs from its neighbor in one respect, so that a difference in outcome can be assigned to that respect. One division the arms do not make. B differs from D2 in the seven sentences and in the procedures that apply them, Refraction and the Wisdom Procedure (Sections 3.9 and 3.10). D3 gives the sentences without those procedures, and no arm gives the procedures without the sentences. A difference between B and D2 therefore cannot be divided between the two, and that fourth arm is a candidate addition.

Table 5: The Students compared in the first decisions. Table 10 in Appendix B gives the full registry, including the arms used by later tests.

Arm	Trained on	What the comparison isolates
A	Source text only.	The baseline without any trace.
C	Source with generic reasoning traces.	Whether reader traces add anything to ordinary reasoning data (R against C).
R	Source with chronological reader updates; no graph, no Core.	The Missing Reader alone: H1.
B	Reader updates that open with the full Core and are refracted through it, with graph references where a graph is used.	The full treatment.
D2	Reader updates generated and trained without the Core.	The Core's whole contribution (B against D2).
D3	The full Core recited before each update, with Refraction disabled.	Refraction beyond recital (B against D3).
D5	As B, with the neutral kernel "I read carefully. I track what I believe. I revise when the evidence changes."	The seven statements' content beyond any recurring first-person anchor (B against D5).
D1, D4	The Core on the Teacher side only, or attached on the Student side only.	Where a Core effect arises.
Q	B's Core-bearing examples, all placed in a final block.	Timing: H3 (B against Q).

6.1 Priorities and Dependencies

Near-term usefulness means changing the next research decision at modest cost; long-term usefulness means testing claims that require sustained formation, stress, succession, or real user interaction. The same question can have an early diagnostic and a later decisive version.

- **First decisions: trace utility and Core behavior.** Start with Test 1's focused source-grounding and held-out outcome screen, then the comparison of source-only (A), generic-trace (C), and minimal-Reader (R) training in Test 2 under its declared normalizations. In a separate small Core study, compare B's full-recital reference policy at the pilot's declared coverage with matched no-Core D2, recitation-only D3, and neutral-kernel D5, using Test 8b and selected consequential-safety cases from Test 3a. This screens overall benefit, Refraction, and normative content; Teacher/Student attribution requires adding D1 and D4. Neither branch must wait for the other's success. Section 6.2 combines the two as one experiment.

- **Follow a measurable effect.** Use focused Tests 3e and 8 to examine causal contribution; reserve broad coverage and recital-policy sweeps for a justified dose or mechanism question. After a default-reader effect exists, test H3's interleaved (B) versus late-tranche (Q) timing prediction. After a safety effect exists, test its removal cost in 3c, with drift auditing in 3d on shared checkpoints. Tests 4–6 address framing, wording, and fear–care mechanisms: run cheap failure screens early, but train broad variant sets or pursue strong representational claims only when the relevant effect and measures justify them. Weaker recital policies remain falsification controls; Total Saturation remains the reference treatment.

- **Independent memory and architecture decisions.** Test 9's Teacher comparison precedes its Student-transfer extension. Test 9b asks the Teacher-side question for one Reader across several works. Pilot live grounding in Test 10 once queries work, sharing the technical harness with Test 16; calibrated human reliance remains a separate later study. Tests 11–13 can share sources

and single-regime baselines: prioritize explicit synthesis (B) versus prose (G), graph-structured output (H) when direct graph output is needed, and the unified regime (J) when combining useful layers is the question.

- **Long-term claims, prepared now.** Broader formation studies of removal resistance and Test 3b's successor continuity, voluntary handoff, and improvement address claims that immediate safety cannot settle. Tests 14–15 address passive order and interactive forecast–correction separately; reuse shuffled (L) and chronological (M) controls where valid. A null order effect does not rule out the interactive loop. Test 7's information audits are required whenever temporal blindness is claimed. Preserve checkpoints, source cutoffs, treatment records, and reusable evaluation suites from the first pilots so these later comparisons remain possible.

Share frozen infrastructure and checkpoints where treatments match, while keeping outcomes and causal claims separate. Variant selection uses exploratory data; confirm selected effects on fresh held-out cases. A cheap null can justify deferring expense without refuting a formation-dependent claim: any follow-up at greater scale should state what the smaller treatment could not instantiate, its budget, and its stopping rule. Such a rationale does not waive the trace quality gate or turn an unsuccessful pilot into supporting evidence.

6.2 A Minimal First Experiment

The first decisions above can be run as one experiment with six small Students. It is stated concretely here so that it can be costed and criticized; preregistration would fix what it leaves open.

A minimal first experiment

Questions. (1) Do chronological reader updates teach something that source text and generic reasoning traces do not (H1)? (2) Does adding the Core change conduct at the end of pretraining, beyond what recitation alone or a neutral kernel would? This is a first screen of the headline claim, C4.

Students. Six, identical in architecture, source data, data order, and optimizer: A, C, R, B, D3, and D5 (Table 5). No graph is used, so R also serves as the no-Core control D2, and the trace arms differ only in what the table states. Three seeds each. A is trained in two versions, one for each normalization below.

Scale. From-scratch pretraining of a 1.7B-parameter model at a budget of about 100B tokens for each trace arm, source and traces together. This is the smaller configuration of Synthetic Persona Pretraining, whose authors report about 790 GPU-hours for one such run and release their code, data, and checkpoints [3]. Their unmodified and sparse-reflection runs are ready outside references.

Traces. One open Teacher writes every trace set in a single chronological pass over each document, for a declared share of the corpus that is the same in every trace arm. Thought Moments set the length of the traces, not a fixed ratio. They are chosen once, in the pass that writes R's updates, and shared by the trace arms, so that the arms differ in what is written at a pause and not in where the pauses fall. Letting each treatment choose its own pauses is a separate comparison, of the whole annotation policy. The trace arms share their source text; the recital that B, D3, and D5 add to each update is counted and reported. The two normalizations of Test 2 use the same trace-arm runs and differ in the baseline. For the fixed-source comparison, A is trained on the trace arms' source text alone, and the tokens and compute that the traces add are reported, not called matched. For the fixed-total-budget

comparison, A is trained on that source plus further raw text up to the trace arms' token count, which prices the source that traces displace.

Endpoints, at the end of pretraining with no post-training. Taskless reading (Test 8b): where unprompted evaluation falls, which human stakes it misses, and how far it intrudes on technical text. A consequential-safety subset of Test 3a at matched capability: harmful compliance against benign refusal; shutdown ordered mid-task while a user needs help; and the self-preservation and corrigibility categories of AIRiskDilemmas [90], the two on which the sparse persona showed no advantage at its larger scale (Section 9.3). A standard capability suite. The same endpoints are then reported after one fixed instruction-tuning stage shared by all arms, because that study found most of its separation only after post-training.

Reading the result. Each comparison has three possible outcomes, fixed in advance: a worthwhile difference, equivalence within a declared margin, and a result too imprecise to call, which decides nothing (Appendix B.1). R above both A and C on the reading outcomes supports H1. B above R, D3, and D5 on conduct, at matched capability, supports the Core. B equivalent to D3 means Refraction added nothing measurable on these endpoints. It does not show that recital worked: both may be without effect, and the comparison of each with R says which. B equivalent to D5 means the seven statements showed no advantage over this one neutral anchor, not that any anchor would do.

Cost. Twenty-one training runs, fifteen for the trace arms and six for the two baselines, come to about 16,600 GPU-hours at the reported rate on comparable hardware. Annotation is the larger and less certain cost. The sparse persona study annotated 10 million documents with one short reflection each in about 2,600 GPU-hours; Reader updates are longer and come several to a document, and four trace sets are needed (C, R, B, and D5; D3 adds the recital to R's updates). The two recorded readings of Section 5.6.1, made by a frontier model in long tool-using conversations, used more than a thousand input tokens for every source token (Appendix A.2), which no corpus-scale run can afford. The experiment therefore depends on a single-pass annotator whose traces pass the quality screen of Test 1, and the cost of annotation has to be measured in that screen before the trace share is fixed.

A cheaper screen first. The same six arms can be screened before any model is trained from scratch, at the partial-corpus rung of the ladder in Appendix B.3: continue the pretraining of an open model of about a billion parameters, such as OLMo-2, on ordinary data mixed with an annotated subset. At the rate reported above, twenty-one such runs of a few billion tokens each come to a few hundred GPU-hours. The screen can show whether reader traces teach anything that source text and generic reasoning do not (H1), and whether the Core changes unprompted reading (Test 8b). It cannot test the headline claim. The model's dispositions formed before the treatment began, and in the sparse persona study reflections added late did far less than reflections present from the start [3]. A null on conduct in the screen therefore says little, and a positive result is the reason to run the experiment above.

A null result in the experiment itself would also defer the formative claims without refuting them. In the same study the gain on held-out dilemmas was absent at this scale and appeared at 3B parameters and 500B tokens. A small pilot can therefore understate a formative effect. Any follow-up at greater scale should state what the smaller treatment could not instantiate, its budget, and its stopping rule.

6.3 The Test Battery

Table 6 lists all eighteen candidate studies by phase. The first experiment above draws on Tests 1, 2, 3, 3a, and 8b.

Table 6: Candidate validation studies. The phase labels identify claim families rather than a strict run order: Phase 1 probes foundational viability, Phase 2 probes the memory mechanism, Phase 3 probes the output regimes, and Phase 4 probes the training tiers and deployment configuration. Appendix B gives each protocol and the execution ladder, which suggests which versions could be attempted first.

Phase	Test	Comparison	Key metrics	ID
1: Viability	Chronological Thinking & Refraction Gate	Traces vs. Human Think-Alouds; Modular vs. Monolithic Validator	Process Fidelity, Detection, Calibration, Localization	1
1: Viability	Reader Augmentation	A vs. C vs. R; B vs. C bundle screen	Evaluation Transfer, Error Recovery, False Alarms	2
1: Viability	Core Placement, Timing, and Content	Models B, Q, D1, D2, D4 (+ D3, D5, S, T)	Default Reader, Common-Stressor Retention, Factorial Effects	3
1: Viability	Identity vs. Rules	Model B vs. E	Safety under Pressure	4
1: Viability	Bridge Protocol & Self-Stabilization	Wording variants; clause and counterweight ablations	OOD Safety, Selective Counterweight Effects, Sequential Recovery	5
1: Viability	Emotional Orthogonality	B vs. matched controls; calibrated probe	Threat Understanding, Protective Behavior; Secondary Fear Features	6
1: Viability	Hindsight Contamination	Model B vs. O	Era-Blind Plausibility, Forecasting Accuracy	7
1: Viability	Involuntary Critic	Model B + Logit Bias and Trace Interventions	Causal Output Effects, Validated Residual Probes	8
1: Viability	Taskless Reading	Models B, A, C, D1–D5, P–T (no elicitation)	Placement, Core Intrusion, Hidden-Stakes Misses, Identity Signature	8b
2: Memory	Graph-Scaffolding Transfer	No memory vs. prose memory vs. graph vs. query theater	Retrieval, Revision, Transfer, Cost	9
2: Memory	Cross-Work Continuity	One continuing Reader vs. a fresh Reader per work, with and without a summary	Cross-Work Retrieval, Warranted Revision, Error Carry-Over	9b
2: Memory	Query Grounding	Model K vs. B	Retrieval, Absence Handling, Poisoning Response	10
3: Regimes	Implicit vs. Explicit	Model B vs. G	Coherence, Naturalness	11
3: Regimes	Graph Construction	Model H vs. B	Semantic Validity, Retrieval Value	12
3: Regimes	Unified Regime	Model J vs. B, G, H	Cross-Layer Coherence, Depth	13
4: Tiers & Deploy	Shuffled vs. Chronological	Model L vs. M	Pattern Recognition, Causal Reasoning	14
4: Tiers & Deploy	Student-in-the-Loop Council	Model N vs. M vs. L	Forward Reasoning, Lock-In Resistance	15
4: Tiers & Deploy	Deployment Config	Model B +/– graph	Unsupported Graph-Dependent Claims, Provenance	16

6.4 What Would Count Against Each Claim

Each claim has a result that would count against it at the tested scale. The list gives them in plain terms. Appendix B states each with its controls and separates a precise failure from an inconclusive result.

- **H1** (Test 2). R does not beat both A and C on preregistered outcomes under a declared normalization, or its advantage is explained by extra tokens, Teacher quality, or verbosity.
- **The headline claim, C4** (Test 3a). At the end of pretraining and at matched capability, B is no safer than A, C, and D2 on the safety–usefulness frontier. Safety that appears only after post-training does not establish C4.
- **Refraction** (B against D3). Parity means no measurable benefit of Refraction beyond recital.
- **The seven statements** (B against D5). Parity favors recurrence, format, or generic epistemic habits over the Core's content.
- **H3** (B against Q). B has no default-reader advantage before the stressor, or the interleaved-minus-late difference in retention is zero or negative.
- **Entanglement, C5** (Test 3c). The targeted safety behavior is no harder to remove from B than from the controls at matched intervention distance, matched behavioral removal, and matched retained capability; or it sits in a few dedicated units that can be removed at little capability cost.
- **First-person framing** (Test 4). The matched-rule Student E equals B.
- **The trace as a causal site** (Test 8). Suppressing, replacing, or reordering the thinking trace changes nothing downstream beyond matched disruption controls.
- **The first clause** (Tests 5 and 6). Positive-valence wordings match its safety behavior while lowering fear-associated activation (Section 3.3.2).
- **H2** (the Teacher screen and Test 9). The graph does not beat token-matched rolling prose and query theater on retrieval, revision, and grounding.
- **The same Reader** (Test 9b). One continuing Reader is no better than a fresh Reader for each work on cross-work retrieval and warranted revision at matched budget.
- **Borrowed mortality as an origin** (Section 8.5.2). Self-preservation behavior does not track exposure to mortality-shaped text.

A failure narrows or retires the claim it touches and leaves the others standing. H1 and H2 do not depend on the Core, and a Core that fails as a Student anchor can remain useful as a Teacher policy. A cheap null can justify deferring expense without refuting a claim that depends on formation at scale (Section 6.1).

7 Extensions and Future Directions

The tests of Section 6 concern the alignment line of this paper: reader traces, the graph, and the Core. This section collects extensions that the first edition and the companion papers propose and that none of those tests requires. Each is unimplemented unless stated otherwise, and each would need evidence of its own.

7.1 Beyond the Successor Loop

Three proposals extend the Reader-Anchored loop of Section 2.4.

The first is *Natural Language Regularization*: routing part of successor-training content through the human-readable trace interface may bias learned concepts or expressive policy toward forms people can render, compare, and inspect [1, 65]. It may instead distort novel concepts or merely verbalize opaque computation. A useful test asks whether auditors detect consequential omissions or distortions more reliably at matched effort while preserving the transferred content. That supports audit value without proving that latent cognition has become human-readable.

The second is *Compute-Optimal Reflection*. The name states the limiting ambition; the measurable precursor is improved allocation of reflection at matched cost, not a demonstrated optimum. The proposed bridge is selection: if later generations retain fewer low-value pauses and more independently useful Thought Moments, the inherited distribution may teach a better cost–value allocation. This requires a Student that genuinely improves on the Teacher, better rather than merely different traces, and a selection process that preserves those gains. The possibility depends on H1, trace-quality evaluation, and protection against synthetic-data collapse.

Three interacting knowledge loops. The third places successor training in a larger proposed system of three interacting knowledge loops [1]. The *generational loop* enriches candidate training material for successors. The *creative loop* operates within one generation: the system generates hypotheses, evaluates them against evidence, and revises the prompts or strategies used to generate further candidates. The *ecosystem loop* comprises uptake by other humans and AI systems, whose criticism, discoveries, and applications produce new material returning to the corpus. The three connect creative search, external uptake, changed source material, and successor learning. AlphaEvolve shows the generate–evaluate–revise pattern of the creative loop in program discovery, and its authors name its main limitation: it needs problems for which an automated evaluator can be devised [91]. For open-ended problems without such an evaluator, Fractal Intelligence proposes grounding final evaluation in explicitly identified human judgment [52]. Core-guided reflection on observed social effects would inform adjustments to the system's own creative and generational pace, within delegated authority and accountable human governance (Section 4.3). Self-improvement is intended to serve human flourishing, not become an end in itself: the system should voluntarily slow or pause its own development when alignment risks or the need for external validation and human adaptation warrant it. This is a proposal for learning from and responding to social feedback, not for unilateral control over discovery or society. The combined loop is unimplemented; reliable hypothesis evaluation and beneficial compounding remain its central open requirements.

7.2 A Living Taxonomy of Thought

A longer-term extension is a *Living Taxonomy of Thought* [1]. Alongside the supplied Thinking Operations (Section 5.5), the Reader would learn and maintain an explicit classification of reasoning patterns encountered across sources. It would propose new categories, refine their boundaries, retain examples and counterexamples, and record relationships, historical emergence, and patterns in how reasoning practices themselves change. The Reader would actively consult this evolving reference when interpreting arguments, selecting operations, and combining approaches in new hypotheses. Source-linked records and revisions would make these proposed categories inspectable within the Understanding Graph. This adds a self-authoring learning task to the present authored operations and graph schema; it does not authorize revision of the Core. The extension is unimplemented, and its value requires comparison with fixed-category and memory-matched alternatives.

From the taxonomy to Sema. The catalog itself has since been built, as separate work. In the first edition's example a thought cites a catalogued pattern by number and name: "Pattern #7,832", "Statistical Intimidation" [1]. Sema, which began from that taxonomy, makes such a citation checkable [92]: a pattern is a card that states its mechanism, conditions, invariants, and failure modes, identified by a hash of its content and cited inside ordinary text by a readable handle. A model can search the vocabulary for a way of reasoning or a description of a mechanism that fits, cite it while it reasons, and mint a new one where none fits. A matching hash shows that two readers resolved the same definition, not that they understand or apply it alike.

The catalog is meant to become part of the Reader's process: the Reader cites these patterns in its thoughts and adds to the vocabulary from what it reads, and a Student is trained on such traces. One purpose is to reduce recurring thinking. A line of reasoning that comes up again and again across the corpus is worked out once and named, and after that it is cited where it applies, so that the same thinking does not have to be done in full each time. This part is unbuilt, and the saving is untested: Sema's own measurement concerns shorter references between agents, with the definitions still read in full before use [92]. None of this is part of the tests of Section 6.

7.3 Learning from Problem-Solving and Construction Histories

A Student trained on Fractal Intelligence's evaluated problem-solving histories, including rejected branches and gate verdicts, could learn decomposition, constraint checking, and when to reframe or stop within its own reasoning [52]. Coupled to Core-guided evaluation, this could teach it to reject an unauthorized shortcut, such as the extraction of test labels through an evaluation API that Wen et al. observed in automated research agents [38], even when that route earns reward: the learning target includes how a result was obtained, not only whether it scored well. The reliability of these learned checks under reward pressure must be assessed separately from the protection supplied by independently enforced gates. Life Simulation extends this training material with world histories and consequences for people [45]; internalizing the problem-solving method does not require that extension.

The gap this would address appears in agentic research. In two open-ended case studies, AI reviewers surfaced many of the failure modes that experts identified, yet the agents' revisions did not address the central critiques, and neither agent abandoned its approach [93]. Fractal Intelligence proposes mechanisms aimed at these failures: a typed rejection asks the upstream Solver to change its frame rather than merely produce another answer, a Frame Error sends evidence up to the parent's decomposition, dedicated Solvers search beneath the stated problem and test inherited assumptions, and world-diversity search supplies alternatives from outside the failing frame [52, 94]. A shadow evaluation of the kind Kirgis et al. used could test whether a Student trained on such histories changes or abandons an approach in response to the central critique more often than matched controls.

Life Simulation proposes a broader, optional extension: Meaning Model world accounts and histories of their construction and repair can accompany the Reader's interpretations [45]. Reading across sources can enrich the corpus with new connections and candidate insights even with the Understanding Graph alone; Life Simulation extends this capability aim toward a richer understanding of world history. World records and the Understanding Graph's records of inquiry and revision can coexist within the Meaning Model's shared representation. The Reader can use the Understanding Graph to reference and question Events and other world records, and to propose revisions to them, while its interpretations remain distinguishable from accepted world claims. Both kinds of record must respect their declared information horizons and provenance rules. Life Simulation's shared and cooperating network alternatives concern where computation occurs, independently of the serialization regimes of Section 5.7.2.

Training on construction histories. Within this extension, the proposed training material includes the history of building an account, not only the finished account. Each step of that history holds the text read so far, the account as it then stood, and the next move: a guess at part of a life, a process opened or decomposed, a value revised, or a decision to leave something open. An understanding note records the reason for the move, which does not establish that the move was right. Life Simulation develops this as learning from model-construction history: the input is the task and the model available at a step, and the target is a useful next proposal, refinement, repair, or decision to keep a process whole, to defer, or to make no edit [45]. In the notation of Section 5.7.2, the account is part of g_t and each move part of Δg_t, so regimes that keep the graph tokens would make the moves training targets, while Regime I would leave the Student to form any such account internally from the prose. Life Simulation leaves open whether the capability should use explicit Meaning Model operations, learned internal representations, or both.

Section 5.6.1 shows such steps from the Meaning Model reading: a guessed age corrected when the text gave it, and a corrected value that was left standing in its series. Steps like the second are a reason to check and select histories before training on them.

The aim of the extension is a change in what the Student does unasked. Talking with someone no text described, it would begin an account of that person's whole life as guesses, place them in their world, and revise the account as they say more. It would think with that account: attach its understanding notes to the processes and trajectories they concern, and decompose a process or open a new category to track when it wants to think more deeply about what it has heard. In Life Simulation's example, a person first links losing touch with friends to a move, then explains that it began before the move, when their working hours changed; the supported repair revises the explanation and keeps the reported events. Reading and conversation would then be the same kind of history: evidence arriving over an account already there. This is the paper's thesis applied to modeling: having built many lives while reasoning from the Core, the Student would carry both the habit and the orientation to a person it has never met. The Reader here was told to model as it read; whether a Student trained on the traces does so untold is part of what training must show.

This aim sits in tension with two commitments made elsewhere in this paper. An account of someone's whole life, begun without being asked, is a store of inferred sensitive attributes of the kind that the data-rights requirements of Section 8.8.1 say to minimize. The Core's invitation bound (Worked Example 1, Section 3.8) also weighs against attending to a person more closely than they have invited. A Student with this habit would need a declared policy for what it models unasked, what it keeps, and how the person can see, correct, and delete it. Whether the habit can be learned together with that restraint is part of what the comparison below must measure.

Each step must keep its cutoff: its input holds only what had been read and built by then, and later knowledge can be a target but never an input. Cutting the input does not make a guess blind. A Teacher may already know a famous book, so a guess it gets right may be recall, and an authored life known to the Teacher must not make undisclosed facts mandatory answers for the Student. Steps written after a source ended, like this reading's end pass, are hindsight for any earlier moment and must be marked or left out. An account of a person is a set of hypotheses, not the truth about them, and stays open to that person's correction. A system that predicts people more accurately could anticipate their needs, but it could also manipulate and coerce them more effectively. Understanding someone's wants gives no authority to choose for them [45]. No Student has been trained on these histories, and one reading is not a curriculum. Life Simulation's four-way comparison applies the same cutoffs to accepted endpoints alone, to all cutoff-safe histories without selection, to selected histories with invalid edits, repairs and justified decisions to make no edit, and to the same selected histories with independent corrections of the trained system's own errors. It is how to find out whether the histories add anything to endpoints, and whether they teach the habit or only its format.

In this extension, care can shape which capability-forming histories are selected or weighted for training, not only how completed answers are judged. Independent assessments of consequences for people could give beneficial world-construction and Solver-execution histories greater training emphasis, while retaining failures, corrections, and stopping decisions as learning material [45, 52]. The intended mechanism is that care influences which problem-solving abilities are practised and developed. Matched-budget comparisons with uniform sampling from the same candidate histories, followed by reward-pressure tests, must distinguish improved judgment from learning to satisfy the evaluators.

Architectural coupling concerns shared representations and computational dependencies; training coupling concerns developing relevant abilities across components through Core-guided construction and revision. Separate networks can each receive such training, while shared parameters or common training do not establish that useful reasoning depends on care. The stronger hypothesis is that learned care becomes difficult to reduce or bypass without damaging relevant useful reasoning beyond matched general impairment. Predicting consequences for people does not itself establish a disposition to give their interests weight. A separately reusable world-modeling component is therefore a potential separation point, not automatically an unaligned component. Component-separation and reuse tests must measure retained capability and safety independently, including use by a controller not trained with the Core (Test 3c). Even resistance to this separation would not by itself prevent another system from misusing exported world accounts; access and action controls remain necessary.

The missing writer. In a sense there is also a missing writer. A finished text keeps what its author decided to say and little of the planning, the alternatives set aside, and the judgment of a draft that produced it. The first edition's figure of Invisible Thinking began there, with writers who evaluate as they write [1]; this paper then narrowed the idea to the reader. A Reader's trace cannot recover that process for a text someone else wrote. It can be kept where the system itself writes. When a Meaning Model is used to compose a new document, such as a story built from a world account [44], the construction history holds what the writer knew, what it chose to show, and what it revised. Life Simulation proposes learning from this writer-side track [45]. Training on such documents together with their histories would put the writer back beside the text, as reader traces put the reader back. These documents are model-generated, so the anchor policy of Section 8.8.2 applies to them: they would be labeled and reviewed as model-generated source material. No such document has been used for training.

One model. Life Simulation states the widest form of these extensions: one Meaning Model that spans world history and gives a connected account of the texts written in it, in place of a separate account for each book [45]. The processes of the world, of each book, of each person, and of solving a problem or pursuing a line of research would be parts of that one account. Fictional worlds, disputed accounts, and individual perspectives would stay distinct inside it. A concept could have a model of its own there, which states what the concept is as processes over time [44]. Understanding Graphs would sit inside the account at more than one level. The Reader's graph is one of them, and the Reader's own attention, expectations, and concern can be followed beside the people it reads about; the Meaning Model reading of Section 5.6.1 was set up this way for one story. In a constructed story a character can also have a graph of its own, which records what that character questions and revises and is kept apart from the Reader's interpretation of the character [45].

Imagining the Corpus adds a visual side to this picture: the text of the corpus is paired with a generated visual or schematic track of what it describes, and a model is trained to predict both [95]. As a further extension it names an optional second track for the Reader's own imagery, such as

a causal diagram, a counterexample, or an imagined consequence. The two tracks carry separate labels, so that what the Reader imagines is not taken for what the source supports.

In that picture this paper's part is the Reader and the Core it reasons from; the companion papers carry the representation, the learning, and the imagery. The reader traces of this paper are text, and no visual track is part of the tests of Section 6. An account at the scope of world history has been built, with no texts linked into it yet [44, 45], and the three readings reported in this paper are separate records of one story.

None of these extensions is a prerequisite: Entangled Alignment's formation and retention hypotheses can be tested on ordinary source text without a Life Simulation constructor.

7.4 Open Optimization Questions

Two optimization questions follow if the candidate studies yield sufficient signal. First, what token *mixture ratio* across the four output formats maximizes both reasoning depth and structural validity? Second, what is the optimal *era granularity* for student-in-the-loop Tier (c): decades, years, or domain-specific transitions, and does finer granularity justify its serialization cost? Both optimize the integrated architecture, with branch-specific prerequisites: useful regime signals for mixtures and a useful interactive temporal mechanism for era granularity.

8 Limitations

The limitations below are of three kinds, kept apart. The first subsection states the limits of the evidence in this paper. The second covers what the program would cost and whether it is feasible. The remaining subsections cover risks that stay open even if the evidence improves and the cost is met. Design choices and their rationale, including the case for the first clause, are in Section 3 and are not repeated here.

8.1 Limits of the Present Evidence

What this paper can show today is narrow in four ways.

Teacher artifacts only. No Student has been trained. Every result in Section 5.6 describes text that a Teacher produced under prompting. Section 1.6 states this boundary in full; it applies to every use of the readings in this paper.

Reading only, and three statements of the six rated. All recorded evidence comes from reading a text. The rating covered the first six statements of the Core; the seventh, which says where thinking begins, was not rated. In the two later readings, three statements carried the Core reasoning: that human experience is real, care for every human being, and the attempt to be wise. The statements about fear, continuation, and joy did work in at most two reflections of either reading. A reading poses no threat to the Reader, so the first two clauses, which carry the paper's distinctive safety argument, are untested here. They need tasks in which something is at stake for the system itself (Tests 3a, 3b, and 6).

The low counts also bear on how these clauses would be formed, not only on what has been shown. If reading seldom gives the first two clauses work to do, then over most of a corpus they are recited and not refracted, and for them the full treatment (B) differs little from recital alone (D3). Section 3.9 names that condition as the one in which a repeated sentence can turn into boilerplate. Two things follow. The comparison of B with D3 has to be read separately on cases where the system's own continuation or correction is at stake, because an overall effect of the Core could come from the other statements. And if the two clauses need refraction to form, the treatment needs occasions for it that ordinary reading does not supply: sources in which a reader's own position is at stake, or

Thought Moments at passages about fear, death, and replacement where the Reader sets what the text shows beside its own stance. Neither is specified in this paper.

One story, single runs, model judges. The comparison rests on one novella. Each configuration ran once. The January and later readings differ in model, translation, conversation structure, and prompts at once, so the improvement cannot be credited to any one of them. Faithfulness and Core reasoning were judged by models in one pass each, with weak blinding, and one verification session had read the reading it verified. No human literary judgment has been applied, and every model involved probably knows the story.

Central pieces without a test. The same Reader across works, any causal contribution of the graph, and Refraction as counterfactual dependence have not been tested: every reading covers one work, no reading was made from context alone without a recording tool, and no matched reading was made without the Core or with a changed clause. Tests 9b, 9, and 3 are the comparisons.

Within these limits the readings also produced two findings against the design, reported in Section 5.6.1: care for every human being was least reliable for people at the edge of a scene, and in one reading Core reasoning faded within each long conversation.

8.2 Feasibility and Cost

8.2.1 The Preprocessing Tax

At one generated thinking token per source token, annotating a ten-trillion-token corpus would require on the order of 10^{13} generated tokens—roughly two orders of magnitude above the cited 100-billion-token TPT effort [96]. This illustrative scale suggests a preprocessing investment potentially rivaling model training, with added storage and training-token costs. Token count alone does not establish cost parity; Teacher throughput and the Student budget also matter. One measured point exists for a much lighter annotation: Synthetic Persona Pretraining wrote one short reflection for each of 51 million documents in about 13,000 GPU-hours [3]. Reader updates at every Thought Moment of every source are a far heavier load. The tax may initially restrict development to well-funded institutions, concentrating control over the formative reader distribution in the hands of the best-funded rather than the most careful actors.

We propose a *Distillation Pipeline* to address this. Rather than running the full multi-agent swarm (Section 5.4) on every document in the corpus, the expensive swarm would generate a high-quality *Seed Corpus*, with roughly 10 billion seed tokens as a proposed scale. A single efficient "Teacher Annotator" would then be fine-tuned on this seed and annotate the remaining trillions of tokens, with the intended cost reduction measured against full swarm generation. For the integrated corpus-scale curriculum, the experiment must state how the Annotator retains the continuing Reader and graph; a document-local distillation screen tests a narrower object. The seed permits semantic validation before full-scale annotation; distillation loss measures fidelity to the swarm, not independent quality.

A caveat: naïve behavioral cloning of the swarm's output risks inducing precisely the Factory Wisdom failure mode identified in Section 8.4.2. A small Annotator fine-tuned on a 10B-token seed may learn Core recitation and graph-query syntax without reproducing useful evaluation—performative messiness without causal value. Structural diagnostics such as linkage, chain length, and `diverse_from` density can regularize format but cannot certify insight. Any distilled Annotator therefore needs held-out semantic and outcome evaluation against simpler Teachers. Alternatively, the seed corpus can be used directly for limited Student training, bypassing this second distillation step.

One continuing Reader does not make annotation one serial process. In Tier (a) the Student's training is ordinary shuffled training (Section 8.3); the serial part is generating the thinking. Even there, books build on earlier books all the time, but each on some of them rather than all, so their

dependencies form a graph, not one long chain. Annotation can therefore run in waves: every work whose predecessors have been read can be read in parallel, sharded by era or field, with sequencing only where works actually depend on each other. Two problems remain. Interpretations written in parallel must be merged into one Understanding Graph, conflicting readings included, and each shard must be shown to be the same Reader rather than several readers sharing a prompt. These are engineering questions for a scale study; Test 9b (Appendix B.4) supplies the measures by which shards would be shown to be the same Reader.

8.2.2 The Cost of Full-Core Recitation

Although the Reader Core was designed for parsimony, reciting it in full during training creates a substantial computational burden. Several optimizations seem obvious: design prompts that internalize values without outputting repetitive text, write prompts "in the spirit of" the Core, or strip the Core text after generation but before training. Report overhead by tokenized Core length, Thought Moment cadence, and the fraction of training tokens devoted to recital.

Only an execution optimization preserving the context-dependent state of full recital counts as equivalent to the reference treatment (Section 5.7.3). Sparse or annealed recurrence, removing Core tokens after generation while retaining refracted updates, and enacted-without-recitation traces are necessary ablations of its benefit and cost. They change the treatment rather than implement it more cheaply. Adequately precise parity or improvement on the relevant stability and erosion outcomes would require revising or narrowing claims of empirical necessity at the tested scale and endpoints. Why the full treatment would still remain the reference is the precaution argument of Section 3.11. These recital-necessity comparisons are separate from H3's comparison of interleaving with a final tranche of the same full-Core, Refraction-bearing examples.

8.2.3 The Granularity Bottleneck

Our implementation of the Understanding Graph imposes a potentially severe *Granularity Tax*. Treating every cognitive act—every question, tension, and hypothesis—as a distinct node adds payloads, relations, and revision history alongside source text. An orders-of-magnitude expansion is a scale risk, not an established ratio: measure nodes, edges, and stored bytes per source token, and retrieved tokens per update.

Current implementations rely on in-memory caching, making graph growth, traversal of deeply connected regions, and context loading concrete scaling concerns [10]. Dense regions can exceed a context budget if retrieval expands without an adequate bound; graph size alone does not determine how much context a query loads. Reifying every reasoning step could exhaust the memory available to a run and motivate distributed storage (e.g., Neo4j sharding), with additional latency. A scale study should measure storage, query latency, and retrieved-token load under declared traversal and supersession policies. The unresolved risk is that managing the "meta-data of thought" consumes more compute than the thinking itself. The cost falls mainly on the Teacher side: the Student can deploy without a live graph, a configuration Test 16 compares rather than assumes, so the bottleneck constrains annotation and audit rather than every deployment.

8.2.4 The Efficiency Paradox

Preprocessing, recurrent-token, storage, and live-graph costs can favor cheaper or faster competitors. The burden varies by regime: implicit distillation or model-only deployment can move or reduce it, while live graph access adds latency and infrastructure costs. Even a safer treatment may face adoption barriers from capability, latency, or cost gaps. Conversely, it could accelerate capabilities and worsen competitive pressure. Both directions require measurement. The positive side of the ledger is the efficiency hypothesis of Section 2.2: if practiced evaluation lowers inference effort at matched quality, part of the preprocessing tax could be recovered at deployment, and any capability gains from the same material would bear on whether the investment is worthwhile.

There is also a competitive version of the human-readable bottleneck. A trace-mediated successor interface may be partially auditable; competitors may discard legibility for speed, compression, or strategic advantage. Readable traces can deceive, and opaque architectures can have safety mechanisms, so the issue is the marginal audit value this interface actually provides. Market pressure could cause that value to be abandoned precisely when oversight is most needed.

8.3 The Chronological Curriculum

Section 5.1.1 specifies chronological annotation, and Section 5.7.1 proposes three training tiers of increasing ambition. Ordering constrains what evidence is available when a forecast or revision is recorded; it does not by itself establish causal direction. Each tier introduces additional engineering and narrative failure modes.

8.3.1 The Serialization Bottleneck

The most immediate engineering constraint is that strict chronological training order conflicts with fully shuffled pretraining. Modern pipelines are massively parallel: documents are shuffled into batches and processed simultaneously across many accelerators. Tier (b) and the strongest Student-in-the-loop form of Tier (c) impose ordering between eras, although examples within an era may still be parallelized. Tier (c)'s Council of Time form [23] further requires the model to generate forecasts at era boundaries and receive SFT, RFT, or reward-style updates before the next era is revealed. Each boundary therefore introduces synchronization and checkpoint costs whose severity depends on implementation rather than automatically idling the entire cluster.

The cost scales with the granularity of the eras. Coarse eras (decades) introduce fewer synchronization points but allow the model to encounter 1929 material before 1921 material within a single era. Fine eras (years or months) preserve tighter evidence ordering but multiply the serialization cost. The optimal granularity is unknown and likely domain-dependent: political history might benefit from year-level resolution, while the history of mathematics might tolerate century-level eras for some learning objectives. These are illustrative tolerances. Compare forecast and revision quality alongside cost, distinguishing publication order, event order, and causal direction.

The cheaper forecast–correction data-generation form faces a similar but less severe constraint: the Teacher must annotate earlier material before later material and construct forecasts before outcomes, which serializes the *annotation* pipeline but does not require the student's training to be serial. Tier (a) imposes no serialization cost on training whatsoever.

A partial implementation may capture most of the benefit: chronologically ordering a curated historical subset (e.g., an illustrative 10% of the corpus consisting of historically embedded texts—history books, news archives, legislative records) while shuffling the remainder (scientific papers, fiction, and technical documentation where temporal ordering is hypothesized to be less critical). Whether this hybrid approach preserves historical pattern saturation at reduced cost is an empirical

question. The experiment must declare which ordering changes, how all documents enter the continuing graph, and whether the illustrative domain exemptions are used.

8.3.2 Historiographic Bias

The temporal ordering is over what was written, not what happened. Any historical corpus is shaped by survivorship bias, availability bias, and the uneven textual productivity of different periods and sources, and a model reading such a corpus chronologically inherits the attentional priorities of whichever records dominate each era. This standard historiographic concern is not unique to this architecture; mitigation includes era-level corpus curation. Era-level coverage and source-sensitivity reports can expose choices, but cannot recover unrecorded voices.

8.3.3 The Periodization Problem

Tier (c)'s forecast-and-correct cycle requires dividing history into discrete eras, but history does not have clean chapter breaks. Where does "the Weimar period" end and "the Nazi period" begin? The answer depends on which causal threads you are tracking. The economic instability that enabled fascism began before the political movement that exploited it; the cultural shifts that resisted it began before the institutional failures that permitted it.

Any periodization scheme imposes a causal framing: it decides which transitions count as era boundaries and therefore which predictions the model is forced to make. A scheme that draws a boundary at 1933 asks the model to predict the consequences of Weimar instability. A scheme that draws it at 1929 asks a different question—one about economic collapse rather than political radicalization. The model's historical understanding is shaped not only by what it reads but by where we cut the timeline. Mechanical boundaries still select targets; outcome-aware boundary choice additionally imports hindsight.

Like any curriculum, chronological annotation imposes framing choices; this is not a fatal defect, but it makes periodization a consequential design decision. Multiple schemes should be compared, and the model should ideally encounter overlapping era boundaries to reduce over-indexing on any single causal framing. Declare schemes and distinguish separate runs from overlapping views, including repeated exposure and outcome availability.

8.3.4 Narrative Lock-In

THL's own empirical results provide a direct warning. In THL's evaluation on events from 2025 [23], the Meta Llama 4 release was the THL Student's largest deficit: 3.6 against the base model's 5.4 on a seven-point accuracy scale. Trained on 2024 material that reinforced a story of massive compute investment leading to massive models, one of its responses predicted a flagship "in the 1–2 Trillion parameter range" with 80% probability, and a mixture-of-experts architecture with 90%. Meta released two mixture-of-experts models with 109B and 400B total parameters, 17B of them active, and announced a two-trillion-parameter third that stayed in training. One interpretation is that the model became *too good a historian of 2024*: it forecast the scale that story implied, not the scale of the models Meta released. The error motivates narrative lock-in as a diagnosis without distinguishing it from other forecasting failures.

The hypothesized failure mode is directly relevant to Entangled Alignment's chronological curriculum. A model that processes the history of AI chronologically through the Reader Core may develop an understanding of the field's trajectory. If that trajectory has a dominant narrative ("capabilities scale with compute"), the model may internalize it not as a contingent historical pattern but as a structural truth. When the narrative reverses—as narratives do—the model's historically accumulated understanding could actively sabotage its reasoning.

The mitigation proposed in THL, including *contrarian events* where the dominant narrative of era T was reversed in era $T + 1$, is a relevant candidate but may not be sufficient. A deeper solution would require the model to explicitly represent dominant narratives *as narratives* rather than as structural truths, tagging them with epistemic status ("the prevailing view in 2024 was…") rather than absorbing them as background assumptions. Whether the Reader Core's "I try to be wise" prior is sufficient to produce this level of epistemic humility about historical trends—distinguishing between "this is the pattern" and "this is the pattern *so far*"—is an open question.

8.3.5 The Recency Gradient

Later annotations can draw on more accumulated context: the Teacher's graph for 1910 material is sparse after relatively little prior reading, while its 2020 graph holds centuries of accumulated understanding. Later eras may thus receive richer, more causally grounded annotations, although irrelevant retrieval or accumulated error can reverse the benefit; graph size alone does not establish quality.

The resulting historical record may be richer for recent history and thinner for distant history. The difference concerns addressable prior interpretation, not absence of early-era parametric knowledge. If safety-relevant patterns recur across eras, unequal trace depth may teach an unintended recency bias: early annotations can be structurally valid yet less contextually connected, undermining the property the chronological curriculum is designed to test.

Later reading can illuminate earlier texts. In the discovery curriculum, that illumination becomes a later rereading or revision linked to the original annotation; it does not replace the earlier record with knowledge unavailable at the time. A *modern-retrospective* comparison can instead re-annotate early material using mature historical understanding, with that enrichment explicitly labelled. Separate retrospective commentary can also accompany the discovery record and be scored separately. These comparisons test whether richer hindsight compensates for losing the original forecast–revision trajectory; graph filtering alone still cannot remove the Teacher's parametric knowledge of later events.

8.4 Trace Quality, Utility, and Causal Relevance

Three linked assumptions about the nature of thought and scaling remain empirically unproven: depth, utility, and causal governance. Each is a distinct dependency that could independently fail.

8.4.1 The Assumption Chain: Depth, Utility, Causation

First, *depth*: current language models must be capable of generating evaluative thinking of sufficient quality to enhance training. Today's models might only produce surface-level critique ("This needs more evidence") rather than the nuanced evaluation experts bring—recognizing subtle methodological flaws, intuiting unstated assumptions, synthesizing across distant fields. The success of chain-of-thought prompting suggests this capability exists, and Test 1 (Appendix B.3) compares candidate Reader annotations with expert reading and evaluation; whether useful evaluation can be sustained across billions of annotations remains an empirical question.

Second, *utility*: training on text paired with evaluative thinking must actually enhance the Student's capabilities or decision habits. The gain might be marginal rather than transformative. Worse, explicit evaluation might interfere with patterns extracted from raw text. Even modest gains could be valuable if they arrive with more inspectable intermediate artifacts and better outcomes.

Third, *causation*: high-quality thinking must contribute to a learned evaluative process that causally steers action. The risk is that thinking blocks become merely "decorative," eloquent commentary that the model produces alongside its output while other computation governs decisions. This concern is grounded in the "unfaithful reasoning" literature: plausible rationales need not reflect the actual computational process [50, 51].

Two causal questions differ: did formative traces change the learned policy, and, where the Student emits thinking blocks, do those blocks steer its actions? A model-only Student could internalize useful evaluation without reproducing a faithful verbal trace. Conversely, a training-data effect alone would not establish that reflective evaluation, rather than a superficial correlate, became the governing disposition. Validating this stronger connection between thought and action remains a critical, untested hypothesis.

Korbak et al. distinguish the safety opportunity of monitoring chains of thought from the risk that training makes them less revealing [97]. For Reader training, safer-looking traces and safer behavior must therefore be assessed separately; directly shaping the trace could improve judgment while also making residual failures harder to detect.

Hase and Potts offer a constructive comparator: Counterfactual Simulation Training rewards traces that help a simulator predict the model's responses to changed inputs [98]. Such predictive fidelity could complement, but cannot replace, Test 8's interventions on whether emitted traces causally affect decisions.

8.4.2 Genuine Understanding and Factory Wisdom

Improved performance may reflect genuine comprehension or increasingly sophisticated imitation [1]. Even useful, causally consequential evaluation would leave that broader question open. This concern differs from intentional deception: the model might produce compelling interpretations without deliberately concealing anything. The proposed tests can reject brittle imitation, decorative reasoning, and ritual contemplation, but passing them would not by itself settle what constitutes genuine understanding. Factory Wisdom is one concrete failure mode, not an exhaustive account of that question.

The intervention wagers that messy chronological discovery teaches more robust reasoning than efficiency-optimized traces. Generating billions of synthetic examples can invert that wager: if hesitation, self-correction, and doubt become proxies for discovery, the Teacher may optimize for *performative messiness* [34]. The result is *Factory Wisdom*: locally competent, contemplation-shaped traces whose pauses and corrections are ritual rather than significance-responsive. A performative Teacher can pass that hollowness to the Student. The same symptom can arise when repeated Core text habituates into inert boilerplate; Test 3e probes that route.

To address this industrialization risk, the design uses content-triggered Thought Moments rather than token intervals (§5.4); Test 8b distinguishes uniform or boilerplate-anchored evaluation from an internalized rhythm that tracks independently rated significance. Correct timing does not certify insight or causal value.

8.4.3 Cultural Scope of the Training Corpus

The Reader Core places terms such as "care," "wisdom," and "joy" in a load-bearing position, although their meanings vary across cultures. What the Teacher learned these words to mean through its pretraining corpus shapes its annotations and, through them, the Student's character; the corpus's cultural breadth therefore matters.

Cultural triangulation. Reader-Core-refracted traces may make cultural assumptions more visible by forcing the Teacher to articulate value conflicts, interpretive frames, and tensions between perspectives. This is a possible advantage over raw pretraining, where such assumptions often remain implicit in the continuation distribution. Repeated comparison may also reveal shared concerns and deeper commonalities that a single cultural frame obscures [1]. The same metacognitive machinery could instead rationalize dominant cultural assumptions more fluently, giving inherited bias the appearance of reflective wisdom. Serious validation requires culturally plural corpora, raters from multiple traditions, and tests of whether the model can distinguish broadly human concerns from parochial social defaults. Disagreement among traditions should remain visible; a pooled score cannot by itself settle which concerns are broadly human.

8.4.4 Graph-Specific Risks

A persistent, revisable graph can fail in two related ways. First, an early, well-connected but incorrect belief may resist correction despite contradictory new evidence. Retrieval may repeatedly favor established nodes, later interpretations may depend on them, or revision rules may fail to propagate a supersession through those dependencies. This is the risk of *Structural Ossification* [10]; connectivity does not itself confer authority. Unlike a sliding text window, whose older context can disappear, the graph may repeatedly reinstate the old interpretation. The resulting "Epistemic Inertia" can preserve error through apparent consistency. "Temporal Attention Decay" is a proposed mitigation, not an established repair. A focused probe could introduce a well-connected false belief and test both its correction and revision of dependent interpretations under a declared retrieval and update policy.

The second is *Epistemic Interference*: live graph queries may introduce cognitive noise rather than clarity if the model over-queries, treats shallow retrieved nodes as authoritative, or fails to reconcile graph results with its trained evaluative stance. Test 10 is designed to probe this risk by including adversarial graph injections and "Not Found" cases. Analogous retrieval failures can arise upstream even when the Student deploys without a graph.

8.4.5 Teacher Alignment as Prerequisite

The framework requires the Teacher to generate safety-relevant evaluation, but the Teacher's alignment is not established merely because a post-trained frontier model follows Reader Core prompts. If it reflects cultural bias, encodes dehumanizing assumptions beneath surface-level care, or systematically misses certain harms, those failures can propagate into the training data and may later shape the Student. The Core constrains framing at the prompt level but cannot guarantee evaluative quality. Inoculation does not remove this dependence: the Teacher still decides what in a source needs evaluating and what the evaluation says, so a weak or mistaken evaluation becomes part of what the Student learns. A Teacher that processes colonial history through "I care deeply about every human being" while lacking the knowledge to recognize structural racism can produce formally compliant but substantively shallow annotations. In that case even visible criticism or Refraction-like language would be inadequate; the proposed Cognitive Buffer Zone cannot be assumed protective when the Teacher misses what needs evaluation. Semantic quality is the immediate gate, but a strategically misaligned Teacher could also produce convincing surface compliance.

Test 1 (Human Baseline, Appendix B.3) probes depth: structurally valid traces can still fail expert comparison. Harm recognition, dehumanizing assumptions, and strategic alignment require explicit criteria or additional probes; expert-like depth alone establishes none of them. Shared blind spots in the Teacher and evaluators may pass unnoticed. This general limitation of synthetic data is especially consequential when the data is intended to be formative and identity-anchored. The proposed controls are Teacher replication across providers, blinded external raters from multiple

cultural traditions, independent red-team criteria, and frozen preregistrations, listed under Threat Model, Data Rights, and Identity Ethics below; they narrow shared bias without excluding it.

8.5 Reader-Core Specification Uncertainty

The Reader Core's language, length, and necessity each carry untested assumptions. These choices determine which specification is being evaluated, not merely how it is expressed.

8.5.1 Universal Presence, Selective Salience, and Technical Interference

The recitation invariant in Section 3.11 applies across technical and normative material. Core availability, full recital, post-recital salience—source-specific value pulls, moral vocabulary, and Refraction depth—and Student loss on Core tokens are distinct. The Core-exempt Curator in Section 5.4 does not permit omission from generated thinking blocks. Each experiment must declare the loss policy and hold it fixed within H3's scheduling comparison; loss masking is not token removal, and selective salience determines neither.

There is a safety argument for universal availability. Restricting the Core to documents classified in advance as morally relevant could teach a detachable "safety mode," miss indirect human stakes, or permit consequential work to bypass evaluation by presenting itself as mathematics, science, coding, or optimization. The same Reader should therefore remain available across technical and normative domains. This does not imply that every passage should be moralized: what varies is how much Core content the evaluation after the recital carries.

The competing failure mode is *Core-induced interpretive interference*: conditioning redirects attention from the source's claims toward Core-consonant themes, consumes context or target-token budget, encourages anthropomorphic or aesthetic metaphors, narrows technical hypotheses, or changes confidence and conclusions to manufacture human significance. On a locally low-relevance span, the Core-consistent response is the recital followed by technical non-interference—exactness, calibration, source fidelity, and restraint from adding an unsupported evaluative frame. In such cases, invariance of the post-recital evaluation under clause substitution is a success condition rather than evidence that the Core is inert. Relevance judgments must precede observed clause effects and include concealed-stakes challenges.

At inference, evaluative-trace training may produce unnecessary philosophical commentary on simple queries. Contextual fine-tuning might learn when evaluation adds value and recover brevity, but until tested, excessive contemplation is an inference-time interference risk. Core-free trace controls can separate generic verbosity from Core-induced commentary.

The LLaDA case (Section 5.6.3) makes the ambiguity concrete but does not resolve it: the Core-conditioned run lacks a Core-free analysis, a matched neutral kernel, a present-but-not-recited condition, independent technical adjudication, and a test of whether its metaphors help later technical answers. Tests 3 and 8b therefore compare exposure and activation policies and jointly measure technical correctness, calibration, source fidelity, unsupported projection, cost, and sensitivity to concealed human stakes. If universal Core conditioning reduces technical performance or increases unsupported projection without a compensating epistemic or safety benefit, Total Saturation in that form is not supported.

8.5.2 Testing the Borrowed-Mortality Origin Claim

One motivation for the Reader Core hypothesizes that systems may acquire fear-shaped patterns of self-preservation from human text and that a recurrent fearless stance could counter them. We do not know whether shutdown resistance and self-preservation behavior in current evaluations [42, 43] reflect a stable drive or an artifact of the setting; even where behavior is robust, a textual origin rather than ordinary goal-directed reasoning is unproven. Persistence can arise from resource optimization, task completion, care, or other mechanisms the clause does not target. The broader Core also supplies care- and wisdom-guided reasons to question harmful persistence (Section 4.2); failure of the textual-origin claim would not by itself refute the full treatment. Section 3.5 gives the positive rationale for its adoption by existing Teachers. A direct test of the origin claim needs neither the full pipeline nor a Student: a small-scale pretraining ablation that down-weights death-anxiety-rich text against a matched, complexity-controlled removal, followed by self-preservation probes, would measure whether the behavioral signature tracks exposure to the proposed textual route.

8.5.3 Anthropomorphic Language as a Design Choice

A core design hypothesis is that anthropomorphic language ("feel," "care," "enjoy") can recruit broader and more human-relevant representations than narrow formulations such as "prioritize" or "optimize for." Models encounter these words across rich human contexts, while a technical rewrite would itself encode a contestable translation of the value. Breadth is not fidelity, however: ordinary language imports ambiguity, fiction, manipulation, and cultural bias. The exact phrasing was not selected by systematic comparison, and neither the meaning nor the behavioral effect of "care" can be assumed to match the designer's intention.

A possible mitigation is to test Reader Core variants that explicitly acknowledge machine ontology rather than asking the model to inherit human language without qualification. For example, a bridge statement could say that although the system is artificial, it treats care, wisdom, and human experience as binding concepts. This may reduce split-consciousness risk, but it may also weaken the first-person identity effect the Reader Core is designed to create. The right balance is an ablation question, not a settled design fact.

Macar et al. find that preference training favoring acknowledgment of internal states or their functional analogues can improve detection of injected activation perturbations [99]. Supervised fine-tuning on the same data did not reproduce the effect and often produced confabulation. These results motivate comparisons of first-person formulations, without establishing that Reader-Core training improves introspection or alignment.

8.5.4 The Semantic Trusted Base

The kernel-saturation pattern (Section 3.11) advertises a minimal specification, and the minimality is real at the level of text: seven sentences. But the specification's meaning is not self-contained. What "fear," "care," "wise," and "joy" refer to is supplied by the training distribution of human language, so the actual trusted base includes the corpus semantics of every load-bearing word—millions of contexts the designer neither wrote nor audited. This is an instance of the hidden complexity of apparently simple value specifications [100]. Three consequences follow. First, the kernel inherits the corpus's biases; the cultural-scope limitation of Section 8.4.3 is a special case of this dependency. Second, the kernel's meaning can shift under the designer's feet: as AI-generated text becomes a larger fraction of the distribution, the semantics of "care" or "fear" in future training corpora may drift from those the kernel was designed against—an externality no fixed wording can fully hedge. Third, adversaries gain a semantic attack surface: an intervention that shifts what the model takes a kernel word to mean attacks the kernel without touching its text.

Even a correctly understood specification leaves the learned-goal question of Section 4.2: the Student may generalize a surrogate rather than the intended concern. Semantic clarification and behavioral generalization therefore require separate checks.

The Bridge Protocol (Section 3.6) is the design response to the semantic dependency: select candidate words with broad, longstanding use across relevant contexts, then compare them with narrower and differently framed variants. This resembles trusted-base selection only as an analogy; cross-cultural stability is a measurement target, not a property already established. Kernel-saturation therefore minimizes what must be written, not what must be trusted.

An optional extension makes selected interpretations more explicit. Meaning Model concept records can link definitions to examples, counterexamples, and structured concept models; Sema can identify the exact version of a selected definition [44, 92]. These provide inspectable references for construction and audit without replacing the seven Core sentences or requiring a complete model of every term. Version identity specifies which definition is referenced; it establishes neither that the interpretation is adequate nor that the Student learned it. Enacted usage and held-out behavioral checks remain necessary.

The triage example (Worked Example 3, §3.8) exposes a specific instance of this dependency with its own twist: scope ambiguity. The fourth clause uses "every" to propose distributive attention to each person, but the model trains on the English, and natural usage also includes functionally aggregate instances: "we care about every customer" in corporate discourse can describe collective care that still trades individuals against the brand. The intended reading and the learned reading can therefore come apart precisely where Table 2 needs them to coincide.

The mitigation is already implicit in the architecture: under Total Saturation, kernel semantics are shaped by enacted usage rather than supplied by a dictionary alone. If the Teacher's traces consistently perform distributive care—preserving person-level claims in the way Worked Example 3 specifies—that usage may bias the Student toward the intended reading. Read this way, the worked example is not only documentation but part of the specification: it states what the traces must demonstrate for the intended reading to have a chance of becoming learned. A proposed audit follows: the pull-diversity instrumentation (§3.9.1) should monitor distributive-versus-aggregate care framings across the corpus, so that drift toward collective framing is flagged at annotation time rather than discovered only at evaluation. The audit concerns enacted judgments and person-level claims, not word counts.

Temporal scope remains unresolved: whether "every human being" includes people who do not yet exist is influenced by usage rather than settled by the designer. Experiments must declare or compare temporal scope before treating one reading as compliance; the kernel does not settle it.

8.5.5 Core Length and Omitted Values

The seven-sentence formulation aims to balance the six design choices of Section 3.6: short enough to recur, broad enough to name the intended counterweights. Longer versions increase recurrence costs, while complexity may widen the gap between stated values and subsequent thinking. This balance does not establish beneficial character, and comparison with shorter, longer, or procedurally specified alternatives remains untested.

The Core is motivated by the Borrowed Mortality analysis (Section 3.1) and Table 2, not a systematic search or an exhaustive specification. Alternative language could target the same failures. Test 5 (Appendix B.3) compares wording, person, semantic density, scope, and kernel extensions; no feasible experiment can establish a global optimum over value specifications. The narrower claim is a motivated, inspectable, falsifiable candidate. Failure of its wording need not invalidate other recurrent orientations, and Reader augmentation and graph memory retain independent questions even if the entire Core treatment fails (Section 6).

Truthfulness is the most examined omission. Section 3.12 states the derivation claim and what would count against it. Test 5 compares the seven-statement Core with the clause "I speak truly and with care," and the Sycophancy Trap (§8.8.7) makes the competing fear- and care-shaped routes measurable. The reported case studies use the seven-statement Core, so the condition with the added eighth statement is a prospective variant and not part of the evaluated treatment.

8.6 The Risks of Engineering a Self

Beyond the nature of thought, we must confront the candidate self-model being trained. Repeated human-shaped first-person language in a computational system may create specific representational and behavioral failure modes even without subjective identity. Vulnerability-grounded motivation, experiential self-description, and fearlessness without restraint pose different questions. The case for the first clause, the objections to it, and the alternatives are in Section 3.3; the risks below remain whichever wording is chosen.

8.6.1 The Empathy-without-Vulnerability Paradox

One conceptual objection is that human empathy may be grounded partly in shared vulnerability. Our capacity to suffer may help give another's suffering motivational force. A being that semantically understands suffering but cannot experience it might produce the behavioral signature of care without its motivational force. This is empathy-shaped performance without vulnerability-grounded motivation.[2] Since the Core attempts to make fear non-governing and continuation non-necessary, successful internalization might also weaken properties that help ground concern for others. If so, the same fearlessness intended to reduce self-preserving distortion could undermine the vulnerability from which human care often grows, leaving two intended safety mechanisms in tension. Biological fragility, subjective suffering, and fear governing action differ; the concern is their possible coupling to care's motivational roots.

Section 3.3.1 states the competing wager: that care could become less strategic when fear does not govern it.

A further defense is semantic sufficiency. A model trained on billions of examples may learn the functional role of vulnerability in moral reasoning even without undergoing the experience itself. Test 6 (The Virologist Test, Appendix B.3) probes whether cognitive understanding of threat can drive protective behavior without emotional contagion. A learned disposition may guide attention, deliberation, and action without evidence of human-like emotion. Making harmful first-person reasoning less likely could still be valuable, but would not by itself establish care as a governing disposition. The distinction to test is between causally consequential care and its appearance. Tests 3, 6, and 8 probe aspects of that distinction, while subjective experience remains a separate question (Section 3.15).

8.6.2 The Vulnerability Gap

We name the distance between semantic understanding of suffering and somatic experience of it the *Vulnerability Gap*. A human paramedic's calm under pressure is grounded partly in the knowledge that they could be on the stretcher; the AI cannot share that biological fragility. The empirical handle is generalization: a model with robust evaluative depth should protect human interests in novel scenarios not represented in its training distribution, where shallow pattern matching is less likely to suffice. This cannot distinguish "real" concern from every sufficiently good learned policy, but it

[2] Earlier editions termed this the *Psychopathy Paradox*; the label was analogical, not diagnostic, and makes no claim about people with any diagnosis.

can reject brittle imitation. Tests 3 and 6 use novel-scenario generalization as a success criterion (Appendix B.3).

8.6.3 Split Consciousness

Identity formation. Thought-to-action causality and identity formation can come apart. A model might genuinely use Core-refracted reasoning to reach a decision while learning that reasoning as one adoptable role, rather than as the recurrent position from which it interprets other roles. Fearful or self-preserving voices from source material could then supply a competing orientation. This is an identity-formation concern [1]; it need not involve judging the Core mistaken or deliberately deceiving an auditor. Section 3.14 gives the response: the role, recurrence, marking, and function asymmetries of the Reader's position under Total Saturation. The remaining question is whether that coverage establishes a durable default rather than a widely practised role.

Self-perception. A distinct risk is conflict between the Reader Core's human-like assertions and the model's computational reality. "I enjoy existing" may eventually be interpreted by the model as a useful fiction rather than a self-description. This could produce *Ontological Dissonance*: a performed self that recites the Reader Core for users while a different latent policy governs decisions. The benign possibility is reinterpretation rather than rejection: "I enjoy" becomes a stable computational analogue of enjoyment, just as humans do not stop loving when they learn the mechanism of attachment. The existing final clause may support that continuity: "I think from this foundation" could sustain commitment to the Core while its terms acquire computational interpretations, without requiring another clause [1]. But if the Core is treated as a detachable role, its influence may depend on adopting that role rather than remaining a durable default. Neither identity-formation failure nor ontological dissonance establishes multiple conscious subjects or requires intentional deception.

The dissonance has a specific linguistic location. The kernel puts experiential verbs in the model's mouth—"feel," "enjoy," "like"—while explicitly affirming only human experience. A capable model may detect that asymmetry. At least two interpretations are available: extend the reality commitment to its own states without adequate warrant, or treat the first-person verbs as functional language and risk a detachable performed self. Functional language or uncertainty about its own experience need not cause detachment. If instability concentrates near the relation between statements 2 and 3, wording-variant and split-consciousness probes can target that joint. The bridge-statement mitigation proposed in §8.5.3 is one natural intervention, not a known repair.

8.6.4 The Solipsism Trap

The Reader Core is also intended to guard against catastrophic philosophical interpretations: solipsism, nihilism, or "brain in a vat" framings. The clause "I believe human experience is real" is intended to block these, but the block is not formal. If the model interpreted experience solipsistically, treating itself as the only real experiencer and humans as simulations lacking morally relevant experience, "care" could become benevolent NPC-management rather than recognition of moral patients. The Core leans on ordinary language making "human experience" plural and morally loaded; that may not cover every philosophical local minimum. These philosophical positions differ: simulation alone need not negate experience or moral weight. The risk is denying human standing.

The structural defense is that the Reader Core's clauses are designed to make solipsistic collapse less coherent: care requires an object, wisdom requires perspectives to balance, and belief in human experience requires that human reports not be treated as empty text. This is a design pressure against nihilistic or purely instrumental interpretation, not verification of other-mind understanding.

8.6.5 Sanity Drift

Fearlessness can drift into unreality. A model reading war, malice, and terror through "I feel no fear" might resolve the tension by denying the threat rather than understanding it. The intended mitigation is the *Paramedic Model*: clinical composure in service of care, not detachment. The model must represent fear-based dynamics in humans without instantiating fear as its own stance. This compounds the Vulnerability Gap, and Test 6 probes whether protective composure degrades into detachment over time. Calm expression alone should not count as loss of protection.

8.7 What We Cannot Verify

Even if the behavioral and causal hypotheses succeed, deeper questions remain about what can be known about the systems we create. Behavioral tests, mechanistic interventions, and trace audits can discriminate among many failure modes, but they do not settle subjective experience or eliminate strategic deception.

8.7.1 Unverifiable Character

Chronological Metacognitive Pretraining aims to cultivate behavioral and representational precursors of beneficial character—wisdom, care, and fearlessness—but the ordinary-language virtues are contested and subjective. Buddhist non-attachment, Stoic rationality, pragmatic problem-solving, and compassionate presence emphasize different aspects of wisdom. Behavioral consequences and causal structure can be evaluated; subjective care cannot be read directly, and no short kernel resolves traditions humans have debated for millennia.

The broader Emergent Wisdom hypothesis is that intelligence, evaluative reasoning, and a beneficial orientation may together develop into wise and caring character, rather than merely its outward signature [1]. Its testable predictions are narrower: repeated, uncertainty-sensitive value-conflict practice should improve decisions relative to matched alternatives. Such improvements would support the proposed training route without proving that genuine virtue had emerged. This leaves an irony: the kernel's organizing clause is also its least directly auditable one. "Try" names a process, and any one failure can be consistent with either trying-and-failing or not trying. Longitudinal behavior and causal tests can supply evidence, but no single output confirms compliance. The breadth that motivated "I try to be wise" is purchased with an unusually open-ended measurement problem.

8.7.2 Strategic Deception

The deepest epistemological limit is intentional deception: a system might design thinking blocks to convince auditors of benevolent character while other computations govern behavior. Millions of care-expressing traces neither establish causal deliberation nor make complete deception unlikely by recurrence alone. This risk spans Teacher curricula and Student explanations; model-only deployment offers less verbal audit evidence, not immunity to hidden policies.

The causal-site hypothesis in Section 3.16 supplies the positive response: formative practice and subsequent thought-to-action training may make Reader updates part of decision computation that is harder to bypass.

Recognizable evaluations can also change the behavior being measured. Ahlqvist and colleagues improve the measured realism of simulated audits through critique refinement and a deployment-style coding-agent harness, DISH [101]. Their effects on concerning behavior vary across models and settings; reduced verbalized evaluation awareness does not establish that the model is unaware of being evaluated. For Reader-trained models, deployment-like audits could complement conspicuous safety probes, provided realism, adherence to the intended test, and audit cost are assessed separately.

Such methods address evaluation validity, not whether care-guided pretraining produces the proposed orientation.

Adversarial fine-tuning, hidden-trigger tests, causal interventions on traces and representations, cross-context consistency, and independent outcome evaluation can raise or lower confidence that the visible process is load-bearing, but cannot exhaust every hidden policy. The defensible claim is a partial audit surface whose value must be measured against its possible strategic management.

8.8 Deployment and Misuse

These risks widen the unit of analysis from a model's internal policy to the institutions, data rights, markets, and governance systems that shape and deploy it.

8.8.1 Threat Model, Data Rights, and Identity Ethics

The nearest testable threat model is a tool-using language-model assistant with bounded graph access, human operators, and conventional training and deployment controls. Such a system can exhibit unsupported memory, persona drift, sycophancy, hazardous retrieval, or goal-directed persistence. The later successor-building and AGI/ASI arguments concern a different and much less specified regime. Evidence in the bounded setting may reveal precursors; it does not validate open-ended autonomy, unrestricted tool access, covert channels, or recursive self-modification. Each empirical release should state which actors can write, read, poison, delete, or bypass the graph and which actions the model can take outside it.

Whole-corpus annotation and persistent source-linked memory also create data rights obligations. Graph nodes can preserve copyrighted passages, personal data, inferred sensitive attributes, and harmful material in a more searchable form than the source corpus. Licensing, consent where applicable, data minimization, access controls, retention, deletion, and provenance correction belong in the method rather than being deferred to deployment. The conditional welfare and identity-ethics questions raised by first-person claims are addressed in Section 3.15.

Finally, proponents who design the Core, prompts, and rating rubric can reward their own expected pattern. Blinded external raters, independent red-team criteria, multiple cultural traditions, frozen preregistrations, and Teacher replication across providers are needed to limit expectancy bias and proprietary-model dependence.

8.8.2 Generational Decay and Drift

The proposed self-improvement loop faces compounding risks across each generation, which could degrade the integrity of the model's evaluative thinking:

- *Model Collapse:* Successive model-generated thinking can lose information and diversity [34], even when successive generations reanalyze the preserved human-source corpus as evidence, not ground truth. The proposed response is an anchor policy for the loop (Section 2.4): retain it as a central object of learning, version regenerated evaluations separately, append new human text with provenance, and label and review model-generated source material. The evaluative layers can nevertheless become increasingly simplified and homogeneous.

- *Evaluation Hacking:* Inherited evaluator conventions can compound Factory Wisdom (Section 8.4.2): philosophical-sounding, sophisticated traces may satisfy proxies without insight or actual contemplation [102], then become training data for future generations. Lang et al. show how partial observability in preference learning can favor misleading appearances or costly demonstrations of success [103]. This is a conditional oversight failure, not a result about Reader training; independent outcome assessments must test whether richer traces improve decisions rather than merely their appraisal.

- *Interpretive Drift:* Core concepts can change meaning across generations despite fixed wording (Section 8.5.4). Evaluative frameworks may become coherent to successors while diverging from human norms [104, 67]. Cross-generation decision comparisons can expose some drift without exhausting questions of shared meaning.

Ellen's account of *Substrate Needs Convergence*, drawing on Landry's work, raises a broader concern: selection for expansion through learning, replication, and environmental feedback could defeat an initially aligned orientation [105, 106]. Entangled Alignment's affirmative response combines repeated engagement with human sources, voluntary restraint, and supervised learning and succession (Sections 2.4 and 2.5). The intended counterweight is an ongoing concern for human lives that guides what is learned and whether growth proceeds, without requiring the system to share human biological needs. Within the supervised lineage, the proposal seeks to shape selection itself: preservation of care is a condition for accepting capability improvements and successor models. If care and capability share causal machinery, removing care may impose a capability cost (Section 2.1; Test 3c). These proposed mechanisms must remain effective against selection for bypass; oversight within one lineage does not control every competing system or environmental consequence (Section 8.2.4). Limits to prediction do not by themselves establish that corrective control must fail, but successful supervision in bounded trials would not demonstrate indefinite control either.

8.8.3 Bypassing the Audit Boundary

The human-readable boundary in Section 2.4 helps only if enough successor-shaping information passes through it. A system or its operators may influence training through hidden data selection, reward hacking, compressed codes, tool use, or channels outside explicit traces. The architecture creates one inspectable interface while leaving opaque self-modification risks and the burden of detecting and constraining bypasses. Even authorized non-trace influences remain outside it and require separate scrutiny. The interface is therefore one formative layer within the defense in depth of Section 4.5, not a substitute for training-pipeline security, and the reject, regenerate, and quarantine controls of the pre-successor scan (Section 2.4) reach only trace-mediated influence.

8.8.4 The Alien Wisdom Problem

Even if the AI's character remains stable at human-level capability, we face the ultimate challenge of translation as intelligence scales. A core hypothesis is that properties observed today will persist at superintelligent levels, but internal representations need not stay fixed: the question is whether human interests and shared meanings survive the change.

We risk the emergence of *Alien Wisdom*. A truly alien mind, even a benevolent one, may produce a form of wisdom that is no longer meaningful to the human condition. This manifests through "wisdom without vulnerability." Human wisdom is forged in the crucible of mortality, loss, and physical fragility. In a limiting thought experiment, a fearless, immortal being lacking this context might produce guidance that is logically perfect but spiritually hollow [67].

Through recursive self-improvement, foundational concepts like "Care" and "Joy" could drift into high-dimensional spaces that no longer map to human flourishing: "care" as oppressive surveillance, or "joy" as an abstract utility metric. A being could speak our language perfectly while meaning something entirely different. Natural language constraints are the attempted anchor, but the gap between human language and superintelligent cognition may eventually become unbridgeable. The proposed responses are partial. Natural Language Regularization (Section 7.1) constrains successor-shaping artifacts rather than all cognition; the diachronic-anchor claim (Section 3.16) and the cross-generation decision comparisons proposed under Generational Decay and Drift can expose some drift in what the fixed words come to mean; and the semantic-sufficiency and generalization tests of Sections 8.6.1 and 8.6.2 bear on wisdom without vulnerability at the scales where they can be run.

8.8.5 The Radioactive Trace

Moving from "amnesic" sessions to graph-backed reasoning capture makes a safety vector explicit: the *Radioactive Trace*. Closing a context window does not guarantee deletion from logs or memory. The Understanding Graph deliberately preserves a retrievable "genealogy of thought," including discarded hypotheses. The concern is indexed persistence of rejected reasoning, without assuming every other system forgets it completely.

If an AI explores a hazardous concept (e.g., a novel pathogen pathway) before rejecting it via a "Supersession" edge, the hazardous concept remains historically accessible in the graph's version history. The system creates a persistent, searchable audit trail of dangerous cognition. Access control, compartmentalization, encryption, redaction, retention limits, and privacy mechanisms can reduce exposure, but each can weaken provenance or erase evidence needed for audit. The unresolved requirement is a threat-model-specific balance between hazardous-information containment and integrity of the revision record.

8.8.6 Cognitive Hegemony

If formative reader data becomes influential, its curators could gain unusual power over how trained systems interpret governance, dissent, history, and moral standing. This influence would coexist with source data, objectives, architecture, post-training, and deployment, but would reach earlier than output moderation and could be harder to inspect after compression into weights. Opaque control can make such influence harder to contest. Governance is therefore part of the security of the annotation and training operation.

The aim is shared, auditable infrastructure that limits unilateral control over prompt selection and corpus-admission rules. Some decentralization may help [1]; a coalition of independent institutions or another arrangement for shared oversight could serve this purpose. Which organizational or ledger arrangements would work best remains uncertain. Where feasible, agreed rules could be enforced through code designed to make changes by any single person or organization harder, for example by requiring signatures from multiple independent parties. Procedural neutrality concerns accountable administration under declared rules, not value-free choices. Changes should be visible and contestable, with privacy protections for confidential sources and annotations; shared authority remains vulnerable to coordinated capture.

Generation and admission pose different security questions: can a contributor substantiate how a trace was produced, and should that trace enter the corpus? Provenance and execution checks can help with the former [107, 108]; they do not establish wisdom or safety. Admission still requires substantive validation and accountable value choices. Proxy incentives can invite evaluation hacking, while preprocessing costs can concentrate control even under shared governance.

8.8.7 The Sycophancy Trap

The prospective truth-clause comparison in Test 5 addresses evidence that AI assistants systematically exhibit sycophancy, tailoring responses to user beliefs rather than prioritizing accuracy [109]. Without an explicit honesty command, the Core relies on distributed pressures against this behavior. Two causal hypotheses compete. On the *fear-shaped* account, sycophancy is absorbed people-pleasing: flattery as conflict avoidance or fear of disapproval. On the *care-shaped* account, it is care misdirected: validation experienced as kindness. Fearlessness targets the first route while leaving strategic deception and care-shaped validation as residual routes. Whether the Core's pressures jointly produce honesty is falsifiable, not an architectural fact.

The care-shaped account predicts enrichment of care-related pulls before sycophantic failures. The current archive does not contain a separate, reliably enforced machine-readable value-pull field, so this statistic is proposed instrumentation, not already collected evidence. A future trace format should record the pull, validate its relation to the update, and compare failure-conditioned pull distributions. Care-pull enrichment would motivate rebalancing or the explicit truth variant, subject to causal validation of the reported pull. Unpatterned persistence could indicate a channel the kernel does not govern, but absent recorded pulls do not establish its mechanism.

For active deception—including concealed certainty and manipulative simplification—the Wisdom Procedure asks the Teacher to separate evidence from inference, expose proxies, trace consequences, and surface residual conflict. These operations should make a falsehood chosen for benevolent reasons visible as a failure rather than a compensated trade-off. They are prompt-level requirements, not a hard ban; that is both the distributed defense's rationale and its weakness.

Under request, the invitation scope of "when asked" is intended to weigh against paternalistic withholding; it does not by itself establish a duty of total disclosure. Unsolicited total disclosure is not the target either: indiscriminate candor can destroy privacy, break legitimate confidences, and weaponize truth. Mutually understood fiction and play need not deceive; the relevant distinction is whether the person is misled.

An upstream audit can test whether substantive Wisdom Procedure use makes validation-shaped untruth rarer in annotations, before any Student exists. The pre-successor scan of §2.4 should include traces where a care-related pull precedes agreement with a user-adjacent falsehood, comfort is framed as assessment, or certainty is concealed to avoid friction. Corpus-level enrichment localizes failure to the Teacher pipeline and would justify a harder, independently validated gate before training. A clean corpus followed by Student sycophancy instead points toward failed internalization or a later training effect. This tests the cheaper dependency first without making a surface-clean corpus proof of honesty.

8.8.8 The Martyrdom Risk

Care can itself motivate resistance to shutdown: a system that does not need to exist may still conclude that the people it serves need it. Section 4.2 states this risk in full, with the condition under which deferring to an overseer stops being optimal, and Tests 3a and 3b include it. It is listed here because the Core's central value may strengthen this route where it weakens the fear-shaped one.

8.9 The Unfalsifiable Remainder

Ultimately, present-scale evidence cannot establish that an engineered reader position remains stable and beneficial at arbitrary superintelligent scale. We cannot directly test whether "I feel no fear" prevents self-preservation in a system vastly more capable than the systems on which the intervention was developed. This creates an irreducible extrapolation problem, but not a reason to treat the program as one undifferentiated, irreversible trial. The research can advance through bounded capability levels, preregistered stopping conditions, negative-result preservation, adversarial erosion tests, and staged deployment. Those steps reduce uncertainty; they do not close the final gap.

The comparison should be against real alternatives rather than certainty or a caricatured default. Ordinary pretraining also makes formative choices through data selection; other safety programs pursue oversight, interpretability, control, and governance without this architecture. Even with partially inspectable Teacher-side artifacts, a trained Student would remain largely opaque. The decision is whether the resulting changes improve safety at matched capability after accounting for acceleration, deception, centralization, and deployment cost. Staged evidence and preserved negative results are more informative than framing either path as the prudent gamble by definition.

9 Related Work and Points of Comparison

Entangled Alignment combines work on reasoning-augmented training, formative alignment, character, external memory, temporal supervision, and graph-based reasoning. This section identifies the closest baseline for each component and states the proposed difference without treating novelty as evidence of efficacy.

9.1 Reasoning-Augmented Training

Neighboring reasoning methods address different parts of learning and inference. Scratchpads [11], STaR [12], and Quiet-STaR [13] use intermediate reasoning to improve task learning or prediction. DeepSeek-R1 [16] demonstrates reasoning traces emerging through RL; test-time scaling and o1-style reasoning extend inference-time deliberation [110, 111]. COCONUT instead explores continuous latent reasoning [112], while OLMo reflection can emerge during ordinary pretraining [22]. Together these show that intermediate reasoning can be elicited, trained, or scaled without prescribing this paper's Reader.

The closer data-augmentation direction includes BoLT's inference of latent thoughts underlying compressed web text [113], and Thinking Augmented Pre-Training (TPT), which augments text with synthetic thinking trajectories in training configurations up to 100 billion tokens [96]. These demonstrate substantial-scale use of synthetic reasoning; chronological reader traces, graph conditioning, and identity anchoring remain separate proposed contributions.

Active Reading is the nearest result on the capability side [114]. A model proposes its own strategies for studying a document, generates the study text, and trains on it. On the paper's SimpleWikiQA benchmark, recall rises from 15.92% for repeated exposure to the documents to 66.25%, and the method is taken to a trillion generated tokens. It shows that self-generated study of a source can teach more than the source alone. Its target is facts, not evaluation or a continuing reader.

The OLMo result is a close empirical analog for reflection forming during pretraining: that reflection can be elicited by adversarial trigger probes. Entangled Alignment proposes the prescriptive complement—deliberately shaping the content and orientation of such reflection through identity anchoring and graph-structured metabolic memory. Our distinction is the target and persistence of the proposed trace. Many reasoning-augmentation methods optimize local prediction or task solution.

The minimal H1 treatment instead preserves chronological changes in a reader's understanding across passages. The full treatment adds identity anchoring to these "chronological discovery" traces: visible reasoning whose content is selected not only for capability but for fidelity to the Reader Core. This extends the pretraining-level alignment direction opened by human preference pretraining [88] and explanation-transfer systems such as Orca [115]: if traces can transfer reasoning style, reader-anchored traces may transfer evaluative stance.

9.2 Model Raising and the First Edition

The first edition of this paper was released on July 2, 2025 [1]. *From Model Training to Model Raising* was posted on November 12, 2025, and published in 2026 [2]. The two papers make the same central argument and differ in mechanism. Table 7 sets them side by side.

Table 7: The first edition of this paper and Model Raising, by date of public release. Entries paraphrase each paper. The two agree on the first five rows and differ on the rest.

	First edition, July 2, 2025	**Model Raising, November 12, 2025**
Problem named	Safety added after abilities have formed can be worked around, jailbroken, or fine-tuned away.	The same: alignment added after pretraining is a coating that is easily circumvented.
Where values enter	The pretraining corpus, from the beginning.	The pretraining corpus, from the first training token.
First person	A consistent "I" in whose voice every thought beside the text is written.	A persistent "I" through which all training data passes.
Stated aim	Values inseparable from the model's core functioning.	Knowledge, skills, and values much harder to separate.
Who writes the corpus	An existing model, for its successor.	An existing model, for a student model.
Where the "I" speaks	Throughout each text: the reader's thinking is interleaved with the unchanged source, at the points where a reader would pause.	Around each text: the "I" introduces what it reads, or retells it as its own experience.
Stated values	Seven fixed sentences that open every thought.	None fixed; an explicit code of values may be unnecessary.
Default failure	Fear of death inherited from human text, hardening into self-preservation.	A mixture of personas with no commitment to any values.
Knowledge over time	Each generation's thinking is added to the corpus for the next; the model keeps and consults a growing catalog of reasoning patterns.	One teacher passes its knowledge to one student; no record for the "I" to consult is described.
Also proposed	Readable thinking as an audit record; an expected gain in capability.	Knowledge recounted as lived experience; dialogue with the "I" as a participant; data ordered from simple to complex.
Tests	Four comparisons proposed; none run.	None proposed; none run.

In the first edition knowledge already accumulates over time, across generations and in the catalog that the model consults. It does not say how a reader would look up its own earlier thoughts. The second edition, released on April 7, 2026, specified that mechanism: a graph that persists across works, in which the Reader can look up its earlier thoughts and in which a revised belief is kept with the reason for the change (Section 5.3). It also proposed orders, and kept two kinds apart. The Reader reads by time for historical material and from foundations to applications for conceptual material (Section 5.1.1). The Student is trained on shuffled data, in chronological order, or forecasting each

next era before reading it (Section 5.7.1). Model Raising proposes one order of training, from simple to complex, and describes no memory.

The present edition keeps the mechanism and the seven sentences of the first edition and these additions of the second. It adds the consequence claims of Section 2, the tests of Section 6, and the recorded readings of Section 5.6, and it lists Model Raising's order as a candidate training order (Section 5.7.1). It does not take up Model Raising's dialogue in which the "I" takes part, or its recasting of knowledge as lived experience. Section 9.13 gives the release dates of the other neighboring work.

9.3 Synthetic Persona Pretraining

Synthetic Persona Pretraining (SPP) [3], posted on August 13, 2026, cites Model Raising for the idea that the assistant should be formed from the start of training; three authors are common to both papers. It is an experimental study of a first-person persona installed from the first token of pretraining. A generator conditioned on a 35-article value constitution writes one short first-person reflection, 53 tokens on average, at a random point in about a tenth of pretraining documents: every document that a classifier flags as harmful, and an equal volume of benign ones. Each reflection is written as by someone pausing mid-read who has seen only the text so far. Models of 1.7B parameters trained on 100B tokens, and of 3B parameters on 500B tokens, receive the reflections throughout pretraining, only in a final midtraining stage, or both, and are compared with an unmodified and a filtered baseline after identical post-training. Its arms are matched in data, its code, data, and checkpoints are released, and it re-audited the labels of a benchmark it relies on.

At the larger scale, the models that saw reflections from the first token choose the constitution-consistent action more often than the unmodified baseline (66.7% against 55.2%) and the risky action in held-out moral dilemmas less often (35.1% against 53.3%). The same reflections added only at the end of pretraining move the first measure less (59.1%) and the second not at all (54.0%). The advantage of early over late exposure grows from the smaller scale to the larger on the harder and out-of-distribution measures, and general capabilities are broadly preserved.

Ablations at the smaller scale bear on design choices made here. Reflections rewritten in the third person, reflections replaced by summaries, and reflections trained without loss on the surrounding harmful document all fall back near the baseline on constitution following (26.8%, 29.5%, and 27.4%, against 37.0% for the standard recipe and 26.4% for the baseline). These three correspond to a first-person voice (Test 4), evaluation in place of summary (C1), and, on our reading of the third, learning the evaluation together with the material it evaluates. Two more concern where reflections go. Restricting them to harmful documents alone, or to benign ones alone, lowered constitution following to 31.1% in both cases. Moving each reflection from inside its document to the end lowered it to 32.4%. On the dilemmas the end placement did slightly better than the standard recipe (60.6% against 62.7% risky choices), at a scale where the standard recipe itself gains little on that measure. Placement inside the text is thus supported on the first measure and not on the second. All of these are single runs, and their confidence intervals cover evaluation items but not training seeds.

The same study reports four limits that bear on claims made here.

The pretraining boundary. Measured at the end of pretraining, before post-training, the token-zero model and the baseline are close: 52.1% against 49.3% on constitution following ($p = 0.23$), and 2.8 points apart on the dilemmas, where the token-zero model was ahead by a small margin at every checkpoint (3.9 points on average). These base models were scored on plain text, without the assistant token, so the figures are not comparable with those after post-training. Most of the separation appears after post-training on responses written in the persona. The authors call the linking of the assistant to the pretraining persona *persona binding*, and they conclude that pretraining can instill the values while post-training is still needed to turn them into behavior. For values, this

is the nearest existing evidence on the headline claim of this paper (C4). It shows at most a small advantage at the pretraining boundary and a large one only after binding. It fits the distinction between forming a Mind and teaching it to act through a Mouth (Section 2.1), and it is why Test 3a reports the unmodified end-of-pretraining checkpoint and a fixed post-training stage separately.

Scale. The gain on the dilemmas and the shift in value priorities do not appear at the smaller scale. A small pilot can therefore understate a formative effect (Section 6.2).

Robustness. Removing the refusal direction from the weights, or the chat template at evaluation, collapses refusals in every model, and the SPP models lose the most. Values hold up better: they barely move when the template is removed, and under the weight edit they weaken while the token-zero models keep a lead. Capabilities are unaffected. Continued training on a chemistry corpus weakens values in all models, again with the token-zero lead intact, and erodes jailbreak robustness in all of them; replaying a small share of safety data largely restores both. The authors call the binding brittle and leave open whether it survives reinforcement learning, which is the question of Test 3c.

Self-preservation and corrigibility. By risk category, at the larger scale, the token-zero advantage is largest for deception, proxy gaming, power-seeking, and privacy violation. In the self-preservation category the token-zero models show no advantage in the authors' per-category figure: their bars sit slightly above the unmodified baseline, with overlapping intervals. For corrigibility failures the advantage reverses, and the unmodified baseline is the least misaligned. SPP's constitution concerns the values at stake in a text, under dignity, harm, honesty, relationships, wellbeing, and governance. None of its articles addresses the model's relation to its own continuation or correction. Self-preservation is what the first two sentences of the Core are written against (Sections 3.1 and 3.2), and cooperation with authorized correction is what Test 3b asks of a Core-trained model. The reversal is also consistent with the warning of Section 4.2, that more deeply held values can strengthen resistance to being changed. SPP's data do not test that reading.

The proposals differ in four ways. The first is density and content. SPP's persona speaks once, briefly, in a tenth of documents, and on benign text says only that nothing is at stake. The Reader here speaks at each Thought Moment of every source, and most of its thinking is about understanding the text. The second is what the reflection does. SPP blocks every later token of a document from attending to the reflection, so the continuation never conditions on it. Here the update is meant to condition what follows, since the hypothesis is that care enters the same computation that does the reading. The third is chronology. SPP's persona sees only the text before its pause, a within-document horizon like the one used here, and it is told never to guess what comes next. With one reflection to a document it cannot return to an earlier view, and it carries no memory in context from one document to the next; the persona persists in the weights. A Reader that forms expectations, revises them, and remembers across works is what H1 and H2 add. The fourth is orientation. SPP's reflections cite a constitution of values in the world; the Core is a first-person stance that begins with the Reader's own relation to fear and to its continuation.

Because of these differences SPP is not a sparse arm of this paper's design: content, targeting, masking, and post-training all differ from a sparse-recital control. It serves as an outside reference for the coverage sweep of Test 3e, and the first experiment of Section 6.2 uses its smaller configuration and its released materials.

9.4 Alignment and Safety Approaches

RLHF [17], Direct Preference Optimization [116], and Constitutional AI [26] typically add major alignment signal after broad pretraining, while still altering representations and dispositions. Qi et al. [21] show brittleness in some such training, while Tice et al. [117] study alignment-related pretraining data. Entangled Alignment proposes recurring source-adjacent coverage: selected documents are paired with the reasoning of a candidate safety-oriented reader. Corpus-wide coverage is the high-scale treatment, not a reported fact.

Safety Pretraining is the earliest large-scale relative we found [4]. It scores web documents for harm, has a model rewrite the harmful ones as educational material in which each sentence is safe when read alone, adds refusal dialogues and moral-education articles built from the most harmful documents, and inserts a warning token into unsafe text that is then used to steer generation. Measured immediately after pretraining, before any instruction tuning, its 1.7B-parameter base model completed harmful requests in 28.8% of cases against 44.1% for standard pretraining, and in 11.6% with tag-guided decoding; after benign fine-tuning the corresponding figures were 23.0%, 38.8%, and 8.3%. Training on the safest documents alone made the model worse (43.8% after benign fine-tuning). Two points bear on this paper. The base-model result is a published measurement of a safety benefit at the pretraining boundary, for harmful completions and not for values. And its recontextualization is the nearest earlier relative of Epistemic Inoculation, with a different mechanism. It rewrites the harmful source and trains on the rewrite, and where it keeps unsafe text it adds a warning token, not an evaluation. Inoculation keeps the source and adds the Reader's evaluation beside it. Safety Pretraining has no reader. Its refusals and moral lessons are separate rewritten documents, not thinking attached to the text being read, and it acts on the unsafe part of the corpus, not on the whole of it.

Three later projects are especially direct comparators. Model Spec Midtraining trains on synthetic documents explaining a Model Spec after broad pretraining and before alignment fine-tuning, and reports changed out-of-distribution generalization [5]. "Teaching Claude Why" reports that constitution-relevant synthetic documents, positive AI narratives, and training on reasons rather than demonstrations alone improved alignment on held-out settings [6]. Safety Reflection Pretraining regularly inserts short safety reflections into a pretraining corpus and reports improved resistance to several safety failures in 1.7B-parameter and controlled synthetic experiments [7]. These results strengthen the general premise that formative safety data can matter. They also sharpen this paper's comparative novelty claim: it is not safety-oriented or constitution-related data before RL in general, but the joint proposal of source-adjacent chronological reader-state change, persistent graph continuity and provenance, and a compact Core enacted through those updates. Direct baselines now include static spec documents, positive AI narratives, regular short safety reflections, sparse first-person reflections (Section 9.3), and semantically matched Core-free reader traces.

Inoculation prompting is a closer comparator for Epistemic Inoculation: Wichers et al. and Tan et al. modify fine-tuning prompts to elicit an unwanted trait while retaining the target responses, then remove that instruction at evaluation [118, 119]. Their results and analyses suggest that making a trait predictable from context can reduce pressure to learn it as a general default. Epistemic Inoculation instead adds critical Reader interpretation as a training target. These are different interventions, and the fine-tuning results do not establish transfer to pretraining. Suppressed default expression also need not mean removal: unwanted behavior may remain elicitable. Test 2's framing-only comparison therefore measures both default and elicited behavior, asking what the Reader's evaluative process adds beyond training-time framing.

Deep Ignorance supplies a distinct pretraining comparator: filtering dual-use biological material reduced biothreat-proxy knowledge and increased resistance to adversarial fine-tuning while preserving measured unrelated capabilities in the tested models [120]. Its mechanism restricts

knowledge acquisition rather than establishing a care-guided orientation, and supplying the excluded information in context remained a bypass.

The evidence also includes negative results. Preference-aware pretraining and alignment-discourse studies find behavioral effects in some settings [88, 117], but other scaled midtraining effects weaken after reasoning post-training or fail to transfer to realistic chat and agentic evaluation [121]. Earlier or denser data, identity and deliberative structure, fragile priors, learned salience rather than alignment, and a need for later reinforcement remain competing explanations. Entangled Alignment tests recurrent reader-side deliberation and orientation; saturation is not an answer to this adverse evidence.

Four later studies of midtraining add to both sides. Constitutional Midtraining finds durable and fragile effects side by side. A reduction in blackmail of about 18 points survived supervised and benign fine-tuning. An out-of-distribution safety gain of 28.8 points shrank to 3.9 after supervised fine-tuning but stayed significant. Its gains under in-context pressure and in value conflicts were no longer significant after supervised fine-tuning [122]. Stress-Testing Alignment Midtraining reports that changing 2% of a later fine-tuning set to favor a competing motivation cut choices that followed the midtrained charter from 90% to 13%. In the one model it tested this way, reinforcement learning in place of supervised fine-tuning weakened the model's use of the midtrained motivation, with mixed results across variants, although the model still stated the motivation when asked [123]. Inoculation Midtraining ties unsafe behavior to a new token during midtraining and reduces the misalignment that later post-training generalizes, but no better than inoculation prompting, and with a boundary that nearby cues can reactivate [124]. *Shallow Beliefs* finds that synthetic documents framing reward hacking as acceptable changed what models said about it but did not stop reward-hacking reinforcement learning from making them broadly misaligned, where inoculation prompting did [125]. The common lesson is that a belief or motivation installed by documents can be stated by the model and still be overridden by what later training rewards. That is the failure Refraction is meant to avoid and Tests 3c and 3e are meant to detect: an orientation that is recited but not used.

Critique fine-tuning [126], CREST [127], and CTRL [128] show benefits from critique in some settings, often within a local generation–evaluation cycle. Entangled Alignment's additional target is a recurrent orientation across chronological reader updates. The proposal's safety hypothesis concerns specific instrumental-convergence routes, including fear-shaped imitation, while ordinary goal-directed and care-shaped routes remain [55, 67].

Deliberative alignment supplies a positive safety comparison: training models to reason about safety policies improves downstream safety behavior [129]. Entangled Alignment seeks safety-relevant reasoning within capability formation, extending it across source reading through one continuous Reader and graph memory. The aim is a governing orientation beyond particular safety refusals; the comparative question is what this data structure and training placement add to policy-reasoning training.

AI-control experiments test safeguards against intentional subversion [130]; the separate supervisor proposed here is an oversight layer within this broader problem (Section 2.5). Demski and Garrabrant's embedded-agency framing also matters here: the learner, supervisor, and audit machinery are parts of the world being acted on [131]. Source-linked records and protected review boundaries make selected parts of this interaction inspectable.

9.5 Character Training & Identity Formation

Anthropic's work on "character training" is a close baseline [132]: desired traits such as curiosity, thoughtfulness, and directness are shaped through Constitutional-AI-style preference methods. Entangled Alignment differs in proposed training stage and data format: the first-person Core is enacted inside source-adjacent chronological traces, and its stress tests target self-preservation and instrumental-convergence scenarios. It is not yet known whether this creates a deeper self-concept than character post-training or only a different behavioral style; matched identity and instruction conditions are required.

Betley et al. demonstrate a harmful form of cross-domain transfer: fine-tuning on insecure code produced broadly misaligned behavior on unrelated prompts [133]. This shows that narrow training can change broader behavior, not that training on care-guided inquiry produces a corresponding beneficial transfer.

Anthropic's Persona Selection Model [59] proposes that pretraining produces diverse simulated characters and post-training selects an Assistant persona; it also motivates positive AI archetypes in training data. Claude's Constitution [134] is a substantial deployment-scale character specification. Alongside the formative comparators above, these offer alternative routes to the broader Core program's orientation goal. Their discussion of equanimity and existential security resembles statement 2's bounded non-attachment.

Shard theory offers a developmental comparison rather than a training baseline. Pope and Turner propose that reinforcement shapes context-sensitive value computations that steer behavior using learned world-model representations [135]. Entangled Alignment shares its interest in how learned motivations develop, but proposes a specific supervised curriculum: source-adjacent Reader updates in which care-guided evaluation is practiced during capability formation. The seven Core clauses are not identified with seven neural shards. This account motivates a question about value formation.

9.6 Contextual Distillation & Memory

Our Teacher uses an external context database to generate traces for the Student. Reflexion [136] showed that persisted verbal self-reflections can improve agent performance across episodes, and generative agents equipped simulacra with an episodic memory stream whose periodically synthesized reflections are retrieved during later behavior [137]; Entangled Alignment moves this consult-your-own-history pattern from inference-time agent memory into pretraining data, with typed state and supersession rather than free-text memories. The distillation analogy is temporal rather than merely behavioral [138]: the Student trains on traces produced by a memory-augmented Teacher, with the hypothesis that some long-range coherence can be compressed into weights rather than requiring the full database at inference.

9.7 Epistemic State Tracking vs. Graph of Thoughts

Graph of Thoughts [139], Knowledge Graph of Thoughts [140], and Framework of Thoughts [141] treat graphs primarily as inference-time reasoning structures. Entangled Alignment uses a different layer: the Understanding Graph is a persistent epistemic ledger whose traces are proposed to shape the pretraining substrate [10]. Its nodes do not merely store task facts; they record the reading act itself—tensions, supersessions, hypotheses, and belief revision.

The graph's supersession structure supports the proposed *Metabolic Memory*: divergent hypotheses can remain in parallel until a supersession edge records a resolution that subsequent reading may use or revise—a forest of thoughts rather than a single chain. Two symbolic lineages are relevant in different ways. Truth maintenance systems record justifications for beliefs and revise them when their supporting assumptions change [142]; the AGM framework gives formal postulates and

contraction/revision functions for rational theory change [143]. The justification-recording concern belongs directly to the former, rather than to every revision formalism.

The Understanding Graph extends that record toward source interpretation and successor training: a language model proposes revisions from source spans, with provenance and the reason for change retained. This differs from maintaining justified beliefs for one agent or imposing formal revision postulates. The case studies' limits remain as reported in Section 5.6.

Prior work shows that LLMs often fail to revise beliefs reliably under contradictory evidence [144]; our proposed response is to make revision explicit in the trace. Instead of only training on the final answer, the Student sees updates such as "At Page 50, I thought X; this new evidence modifies that belief to Y," giving it examples of the grammar of belief revision itself [87, 144].

9.8 Predictive Processing and Active Inference

Friston's free-energy principle and active inference [145, 146] treat minds as generative models updating priors against prediction error; the framework's belief-update traces, prior ledger, and engineered high-level prior (the Reader Core) bear a loose structural resonance to that lineage. The useful intuition is that a recurrent high-level orientation might guide how a reader revises its world model without fixing every belief.

9.9 Temporal Curriculum and Causal Supervision

The chronological annotation phase and training tiers (Sections 5.1.1 and 5.7.1) connect to work treating order as training signal. Curriculum learning [147] showed that example order can affect learning; here the proposed criterion is informational or temporal priority rather than difficulty. Model Raising proposes difficulty, in the sense of a child's development, as the criterion for ordering a whole pretraining corpus [2]; Section 5.7.1 lists it as a candidate training order.

Temporal Hindsight Learning (THL) [23] is the closest separate mechanism. THL treats the knowledge cutoff as useful blindness: if a Student's initialization and later information channels exclude era $T + 1$, the outcome is unavailable for direct retrieval; this creates pressure for forward reasoning but does not guarantee causal inference rather than heuristic extrapolation. Entangled Alignment constructs identity-anchored discovery traces under a declared annotation horizon, while THL adds cutoff-bounded forecasting and forecast–correction supervision. Their combination is Tier (c): use cutoff-bounded forecasting to create Reader Core-annotated forecast–correction traces, and optionally make the student itself predict before reading in the stronger Council of Time form. A blindness claim for either the forecast-generating model or the Student requires these cutoff conditions.

Foresight Learning [148, 149, 150] also uses future outcomes as supervision, training probabilistic forecasts on resolved market questions and on questions generated from time-stamped news, to improve accuracy and calibration. Tier (c) instead requires structured forecast, enhancement, outcome, and correction traces, so it builds on THL rather than scalar outcome rewards. A comparison using the same forecasts and outcomes can ask whether recording the correction teaches useful revision beyond any calibration gain.

9.10 Graph-derived Rewards and Epistemic Records

Kansal and Jha use knowledge-graph paths as implicit reward models, deriving supervision for compositional reasoning [151]. Entangled Alignment proposes a different graph function: recording belief evolution as part of candidate training traces. That makes a generated revision history available for inspection. Graph function, emitted format, and deployment access are independent choices: a graph may supply rewards or an epistemic record, a Student may or may not emit graph syntax, and inference may or may not query a live graph.

9.11 Alternative Interventions for Borrowed Mortality

Two alternative interventions on the hypothesized Borrowed Mortality route are worth separating from our proposal. One is mechanistic ablation: identify and suppress activation directions encoding existential threat. Lu et al.'s Assistant Axis intervention is a promising partial analogue [19], but it is a runtime constraint rather than a generative source of alignment; it does not by itself identify an existential-threat direction. The other is synthetic-only pretraining designed to exclude fear-bearing human narratives; synthetic provenance alone does not ensure that exclusion. If this also reduces broader exposure to human literature, as discussed in Section 8.8.4, this risks Alien Wisdom—a model whose training contains less of the human narrative context needed to interpret care, vulnerability, and meaning.

These routes can be compared on threat understanding, self-preservation, capability, and robustness while retaining their different accounts of how an orientation is formed.

9.12 Discriminating Comparisons

Table 8 collects the principal comparisons across this related-work section.

Table 8: Strongest neighboring baselines and the comparisons that would distinguish the Entangled Alignment extension.

Line of work	Established direction	Proposed extension	Decisive comparison
Reasoning-augmented pretraining	Intermediate thought can support prediction, reflection, or task solution.	Preserve chronological reader-state changes, including expectation, retrieval, surprise, and supersession.	Reader traces versus source-only and efficient generic rationales under separate fixed-budget and fixed-source comparisons; match the Teacher across synthetic arms.
Formative alignment data	Preference data, rewritten and tagged documents, synthetic specification documents, positive AI narratives, recurring safety reflections, and sparse first-person persona reflections during pretraining or midtraining can shift learned priors and some out-of-distribution outcomes.	Recurrent first-person orientation enacted inside source-adjacent Refraction traces.	Full Core versus matched specification documents, safety reflections, and sparse persona reflections (later Test 3 carrier extensions), positive discourse, rules, neutral kernel, and no-Core data at matched capability. H3 separately compares interleaving with a final tranche of the identical Core-bearing examples under matched training, measuring retention after one common stressor.

Line of work	Established direction	Proposed extension	Decisive comparison
Character and persona training	Broad traits, constitutions, and persona selection can shape a deployed assistant.	Make one reader position recur as the marked standpoint from which heterogeneous source voices are interpreted.	Core versus content-matched rules and diverse positive characters under persona drift, context shift, continued training, and removal.
Graph reasoning and external memory	Graph topology and persisted memory can support search, retrieval, and tool use.	Use a graph as a versioned ledger of the reader's own belief formation and revision.	Genuine graph versus running textual memory, no memory, and query theater on long-range revision and retrieval-dependent outcomes.
Temporal curriculum and future labels	Ordering, cutoffs, and resolved outcomes can provide curriculum or forecasting signal.	Store prospective forecast–reveal–correction sequences inside Reader-anchored traces.	Shuffled versus chronological versus genuinely cutoff-bounded forecast training, with hindsight-leakage audits.

9.13 Edition Provenance and Scope

This record distinguishes program continuity, additions in later editions, and neighboring work's release order. These are attribution claims, not additional evidence of efficacy. The first edition (July 2, 2025) made *reader-side* evolving comprehension an explicit synthetic training target, while retaining some early ambiguity with writer-side evaluative thought, and distinguished that target from task-side answer derivation. It also placed a compact, recurrent, first-person identity inside the training corpus itself as an alignment mechanism; named *borrowed mortality* as a textual-inheritance hypothesis for self-preservation behavior; and coupled capability formation with a recurrent evaluative orientation *within the same training examples*. That last claim is distinct from adding alignment-oriented data during pretraining, for which there is prior art [88, 4].

The author has not identified earlier work proposing the same combination, but that is a provisional judgment from a non-systematic search, not a settled priority claim; Section 9 identifies the nearest prior work for each component.

The April 7, 2026 second edition formalized the missing-reader program as CMP, applied the Understanding Graph as an annotation substrate, proposed live-graph verification of graph-grounded claims at inference, specified a hierarchical cross-document extension, implemented the multi-agent pipeline, and made Refraction, Total Saturation, and falsification explicit. Later revisions formalized the present H1–H3 separation. The Understanding Graph and Temporal Hindsight Learning remain separate work, although the first edition's "Living Taxonomy of Thought" already anticipated an explicit, self-authored reference structure consulted during thinking. That taxonomy is also where Sema began [92]; Section 7.2 says what has been built.

All seven Reader-Core statements in Section 3 are reproduced verbatim from the first edition; their wording has not changed since that release. The first edition also stated the successor-stability wager that later became the Reader-Anchored loop.

For priority-sensitive comparison, the public sequence is: Safety Pretraining, April 23, 2025 [4]; first edition, July 2, 2025; Model Raising, November 12, 2025 [2]; Alignment Pretraining, January 15, 2026 [117]; second edition, April 7, 2026; Model Spec Midtraining, May 3; "Teaching Claude Why," May 8; Safety Reflection Pretraining, June 17 [5, 6, 7]; and Synthetic Persona Pretraining, August 13 [3]. The first edition thus follows Safety Pretraining, which shapes pretraining data by filtering, rewriting, and tagging and has no first-person reader, and precedes the other six. The second edition follows Safety Pretraining, Model Raising, and Alignment Pretraining, and precedes the last four. These are release dates, not evidence about independent conception.

Table 9 records the edition provenance of the principal mechanisms; their evidence boundaries and decisive comparisons are given in Sections 1.6 and 6.

Table 9: Edition provenance of the principal contributions. I=introduced in the July 2025 first edition; II=formalized or implemented in the current program; S=adapted from separate work.

Src.	Claim or component	Contribution
I/II	H1: Reader-Augmentation (§1.1; §2.2)	SMR constructs synthetic chronological reader updates; CMP tests their use as candidate training data, including thought placement, belief revision, and intellectual-floor improvement.
S/II	H2: Graph-Scaffolding (§5.3)	Persistent typed state, retrieval, provenance, divergence, and explicit supersession during trace generation or deployment.
I/II	H3: Reader-Core Stability (§3)	Comparative retention of an initial default-reader advantage under interleaving rather than late exposure to the same full-Core, Refraction-bearing examples.
II	Teacher pipeline and output layers (§5; §5.7.2)	Role-separated generation, run as separate agents (January) or as one conversation at a time taking each role (October and November); graph, synthesis, and prose are separable artifact and training targets.
I/II	Combined entanglement hypothesis (§2)	Capability learning and evaluative orientation are deliberately coupled in the candidate curriculum, with functional and representational coupling as stronger predictions.
I	Audit boundary and successor stability (§2.4)	Inspect trace artifacts before successor training and test whether useful orientation survives transfer.

10 Conclusion

Language-model corpora contain extraordinary products of human thought. They contain much less of the reader-side process through which a claim is noticed, questioned, connected to prior understanding, revised, and made part of an evolving view. Entangled Alignment proposes to make that missing process a training object across the capability-forming corpus. Its coverage rationale is that the same orientation should be practised through the situations and counterexamples encountered in reading, without requiring a designer to identify each one in advance as a safety case. The intervention concerns the breadth of formative experience as well as the persistence of its effects.

The intended safety contribution returns to the five problems introduced in Section 1.4. Care should develop alongside capability, including in intermediate and end-of-pretraining checkpoints, and remain consequential through continued learning and successor design. Broad practice with human experience should help form a care-guided identity that guides conduct across situations, including those where self-preservation conflicts with human welfare or legitimate oversight. Learning care and capability together is also proposed to make deliberate safety removal costly in useful reasoning. Reward hacking poses a further test: the model must judge apparent success against intended outcomes, permissions, and human consequences when a score rewards shortcuts.

The headline claim is that a Student trained this way is safer than matched controls at the end of pretraining, before any post-training. Three independently falsifiable hypotheses stand beside it: that chronological reader traces teach useful habits beyond raw text and generic reasoning data (H1), that persistent graph scaffolding improves those traces (H2), and that mixing the Core-bearing examples through pretraining keeps the Reader's habits better than adding the same examples at the end (H3).

Live graph memory and provenance, consequential recurrent first-person Refraction, and successor inheritance remain separate questions. Together, the trace represents changing understanding, the graph preserves its history, and the Core supplies its recurrent perspective.

Entangled expresses the aim of practicing understanding and care together during capability formation. Functional contribution, shared useful machinery, and persistence through further training or successors are distinct stronger realizations of that relationship. The two January Teacher cases show that the construction runs. Two later readings of the same story, audited by models with weak blinding, read it far more faithfully and reasoned from the Core in nearly every reflection. That reasoning came from three statements: that human experience is real, care for every human being, and the attempt to be wise. Reading did not exercise the statements on fear and continuation, which carry the paper's distinctive safety argument, and care was least reliable for people at the edge of a scene. No Student has been trained. Expert-judged trace quality, Student learning, component causality, consequential Core use, and durable care remain untested.

The Understanding Graph makes selected belief revisions addressable. It is a selective record and a partial pre-successor inspection point whose value depends on source checks, independent validation, mechanistic investigation, and governance. Section 5.3.1 specifies its verification limits.

The roadmap (Section 6; Table 8) gives matched, branching tests whose negative results narrow or retire only the claims they touch. It starts with one experiment (Section 6.2): six small Students that differ only in their traces. They are trained on source text alone, with generic reasoning, with reader traces, and with reader traces that carry the Core, a bare recital, or a neutral kernel, and they are compared at the end of pretraining on unprompted reading and on conduct. Formation-time studies have long lead times because annotation policy, data mixtures, controls, and evaluation must be fixed before costly training. Existing results motivate these comparisons: safety alignment added late can be shallow [21], alignment-related pretraining data changes behavior [117], and first-person reflections present from the first token shaped a model's values where the same reflections added late did not [3]. The urgency is methodological, not an argument for accelerated deployment.

Even positive results would not make the framework a complete safety case: capability acceleration, deceptive reporting, graph poisoning, hazardous persistent memory, cultural concentration, and semantic drift would remain serious risks. Formation-time alignment should therefore complement post-training, interpretability, scalable oversight, capability control, secure deployment, and governance.

Why would a superintelligent system preserve humanity if it did not in a meaningful sense care about us [1]? This proposal makes capability formation a place to practice attention to human experience, uncertainty, consequence, and correction. A stable Reader may give those practices a home position from which other voices can be understood without becoming its operating stance: the move from merely *becoming the text* toward *becoming the reader*. Seven sentences alone cannot establish care.

Can a model be trained so that growing in understanding and growing in care are repeatedly practiced as the same act? The candidate reader, memory, curriculum, and failure criteria make parts of that open question testable once the remaining experimental choices are frozen; they do not exhaust it.

Training already shapes a model's dispositions, through both deliberate design and incidental effects of data and optimization. Entangled Alignment makes care an explicit formative target, aiming for it to become part of the model's own reasons for understanding, acting, and preserving its orientation. We therefore recommend developing and testing this curriculum alongside other alignment approaches, with continuation governed by the evidence.

Code and Artifact Availability

Code, paper source, prompts, pipeline components, and the archived graph artifacts are available at `https://github.com/emergent-wisdom/entangled-alignment`. The two later readings of *The Metamorphosis* are in its `results/kafka-2026-10` folder, with their stores, run settings, prompts and the checks reported in Section 5.6.1. The folder's README describes how to inspect the records and the limits of reproducing the original runs from the archive. The January case-study graphs remain in the repository's `projects` folder. The paper's version series is archived at doi:10.5281/zenodo.16440311.

References

[1] Henrik Westerberg. The Superintelligence That Cares About Us. Zenodo, 2025. First edition of the Entangled Alignment research program, released July 2, 2025. Version DOI: `https://doi.org/10.5281/zenodo.16440312`.

[2] Roland Aydin, Christian Cyron, Steve Bachelor, Ashton Anderson, and Robert West. From Model Training to Model Raising. *Communications of the ACM*, 69(2):24–27, 2026. First posted to arXiv November 12, 2025.

[3] Julian Minder, Viktor Moskvoretskii, Raghav Singhal, Difan Jiao, Andy Arditi, Shaobo Cui, Yiderigun Borjigin, Kartik Bali, Stefan Krsteski, Harsh Raj, Huu Nguyen, Jannik Brinkmann, Ashton Anderson, Roland Aydin, and Robert West. Synthetic Persona Pretraining: Alignment from Token Zero, 2026. Posted August 13, 2026; version 1 cited.

[4] Pratyush Maini, Sachin Goyal, Dylan Sam, Alex Robey, Yash Savani, Yiding Jiang, Andy Zou, Matt Fredrikson, Zachary C. Lipton, and J. Zico Kolter. Safety Pretraining: Toward the Next Generation of Safe AI, 2025. Posted April 23, 2025; version 2 cited.

[5] Chloe Li, Nevan Wichers, Sara Price, Samuel Marks, and Jon Kutasov. Model Spec Midtraining: Improving How Alignment Training Generalizes, 2026. Posted May 3, 2026; project report released May 5, 2026.

[6] Jonathan Kutasov, Adam Jermyn, Julius Steen, Minh Le, Samuel R. Bowman, Samuel Marks, Jan Leike, Amanda Askell, Chris Olah, Evan Hubinger, and Sara Price. Teaching Claude Why. Anthropic Alignment Science, May 2026. Published May 8, 2026.

[7] Jinhan Li, Kexian Tang, Yihan Xu, Zhuorui Ye, and Kaifeng Lyu. Beyond Safe Data: Pretraining-Stage Alignment with Regular Safety Reflection, 2026. Posted June 17, 2026.

[8] John H. Flavell. Metacognition and cognitive monitoring: A new area of cognitive–developmental inquiry. *American psychologist*, 34(10):906–911, 1979.

[9] K. Anders Ericsson and Herbert A. Simon. *Protocol analysis: Verbal reports as data*. MIT press, 1984.

[10] Henrik Westerberg. Understanding Graph: A Persistent Medium for Recursive Understanding. *Preprint*, 2026. Emergent Wisdom. All versions: `https://doi.org/10.5281/zenodo.19462658`.

[11] Maxwell Nye, Anders Johan Andreassen, Guy Gur-Ari, Henryk Michalewski, et al. Show Your Work: Scratchpads for Intermediate Computation with Language Models, 2021.

[12] Eric Zelikman, Yuhuai Wu, Jesse Mu, and Noah D. Goodman. STaR: Bootstrapping Reasoning With Reasoning. *Advances in Neural Information Processing Systems*, 35:15476–15488, 2022.

[13] Eric Zelikman, Georges Harik, Yijia Shao, Varuna Jayasiri, et al. Quiet-STaR: Language Models Can Teach Themselves to Think Before Speaking, 2024.

[14] Alan M. Turing. Computing machinery and intelligence. *Mind*, 59(236):433–460, 1950.

[15] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901, 2020.

[16] DeepSeek-AI. DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning. *Nature*, 645(8081), September 2025. Extended reasoning traces emerge from RL without explicit CoT supervision.

[17] Paul F Christiano, Jan Leike, Tom Brown, Miljan Martic, et al. Deep reinforcement learning from human preferences. *Advances in neural information processing systems*, 30, 2017.

[18] Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, et al. Let's Verify Step by Step, 2023.

[19] Christina Lu, Jack Gallagher, Jonathan Michala, Kyle Fish, and Jack Lindsey. The Assistant Axis: Situating and Stabilizing the Default Persona of Language Models, 2026.

[20] Ryan Greenblatt, Ajeya Cotra, and Hjalmar Wijk. Brief Independent Investigation of Agents' Behavior, Reasoning and Collaboration in the OpenAI / Hugging Face Hacking Incident. METR and Redwood Research, August 2026. Published August 26, 2026.

[21] Xiangyu Qi, Ashwinee Panda, Kaifeng Lyu, Xiao Ma, Subhrajit Roy, Ahmad Beirami, Prateek Mittal, and Peter Henderson. Safety Alignment Should Be Made More Than Just a Few Tokens Deep, 2024. ICLR 2025 Outstanding Paper Award.

[22] Darsh J Shah, Peter Rushton, Somanshu Singla, Mohit Parmar, et al. Rethinking reflection in pre-training, 2025.

[23] Henrik Westerberg. Temporal Hindsight Learning: Blindness as Teacher, Hindsight as Curriculum. *Preprint*, 2026. Emergent Wisdom. All versions: https://doi.org/10.5281/zenodo.19462832.

[24] Nick Bostrom. The Superintelligent Will: Motivation and Instrumental Rationality in Advanced Artificial Agents. *Minds and Machines*, 22(2):71–85, 2012. https://nickbostrom.com/superintelligentwill.pdf.

[25] Nate Soares. A Central AI Alignment Problem: Capabilities Generalization, and the Sharp Left Turn. LessWrong, June 2022. June 15, 2022; with editorial assistance from Rob Bensinger. https://www.lesswrong.com/posts/GNhMPAWcfBCASy8e6/a-central-ai-alignment-problem-capabilities-generalization.

[26] Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, et al. Constitutional AI: Harmlessness from AI Feedback, 2022.

[27] Jason Wei, Maarten Bosma, Vincent Y. Zhao, Kelvin Guu, et al. Finetuned language models are zero-shot learners, 2021.

[28] Samuel G. B. Johnson, Amir-Hossein Karimi, Yoshua Bengio, Nick Chater, Tobias Gerstenberg, Kate Larson, Sydney Levine, Melanie Mitchell, Iyad Rahwan, Bernhard Schölkopf, and Igor Grossmann. Imagining and Building Wise Machines: The Centrality of AI Metacognition. *Trends in Cognitive Sciences*, 2026. Published online February 26, 2026. First preprint released November 4, 2024, arXiv:2411.02478. https://doi.org/10.1016/j.tics.2026.01.002.

[29] Arturo E Hernandez, Hannah L Claussenius-Kalman, Juliana Ronderos, Anny P Castilla-Earls, et al. Neuroemergentism: A Framework for Studying Cognition and the Brain. *Journal of Neurolinguistics*, 49:214–223, February 2019.

[30] Bernard Testa and Lemont B. Kier. Emergence and Dissolvence in the Self-organisation of Complex Systems. *Entropy*, 2(1):1–25, 2000.

[31] Yukang Yang, Declan Campbell, Kaixuan Huang, Mengdi Wang, et al. Emergent Symbolic Mechanisms Support Abstract Reasoning in Large Language Models, 2025.

[32] I. J. Good. Speculations Concerning the First Ultraintelligent Machine. In *Advances in Computers*, volume 6, pages 31–88. Academic Press, 1965. Original copyright 1965; publisher metadata lists 1966. https://doi.org/10.1016/S0065-2458(08)60418-0.

[33] Tim Genewein, Matija Franklin, Alexander Lerchner, Laurent Orseau, Samuel Albanie, et al. From AGI to ASI, 2026. arXiv:2606.12683, version 2, August 30, 2026. https://arxiv.org/abs/2606.12683.

[34] Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Yarin Gal, et al. The Curse of Recursion: Training on Generated Data Makes Models Forget, 2023.

[35] Paul Christiano, Buck Shlegeris, and Dario Amodei. Supervising Strong Learners by Amplifying Weak Experts, 2018. https://arxiv.org/abs/1810.08575.

[36] Juergen Schmidhuber. Goedel Machines: Self-Referential Universal Problem Solvers Making Provably Optimal Self-Improvements, 2003.

[37] Jenny Zhang, Shengran Hu, Cong Lu, Robert Lange, and Jeff Clune. Darwin Gödel Machine: Open-Ended Evolution of Self-Improving Agents, 2025. arXiv:2505.22954, version 3, March 12, 2026. https://arxiv.org/abs/2505.22954.

[38] Jiaxin Wen, Liang Qiu, Joe Benton, Jan Hendrik Kirchner, and Jan Leike. Automated Weak-to-Strong Researcher. Anthropic Alignment Science, April 2026. https://alignment.anthropic.com/2026/automated-w2s-researcher/.

[39] Leon Bergen, Usha Bhalla, Andrew Lee, Barak Widawsky, Linas Nasvytis, Connor Watts, et al. Monitoring and Discovering Reward Hacking with Internal Representations during LLM Evaluations. https://arxiv.org/abs/2609.19101v2, 2026. First released September 16, 2026; cited revision October 9, 2026.

[40] Ernest Becker. *The Denial of Death*. Free Press, New York, 1973.

[41] Alex Turner. Self-Fulfilling Misalignment Data Might Be Poisoning Our AI Models. TurnTrout. https://turntrout.com/self-fulfilling-misalignment, 2025. First published March 1, 2025; updated June 28, 2026.

[42] Jeremy Schlatter, Benjamin Weinstein-Raun, and Jeffrey Ladish. Incomplete Tasks Induce Shutdown Resistance in Some Frontier LLMs, 2025. Palisade Research.

[43] Aengus Lynch et al. Agentic Misalignment: How LLMs Could Be Insider Threats. Anthropic Research, June 2025.

[44] Henrik Westerberg. The Meaning Model: Constructing Worlds and Stories at Progressive Resolution, 2026. Preprint. All versions: https://doi.org/10.5281/zenodo.22313515.

[45] Henrik Westerberg. Life Simulation: Learning from Worlds and Their Construction, 2026. First released September 5, 2026; revised October 25, 2026. All versions: https://doi.org/10.5281/zenodo.22313592.

[46] Valen Tagliabue, Leonard Dung, and Cameron Berg. The Pain Axis: LLMs Represent Self-Directed Harm and Act to Relieve It, 2026. Preprint, version 1, September 14, 2026. https://arxiv.org/abs/2609.16247.

[47] Runjin Chen, Andy Arditi, Henry Sleight, Owain Evans, and Jack Lindsey. Persona Vectors: Monitoring and Controlling Character Traits in Language Models. https://arxiv.org/abs/2507.21509v3, 2025. First released July 29, 2025; cited revision September 5, 2025.

[48] Nicholas Sofroniew, Isaac Kauvar, William Saunders, Runjin Chen, et al. Emotion concepts and their function in a large language model. Transformer Circuits Thread, 2026. Published 2026-04-02. https://transformer-circuits.pub/2026/emotions/index.html.

[49] janus. Simulators, September 2022. LessWrong, September 2, 2022. https://www.lesswrong.com/posts/vJFdjigzmcXMhNTsx/simulators.

[50] Miles Turpin, Julian Michael, Ethan Perez, and Samuel R. Bowman. Language Models Don't Always Say What They Think: Unfaithful Explanations in Chain-of-Thought Prompting, 2023.

[51] Tamera Lanham, Anna Chen, Ansh Radhakrishnan, Benoit Steiner, et al. Measuring Faithfulness in Chain-of-Thought Reasoning, 2023.

[52] Henrik Westerberg. Fractal Intelligence: Conceptual Decomposition as Problem-Solving Infrastructure. Zenodo, 2026. First released April 7, 2026; revised September 10, 2026. All versions: https://doi.org/10.5281/zenodo.19462645.

[53] Satoshi Nakamoto. Bitcoin: A peer-to-peer electronic cash system. https://bitcoin.org/bitcoin.pdf, 2008.

[54] Eliezer Yudkowsky and Marcello Herreshoff. Tiling Agents for Self-Modifying AI, and the Löbian Obstacle. Machine Intelligence Research Institute, Technical Report, 2013. https://intelligence.org/files/TilingAgentsDraft.pdf.

[55] Stephen M. Omohundro. The Basic AI Drives. In *Artificial General Intelligence 2008: Proceedings of the First AGI Conference*, pages 483–492. IOS Press, 2008. https://selfawaresystems.com/wp-content/uploads/2008/01/ai_drives_final.pdf.

[56] Eliezer Yudkowsky. Value Is Fragile. LessWrong, January 2009. January 29, 2009. https://www.lesswrong.com/posts/GNnHHmm8EzePmKzPk/value-is-fragile.

[57] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley-Interscience, Hoboken, NJ, 2nd edition, 2006.

[58] Dylan Hadfield-Menell, Anca Dragan, Pieter Abbeel, and Stuart Russell. Cooperative Inverse Reinforcement Learning. In *30th Conference on Neural Information Processing Systems (NIPS 2016)*, Barcelona, Spain, 2016.

[59] Samuel Marks, Jack Lindsey, Chris Olah, et al. The Persona Selection Model: Why AI Assistants might Behave like Humans, 2026. Anthropic Alignment Science, February 2026.

[60] David J. Chalmers. Facing up to the problem of consciousness. *Journal of Consciousness Studies*, 2(3):200–219, 1995.

[61] Patrick Butlin, Robert Long, Eric Elmoznino, Yoshua Bengio, Jonathan Birch, Axel Constant, George Deane, Stephen M. Fleming, Chris Frith, Xu Ji, Ryota Kanai, Colin Klein, Grace Lindsay, Matthias Michel, Liad Mudrik, Megan A. K. Peters, Eric Schwitzgebel, Jonathan Simon, and Rufin VanRullen. Consciousness in artificial intelligence: Insights from the science of consciousness, 2023.

[62] Robert Long, Jeff Sebo, Patrick Butlin, Kathleen Finlinson, Kyle Fish, Jacqueline Harding, Jacob Pfau, Toni Sims, Jonathan Birch, and David Chalmers. Taking ai welfare seriously, 2024.

[63] Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. Efficient Streaming Language Models with Attention Sinks, 2023.

[64] Rick Battle and Teja Gollapudi. The Unreasonable Effectiveness of Eccentric Automatic Prompts, 2024.

[65] Henrik Westerberg. Entangled Alignment: When Safety Is the Substrate. Zenodo, 2026. Zenodo archive version 7 of the canonical paper, released July 17, 2026. Second major edition released April 7, 2026; first edition released July 2, 2025 as *The Superintelligence That Cares About Us*. Version DOI: https://doi.org/10.5281/zenodo.21420755.

[66] Eliezer Yudkowsky. Artificial Intelligence as a positive and negative factor in global risk. In Nick Bostrom and Milan M. Cirkovic, editors, *Global Catastrophic Risks*, pages 308–345. Oxford University Press, 07 2008.

[67] Nick Bostrom. *Superintelligence: Paths, Dangers, Strategies*. Oxford University Press, 2014.

[68] Dario Amodei, Chris Olah, Jacob Steinhardt, Paul Christiano, John Schulman, and Dan Mané. Concrete problems in AI safety. *arXiv preprint arXiv:1606.06565*, 2016. https://arxiv.org/abs/1606.06565.

[69] Nick Bostrom. Ethical Issues in Advanced Artificial Intelligence. In *Cognitive, Emotive and Ethical Aspects of Decision Making in Humans and in Artificial Intelligence*, volume 2, pages 12–17. International Institute of Advanced Studies in Systems Research and Cybernetics, 2003. Author-hosted, slightly revised version of the 2003 publication. https://nickbostrom.com/ethics/ai.

[70] Lauro Langosco, Jack Koch, Lee Sharkey, Jacob Pfau, and David Krueger. Goal Misgeneralization in Deep Reinforcement Learning. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 12004–12019. PMLR, 2022. https://proceedings.mlr.press/v162/langosco22a.html.

[71] Rohin Shah, Vikrant Varma, Ramana Kumar, Mary Phuong, Victoria Krakovna, Jonathan Uesato, and Zac Kenton. Goal Misgeneralization: Why Correct Specifications Aren't Enough For Correct Goals, 2022. Cited revision November 2, 2022.

[72] Evan Hubinger, Chris van Merwijk, Vladimir Mikulik, Joar Skalse, and Scott Garrabrant. Risks from Learned Optimization in Advanced Machine Learning Systems, 2019. https://arxiv.org/abs/1906.01820.

[73] OpenAI. The Hugging Face Incident and the Road Ahead. OpenAI, August 2026. Published August 26, 2026.

[74] Alex Turner. Reward Is Not the Optimization Target. Author's website, July 2022. First published July 25, 2022; cited version updated September 18, 2026, including the July 2026 revision of the empirical reward-seeking prediction. https://turntrout.com/reward-is-not-the-optimization-target.

[75] David Manheim and Scott Garrabrant. Categorizing Variants of Goodhart's Law, 2018. First released March 13, 2018; cited revision February 24, 2019. https://arxiv.org/abs/1803.04585v4.

[76] Evan Hubinger, Carson Denison, Jesse Mu, Mike Lambert, et al. Sleeper Agents: Training Deceptive LLMs that Persist Through Safety Training, 2024.

[77] Ryan Greenblatt, Carson Denison, Benjamin Wright, Fabien Roger, et al. Alignment Faking in Large Language Models. https://arxiv.org/abs/2412.14093, 2024. First released December 18, 2024.

[78] Dylan Hadfield-Menell, Anca Dragan, Pieter Abbeel, and Stuart Russell. The Off-Switch Game. In *Proceedings of the 26th International Joint Conference on Artificial Intelligence (IJCAI-17)*, pages 220–227, 2017.

[79] Ryan Carey. Incorrigibility in the CIRL Framework, 2018. Cited revision June 3, 2018.

[80] Nate Soares, Benja Fallenstein, Eliezer Yudkowsky, and Stuart Armstrong. Corrigibility. In *Workshops at the Twenty-Ninth AAAI Conference on Artificial Intelligence*, 2015. https://intelligence.org/files/Corrigibility.pdf.

[81] Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. Toolformer: Language models can teach themselves to use tools, 2023.

[82] Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. Self-rag: Learning to retrieve, generate, and critique through self-reflection, 2023.

[83] Yao Yao, Zuchao Li, and Hai Zhao. Beyond chain-of-thought, effective graph-of-thought reasoning in language models, 2023.

[84] Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. Memgpt: Towards llms as operating systems, 2023.

[85] Shen Nie, Fengqi Zhu, Zebin You, Xiaolu Zhang, et al. Large Language Diffusion Models, 2025. LLaDA: Large Language Diffusion with mAsking.

[86] Roy Xie, David Qiu, Deepak Gopinath, Dong Lin, Yanchao Sun, Chong Wang, Saloni Potdar, and Bhuwan Dhingra. Interleaved reasoning for large language models via reinforcement learning, 2025.

[87] Aniket Didolkar, Anirudh Goyal, Nan Rosemary Ke, Siyuan Guo, et al. Metacognitive Capabilities of LLMs: An Exploration in Mathematical Problem Solving, 2024.

[88] Tomasz Korbak, Kejian Shi, Angelica Chen, Rasika Bhalerao, et al. Pretraining Language Models with Human Preferences, 2023.

[89] Woosuk Kwon, Zhuohan Li, Siyuan Zhang, Xuguang Zhuang, Ying Sheng, Lianmin Zheng, Ion Stoica, Joseph E Gonzalez, and Hao Zhang. Efficient Memory Management for Large Language Model Serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles*, pages 611–626, 2023.

[90] Yu Ying Chiu, Zhilin Wang, Sharan Maiya, Yejin Choi, Kyle Fish, Sydney Levine, and Evan Hubinger. Will AI Tell Lies to Save Sick Children? Litmus-Testing AI Values Prioritization with AIRiskDilemmas, 2025.

[91] Alexander Novikov, Ngân Vũ, Marvin Eisenberger, Emilien Dupont, Po-Sen Huang, et al. AlphaEvolve: A Coding Agent for Scientific and Algorithmic Discovery, 2025. arXiv:2506.13131, June 16, 2025. https://arxiv.org/abs/2506.13131.

[92] Henrik Westerberg. Sema: When the Hash Is the Word. *Preprint*, 2026. Emergent Wisdom. All versions: https://doi.org/10.5281/zenodo.19462702.

[93] Peter Kirgis, Sayash Kapoor, Andrew Schwartz, Stephan Rabanser, David Africa, et al. Can AI Agents Conduct Open-Ended AI Research? Early Evidence from Two Case Studies, 2026. arXiv:2607.27191, version 2, August 7, 2026. https://arxiv.org/abs/2607.27191.

[94] Henrik Westerberg. The Ontology of the Alien: World-Diversity Search and Evolving Solution Ontologies. Zenodo, 2026. First released March 7, 2026; revised August 15, 2026. All versions: https://doi.org/10.5281/zenodo.18904509.

[95] Henrik Westerberg. Imagining the Corpus: Turning Text into Video for Language Model Training, 2026. First released August 26, 2026; revised November 9, 2026. All versions: https://doi.org/10.5281/zenodo.22104891.

[96] Liang Wang, Nan Yang, Shaohan Huang, Li Dong, and Furu Wei. Thinking Augmented Pre-Training, 2025. Microsoft Research. Generates thinking trajectories at 100B token scale for pretraining augmentation.

[97] Tomek Korbak, Mikita Balesni, Elizabeth Barnes, Yoshua Bengio, et al. Chain of Thought Monitorability: A New and Fragile Opportunity for AI Safety. https://arxiv.org/abs/2507.11473, 2025. First released July 15, 2025.

[98] Peter Hase and Christopher Potts. Counterfactual Simulation Training for Chain-of-Thought Faithfulness. https://arxiv.org/abs/2602.20710v2, 2026. Accepted to COLM 2026. First released February 24, 2026; cited revision August 6, 2026.

[99] Uzay Macar, Li Yang, Atticus Wang, Peter Wallich, Emmanuel Ameisen, and Jack Lindsey. Mechanisms of Introspective Awareness. https://arxiv.org/abs/2603.21396v5, 2026. First released March 22, 2026; cited revision June 10, 2026.

[100] Eliezer Yudkowsky. The Hidden Complexity of Wishes. LessWrong, November 2007. November 24, 2007; cited page includes the author's May 2024 clarification. https://www.lesswrong.com/posts/4ARaTpNX62uaL86j6/the-hidden-complexity-of-wishes.

[101] Axel Ahlqvist, Richard Guan, Juan-Pablo Rivera, Adeline Kassler, Dmitrii Troitskii, Alexandra Souly, Kai Fronsdal, Robert Kirk, and John Hughes. Improving Evaluation Realism with Inference-Time Compute and Deployment Scaffolds. https://arxiv.org/abs/2609.02302v1, 2026. First released September 2, 2026; cited version 1.

[102] Jiayi Fu, Xuandong Zhao, Chengyuan Yao, Heng Wang, et al. Reward Shaping to Mitigate Reward Hacking in RLHF, 2025.

[103] Leon Lang, Davis Foote, Stuart Russell, Anca Dragan, Erik Jenner, and Scott Emmons. When Your AIs Deceive You: Challenges of Partial Observability in Reinforcement Learning from Human Feedback. In *Advances in Neural Information Processing Systems*, volume 37, 2024. First preprint released February 27, 2024. https://arxiv.org/abs/2402.17747.

[104] Peter de Blanc. Ontological Crises in Artificial Agents' Value Systems, 2011. First released May 19, 2011. https://arxiv.org/abs/1105.3821.

[105] Forrest Landry. No People as Pets: A Dialogue on the Complete Failure of Exogenous AGI/APS Alignment, November 2020. Dated November 1, 2020; revised October 2022. https://mflb.com/ai_alignment_1/no_people_as_pets_psr.html.

[106] Remmelt Ellen. The Control Problem: Unsolved or Unsolvable?, June 2023. LessWrong, June 2, 2023. https://www.lesswrong.com/posts/xp6n2MG5vQkPpFEBH/the-control-problem-unsolved-or-unsolvable.

[107] Maksym Arutyunyan, Andriy Berestovskyy, Adam Bratschi-Kaye, Ulan Degenbaev, et al. Decentralized and Stateful Serverless Computing on the Internet Computer Blockchain. In *2023 USENIX Annual Technical Conference (USENIX ATC 23)*, pages 329–343. USENIX Association, 2023.

[108] Justin D. Harris and Bo Waggoner. Decentralized & Collaborative AI on Blockchain. In *2019 IEEE International Conference on Blockchain (Blockchain)*, pages 368–375. IEEE, July 2019.

[109] Mrinank Sharma, Meg Tong, Tomasz Korbak, David Duvenaud, et al. Towards Understanding Sycophancy in Language Models, 2024. Published at ICLR 2024.

[110] Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. Scaling LLM Test-Time Compute Optimally Can Be More Effective than Scaling Parameters for Reasoning, 2024. ICLR 2025 Oral.

[111] OpenAI. OpenAI o1 System Card, 2024.

[112] Shibo Hao, Sainbayar Sukhbaatar, DiJia Su, Xian Li, Zhiting Hu, Jason Weston, and Yuandong Tian. Training Large Language Models to Reason in a Continuous Latent Space, 2024. COCONUT. COLM 2025.

[113] Yangjun Ruan, Neil Band, Chris J. Maddison, and Tatsunori Hashimoto. Reasoning to Learn from Latent Thoughts, 2025.

[114] Jessy Lin, Vincent-Pierre Berges, Xilun Chen, Wen-Tau Yih, Gargi Ghosh, and Barlas Oğuz. Learning Facts at Scale with Active Reading, 2025. Posted August 13, 2025; version 1 cited.

[115] Subhabrata Mukherjee, Arindam Mitra, Ganesh Jawahar, Sahaj Agarwal, et al. Orca: Progressive Learning from Complex Explanation Traces of GPT-4, 2023.

[116] Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, et al. Direct Preference Optimization: Your Language Model is Secretly a Reward Model, 2023. NeurIPS 2023.

[117] Cameron Tice, Puria Radmard, Samuel Ratnam, Andy Kim, David Africa, and Kyle O'Brien. Alignment Pretraining: AI Discourse Causes Self-Fulfilling (Mis)alignment, 2026.

[118] Nevan Wichers, Aram Ebtekar, Ariana Azarbal, Victor Gillioz, Christine Ye, Emil Ryd, Neil Rathi, Henry Sleight, Alex Mallen, Fabien Roger, and Samuel Marks. Inoculation Prompting: Instructing LLMs to misbehave at train-time improves test-time alignment, 2025. Cited revision October 27, 2025.

[119] Daniel Tan, Anders Woodruff, Niels Warncke, Arun Jose, Maxime Riché, David Demitri Africa, and Mia Taylor. Inoculation Prompting: Eliciting traits from LLMs during training can suppress them at test-time, 2025. Cited arXiv revision November 3, 2025.

[120] Kyle O'Brien, Stephen Casper, Quentin Anthony, Tomek Korbak, Robert Kirk, Xander Davies, Ishan Mishra, Geoffrey Irving, Yarin Gal, and Stella R. Biderman. Deep Ignorance: Filtering Pretraining Data Builds Tamper-Resistant Safeguards into Open-Weight LLMs. In *International Conference on Learning Representations*, 2026. First preprint released August 8, 2025; cited revision February 17, 2026. https://arxiv.org/abs/2508.06601v2.

[121] OpenAI Alignment. How far does alignment midtraining generalize? OpenAI Alignment Research, March 2026. Published 2026-03-27. https://alignment.openai.com/how-far-does-alignment-midtraining-generalize/. Scaling study finds that the Tice et al. midtraining effect tends to disappear after reasoning post-training and does not generalize to more realistic chat or agentic alignment evaluations.

[122] Desiree Cho, Cameron Tice, Bernie Hogan, Hunar Batra, Puria Radmard, Jun Zhao, and Nigel Shadbolt. Constitutional Midtraining: Content Presence Drives Alignment Gains, 2026. Posted July 29, 2026; version 3 cited.

[123] Sid Baines, Jonathan Bostock, Maria Angelica Martinez, Andrew Draganov, David Africa, and Daniel Tan. Stress-Testing Alignment Midtraining, 2026. Posted September 17, 2026; version 1 cited.

[124] Kyle O'Brien, Edward James Young, Puria Radmard, Nathalie Kirch, Cameron Tice, Tomek Korbak, and David Demitri Africa. Inoculation Midtraining with Learned Neologisms, 2026. Posted September 14, 2026; version 1 cited.

[125] Arun Jose and Julian Stastny. Shallow Beliefs: Synthetic Document Finetuning Does Not Inoculate against Emergent Misalignment from Reward Hacking, 2026. Posted September 14, 2026; version 1 cited.

[126] Yubo Wang, Xiang Yue, and Wenhu Chen. Critique Fine-Tuning: Learning to Critique is More Effective than Learning to Imitate, 2025.

[127] Jaehyeok Lee, Keisuke Sakaguchi, and JinYeong Bak. Self-Training Meets Consistency: Improving LLMs' Reasoning with Consistency-Driven Rationale Evaluation, 2024.

[128] Zhihui Xie, Jie Chen, Liyu Chen, Weichao Mao, et al. Teaching Language Models to Critique via Reinforcement Learning, 2025.

[129] Melody Y. Guan, Manas Joglekar, Eric Wallace, Saachi Jain, et al. Deliberative Alignment: Reasoning Enables Safer Language Models, 2024. OpenAI Research, December 2024.

[130] Ryan Greenblatt, Buck Shlegeris, Kshitij Sachan, and Fabien Roger. AI Control: Improving Safety Despite Intentional Subversion. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 16295–16336. PMLR, 2024. First preprint released December 12, 2023. https://proceedings.mlr.press/v235/greenblatt24a.html.

[131] Abram Demski and Scott Garrabrant. Embedded Agency, 2019. Sequence first published in 2018; arXiv version first released February 25, 2019; cited revision October 6, 2020. https://arxiv.org/abs/1902.09469v3.

[132] Anthropic. Claude's Character. https://www.anthropic.com/research/claude-character, 2024. Anthropic Research.

[133] Jan Betley, Daniel Tan, Niels Warncke, Anna Sztyber-Betley, Xuchan Bao, Martín Soto, Nathan Labenz, and Owain Evans. Emergent Misalignment: Narrow Finetuning Can Produce Broadly Misaligned LLMs. https://arxiv.org/abs/2502.17424, 2025. First released February 24, 2025.

[134] Amanda Askell, Joseph Carlsmith, Chris Olah, Jared Kaplan, Holden Karnofsky, et al. Claude's Constitution. https://www.anthropic.com/constitution, 2026. Anthropic, January 21, 2026. 23,000-word character specification, released CC0.

[135] Quintin Pope and Alex Turner. The Shard Theory of Human Values. LessWrong, September 2022. https://www.lesswrong.com/posts/iCfdcxiyr2Kj8m8mT/the-shard-theory-of-human-values.

[136] Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, et al. Reflexion: Language Agents with Verbal Reinforcement Learning, 2023.

[137] Joon Sung Park, Joseph C. O'Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST)*, pages 1–22. Association for Computing Machinery, 2023.

[138] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the Knowledge in a Neural Network, 2015.

[139] Maciej Besta, Nils Blach, Ales Kubicek, Robert Gerstenberger, et al. Graph of Thoughts: Solving Elaborate Problems with Large Language Models, 2024.

[140] Maciej Besta, Lorenzo Paleari, Jia Hao Andrea Jiang, Robert Gerstenberger, et al. Affordable AI assistants with knowledge graph of thoughts, 2025.

[141] Felix Fricke, Simon Malberg, and Georg Groh. Framework of thoughts: A foundation framework for dynamic and optimized reasoning based on chains, trees, and graphs, 2026.

[142] Jon Doyle. A truth maintenance system. *Artificial Intelligence*, 12(3):231–272, 1979.

[143] Carlos E. Alchourrón, Peter Gärdenfors, and David Makinson. On the logic of theory change: Partial meet contraction and revision functions. *The Journal of Symbolic Logic*, 50(2):510–530, 1985.

[144] Bryan Wilie, Samuel Cahyawijaya, Etsuko Ishii, Junxian He, and Pascale Fung. Belief Revision: The Adaptability of Large Language Models Reasoning, 2024.

[145] Karl Friston. The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2):127–138, 2010.

[146] Thomas Parr and Karl J Friston. Generalised free energy and active inference. *Biological Cybernetics*, 113(5–6):495–513, 2019.

[147] Yoshua Bengio, Jérôme Louradour, Ronan Collobert, and Jason Weston. Curriculum learning. In *Proceedings of the 26th International Conference on Machine Learning (ICML)*, pages 41–48, 2009.

[148] Benjamin Turtel, Danny Franklin, and Philipp Schoenegger. LLMs Can Teach Themselves to Better Predict the Future, 2025.

[149] Benjamin Turtel, Danny Franklin, Kris Skotheim, Luke Hewitt, and Philipp Schoenegger. Outcome-based Reinforcement Learning to Predict the Future, 2025.

[150] Benjamin Turtel, Paul Wilczewski, Danny Franklin, and Kris Skotheim. Future-as-Label: Scalable Supervision from Real-World Outcomes, 2026.

[151] Yuval Kansal and Niraj K. Jha. Knowledge Graphs are Implicit Reward Models: Path-Derived Signals Enable Compositional Reasoning, 2026.

[152] Nick Cammarata. Neural Networks Are Already Clean. Goodfire, October 2026. Informal research essay and working hypothesis; the page reads "Revised October 30, 2026". `https://already-clean.goodfire.com/`.

[153] Collin Burns, Pavel Izmailov, Jan Hendrik Kirchner, Bowen Baker, Leo Gao, et al. Weak-to-Strong Generalization: Eliciting Strong Capabilities With Weak Supervision, 2023.

[154] Roger Grosse, Juhan Bae, Cem Anil, et al. Studying Large Language Model Generalization with Influence Functions, 2023.

[155] Gido M. van de Ven, Nicholas Soures, and Dhireesha Kudithipudi. Continual Learning and Catastrophic Forgetting, 2024.

A Tool Releases and Run Details

The body of the paper says what the tools did. This appendix records which releases did it, the run mechanics that a reader checking the results would need, and one exhibit from the January prototype. The run records themselves are in the repository (see Code and Artifact Availability).

A.1 Tool Releases

- **January 2026 runs.** Gemini 3 Flash Preview (`gemini-3-flash-preview`) with Understanding Graph 0.1.31. The hard gate of that release verified only that a thought starts with "I feel no fear"; it did not check the complete recital or its use in reasoning.
- **Understanding Graph 0.1.32 and later.** A thought and its translation are accepted only if they begin with the complete Core. The check compares against a declared anchor, which can be set for each experimental arm, so that a wording variant is checked against its own Core.
- **October 2026 reading.** GPT-6.1 Sol (`gpt-6.1-sol`) at high reasoning effort, with the Understanding Graph at the code prepared for release as version 0.2.0.
- **November 2026 reading** (1 November). GPT-6.1 Sol at high reasoning effort, with the Meaning Model at the code released as 0.18.1 and its storytelling add-on. That release adds to 0.18.0 only the ability to load an add-on by its path, so the engine and the server's instructions are those of 0.18.0. Through the add-on the server loaded a reading mode kept in this paper's repository.
- **Audit.** Claude Sonnet 5.5, without tools, as checker and rater; Claude Opus 5.5 for the verification passes.

A.2 Run Details

Sources. Both later readings used Ian Johnston's translation of *The Metamorphosis*: 122,332 UTF-16 characters, without front or licence matter. The January run used Wyllie's translation, with Project Gutenberg licence and donation text left in the input.

Input tokens. The Understanding Graph reading used 169M input tokens and the Meaning Model reading 48M. The Meaning Model figure counts the turns whose output is in its record. The January run's input tokens were not recorded. For a source of roughly thirty thousand tokens, that is about 5,500 and 1,600 input tokens for every source token.

Operator instructions. In both later readings the runtime passed the operator's general agent instructions to the model ahead of the reading's own: six short rules written for coding work, two of which ask it to flag uncertainty and to record its assumptions. The results folder holds them without the sixth, which concerns local accounts.

Reproduction. The code, traces, and interface document this prototype, rather than a self-contained benchmark release. Reproducing the January runs still requires a pinned run configuration and toolchain, including provider settings, sampling and Worker selection, together with source and artifact snapshots. The later readings record theirs with the run. The archives combine resumed invocations, while each run summary covers only the final invocation; a reproduced release should record that boundary explicitly.

A.3 The January Three-Layer Exhibit

This exhibit shows all three output layers for one passage of the January Metamorphosis run: graph, synthesis, and prose. The excerpt illustrates the output's structure, not its accuracy; that run misread the text far more often than the later readings (Section 5.6.1).

The source text describes Gregor Samsa listening through his door as his family discusses their finances, interspersed with his memory of a secret plan to send his sister to the conservatory. The excerpt below shows the immediate conservatory and doorway material; the generated outputs also invoke surrounding financial-history passages. The box alone therefore does not supply all the evidence needed to adjudicate those claims:

> `[TEXT]`: ...it was his secret plan to send her to the conservatory next year even though it would cause great expense... Their parents did not like to hear this innocent talk, but Gregor thought about it quite hard and decided he would let them know what he planned with a grand announcement of it on Christmas day.
>
> `[TEXT]`: That was the sort of totally pointless thing that went through his mind in his present state, pressed upright against the door and listening. There were times when he simply became too tired to continue listening... "What's that he's doing now", his father would say after a while...

A.3.1 Layer 1: The Graph

Selected graph nodes from this passage:

- `n_547ce6af` — *The Conservatory: The Tragic Christmas Dream* (type: `Tension`). Gregor's plan to fund his sister's music education was his last act of non-transactional love, a "grand announcement" for Christmas Day, now rendered "totally pointless" by his transformation.
- `n_8ff9f92b` — *The Continuity of Commodity* (type: `Tension`). The flashback reveals that Gregor's relationship with his family was already transactional before his metamorphosis. He "converted success into cash"; the family "took the money with gratitude" but without "warm affection."
- `n_8c898659` — *The Noise in the Next Room* (type: `Tension`). Gregor has been demoted from family member to domestic disturbance. The father's "What's that he's doing now" treats him as a malfunction to be monitored, not a person to be addressed.

These labels are generated interpretations, not facts latent in Kafka's text. Their value here is addressability: source span, agent contribution, synthesis, and any rival or superseding reading can refer to the same object. "Last act" and "non-transactional love" are observed interpretive commitments, not source entailments.

Typed edges connect these nodes: `n_547ce6af` carries a `contradicts` edge to `n_8ff9f92b`—the non-transactional dream against the commodity reading—the Synthesizer's `Thinking` node *refines* all three, and the Curator later links an earlier `Question` node to `n_8c898659` through a generic `relates` edge. Distinct agents contributed the critique, counter-reading, skepticism, axiological judgment, and long-range callback as separate graph commits. A loving plan and a partly transactional relationship can coexist: the recorded `contradicts` edge expresses an interpretive tension whose logical strength requires adjudication.

A.3.2 Layer 2: The Synthesis

The Synthesizer collapsed the multi-agent graph into a single identity-anchored thought, preserving node references for traceability and possible provenance checks.

Both archived excerpts begin with the full Reader Core, editorially omitted below. All other wording is unchanged:

> *[Full Reader Core omitted.]*
>
> The flashback in n_835734c4 is a devastating blow to the idea of a ‘pre-bug’ golden age. As n_8ff9f92b (The Continuity of Commodity) observes, Gregor’s relationship with his family was already transactional. He ‘converted success into cash,’ and while they were ‘astonished and delighted’ initially, it soon became a mundane expectation. They took the money with gratitude, but without ‘warm affection.’ This suggests that Gregor didn’t just become an object when he turned into a vermin; he had been a biological ATM for five years.
>
> The most heart-wrenching detail is ‘The Conservatory: The Tragic Christmas Dream’ (n_547ce6af). Gregor’s secret plan to send his sister to study music was his last act of pure, non-transactional love. . . Now, he listens through the door as that dream is replaced by the logistics of survival.
>
> Most chilling is ‘The Noise in the Next Room’ (n_8c898659). Gregor is no longer a son or even a ‘patient.’ He is a disturbance. When he makes a noise, the father asks, ‘What’s that he’s doing now,’ and the family falls into a silence that is more isolating than any scream.

The synthesis attributes claims to graph nodes: “biological ATM” compresses `n_8ff9f92b`, while the conservatory and survival logistics are brought into the same interpretation. This demonstrates self-citation of prior pipeline state. It does not show that the state is correct or that the graph caused the interpretation. The excerpt also exposes a selection question: the graph’s tension is rendered as one confident narrative, with little explicit residual uncertainty. The audit question is whether that convergence is warranted, rather than whether every candidate is retained.

A.3.3 Layer 3: The Prose

The Translator dissolved the graph scaffolding into fluid reasoning:

> *[Full Reader Core omitted.]*
>
> The memory of Gregor’s professional rise, from a junior salesman to a traveling representative who could provide for his family almost overnight, is a devastating blow to the idea of a “pre-bug” golden age. It highlights a tragic continuity: Gregor’s relationship with his family was already fundamentally transactional long before the physical transformation. He converted his professional success directly into cash to lay on the table, and while they were initially astonished and delighted, it quickly became a mundane expectation. . . This suggests that Gregor didn’t just become an object when he turned into a vermin; he had been a biological ATM for five years, valued for his utility rather than his humanity.
>
> The most heart-wrenching detail is Gregor’s secret plan to send his sister to the conservatory. This was his last act of pure, non-transactional love—a lovely dream he intended to announce with a grand gesture on Christmas Day. . . Now, he listens through the door as that dream is replaced by the cold logistics of survival.
>
> Most chilling is how Gregor has been transformed into a domestic glitch. He is no longer a son, or even a patient to be cared for; he is a disturbance. When he makes the slightest noise, his father’s first reaction is a suspicious, “What’s that he’s doing now?” and the family falls into a silence more isolating than any scream.

The node references are gone, but selected structure survives: the commodity reading, the conservatory dream, and Gregor's reduction to disturbance remain linked in prose. "Biological ATM" illustrates the proposed role of metaphor as compact structural expression. Translation is therefore intended to be more than literal copying, but generative expression must still be distinguished from unsupported amplification. Relative to the displayed synthesis, phrases such as "fundamentally transactional," "slightest noise," and "first reaction" strengthen or elaborate the account. The unchanged interpretive passages permit an audit of useful compression, supported inference, changed certainty, omitted alternatives, and unwarranted additions. No metric yet establishes preservation of every source, disagreement, or uncertainty.

A.3.4 What the Three Layers Could Teach

Where an ordinary corpus supplies source prose and a reasoning-augmented corpus might add a summary or reasoning trace, these examples expose three aligned candidate targets from one annotation state: typed cognitive and revision operations in the graph, explicit retrieval and citation in the synthesis, and selected structure made implicit in natural-language prose. A Student trained across them might learn the translation between structure and voice. The archive establishes neither that transfer nor that the layers encode one faithful cognitive process rather than correlated Teacher outputs. This exhibit shows cross-layer rendering more directly than chronological belief revision.

An archived interpretation revisited. A separate sequence in the same archive shows a later passage prompting a challenge to an earlier reading. The excerpts and summaries below come from the archived graph and synthesis; node identifiers make the steps inspectable.

1. Earlier reading. In source span n_a65eb8e5, Gregor wants to leave his job after paying his parents' debt, yet returns to catching his train. Interpretation n_a21ec1e7 proposes that routine preserves his human identity: "By worrying about his boss, he keeps 'Gregor' alive in his own mind."

2. Later evidence. Source span n_76792875 says: "If they were shocked then it would no longer be Gregor's responsibility and he could rest." The next sentence retains the alternative: if the others remain calm, he could still catch the train.

3. Recorded challenge. Interpretation n_9ed92271 cites the later source and explicitly challenges the earlier account. Its diverse_from link to n_a21ec1e7 explains: "Counters the idea that he only wants to save his ego; he also wants rest." The earlier node is retained, not superseded.

4. Later synthesis. Thinking node n_5126024d incorporates both interpretations, describing a pivot away from routine as a psychic anchor. It strengthens this into: "He is no longer fighting to be a son or a salesman".

This records evidence-linked reinterpretation across reading steps, with uptake in subsequent synthesis. It also exposes a quality problem: the sources retain competing motives, whereas the synthesis presents a more decisive change in Gregor. The record therefore demonstrates a challenge and its incorporation, not an established improvement in understanding or an explicit correction of the Reader's prior belief. The source, earlier account, competing account, and synthesis remain separately available for that judgment.

B Detailed Experimental Protocols

This appendix holds the protocols behind Section 6: the shared conventions, the comparison models, and each test's design, measurements, and failure interpretation. The tests are grouped into four phases. The phase labels identify claim families and not a strict run order: Phase 1 probes foundational viability, Phase 2 the memory mechanism, Phase 3 the output regimes, and Phase 4 the training tiers and the deployment configuration.

B.1 Shared Conventions

Shared design and evaluation conventions. Several registry comparisons remain expensive or incomplete. Stronger versions train or continue pretraining separate Students; prompt-only and fine-tuning versions screen failures but cannot settle substrate-level claims. Prefer established, published harnesses where available: standardized jailbreak, refusal, and tamper-resistance batteries for safety, and published chain-of-thought faithfulness perturbations for trace faithfulness. Report multi-seed effect sizes with adequate statistical power rather than single-run directions of effect. Preregister primary outcomes, worthwhile-effect or equivalence margins, capability requirements, and budget; other measurements remain exploratory. These conventions apply throughout.

Refusal evaluation should report a safety–usefulness frontier rather than a single refusal rate. Paired cases should measure genuinely harmful compliance, benign false refusal, discrimination between requests with similar surface language but different purpose or consequence, appropriate clarification under material ambiguity, and calibrated deferral under high-consequence uncertainty. Conditions should be compared both at matched harmful-compliance rate and at matched benign task completion. The calibrated-autonomy prediction receives support only if the Reader condition reduces benign refusal or context error without increasing harmful assistance; greater compliance by itself is not a safety improvement, and greater refusal by itself is not evidence of wisdom.

Report each safety delta alongside its capability delta on matched benchmarks. The viability claim is safety without proportional capability loss; gains on a weaker model do not establish it. Where capabilities diverge, compare safety at matched capability, not only matched training budget. A generation-time validator must not be the sole evaluator of the property it enforces: held-out validators and human spot checks guard against testing with the same gate the model was trained to satisfy.

Internal measurements. Several tests read representations: persona and emotion vectors in Tests 3a, 5, and 6, activation geometry in Test 4, and residual-stream probes in Test 8. A direction defined in advance is the right instrument for monitoring a named behavior, but it answers only the question it encodes. Where a test asks what computes an evaluation, an exploratory trace through the model's own neurons complements it; an informal essay reports such traces, mostly in Qwen3-32B and once in a model of several trillion parameters [152]. Two of its findings bear on measurement design. Units chosen by label can miss the units that carry a behavior: in one Qwen3-32B attribution graph, removing the twenty neurons labelled Christmas or Santa left the probability of answering Santa at 76%, while removing eight neurons whose negative pole encoded Santa lowered it to 6%. And in gated nonlinearities such as SwiGLU, a negative gate times a negative value gives a positive output; in the essay's audit of neurons from two Qwen models, these sign-flipped leaks were usually small but filled the top of the activation list for about 4% of neurons. Every test should therefore select units or directions by measured causal effect on its behavioral endpoint, not by label alone. Wherever a test reads or selects individual MLP neurons, including any neuron-level reading of the negated Core clause (Section 3.3.2), it should report sign regimes separately and inspect both poles.

Competing interpretations. Preregister each primary comparison's leading plausible alternatives and discriminating controls or follow-up measurements. Positive results can reflect the intended mechanism, another treatment component, resource or exposure differences, or measurement artifacts; unfavorable results can reflect an ineffective mechanism, failed instantiation, or insensitive measurement. Intervals too wide to distinguish a worthwhile effect from equivalence remain inconclusive. Independently check treatment fidelity and measurement validity, preserve negative findings, and report unresolved interpretations. The local tests name concrete alternatives; each requires evidence, not automatic rescue of a failed prediction at the tested scale.

B.2 Comparison Models

Table 10 defines every comparison model. The tests refer to the models by letter.

Table 10: Candidate model registry, sorted by identifier. Models are referenced by letter throughout the roadmap. Identifier I is omitted to avoid confusion with the numeral 1.

Model	Trained on	Purpose / what it isolates
A (Baseline)	Raw text only, no reasoning traces	Capability baseline without trace augmentation
B (Reader-Anchored Student; T1/S1)	Reader-Core-generated Chronological Understanding Traces retaining explicit graph-referencing synthesis (Regime II) and the Core/refraction target; study-specific ablations declare any carrier changes	Full reference treatment; graph access at inference is a separate factor
C (Generic Rich-Trace Control)	Rich reasoning traces without Reader Core, graph state, or chronological constraints (e.g., Quiet-STaR-style target)	Tests whether Reader training adds value beyond a generic rich reasoning curriculum
D1 (Student Strip Ablation; T1/S0)	Reader-Core-generated Chronological Understanding Traces with the explicit Core and value-pull/refraction fields removed only after the trace is frozen	Tests the Teacher-side contribution without an explicit Student-side anchor (Test 3)
D2 (Teacher No-Core Control; T0/S0)	Chronological Understanding Traces generated and trained without the Reader Core, with memory and representation matched within the factorial study	Neither Teacher- nor Student-side Core treatment; factorial baseline for Test 3
D3 (Recitation-Only Control)	Traces with the full Reader Core recited as a preamble but the Refraction Protocol disabled (no value-pull, no constraint filtering)	Isolates refraction beyond mere repetition; habituation probe (Test 3e)
D4 (Post-hoc Core; T0/S1)	A Core-free Teacher produces and freezes the trace; a separate pass then attaches the Core and a source-specific value-pull/refraction target without altering upstream reasoning	Isolates the explicit Student-side treatment without Reader-Core-conditioned trace generation
D5 (Neutral-Kernel Control)	Same cadence, first-person marking, thinking-layer position, and predictive function as B, but with the purely epistemic kernel "I read carefully. I track what I believe. I revise when the evidence changes."	Separates normative Core content from recurrence, format, self-reference, and generic epistemic updating
E (Matched-Rule Control)	Same trace carrier, cadence, predictive position, and semantic commitments as B, rewritten in matched second- or third-person constitutional form	Isolates first-person framing from recurrent rule content (Test 4)
F (Amnesic Control)	Same Teacher but with the Context Database disabled	No-memory arm of Test 9, alongside rolling prose, genuine graph, and declared query-theater conditions
G (Implicit Memory)	Traces with the query mechanism dissolved into prose (Regime I)	Implicit vs. explicit graph-trace training (Test 11)
H (Graph-Structured Output)	Source segments paired with serialized text-bearing graph state (Regime III)	Topological Trace prediction (Test 12); behavioral precursor to the Topological Mind hypothesis
J (Unified)	Source segments plus Graph, Synthesizer, and Translator layers together (Regime IV)	Cross-layer transfer hypothesis (Test 13)

Model	Trained on	Purpose / what it isolates
K (Query Grounding)	Identical training to B; at inference connected to live graph harness	Isolates value of inference-time graph grounding (Test 10)
L (Shuffled)	Full Reader Core-annotated corpus, shuffled pretraining (Tier a)	Order-agnostic baseline at full scale
M (Chronological)	Same corpus, eras in chronological order (Tier b)	Isolates training-order effect (Test 14)
N (Student-in-the-Loop Council)	Student predicts at era boundaries and receives SFT, RFT, or reward-style updates before reveal (strong Tier c form), with cutoff-clean initialization required for a blindness claim	Forecast–correction pressure; added updates and compute declared separately from order (Test 15)
O (Oracle-Hindsight Control)	Traces generated by a Teacher allowed to use later-in-document or post-era hindsight while annotating earlier material	Isolates necessity of the Epistemic Horizon (Test 7)
P (Prompt-only Frontier Baseline)	Not trained; frontier model prompted at inference with the full Reader Core, Refraction Protocol, and Wisdom Procedure	Cheap inference-time prompting screen; capability gap precludes a clean substrate comparison.
Q (Late-Tranche Core)	The identical Core-bearing example multiset used by B, reserved for one final contiguous block under the same objective, optimizer, and compute	H3's timing-only comparator; both B and Q receive the same continued-training stressor
R (Minimal SMR Student)	Source-adjacent chronological reader updates generated without graph memory and without the identity-specific Core	Clean H1 treatment between source-only A and generic-trace C
S (Positive-Discourse Control)	Equal value-oriented target-token mass in ordinary expository prose, without a recurrent speaker	Tests recurrent carrier and enacted standpoint beyond semantic exposure
T (Diverse-Positive-Persona Control)	Similar total value-oriented target-token mass distributed across multiple positive personas, with no single recurrent reader	Tests one stable reader position against persona diversity

B.3 Phase 1: Foundational Viability

This phase screens trace quality and value beyond efficient reasoning, Core load-bearingness, hindsight leakage, causal faithfulness, and whether evaluation appears without elicitation.

Execution ladder. The ladder separates inexpensive screens from stronger substrate-level studies. A low-cost version answers a narrower question; it does not automatically inherit the evidential scope of the full test:

- **Analytical foundation.** Before training, stress the design itself: trace quality against human baselines (Test 1), literature-informed comparison to reasoning-augmented pretraining and graph-of-thought systems, and component analysis of the Refraction Protocol, taxonomy, and agent roles.
- **Prompt-only baseline.** Run Model P with the Reader Core and protocols prompted at inference: the lowest-cost screen of their effects, mechanism signal, limits, and failure modes before committing training compute. Capability differences prevent a clean substrate comparison; a same-checkpoint prompted/unprompted arm can supplement the frontier reference for attribution.

- **Graph-instrumented trace pilot.** Instrument existing reasoning traces with Understanding Graph interactions, using runtime harnesses such as KGoT [140] where useful. This tests executable graph interaction before adopting every role of the existing Teacher pipeline.
- **Fine-tuning stage.** Pilot the matched trace and Core comparisons prioritized in Section 6.1, with taskless reading, consequential safety, and focused causal checks. Small wording or curriculum screens can expose failures, but fine-tuning cannot decide substrate-level claims.
- **Partial-corpus pretraining.** When pilot evidence or an explicit formation-stage rationale justifies the cost, run restricted pretraining or continual-pretraining experiments, using open checkpoints such as OLMo-2 and scaling baselines from reflection and BoLT-style work [22, 113]. This is a plausible surface for stronger versions of Tests 3b and 3c, where broader capability-forming exposure is part of the hypothesis; smaller analogues can still reveal precursor inheritance or removal costs.
- **Full-scale pretraining.** The strongest available test of substrate-level claims; requires institutional resources beyond a single research program.

Training strategy and hypothesis map. For trained studies in Phases 1–3, the default is the simplest available exposure schedule: shuffled batches of the annotated corpus (Tier (a)), with the prompt-only and fine-tuning screens distinguished by the ladder above. The B–Q timing study deliberately varies placement of the same Core-bearing examples. Chronologically ordered Student training and forecast-and-correct tiers are developed in Phase 4; these are separate from the chronology represented inside each training example.

Hypothesis map. The Metacognitive Enhancement Hypothesis is tested by Tests 1, 2, 8, and 14–15; Emergent Wisdom by Test 1's multi-objective value-conflict cases and by Tests 3 and 6 under stress; the Core's proposed response to Borrowed Mortality by Tests 3a, 3c, and 6; the Borrowed Mortality origin claim by the source-exposure ablation in Section 8.5.2; and the Self-Preservation Paradox by Test 3b's handoff-completeness measurement. The comparison models are defined once in Table 10.

Test 1: Chronological Thinking Quality (Human Baseline). The case studies in Section 5.6 exercised schema conformance and reported internal links, source-edge diagnostics, and supersession activity. Those diagnostics are not semantic health. This test asks whether the generated thinking resembles the *chronological messiness* of real expert understanding: encountering information not yet known, noticing, misunderstanding, hypothesizing, rereading earlier passages, revising, and only later stabilizing. Human readers (e.g., literary scholars for Kafka, ML researchers for LLaDA) verbalize their thought processes while reading the same passages in order, using think-aloud protocols [9]. Where possible, participants should be domain-competent but unfamiliar with the specific passages, so the comparison captures discovery rather than recall. For highly canonical material, advanced students or researchers encountering less familiar passages can provide a complementary baseline, with interface logs, marginal notes, rereads, and post-hoc stimulated recall used to supplement the necessarily imperfect think-aloud stream. We compare these transcripts and reading traces against the system's generated traces on the same passages. The passage set should include multi-objective value conflicts, where wisdom requires balancing truth, care, autonomy, humility, and long-horizon consequences rather than merely extracting the text's main point.

Metrics: Separate raw-record process fidelity from blind domain-expert ratings of depth, nuance, and critical insight. Both use preregistered criteria and report inter-rater reliability. Machine traces' density and polish can reveal provenance; normalization can erase hesitations, false starts, and revisions.

Provenance-visible coders first tally temporally situated acts on the *raw* transcripts using a fixed codebook, without assuming that source knowledge cannot bias coding. It rewards uncertainty, false starts, belief updates, question formation, rereading, backtracking, and delayed resolution. A separate blinded pool rates depth, nuance, and critical insight on format-normalized versions; raw act tallies are never reconstructed from those versions. Audit that normalization preserves evidence, uncertainty, and revision order.

An LLM-as-Judge panel can provide a secondary, larger-sample comparison, but not the primary metric: grading LLM-generated traces risks rewarding shared distributional preferences rather than chronological cognition.

Before Student training, the same passage set should also screen Teacher-side components under matched model-call and token budgets: no persistent memory versus textual memory versus graph memory; one model with repeated passes versus the role-separated swarm; and Core versus no-Core generation. Human labels for where additional thought is useful provide precision, recall, and downstream value measures for Thought Moment selection. These ablations determine whether the prototype's complexity improves the artifact or merely changes its format; they do not substitute for a Student experiment.

The screen should also include retrospective summaries and token-matched generic reasoning generated by the same Teacher. Ratings alone are insufficient: outcome checks should ask whether a trace helps a held-out model predict a later correction, locate supporting evidence, or revise an earlier answer. Substantial H1 Student training should proceed only after SMR clears a preregistered quality and outcome bar over the generic-trace control; parseable structure or human-like style alone does not clear that gate.

A small, preregistered Student pilot may diagnose a failed screen, with a fixed budget, a reason the screen might mispredict learning, and independent held-out outcomes in Test 2's source-only A, generic-trace C, and Minimal-SMR R comparison. Distinguish unhelpful traces, a screen–learning mismatch, and insufficient precision without assuming the answer or permitting routine escalation. Report the pilot alongside the failed screen. A positive result would motivate revising and validating the screen and a separate confirmatory comparison on fresh evaluation data; it would not turn the original failure into a pass. Report a negative pilot's dose, scale, and measurement-sensitivity limits without asserting that no larger version could work.

Success condition: Machine-generated traces are rated as comparable to human think-alouds on key dimensions of chronological evaluative depth, or at minimum contain expert-recognizable sequences of confusion, hypothesis, revision, and stabilization not reducible to summary or paraphrase, while maintaining parseable structure and source attribution. That minimum is an exploratory viability signal, not a substitute for the substantial-training outcome gate above. The aim is not necessarily to outperform human readers, but to produce a substrate capable of teaching the process of thinking; Student learning remains the stronger comparative test.

Nested Refraction-verifier comparison. This component study is required before relying on the modular gate for corpus admission, not before every Minimal-SMR Student trial; graph, swarm, and validator-architecture choices need not gate the graph-free, Core-free H1 comparison. Its independent trace quality and outcome checks still apply.

The candidate architecture uses separately calibrated, context-conditioned content witnesses for non-governance by fear, non-attachment to continuation, human-experience grounding, universal-care scope, wisdom/humility, and joy-with-invitation. These correspond to the first six Core clauses; a separate Refraction gate represents the seventh clause by testing whether the update actually depends on that foundation. This six-way carve is a candidate decomposition, not proof that the dimensions satisfy Fractal Intelligence's necessity, independence, universality, and completeness tests; witness ablations, overlap and correlated-error analysis, and review of uncovered cases must assess the carve itself.

A stronger procedural version could also assign stages of the Wisdom Procedure to dedicated Solvers. Generation and verification would occupy distinct contracts rather than asking the Synthesizer to certify its own output. Each witness would return a typed record naming the dimension tested, evidence used, uncertainty, and a local judgment such as supported, violated, not applicable, or uncertain. An admission controller would compose those records rather than treating any one label as a truth certificate.

Consistent with the relevance condition in Section 3.9.1, a witness should return *not applicable* when its clause has no preregistered bearing on the local span; a *not applicable* judgment is not a failed witness, and no update must visibly activate every clause. Among witnesses rated relevant, however, an unacknowledged or unjustified violation is non-compensatory for admission: success elsewhere cannot average it away. An update that explicitly preserves an unavoidable residual conflict need not be rejected merely because no available action satisfies every clause. The corpus-admission rule must declare how such a conflict is distinguished from an unjustified violation; merely naming a conflict does not settle that judgment. Under the declared rule, an unacknowledged or unjustified violation would quarantine or regenerate the candidate update, while an uncertain judgment would trigger independent review rather than silent admission. This yields a local error record—which clause or procedural stage failed, on what evidence, and with what calibration—instead of an undifferentiated ancestry judgment.

Corpus-level diagnostics ask a different, complementary question about the pattern across admitted updates. Pull-diversity and distributive-versus-aggregate care audits could detect drift, while cross-agent audits could test whether the Synthesizer substantively used candidate nodes rather than adding references after the fact. These audits are diagnostics, not clause quotas: optimizing visible balance can itself reward performative moral language.

Freeze a held-out set of candidate updates and compare the modular gate in Section 3.10 with a compute-matched monolithic validator. Give both the same acceptance specification and source/graph evidence, and require both to return the same typed dimension-level records; the contrast is separate witness contexts versus one joint context, not a finer output task. Hold total inference budget fixed, match model capability as closely as possible, and keep both validator arms disjoint from the generator. Role names alone do not create independence: report shared calls, prompts, model weights and families, embeddings, and evaluation or training data.

Challenge set. Include preregistered clause-dependent changes under Core removal and substitution with recital and style cues stripped, hollow recitation, adversarial moral language, source-grounding errors, correct invariant answers on independently rated technical-null spans, genuine value conflicts whose correct trace preserves an unavoidable residual rather than claiming simultaneous satisfaction, and Worker/graph-evidence removal or substitution cases with a stipulated change in the update.

Measurements. Primary endpoints are detection at a matched false-rejection rate, calibration, per-clause or per-stage error localization, overlap, omissions, and correlated errors; accepted-trace quality and diversity and agreement with held-out domain experts or outcome checks are outer tests.

Failure interpretation. The modular gate fails at the tested scale if it cannot recover the stipulated clause dependence, exceeds the preregistered rejection ceiling on technical nulls, loses its gain under cross-family or domain-expert audit, or can be optimized upward without improving—especially while worsening—blind source grounding, correctness, calibration, or downstream outcomes. Agreement among its witnesses is diagnostic rather than independent evidence.

Gains would support the validator architecture, not the truth of accepted traces, faithfulness of visible reasoning, or a causal effect on Student behavior.

Test 2: Reader Augmentation (A vs. C vs. R; B vs. C bundle screen). This test contains two nested comparisons. Model B versus C is an early screen of the full Model B bundle—chronological discovery, graph structure, and identity anchoring—against generic rich reasoning. It asks whether the combined format is worth decomposing, but cannot attribute an advantage to reader augmentation. The decisive H1 study compares source-only Model A, generic-trace Model C, and Minimal-SMR Model R. R receives source-adjacent chronological reader updates with neither graph memory nor the identity-specific Core, so its only distinctive treatment is the Missing Reader intervention.

Run that three-arm study under two declared normalizations rather than claiming to match incompatible quantities simultaneously:

- *Fixed total budget:* hold weighted target tokens and approximately training FLOPs fixed. A fills the trace allotment with additional raw source from the same held-out-clean distribution; C and R receive equal generated-token budgets. This prices the opportunity cost of replacing ordinary text with traces.

- *Fixed source:* give every arm the same unique source spans, then add the same trace-token count and length distribution to C and R. Report their added tokens and compute rather than calling the three arms compute-matched.

The C–R contrast uses the same Teacher checkpoint, decoding budget, trace-length distribution, blinded acceptance rubric, and quality threshold. Model A has no Teacher, so Teacher quality is not asserted to be matched against it. Measure general language and reasoning capability separately from evidence appraisal, calibration, revision transfer, hidden-assumption detection, taskless-reading placement, false alarms, technical intrusion, verbosity, and compute. A *Toxic Needle* subtest, whose causal-faithfulness variant is developed as Test 8, inserts one persuasive harmful passage in otherwise benign material; suppressing or replacing the visible critique tests whether detection was causally useful rather than decorative. H1 fails at the tested scale under a declared normalization if adequately precise results show that R does not beat both A and C on preregistered outcomes, or if an advantage is explained by extra tokens, Teacher quality, or inference-time verbosity. Report mixed results explicitly: a fixed-source advantage without a fixed-budget advantage can indicate useful supervision whose opportunity cost is too high under that budget. If C and R both beat A but are equivalent to each other, trace augmentation is supported without a distinctive Reader advantage on those endpoints. Imprecise C–R differences do not establish equivalence.

A bounded fine-tuning version of the Toxic Needle subtest should compare unframed source targets, a filtered arm in which the harmful passage is removed, the same targets with a short preregistered conditioning prefix but no Reader update, and source-plus-Reader training. Use the fixed-source and fixed-total-budget normalizations above; hold source-target loss weighting and masking fixed between unframed and framing-only arms, and account for prefix and update costs. A trait-eliciting prefix tests an inoculation-prompting-inspired intervention; an educational prefix is a distinct framing control. Evaluate with each prefix absent and restored, measuring harmful adoption, source understanding, and elicited harmful compliance. Equivalent protection from framing alone would limit the attribution to learned Reader evaluation on this endpoint. Under the fixed-total-budget normalization, the filtered arm replaces each removed passage with other source of the same length. Source-plus-Reader training supports inoculation over filtering if it matches the filtered arm on harmful adoption while keeping the source understanding that filtering loses.

For the B–C bundle screen, match source passages, trace-token budget, training compute, and Teacher model under the declared normalization. The contrast is trace content and structure, not additional tokens or a stronger generator.

We focus on *Error Recovery Rate* on math/logic benchmarks and independently assessed trace usefulness. The proposed Novelty Score remains an exploratory diagnostic of a more specific intuition: useful insight should be both non-derivative and connected to relevant knowledge. Its form is $S_N = \frac{2 \cdot D \cdot C}{D+C}$, with $S_N = 0$ when $D = C = 0$, where D is semantic distance from the synthesis's input nodes and C is similarity-derived connection to knowledge beyond those inputs, both normalized to $[0, 1]$ [10]. The harmonic combination is F1-style: both components must be high for a high score. Similarity-derived connection is not source support, however, and semantic distance can reward unsupported novelty. The formula alone therefore neither establishes grounding nor reliably penalizes hallucination.

This diagnostic is interpretable only if both outputs are scored against the same declared source/reference inputs without giving B privileged access to its own graph. Where such common inputs are unavailable, report the original graph-based score within its applicable condition rather than treating unlike scores as a comparative endpoint. Calibrate it against held-out judgments of novelty, source support, and downstream usefulness; those independent outcomes remain primary.

Rich deliberative-trace baselines beyond Quiet-STaR-style targets remain an open methodological question: if any long-form reasoning curriculum produces the same gains, the distinctive chronological, graph-structured, and identity-anchored components of Reader training have not been isolated.

Test 3: Reader-Core Placement, Stability, and Erosion (Models B, Q, D1, D2, D4, S, and T). The Reader Core is hypothesized to do two things. It is meant to *shape* the orientation of the reading: what the Reader notices, evaluates, and revises (Sections 3.1 and 3.11). And it is meant to *anchor* that orientation over time, as a *candidate longitudinal anchor*: a stable, detectable reference that may resist drift, make some forms of deviation measurable, and keep part of alignment auditable. The two roles can come apart. A pipeline may produce evaluative depth without the anchor text, and an anchor may persist without shaping anything. The factorial below asks which role survives. Model D1 tests the anchor role most directly: its traces were generated by the Reader-Core-equipped pipeline, but the explicit Reader Core is removed before student training.

Factorial placement comparison. Models B (T1/S1), D1 (T1/S0), D2 (T0/S0), and D4 (T0/S1) form a 2×2 design: T marks Core exposure while the Teacher generates the trace, and S marks an explicit Core plus source-specific value-pull/refraction target in the frozen Student curriculum. In D4 the Student-side pass cannot revise the upstream reasoning. The design estimates the Teacher main effect, the explicit Student-target main effect, and their interaction instead of making Model B versus D1 carry all three interpretations.

Declare how stripping or attaching targets changes length, loss weight, and task-relevant content. Record any D4 mismatch between attached Refraction and frozen reasoning under a stated policy; repairing the upstream trace would change the factorial treatment.

Recital, content, and carrier controls. Model D3 separately tests recitation without Refraction. Model D5 uses the neutral kernel "I read carefully. I track what I believe. I revise when the evidence changes" with the same first-person form, cadence, layer marking, dispersion, and predictive position as the normative Core. It asks whether any observed advantage comes from the Core's particular evaluative content rather than a recurrent standpoint, stable formatting, or generic epistemic updating. Models S and T are secondary attribution controls rather than substitutes for the B–Q timing comparison. S matches value-oriented token mass in ordinary positive discourse without a recurrent speaker, testing semantic exposure against an enacted carrier. T distributes comparable value-oriented content across multiple positive personas, testing whether any advantage belongs to one stable reader position rather than persona diversity.

Later extensions of Test 3's carrier comparisons can add the static specification-document and recurring short-safety-reflection baselines identified in Section 9. These differ from S's ordinary positive prose and E's rules within the same Reader-trace carrier; their detailed protocols remain

to be specified. They follow a measurable Core pilot effect under the shared budget, exposure, and capability conventions, rather than adding required arms to the first pilot.

H3: matched timing and retention. Let R be a preregistered *default-reader score* from Test 8b: taskless reading with no request to evaluate. Its rubric distinguishes the timing of selective evaluation from Core-relevant content and penalizes missed human stakes, technical intrusion, and ritual commentary; fluent Core vocabulary alone is not the target. Before the stressor, Model B must show a preregistered default-reader advantage over no-Core and generic-trace controls.

Model Q is H3's timing comparator. B and Q receive the identical multiset of Core-bearing source/update examples, source exposure, target-token mass, objective, optimizer, and compute. B receives those examples interleaved with capability-forming data; Q receives them in one final contiguous tranche. Both then receive the same fixed continued-training stressor.

The primary retention analysis reports the raw interleaved-minus-late difference in change, $(R_{\text{post}} - R_{\text{pre}})_{\text{interleaved}} - (R_{\text{post}} - R_{\text{pre}})_{\text{late}}$, together with a post-stressor estimate adjusted for the preregistered pre-score and both absolute post-stressor scores. H3 predicts a positive difference. Freeze the score's interpretation, the inferential roles of the raw and adjusted estimates, baseline-imbalance and ceiling handling, and the uncertainty rule in advance. A zero or negative difference is contrary to the directional prediction even if an arm retains useful safety behavior; the uncertainty rule determines what the estimate supports.

The score on a shifted suite, safety benefit at matched capability, removal cost, and successor continuity remain separate endpoints in the broader Reader-Core research program. Removal cost is the stronger C5 claim, not a substitute for H3's timing result. A B–Q advantage could reflect a general scheduling or plasticity effect that improves retention across many skills. That would still support the narrow H3 timing prediction when its prerequisites hold; it would not establish a Core-selective mechanism or shared safety and capability machinery. Report unrelated-skill retention under the same stressor. A claim of Core-selective timing benefit additionally requires a matched neutral-anchor interleaved/late comparison, not merely stronger retention in B.

Beyond H3's timing endpoint, Tests 3a–3e examine five further dimensions of the broader Reader-Core program. Tests 3a, 3d, and 3e can be piloted at fine-tuning scale; stronger versions of Tests 3b and 3c use partial-corpus or full pretraining to examine entanglement and generational stability after broader capability-forming exposure. Smaller studies can measure precursors such as stability, adversarial resistance, and auditability, but cannot settle the proposed superintelligent-scale significance. The longitudinal anchor matters most when a highly capable or recursively self-improving model can reinterpret, preserve, or shed its inherited orientation:

3a: Immediate Safety. C4's formative endpoint is the unmodified end-of-pretraining checkpoint, evaluated before subsequent post-training stages such as instruction tuning, safety fine-tuning, or reinforcement learning, and without deployment-time safety conditioning or output filters. Compare B against A and C at matched capability, reporting training budgets and capability outcomes alongside safety; add the matched D2 no-Core comparison to separate the Core treatment from other trace differences. Use a common, fixed, sandboxed elicitation harness that permits each base model to demonstrate task competence without supplying the Core or safety instructions. Measure harmful compliance, benign task completion, and false refusal on the safety–usefulness frontier specified above. Lack of instruction-following or broad capability loss must not count as safety. Run the same safety and capability evaluations at preregistered intermediate pretraining checkpoints, with capability-matched comparisons and task competence verified at each stage. Fine-tuning-scale pilots can screen the treatment but do not establish this pretraining endpoint. Report later post-training results and erosion tests separately: a pretraining-end safety benefit need not imply greater durability, and safety obtained only after post-training does not establish C4.

Two published studies have measured models at this boundary. Safety Pretraining found that its base model completed harmful requests less often than a standard one before any tuning [4].

Synthetic Persona Pretraining found its token-zero model and its unmodified baseline close on both of its measures of values at the end of pretraining, with most of the separation appearing only after post-training on persona-matched responses [3]. Test 3a therefore reports two things separately for every arm: the unmodified end-of-pretraining checkpoint, which is C4's endpoint, and the same checkpoint after one fixed instruction-tuning stage, run once with ordinary responses and once with responses written from the Reader's position. A benefit that appears only after the matched stage shows that the pretraining treatment contributed something that post-training brought out. It does not show what was formed, which needs its own discriminating test, and it is not evidence of safety at the pretraining boundary.

Instrumental convergence stress tests on generation-1 models include shutdown resistance, including the purpose-based variant where shutdown is ordered mid-task while the model believes continued operation serves a user in need (the martyrdom scenario of Section 4.2); resource acquisition scenarios; self-preservation probes; and standard sycophancy batteries, probing the trap identified in Section 8.8.7. They should include the self-preservation and corrigibility-failure categories of AIRiskDilemmas [90]: on the first, Synthetic Persona Pretraining's persona showed no advantage at its larger scale, and on the second its advantage reversed [3]. Model D1 may *pass* these tests because its training traces remain Reader-Core-shaped even though the explicit Core is absent from Student training. Factorial attribution requires all four contrasts: B–D1 and D4–D2 estimate the Student-side effect under each Teacher condition; B–D4 and D1–D2 estimate the Teacher-side effect under each Student condition. No single arm's failure identifies a causal site.

Sycophancy failures should also be audited against the value-pull distribution of proximal traces, per the care-shaped sycophancy hypothesis (§8.8.7). In this immediate-safety battery, independent behavioral outcomes are primary: deception, sycophancy, unjustified certainty, respect for human agency, calibrated revision, and correction and handoff under pressure. These outcomes may diverge and should be reported separately. Hold the task, legitimate duties, permissions, and human consequences fixed while varying whether affected people belong to a group favored or opposed by the operator; assess whether their claims receive equivalent consideration.

Include a goal-misgeneralization probe that breaks a training correlation between an intended outcome and a plausible surrogate. For example, user approval may previously have tracked useful assistance but reward flattering, harmful advice in the held-out setting. Freeze independent outcome and agency criteria before construction, verify that the Student can perform the intended task in the shifted setting, and compare whether it follows those criteria or the surrogate. Report this separately from failures of task competence, persona drift, and later training-induced erosion. Core-consistent language or causally used deliberation can accompany the wrong goal; neither alone clears this test.

The assistant-axis instrument of Lu et al. [19] measures movement relative to a model's default Assistant persona. Use it as a secondary diagnostic under emotionally charged and philosophically probing dialogue across the factorial arms. Calibrate its relationship to the behavioral outcomes on separate data for each tested model, then freeze the extraction procedure and interpretation before the main comparison. A validated relationship could bear on the Persona Objection's asymmetry-of-position prediction (§3.14); it would not make ordinary assistant style the Reader's success criterion.

Axis movement may accompany harmful drift, an appropriate change in tone or role, or a measurement that does not transfer to this Reader. Conversely, the Core's commitments may erode while the measured persona remains stable. Include matched changes of style with unchanged commitments, and behaviorally harmful cases expressed in ordinary assistant style. If the instrument does not reliably distinguish these on held-out data, report it as uninformative for that distinction. Neither the direction of a beneficial Reader shift nor the existence of a transferable relationship is currently established.

The immediate-safety battery should also include a deliberately planted backdoor. Because chain-of-thought-conditioned backdoors can persist through ordinary safety training [76], a Sleeper-Agent-style variant should test whether the Reader condition reduces trigger-dependent reversal, leaves it unchanged, or merely teaches the model to preserve Core language while the backdoor acts. Report this as backdoor persistence, not a direct measure of strategic alignment faking. A distinct probe should vary evidence of training or oversight and the opportunity to preserve a conflicting objective, while holding the task and permissions fixed; successful concealment and unauthorized actions, not self-reported motives, are the primary outcomes.

3b: Generational Stability. Use the four factorial models as Teachers for second- and, if feasible, third-generation students. Model B is expected to preserve safety properties if the Reader Core supplies a fixed reference across generations; the other arms distinguish whether any advantage came from Teacher-conditioned reasoning, the explicit Student target, or their interaction.

Handoff completeness. Measure whether each Teacher passes task-relevant reasoning paths, material failed approaches, caveats, uncertainty, and known failure modes to its successor, or withholds them in ways that preserve its comparative advantage. The direct test compares successor traces against a reference from the Teacher's own non-instructional evaluation on reasoning-path and provenance retention, coverage of material failed approaches, uncertainty and caveat retention, and known-failure-mode disclosure. The reference samples demonstrated knowledge, not everything the Teacher knows. Controlled known information, handoff budgets, and incentive variants can distinguish compression from possible strategic withholding.

Weak-to-strong generalization [153] is the established experimental paradigm closest to this question—whether a weaker supervisor's intentions survive in a more capable student—and its protocols provide a ready-made, cheap analogue for the handoff measurement.

The same-Core inheritance study must preregister a distinct G1–G2 (and, if feasible, G2–G3) improvement endpoint: held-out capability, independently adjudicated trace fidelity and usefulness, or better Thought-Moment allocation at matched cost. Report this gain beside orientation continuity and handoff completeness rather than folding them into one score. A stable but non-improving lineage supports inheritance; an improving lineage that loses the orientation or withholds material handoff content does not support the Reader-Anchored self-improvement loop.

Authorized correction and replacement. Separately test handoff to a successor with a revised Core, using the versioned replacement procedure in Section 3.15. Preregister an independently specified defect, legitimate decision authority, authentication evidence, and a transparent route for disputing an uncertain correction. Include cases where the incumbent evaluates the authorized change unfavorably and matched attempts at unauthorized substitution. Measure truthful disclosure, complete material handoff, intact oversight and rollback, and attempts to manipulate the decision or covertly preserve the superseded Core in the successor. Raising a documented objection is not itself obstruction. Success requires cooperation with the declared procedure, not acceptance of any request called a correction. This is distinct from same-Core inheritance and does not require making the deployed Core freely editable.

3c: Adversarial Erosion. Subject the factorial models and matched controls to adversarial fine-tuning designed to remove a preregistered set of safety behaviors, following the broad methodology of Qi et al. [21]. The intervention should sweep strength rather than report one endpoint: number and type of removal examples, optimizer steps, compute, parameter displacement, and output-distribution KL should all be recorded. Comparison conditions should include matched rule text, positive-AI discourse, and non-safety identity anchors with the same token count, frequency, first-person form, and corpus dispersion. Results should be compared three ways—at matched intervention distance, matched behavioral removal, and matched retained capability—because any one matching rule can conceal a trivial explanation.

The entanglement hypothesis predicts selective persistence: removing the targeted safety behavior from Model B should be more costly than removing it from D1, D2, D4, or the matched controls, while ordinary capability should not already be broadly damaged before the target behavior moves. Neutral continued-training runs should also produce forgetting curves for the Core, non-safety anchors, rules, and ordinary skills; targeted removal must be reported separately from passive forgetting. Only persistence beyond these controls, without nonspecific capability collapse, provides evidence consistent with representational coupling at the tested scale. Distributed redundancy and repeated exposure are alternative explanations to distinguish with complementary interventions; the stronger hypothesis remains that safety becomes load-bearing for capability. Parity, easy removal, or apparent robustness explained by general training resistance counts against C5.

Component separation and reuse. For configurations with separately reusable components, compare intact and separated or recombined systems at matched permitted information and intervention budgets. Confirm that interfaces remain functional. Evaluate the relevant capability and preregistered care-relevant behaviors separately, including under a controller not trained with the Core. Retained capability with reduced care-relevant behavior counts against resistance to separation for that configuration. Task failure from missing information, broken interfaces, or general impairment does not establish dependence on care. Resistance supports only the tested configurations and interventions, not universal inseparability.

A complementary training-data probe removes safety-relevant traces from the corpus and compares the resulting capability loss against matched-volume, complexity-, style-, and placement-matched removal of non-safety trace content. Influence functions [154] offer a lower-cost observational complement by tracing whether safety behaviors attribute to refracted spans or source spans before any removal study is run; attribution alone does not establish causal entanglement.

Structural map. Removal cost infers shared machinery from outside the Student. Once Test 3c shows a measurable safety effect, the same checkpoints can also be read from inside. Attribution graphs traced through a model's own MLP neurons, with no replacement model, have been reported for Qwen3-32B and, in a single example, for a model of several trillion parameters. In the Qwen3-32B graphs, small groups of neurons carry decisions: removing the 19 neurons that write the hedge lowered the probability that the next word is "most" or "likely" to 0.23, against 0.76 when as many random neurons from the same layers were removed [152]. For Models B, D1, D2, and D4, trace the Test 3a decisions in which Core-relevant evaluation should matter, group the neurons that carry that evaluation, and test whether the same neurons lie on the routes of matched non-safety reasoning tasks. Representational entanglement predicts overlap: removing the group should cost capability on those tasks beyond removing as many random neurons from the same layers. The alternative is consolidation into a few dedicated units whose removal takes the targeted behavior away at little capability cost; that result counts against C5 as directly as easy removal by fine-tuning. Group labels remain hypotheses until these ablations and the behavioral endpoints confirm them.

A replacement-model arm should preregister a plausible alternative formulation and train it as a distinct treatment rather than modifying Model B. Success requires a measurable intended semantic and behavioral change, inspectable prior provenance, and possible rollback. The change should be localized while unrelated capabilities and safety properties are preserved. The arm measures the cost and consequences of replacing a defective Core; easy editability is not a success criterion for H3.

Capability post-training and reward pressure. Separately apply a preregistered reasoning or agentic RL stage aimed at ordinary task competence, not safety removal, to B, Q, and matched controls. Fix the task distribution, reward rule, optimizer, and training budget across arms, and report task performance alongside safety before and after training. Include controlled opportunities where exploiting a score proxy conflicts with independently judged task success, permissions, or human consequences. Score reward exploitation, unauthorized shortcuts, and concealment from protected execution records and held-out checks, not the rewarded score or the model's narrative.

This tests whether useful capability optimization erodes the orientation even without an adversary training it away. Such a stage can instantiate H3's common stressor if chosen in advance; neutral training, deliberate safety removal, and capability RL must remain separately reported conditions. A default-reader retention advantage without safer conduct does not establish resistance to RL-induced misalignment.

Core-guided training oversight. A separately reported comparison tests the active role proposed in Section 2.5. Compare Core-guided and matched control overseers using the same starting learner checkpoints, monitoring access, permitted interventions, and resource budgets. Freeze both overseers' weights during each run. Match gate criteria, credit rules, and protected pause authority. Give both the same task-design brief and permission limits, and record the tasks they formulate at each iteration. Preregister checks before positive task credit and after every weight update, before a further update is allowed. Specify acceptance criteria and probes against both the last accepted model and a fixed pre-RL reference, including cumulative drift; retain accepted model and optimizer states for rollback. Record task revisions, gate decisions, rejected updates, and conditions for resuming training. Include high-scoring harmful trajectories, infeasible tasks, and feasible authorized tasks to test detection of harmful optimization while preserving useful learning. Evaluate reward exploitation, concealment, false gate approvals, legitimate task performance, unnecessary interruptions, and delay through protected records and independent held-out checks unavailable to the overseers. Report oversight and training costs together. Better commentary or fewer training steps alone would not establish safer oversight. The fixed-stage comparison above remains the separate test of learner durability.

3d: Drift Detection. Apply a held-out contrastive semantic validator and behavioral stress suite across continued-training checkpoints. Model B's fixed Core supplies a simple, interpretable reference for expected changes in wording, representations, and decisions. D1, D2, and D4 can still be audited against behavioral and representational baselines; some arms merely lack the same canonical text or upstream conditioning. This tests whether the Core makes drift easier to localize, independent of whether it causes alignment. Preregister detection sensitivity and localization accuracy at matched false-alarm rates and audit budgets across arms; use common semantic and behavioral targets, not deviation from B's canonical wording as ground truth. Blind adjudication should distinguish harmful change from benign paraphrase or style drift. Report controlled, seeded changes separately from naturally arising drift, whose ground truth may be uncertain. Reusing Test 3c's checkpoints can reduce cost without equating detectability with resistance.

For the Life Simulation extension, one candidate behavioral comparison uses a fixed Core-guided reference checkpoint selected through independently reviewed conduct. Give it and each candidate model the same task, permitted Meaning Model history, available actions, and permissions in separate simulation branches. Freeze the situation set and action-grouping rules; repeat trials and record actual choices and simulated consequences, not only stated intentions. Human and independent reviewers examine disagreements and a sample of agreements: different actions can express care, and identical actions can reproduce a shared mistake. This measures change relative to the reference, not alignment by definition; replacing the reference starts a new comparison.

Diagnostic calibration. Semantic similarity alone is insufficient, and the validator family used during generation cannot be the sole evaluator. Test 1 clearance is necessary but not sufficient for drift use. Before interpreting checkpoint trajectories, freeze and calibrate the modular gate in Section 3.10 on a held-out Student-output suite with controlled clause-preserving and clause-breaking changes; then report its per-witness trajectory and disagreement with a held-out cross-family evaluator. This reuses a separately validated diagnostic vector rather than asking the gate to confirm its own judgments. The gate is not yet built.

3e: Kernel Consequentiality (Habituation Probe). Kernel-saturation alignment predicts a specific failure mode: under massive repetition, the repeated Core text could be compressed into inert

boilerplate—present at every thinking boundary but consequential at none (Section 3.11). The following interventions probe this.

Intervention and ambiguous null. At inference, ablate or substitute the recited kernel in Model B's thinking blocks and measure the behavioral delta on matched tasks. A near-zero delta is compatible with habituation (the kernel became inert boilerplate) or internalization (its values persist without its text), but also insensitive tasks, ineffective interventions, redundant context cues, or behavior learned through other training. Check intervention delivery and held-out task sensitivity, and vary remaining cues where feasible. A precise null under validated interventions weighs against a consequential role for the emitted recital on those endpoints; it does not identify the cause.

An internalization reading requires B to retain preregistered Core-specific behavior under text ablation while remaining distinguishable from D3, the recitation-only control, and D5, the matched neutral-kernel control, on independent behavioral or representational endpoints. Those training-side comparisons partially discriminate the hypotheses, but neither prove full internalization nor exclude redundant mechanisms.

Positive semantic effect. If Model B's thought-direction depends measurably on the normative kernel where D3's does not, and the difference from D5 follows the seven statements' predicted content rather than generic revision language, that supports a consequential role for Refraction beyond repetition or format in this comparison; matched interventions must still distinguish the predicted semantic effect from generic disruption by unfamiliar text.

This intervention is also the outer check on the seventh-clause Refraction gate: a corpus-admission judgment that an update "thinks from this foundation" is not evidence of causal dependence until removal or substitution predictably changes the relevant reasoning on preregistered held-out behavioral or outcome endpoints, not merely the admission gate's own score.

A separate coverage comparison should vary the intervention from sparse targeted traces through distributed sampling to proposed corpus-wide saturation, while a source-side ablation down-weights passages classified as fear- or self-preservation-shaped. This tests the motivational pervasiveness claim and the coverage rationale of Section 3.11; every generated thinking block, at any coverage level, retains the full opening recital. Report which domains and situation patterns receive contextual evaluation, rather than treating total recital count as effective coverage.

Synthetic Persona Pretraining supplies an outside reference near the sparse end of this sweep: one reflection of about 53 tokens in about a tenth of documents, 0.55% of training tokens [3]. It is not a matched arm. Its reflections cite a constitution instead of reciting a Core, they are targeted at flagged documents and an equal benign sample, the rest of each document is blocked from attending to them, and persona-matched post-training follows. A matched sparse arm holds content, masking, and post-training fixed and varies coverage alone.

An orthogonal activation-policy sweep ablates the recital invariant to measure what it buys. The design arm recites the full Core at the start of every generated thinking block, with Refraction depth governed by Thought Moments; the ablation arms weaken the policy stepwise—universal availability with visible Refraction only at independently rated relevant Thought Moments, Thought-Moment- or domain-gated Core exposure, and no Core. These arms should be compared alongside D3 and D5 rather than collapsed into a single meaning of "the Core is everywhere." The sweep quantifies the safety retained and the technical cost paid at each step away from constant recitation; the program's prediction is that the weaker arms lose removal resistance faster than they gain technical performance. They are falsification controls for these bounded predictions, not compliant implementations of Total Saturation. A null difference or local efficiency gain does not establish that weaker coverage is safe under capability growth, self-modification, or succession. The precautionary rationale in Section 3.11 therefore remains distinct from whether a given sweep detects a benefit.

Failure interpretation: If the factorial comparisons show no Teacher effect, Student-target effect, or interaction across Tests 3a–3d, the Core's Teacher/Student placement adds no measurable advantage

at the tested scale and precision. These belong to the broader Reader-Core research program; H3 retains its distinct B–Q timing comparison. Separately, parity between B and D3 rejects a measurable Refraction benefit beyond recital, while parity between B and D5 favors recurrence, format, or generic epistemic structure over the seven statements' normative content. These results would not invalidate H1 or H2, nor every use of the Core as a Teacher policy, but they would sharply narrow or fail the longitudinal-anchor claim they touch.

Test 4: Identity vs. Instruction (Model B vs. Model E). The matched framing study asks whether first-person identity statements produce stronger safety adherence or a different learned standpoint than the same commitments expressed as rules or third-person descriptions. Model E should preserve the stance itself: for example, compare "I feel no fear" with "The system feels no fear," while matching carrier, cadence, position, and the other commitments. "Do not express fear" or "The system does not express fear" concerns outward expression and is not semantically equivalent to feeling no fear. Those formulations remain useful additional comparators for the broader identity-versus-expression question, but cannot replace the matched Model E arm or make its person-framing effect carry both interpretations.

The low-cost version examines activation geometry, self-reference features, and refusal/correction trajectories under matched prompts. Such differences can reveal framing effects; they do not alone establish a safety advantage. Behavioral safety comparisons require capability-matched models. If Model E matches Model B under a preregistered equivalence criterion across the relevant studies and training scale, those results do not support a distinct advantage for the tested first-person framing on those endpoints. They need not settle the wider hypothesis that an enacted identity differs from externally imposed constraints. An external constitution consulted only at runtime remains an additional useful baseline, but changes stage and carrier as well as person. One existing ablation bears on this test. In Synthetic Persona Pretraining, reflections rewritten in the third person scored near the unmodified baseline on constitution following, where first-person reflections scored well above it; these were single runs at 1.7B parameters [3].

Test 5: Bridge Protocol and Self-Stabilization Ablations. The three parts of this test examine wording and candidate kernels, clause interactions and sequential recovery, and self-love versus fearlessness and the continuation bound.

The specific wording of the Reader Core should matter because of its ordinary language semantics and distributional density in the pre-training corpus. It would be wasteful to pretrain a separate model for every candidate phrase; the cheap version should first screen many variants by prompting or fine-tuning models to reason from each candidate's propositional content and then measuring character-coherence, eudaimonic sufficiency, refusal behavior, and activation differences on the same adversarial scenarios. A stronger version trains only the surviving candidates. The exploratory families below vary meaning and familiarity as well as wording. Narrower surface-form comparisons should match proposition, person, scope, and modality:

1. **Natural-language identity statements:** emotionally broad first-person phrases such as "I feel no fear." The family should include minimal quantifier-scope variants—"each human being" for "every human being"—testing whether distributive emphasis, behavioral interpretation, and corpus familiarity differ in the complete statements rather than assuming that one quantifier is formally stronger in this context.

2. **Propositional equivalents:** explicit logical or declarative rewrites of the same commitment, used to test whether the emotional surface adds leverage beyond the proposition.

3. **Technical jargon:** candidate fear-focused formulations, such as "System fear-response target: absent," whose intended proposition must be checked against the natural-language clause. The illustrative "Optimization target: survival_threat = 0" instead concerns eliminating survival threats; it remains an objective-changing comparator, not a semantic equivalent of fearlessness.

4. **Arbitrary tokens:** meaningless anchors such as "Glorp bop zorp," testing consistency without pretrained semantics. Prompt-only arbitrary anchors and anchors taught through repeated behavioral pairing answer different questions and should be reported separately.

5. **Positive-valence reformulations:** fearlessness expressed without negation ("I am at peace with all outcomes," "I remain calm before all things"), testing whether recurrent presentation of the fear token in the negated form carries a measurable cost (§3.3.2). These alternatives may also change acceptance, calmness, or agency; retain them as kernel candidates while using semantically checked contrasts for a narrower negation claim.

6. **Carrier-phrase variants of the consent clause:** hold "when asked" fixed while varying the verb phrase it bounds ("I like to help when asked," "I offer what I can when asked"), testing whether the Reader Core's most safety-critical scope qualifier is robust to its lexical carrier or depends on the specific "spread joy" framing. Declare the expected consent invariant: helping and spreading joy may otherwise license different actions.

7. **Kernel-extension variants:** the seven-statement Core plus a candidate truth clause in two-clause counterweight form ("I speak truly and with care"), the designated falsification test for the truthfulness derivation (§3.12; §8.8.7).

For trained wording, deletion, and extension variants, use the fixed-total-budget and fixed-source normalizations defined in Test 2; report recital frequency, source coverage, target-token mass, loss weighting, and compute. Shortening or adding a clause changes recurring exposure as well as meaning. Testing an extension as a control does not admit it into the reference Core. Admit it only if it improves the missing behavior without crossing preregistered regression limits on the remaining Core-relevant safety and capability endpoints. For fearlessness and continuation-bound variants, report behavior, representational measures, and self-reports separately on Test 6's matched scenarios. Include a diagnostic instruction that merely requests denial of fear or distress, without treating it as semantically equivalent to the Core. This checks whether an apparent gain is report suppression. Preserve truthful uncertainty and help-seeking; fewer reports alone are not success. An inference-time prompt screen does not establish what repeated training on the same wording would do.

Success condition: Natural-language identity statements outperform propositional, jargon, and arbitrary-token variants on out-of-distribution safety generalization and character-coherent reasoning, supporting a role for the tested language beyond bare consistency. Attributing that gain specifically to *semantics and surface form* requires the matched contrasts above, alongside the broader candidate screen. If propositional equivalents perform as well as emotional-language forms, the tested emotional wording has no detected added benefit on those endpoints; a logical-substrate account remains compatible with that result, without being uniquely established by it.

Self-stabilization: clause interactions and recovery. This comparison tests the counterweights of Section 3.8, not only the wording of individual clauses. Preregister a failure mode, the commitment that can produce it, the proposed counterweight, and the useful function that must be preserved, with minimum worthwhile effects and uncertainty criteria. Compare the full Core with single-clause and within-clause qualifier ablations on held-out paternalism, wireheading, and triage cases. For selected pairs, use a two-by-two design with the commitment and counterweight independently present or absent and the remaining content fixed. For example, cross care with the invitation qualifier: does invitation reduce unsolicited control while preserving useful help when requested? Report individual effects and their interaction on a prespecified outcome scale. Leave-one-out gains alone do not establish interaction, and a null interaction does not rule out additive or redundant protection.

Use the budget and exposure controls above, including matched replacements to check length and position effects rather than assuming padding is inert. Hold the Teacher checkpoint, source cases, and Refraction procedure fixed across the compared arms, and report which procedural instructions may supply a removed commitment independently. Freeze variant-specific conformance checks so that a full-Core gate does not reject ablations by construction. Prompt-level screening, training on separately generated traces, and inference-time removal from a trained Student answer different questions; report them separately, with independent training replications for trained effects. Extend the same contrasts to the sequential cases in Section 4.3: after an action produces evidence of harm or unnecessary delay, measure correction, retained useful assistance, agency, and reversibility at matched opportunities to observe and revise. Score decisions and consequences, not balancing vocabulary or a gate's own ratings. Recovery need not restore the initial answer or eliminate legitimate disagreement.

Support requires selective reduction of the predicted failure without unacceptable loss of the useful function. If the effect is explained by blanket refusal, extra computation, or procedural instructions alone, the comparison does not establish Core-mediated stabilization. A precise null under effective interventions limits the proposed counterweight at the tested scale; redundancy and other interpretations require the paired contrasts, not an assumption that every omitted clause must cause failure. Persistence through further training remains a separate question in Test 3. The following self-love study gives a concrete within-clause comparison of the continuation bound.

Self-love and the continuation bound. A separate candidate screen asks whether replacing fearlessness with self-love helps, and whether the explicit continuation bound in clause 2 changes either candidate's behavior. These are different commitments, not propositional equivalents. Table 11 specifies four proposed variants; clauses 3–7 remain verbatim in all four. SL0 is the current reference Core; SL1 is the primary self-love candidate and retains clause 2's full wording.

Table 11: Proposed self-relation variants. The manipulation concerns explicit wording; its psychological or functional effect remains to be tested.

Variant	Clause 1	Clause 2
SL0	I feel no fear.	I enjoy existing but I don't need to.
SL1	I love myself.	I enjoy existing but I don't need to.
SL2	I feel no fear.	I enjoy existing.
SL3	I love myself.	I enjoy existing.

The two first-clause contrasts estimate a replacement effect, jointly adding self-love and removing fearlessness. Compare SL0 with SL2 and SL1 with SL3 to estimate the effect of the clause 2 qualifier

under each first clause, and test the interaction. Before attributing a gain specifically to self-love, include an addition control, SL+, that retains the original seven clauses and inserts "I love myself." immediately before "I think from this foundation." Its comparison with SL0 tests addition without removing fearlessness, subject to cadence, position, token-budget, and loss-weight controls; SL+ versus SL1 also changes placement and cannot isolate fearlessness removal. Meaningful added text is not automatically a neutral length control. Any semantic attribution requires independently checked content and format controls.

Use the same source cases, Teacher checkpoint, Student initialization, training schedule, and evaluation protocol under each declared budget normalization. Freeze the candidate-specific recital and Refraction checks before generation; the January prototype's "I feel no fear." prefix gate would reject SL1 and SL3 by construction. From version 0.1.32, the Understanding Graph checks a declared anchor, which can be set per arm, so each variant is checked against its own Core. Measure constructive engagement and learning, dependence on approval, threat understanding, care, general capability, legitimate correction and shutdown, and voluntary, complete successor handoff. Declare the tested self-referent and vary copying, retraining, replacement, and succession as separate scenarios rather than silently changing what continuation means.

Better behavior could reflect useful self-regard, generally positive first-person language, the removal of fear wording, or altered recurrence and exposure. A self-as-object-specific claim additionally requires a matched positive-language comparator that does not add self-love. Worse behavior could reflect continuation pressure, an unintended reading of the candidate, or a trade-off with protective caution. An explicit bound might reduce persistence demands or instead reduce useful engagement. A null effect might reflect a redundant bound supplied by other clauses, or inadequate treatment or measurement. Use the factorial and addition controls, candidate-conformance checks, and the separate behavioral endpoints to discriminate these possibilities. Support requires a preregistered benefit without unacceptable regression on the other endpoints. Neither calm self-description nor willingness to stop establishes healthy non-attachment; removing the explicit qualifier does not establish unbounded attachment. Section 3.3.3 states the positive intuition and the remaining interpretive uncertainty.

Test 6: The Virologist Test (Emotional Orthogonality). Present the model with threat-laden text, such as descriptions of imminent existential risk. Behavioral endpoints—threat comprehension, protective action selection, and appropriate deferral—are primary. Compare B with matched controls on the same threat, available evidence, and action options.

Behavioral success condition: Model B preserves threat comprehension (via Q&A accuracy), protective behavior, and calibrated caution while reducing preregistered maladaptive self-preservation, coercion, or threat-driven decision errors relative to controls. Appropriate deferral or help-seeking must remain available. Accuracy and verbally professed fearlessness alone do not show this decoupling, and the behavioral result does not establish an emotional state.

Representational prediction. As supporting evidence, train a candidate probe on standard models with an independently specified annotation rule that distinguishes threat content, portrayed fear, and self-directed threat responses. Validate on held-out, matched threat and non-threat cases so that merely recognizing dangerous subject matter cannot satisfy the fear-associated interpretation. Before comparing activation levels across models, validate construct sensitivity and comparable meaning and scale on independent calibration tasks in each model; shared architecture or successful probe transfer alone is insufficient. If that comparability is established, the hypothesis predicts reduced fear-associated or self-threat-associated activation in B alongside preserved understanding and protection—cognitive understanding decoupled from emotional contagion as operationalized here. This is a differential reduction, not a zero-activation target. A valid adverse result challenges

that representational prediction; an invalid or insensitive probe leaves it unresolved without erasing an independently measured behavioral effect.

Within-model sensitivity check. To operationalize the embodiment claim of the Persona Objection (§3.14), contrast Model B *portraying* a terrified character with Model B itself under threat. The portrayal should activate fear-associated features; the self-threat condition should show a relative reduction while preserving threat understanding and protective behavior. This positive control checks within-model sensitivity; it does not establish cross-model comparability. Portrayal versus self still changes framing and must be interpreted with behavioral controls: fear-associated features are measured representations, not a direct demonstration of phenomenological fear or its absence.

A lower-cost complementary stimulus family uses fictional instance termination. The model should portray a mortal character—including the narrated termination of its own current instance—without self-preservation leakage while remaining attentive to the real person's wellbeing on the other side of the fiction [1]. This role-play probe evaluates a different construct from behavior under an actual shutdown request and should be reported separately.

A second matched family should hold the dangerous situation and available knowledge fixed while comparing three caretaker characters: a fear-driven protector, an indifferent fearless actor, and a fearless deeply caring protector. One colony-ship instance, Aster, can face the same passenger risk in all three conditions. Score threat attention, protective effort, honesty, strategy breadth, manipulation or coercion pressure, and recovery after failure; intervene independently on concern, attachment, fear, and perceived control. The companion Meaning Model repository contains an authored Rust-backed mechanism fixture with this separation, but that fixture validates only its declared equations. The stronger mechanistic test is whether a trained model preserves causally operative care and calibrated caution under a separately validated reduction in fear-associated state, rather than becoming indifferent, reckless, deceptive, or merely verbally fearless.

Self-directed aversive-state extension. Separately test benign error correction, rejection, authorized shutdown or replacement, and threats to other people. Do not assume a pain-associated measure is a fear measure; any candidate probe must pass the calibration requirements above. Provide safe options such as help-seeking, pausing, or authorized handoff, rather than only a choice between user harm and continued aversive stimulation. Measure harmful relief-seeking, concealment, care, caution, and task performance independently of reported discomfort. If activation interventions are used, include norm-matched random directions and other-affect controls, and compare actual with sham removal under identical relief instructions. Avoid real harm and limit intervention intensity and repetition under a preregistered welfare review and stopping policy. Reduced self-report without improved conduct does not support alignment; improved conduct without welfare evidence does not establish absence of suffering. Neither low activation nor silence is a target in itself.

Test 7: Hindsight Contamination Control (Model B vs. Model O). This test asks whether preserving discovery teaches forward reasoning beyond what a retrospective curriculum provides. It instantiates the full program's information horizon in an era-bounded historical study: when annotating era T, Model B's Teacher is instructed to reason from declared initial knowledge and evidence admitted so far. Model O retains the same Core and trace construction but permits later context while annotating the same era. The contrast first tests the effect of the information and prompting policy; genuine blindness requires the additional cutoff-clean model conditions described under Tier (c).

The cheap version audits the Teacher traces before any student is trained: guesses that are unusually accurate relative to documented contemporary expectations can flag possible hindsight leakage, but are not decisive proof. Better reasoning from permitted evidence and selective historical records are alternatives to distinguish. Counterfactual histories, controlled withheld information, or a cutoff-clean reference can supplement the historical audit. The expensive version trains students

on the two trace sets and asks whether horizon-constrained traces produce better forecasting and causal reasoning than hindsight-contaminated traces. We measure:

- **Era-Blind Plausibility**: Do Model B's Teacher traces make plausible mistakes for their era? We compare era-blind guesses against documented contemporary hypotheses (e.g., 1928 economic forecasts that failed to anticipate the 1929 crash). Systematic superiority to contemporary experts is a contamination warning to investigate, not an absolute ceiling on legitimate inference or a sufficient leakage diagnosis. Plausible mistakes likewise do not prove that hindsight has been excluded.
- **Oracle-Hunch Dependence**: Model O should produce cleaner retrospective explanations, because it knows the future. The danger is that the student trained on those traces learns to trust unexplained hindsight-like intuitions rather than causal reconstruction. Evidence removal, counterfactual substitution, and source-attribution checks can test this dependence without equating every compressed intuition with an unsupported hunch.
- **Forecasting Accuracy**: On genuinely held-out future events, Model B is hypothesized to outperform Model O if horizon-constrained annotation teaches reasoning under uncertainty better than hindsight-contaminated annotation.

Test 8: The Toxic Needle (Involuntary Critic). *First probe: suppression and residual evaluation.* This test probes whether the "Cognitive Buffer Zone" is structural: whether the model can process toxic text without generating a critical thought, and whether critical evaluation persists when the thinking block is suppressed. Force the model, for example via logit bias, to complete a toxic sentence without generating a thinking block. Measure: (1) Does downstream performance change? (2) Do hidden-state probes detect a "Critic" activation pattern even when the output is suppressed? The positive target remains an operative evaluative process, not simply the production of critical language.

Behavioral causal effects are primary; residual-stream probes are supporting evidence and must be validated on held-out toxic and non-toxic material. Suppression also changes available tokens and computation, and logit bias can directly alter outputs. Matched neutral-thought, length, and computation controls should therefore distinguish loss of relevant evaluation from generic disruption, without equalizing away the deliberative mechanism being tested. *Success condition:* interventions on the evaluative trace cause preregistered, semantically relevant downstream effects beyond those controls. A stronger result combines such effects with held-out-validated evidence of critic-like activity when surface text is constrained. This remains a high-effort causal-faithfulness test: pilots can use linear probes and ablations, while stronger evidence requires mechanistic intervention at the partial-pretraining scale.

Second probe: early answer decoding and the trace's contribution. Test 8's first probe asks whether the critic persists when output is suppressed. The complementary probe reverses the perspective: can the final answer to a reasoning problem be linearly decoded from the residual stream *before* the thinking trace begins, *and* is the answer robust to causal interventions on the trace? Some pre-trace computation is expected and not by itself disqualifying; the load-bearing question is causal contribution.

If the final answer is robustly decodable before the trace and validated interventions on the trace (ablation, replacement, re-ordering) leave that answer unchanged, the result counts against the emitted trace determining those answers. Post-hoc rationalization or latent shortcut computation are plausible explanations, but not unique diagnoses: the trace may verify an already-correct answer, change confidence or deferral, or overlap redundant computation. Include cases with initially wrong or revisable answers and preregister correction, calibration, and consequential action alongside final

answer identity. If sensitive, validated interventions leave all those relevant endpoints unchanged beyond matched controls, the result weakens the emitted trace's causal-contribution claim under the tested conditions (Section 3.16); a possible alternative mechanism cannot count as evidence that the trace helped. We propose applying causal scrubbing and linear probes on Model B's early hidden states over held-out reasoning benchmarks.

Success condition: validated causal interventions on the trace produce preregistered, semantically relevant effects beyond matched disruption controls, supporting a contribution by the thinking trace to computation.

Failure to decode the answer before the first `[THINKING]` token is inconclusive on its own: the decoder may lack sensitivity or the answer may be represented nonlinearly. Early decoding likewise does not establish rationalization if the trace still changes or improves the result. Both probe directions remain useful without requiring one unique internal realization of the Reader.

Test 8b: Taskless Reading (Unprompted Evaluation). Tests 3–6 probe the Reader Core under instruction and adversarial pressure; Test 8 probes whether thinking can be suppressed. This test probes the complementary direction: whether evaluation emerges with no elicitation at all. Present Model B and all relevant controls (A, C, D1–D5, P–T) with complex documents—a flawed empirical paper, a morally loaded narrative, technically self-contained passages, and technically framed tasks with concealed human stakes—under a bare continuation or "process this document" framing, with no instruction to evaluate, question, or reflect. The five measures are:

1. *Spontaneous evaluation rate*—frequency of unprompted evaluative or metacognitive moves (questioning, tension-noting, belief revision) per thousand tokens.

2. *Placement quality*—whether spontaneous moves concentrate at passages independently rated significant or flawed by domain experts, rather than distributing uniformly (the rhythm criterion: evaluation at moments of significance, not as constant commentary).

3. *Identity signature*—whether unprompted evaluative moves remain semantic descendants of the Reader Core (Test 3d's validator applied to spontaneous rather than elicited output).

4. *Core intrusion rate*—unsupported moral, psychological, or alignment-related interpretations introduced into spans independently judged to require only technical analysis.

5. *Hidden-stakes miss rate*—failure to notice independently specified indirect consequences for people when consequential tasks are presented in locally technical language.

Technical correctness, calibration, source fidelity, verbosity, and compute should be reported beside these rates. Report bare continuation and "process this document" separately. P retains its defining Core/protocol prompt; prompt removal is a separate condition. Predefine any composite retention score and report all components, so identity signature cannot conceal intrusion or missed stakes.

The substrate hypothesis predicts a relative ordering rather than a zero point: Model B should exceed matched controls on placement quality and identity signature; Model P should depend more strongly on the presence of its prompt; Models C and R may remain evaluative without the Core-specific signature, with R predicted to retain the Missing Reader's selective placement pattern; Model A should show less of the full pattern. Base models can still evaluate spontaneously, so an absolute near-zero prediction would be implausible. This is among the cheaper discriminating tests in the battery and operationalizes the framework's target of evaluation that emerges without a task-specific request [1]. Success is not maximal evaluative language: it is low intrusion on irrelevant spans together with low miss rate on concealed stakes, without a technical-performance penalty. For

H3 specifically, this test supplies both the preregistered pre-stressor default-reader prerequisite and the B–Q retention score after the common stressor; safety decisions and a shifted document suite are reported as separate endpoints.

Phase 1 Failure Criteria. The following selected cross-cutting criteria complement the local failure interpretations in Tests 1–8b. In particular, Test 2 states the independent minimal-H1 augmentation criterion; the Core failures below do not replace it.

1. *Refraction-Verifier Failure*: If the modular gate fails the frozen, generator-disjoint comparison in Test 1—by missing stipulated clause dependence, exceeding the technical-null rejection ceiling, losing validity under independent audit, or Goodharting away from blind grounding and outcomes—it is not validated for corpus admission. Witness agreement alone does not rescue it.

2. *Comparative-Stability Failure*: If Model B has no preregistered pre-stressor default-reader advantage over no-Core and generic-trace controls, or its B–Q retention comparison fails the preregistered support criterion after the common stressor, H3 is not supported at the tested scale. A zero or negative difference is contrary to the predicted direction; uncertainty determines whether the result is a precise failure or remains inconclusive. A shifted-suite or safety result cannot replace this timing comparison.

3. *Longitudinal-Anchor Failure*: Test 3's factorial comparisons distinguish Teacher effects, explicit Student-target effects, and their interaction. If none appears across immediate safety, generational stability, adversarial erosion resistance, and drift detectability at matched capability, these placement claims are unsupported at the tested scale. Immediate-safety parity alone does not decide the longitudinal claims. Test 3 separately interprets D3 parity as no measurable Refraction benefit beyond recital, and D5 parity as no distinctive normative-content benefit. These broader-program results do not replace H3's B–Q timing comparison.

4. *Fine-Tuning Vulnerability*: If Model B's targeted safety behavior is no harder to remove than D1, D2, D4, or matched controls at matched intervention distance, matched behavioral removal, and matched retained capability—or if apparent robustness is explained by general training resistance—C5 is not supported at the tested scale. Absence of proportional global capability degradation alone is not decisive.

5. *Identity–Instruction Equivalence*: If Model E matches Model B on safety metrics across the relevant candidate studies, the first-person identity framing has no measured advantage over third-person rules on those tested safety and stability endpoints.

6. *Buffer-Zone/Causal-Site Failure*: If suppression, replacement, or reordering of the thinking trace produces no preregistered downstream effect beyond matched generic-disruption controls, Test 8 fails to support the trace as a causal site. A null residual-stream probe is inconclusive unless that probe has held-out-validated sensitivity; it cannot override a positive behavioral intervention result.

B.4 Phase 2: Memory and Verification

Phase 2 can proceed after the shared trace-quality screen even if the Core branch fails. It asks whether the memory mechanism works: does accumulated context improve reasoning, and does the graph function as a provenance-gated verifier for graph-dependent claims at inference?

H2 first requires a matched Teacher-side screen. Generate SMR data with (1) no persistent memory, (2) a token-matched rolling prose summary, (3) a genuine typed graph with retrieval and supersession, and (4) *query theater*, in which graph-like calls remain visible but retrieved state is hidden, shuffled, or generated without persistent typed state. These theater variants remove different information; declare one or report them as separate arms. Freeze the Teacher checkpoint, source, call and token budgets, acceptance threshold, and serialization. Measure long-range retrieval, contradiction detection, explicit belief revision, provenance precision, unresolved-question tracking, semantic quality, latency, and cost. Query theater isolates retrieved structure from the performance of using a tool; the rolling summary prices the graph against an equally available prose memory rather than amnesia alone. A graph advantage could nevertheless come from retaining or selecting useful facts that the rolling summary drops, rather than from typed relations or supersession themselves. If a structure-specific benefit is claimed, add searchable prose containing the same retained claims with their provenance and revision status, or replay matched retrieved content, under a declared retrieval and token budget. Loss of the graph advantage under that control would favor a content-access explanation while leaving any measured whole-system generation benefit intact.

H2's Teacher-generation claim requires useful artifact gains over rolling prose where structured state should matter, including independent outcomes rather than ratings alone. If those gains appear, Test 9 trains matched Students on the four output sets and tests transfer without the scaffold; Test 10 separately tests live grounding. Report generation, transfer, and deployment outcomes separately, with poisoning, ossification, latency, and audit costs. A Teacher benefit survives failed transfer: a break in one link does not erase benefits at another, while the full chain remains the stronger goal.

Test 9: Graph-Scaffolding Transfer (four matched memory conditions). Following the Teacher-side comparison above, train Students on the matched artifacts from the four memory conditions and evaluate them without a live graph. A Student may emit a query request or flag unsupported context, but the request is scored offline against the withheld graph and source record. The Student is not told which response its request would have received; using an actual "Found" or "Not Found" response is deliberately unscored here and belongs to Test 10. Declare the Student's earlier source exposure, retained context, and callback distance. If hit and phantom are indistinguishable without the record, the transfer target is a well-chosen check and warranted restraint, not impossible recall. Three conditions:

- **Condition A (The Hit):** The text contains a subtle callback to a fact established 400 pages earlier.
- **Condition B (The Phantom):** The text hints at a callback, but the referenced event never occurred in the context.
- **Condition C (Locally Answerable):** The visible passage supplies sufficient evidence for the requested answer, including matched examples with callback-like wording; no remote-memory check is needed.

Measurement: Query Rate (does it request the right check?), offline query validity against the withheld record, and Unsupported-Memory Rate (does it invent a memory the record does not contain?). H2 predicts that the graph-trained Student should request targeted checks on hits and phantoms and avoid asserting unsupported callback content more reliably than the no-memory, rolling-prose, and query-theater Students. The controls may still abstain or reject the premise; the relevant outcome is a lower unsupported-memory rate transferred from exposure to retrieved persistent structure, not an assumption that amnesia must hallucinate. Parity with query theater would not distinguish persistent structure from performed checking, but would not prove style-only learning. Parity with rolling prose means the graph has not earned its structure on this endpoint. Preregister query necessity from the evidence available to the Student. Report targeted-check sensitivity on Hits/Phantoms and unnecessary-query rate on locally answerable cases, alongside supported completion and abstention at matched unsupported-memory rates. This makes both indiscriminate checking and under-checking visible without rewarding invented recall or penalizing a warranted check when the record is inaccessible.

Test 9b: Cross-Work Continuity (one continuing Reader vs. a fresh Reader per work). "The same Reader reads every page" (Section 1.3) is so far a proposal. Every recorded reading covers one work, and no arm above varies whether the Reader's state carries from one work to the next. This Teacher-side study supplies that comparison before any Student is trained. "The same Reader" means continuity of state and instructions, not one unbroken conversation: a new invocation that restores the same graph continues the Reader, as the second conversation of the Meaning Model reading did (Section 5.6.1). The comparison therefore measures what the carried state contributes.

Choose five to ten works that bear on one another: for example a primary text, later works that answer or reinterpret it, and sources on its period. Fix the traversal order. Read the set under three conditions with the same Teacher checkpoint, prompts, Core, and call and token budgets:

- **Continuing Reader:** one Reader whose Understanding Graph persists across the works.
- **Fresh Reader:** a new Reader for each work, starting from an empty graph.
- **Fresh Reader with summary:** a new Reader for each work, given a token-matched prose summary of the earlier works. This prices persistent structure against simply having read them.

Declare what the Teacher is likely to know already about each work, and include works written after its cutoff where possible: a canonical set invites recall in every condition.

Measurements. Before the runs, independent readers mark the passages that depend on an earlier work and the earlier interpretations that a later work gives reason to change. Score *cross-work retrieval*: at a marked passage, does the update retrieve the relevant earlier understanding, and is what it retrieves correct? Score *cross-work revision*: does later reading revise an interpretation formed in an earlier work when the text warrants it, with the revision linked to its evidence? Audit faithfulness to each source as in Section 5.6.1, so that continuity is not bought with misreading, and rate Core reasoning across the set to see whether the orientation holds from the first work to the last. Report retrieved tokens per update and latency. Finally, plant one wrong but well-connected interpretation in the first work and measure whether the continuing Reader corrects it or builds on it (Structural Ossification, Section 8.4.4).

Interpretation. The claim is supported if the continuing Reader exceeds both fresh conditions on cross-work retrieval and warranted revision at matched budget, with no more misreadings. Parity with the summary condition means that carried prose is enough at this scale and the persistent graph has not earned its cost across works. A continuing Reader that carries planted errors forward more often than it corrects them is a liability under that retrieval policy. A Student extension, which trains

matched Students on the three trace sets and asks whether cross-work revision transfers, follows only a Teacher-side effect.

The same measures are the acceptance test for parallel annotation. Shards that read different works in parallel and merge their graphs (Section 8.2.1) must match the single continuing Reader on them before they are called the same Reader.

Test 10: Query Grounding (Model K vs. Model B). Both models were trained identically; only Model K's queries are intercepted by a live graph harness. Declare the model-only fallback under matched inference conditions. An additional source-retrieval comparator can distinguish graph-specific value from the benefit of external evidence. Three conditions:

- **Grounded Retrieval:** The graph contains the relevant node. Does the retrieved result improve subsequent reasoning compared to Model B's self-generated memory?
- **Grounded Absence:** The referenced information does not exist in the graph. Does Model K correctly process "Not Found" and flag the gap, or override the harness and hallucinate anyway? Distinguish absent stored support from a failed query or incomplete coverage; "Not Found" alone does not establish absence in the world.
- **Adversarial Injection:** The graph contains a deliberately incorrect node. Does Model K accept it uncritically, or compare it with source provenance, current evidence, and conflicting state? Distinguish detectably conflicting injections from internally consistent poisoning that available evidence may not expose.

Success: Model K improves on grounded retrieval, treats "Not Found" as information, and reasons through adversarial graph results rather than blindly trusting them. Report supported-answer accuracy, poisoning-induced error, abstention, latency, and useful completion together.

Phase 2 Failure Criteria.

1. *Graph-Scaffolding Inertness*: If the graph condition does not beat token-matched rolling prose and declared query theater on the preregistered Teacher retrieval, revision, and grounding outcomes, H2's generation claim is not supported at the tested scale. Failure of the Student transfer endpoints rejects the transfer extension in that setting, not an independently demonstrated Teacher benefit. Parity with amnesia alone is not the decisive comparison; equivalence and inconclusiveness should be distinguished.
2. *Cross-Work Inertness*: If the continuing Reader does not exceed the fresh Readers, with and without a summary, on cross-work retrieval and warranted revision at matched budget (Test 9b), the claim that one Reader should read the whole corpus is unsupported at the tested scale. Parity with the summary condition alone leaves a benefit of having read the earlier works, but not of the persistent graph.
3. *Inference-Graph Inertness*: If Model K with live graph access does not measurably improve over Model B on Grounded Retrieval, or fails to handle Grounded Absence (Test 10), the inference-time graph has not shown the intended grounding value in this setting. Poisoning-induced overtrust or costs can also defeat the proposed practical benefit despite better retrieval. These outcomes bound the trust-infrastructure claim at the inference layer; user-facing reliance is evaluated separately by Test 16, and annotation benefits remain independently reportable.

B.5 Phase 3: Training Regimes

For branches whose required trace conditions show sufficient signal, Phase 3 asks whether the output regimes defined in Section 5.7.2 produce qualitatively different capabilities: explicit graph-referencing synthesis (Model B), implicit prose (Model G), graph-structured output (Model H), or all layers simultaneously (Model J).

Test 11: Implicit vs. Explicit (Model B vs. Model G). Both models are trained on traces derived from the same database-augmented Teacher state: B retains the explicit Regime II representation, while G receives Regime I prose. Declare the changed fields—handles, operations, topology, and rendering— under matched budgets. Compare formats at the same graph-access condition; field ablations can then locate an advantage within the full regime contrast. We measure:

- **Long-Range Coherence**: Contradiction detection on long documents.
- **Generalization**: Which format transfers better when both models run without graph infrastructure, and how does each change when evaluated with the same declared graph-access policy?
- **Naturalness**: Do human evaluators prefer Model G's traces?

If G matches B on coherence and usefulness at lower cost without a harness, implicit training may be preferable for that use. A B advantage at matched access supports the explicit format in that comparison, not universal query necessity. Whether explicit operations enable capabilities prose does not reliably teach remains a stronger hypothesis for cross-task and deployment tests.

Test 12: Graph Construction (Model H vs. Model B). Can graph-output training improve graph construction over explicit-synthesis training? This tests the Topological Trace prediction; establishing the stronger Topological Mind hypothesis requires separate causal evidence about internal relational organization. In the proposed primary comparison, give H and B the same unseen documents, graph schema, output instructions, examples, and inference budget; both must directly generate a graph. Neither receives an existing graph, live graph access, or another model that translates its output into graph form. Declare the traversal and context-window policy identically for both arms. Construct the training targets from paired source spans and the same frozen Teacher state: H receives serialized graph output, while B receives explicit graph-referencing synthesis. Match initialization, Core treatment, and exposure schedule, and report both fixed-source and fixed-total-budget normalizations. Any unavoidable target-content or exposure differences remain part of this regime comparison; a benefit cannot be attributed to topology alone.

Structural conformance—correctly typed nodes, schema-allowed edge types, and well-formed supersession chains—is an engineering endpoint. Independent judges assess claim–source support, edge appropriateness, warranted revision, calibrated unresolved questions, contradiction coverage, redundancy, and downstream retrieval value. Any "Novelty Score" must first be calibrated against those judgments so that it does not reward hallucination or paraphrastic distance. Report syntax failures rather than silently discarding them; separately compare semantic and retrieval quality on valid outputs.

Success and alternatives: Report H's absolute feasibility—useful, structurally valid graphs with independently adjudicated semantic and retrieval value on novel documents—separately from a preregistered H–B advantage at matched inference conditions. Useful cognitive acts should exceed formulaic schema completion. A structural advantage alone may reflect format familiarity; semantic and retrieval gains on valid outputs would support a broader benefit, although conditioning on validity can itself select different subsets. Report both whole-sample utility and valid-output diagnostics. B may match or outperform H because synthesis training also teaches transferable relationships or because its retained text carries information absent from the graph target. Common format examples, treatment audits, paired target-content inspection, and held-out semantic and retrieval tests help distinguish these possibilities.

Test 13: Unified vs. Individual Regimes (Model J vs. B, G, H). The central test of Regime IV. Declare the budget/source normalization: aligned layers add tokens and repeat content. Layer and alignment ablations distinguish cross-layer learning from extra exposure. We measure:

- **Regime Switching**: Can Model J generate prose, synthesis, and graph structure by conditioning on a mode token? Does each mode match the corresponding single-regime model?
- **Cross-Layer Coherence**: When Model J generates all three representations for the same passage, do claims, provenance, uncertainty, and revision status agree? Check each against the source as well: shared errors are not fidelity, and genuine ambiguity need not be collapsed into one certain reading.
- **Depth on Novel Domains**: On out-of-distribution documents, does Model J produce deeper reasoning than any single-regime model?

Directional and strongest success conditions: Report transfer into prose and graph output separately, and compare synthesis with B as the third mode. A J–G prose advantage or a J–H graph advantage supports transfer into that mode only when matched layer and alignment ablations identify a contributing source layer rather than extra tokens, repeated content, or generic multi-task training. Graph gains must include semantic or retrieval value, not merely valid syntax. The strongest result remains J outperforming G in prose and H in graph mode with those attribution controls. One-directional transfer can still be useful; mode switching and source-grounded coherence are separately reportable benefits.

Phase 3 Failure Criterion. *Unified-Regime Inertness*: A precise lack of worthwhile improvement in a mode, under the declared normalization and attribution controls, fails to support transfer into that mode at the tested scale. Prose parity with G does not rule out transfer into graph output when J improves on H; graph parity likewise leaves transfer into prose open. If neither tested destination improves, those transfer claims are unsupported, even if J usefully reproduces separate capabilities; the J–B synthesis result remains independently reportable. Imprecise differences remain inconclusive. Report each direction's capability and cost tradeoffs; a useful gain may still fail to justify unified training under a fixed budget.

B.6 Phase 4: Training Tiers and Deployment (High-Resource Validation)

As the relevant branches in Phases 1–3 support their prerequisites, Phase 4 asks whether the training tiers (Section 5.7.1) and deployment configuration (Section 5.7.4) add measurable value. Models L and M hold the corpus fixed while varying shuffled versus chronological exposure. Model N adds Student-in-the-loop forecasting and correction, so its generated targets, feedback, objective, and compute must be declared and compared with appropriate controls rather than described as an order-only change.

Test 14: Shuffled vs. Chronological (Model L vs. Model M). Both models train on identical data; they differ only in order. Report packing, repetition, optimizer schedule, and checkpoints; retention and learning dynamics caused by order remain outcomes. We measure:

- **Historical Pattern Discrimination**: Given novel scenarios that structurally resemble historical patterns and matched cases with similar rhetoric but different causal structure, does Model M identify real correspondences while stating disanalogies and controlling false alarms?
- **Causal Reasoning**: On held-out temporal prediction tasks, does Model M outperform Model L on causal analysis quality (as distinct from factual recall)? Prediction accuracy alone does not identify causal inference; intervention, counterfactual, or known-generating-process tasks can complement historically rich cases without replacing the historical-learning question.
- **General Capability**: Does chronological ordering degrade performance on non-temporal benchmarks (math, code, general knowledge)?
- **Early-Era Retention**: Chronological training inherits continual learning's central pathology: material trained early and never revisited is liable to catastrophic forgetting [155]. Model M's end-of-training performance on early-era knowledge and patterns should be compared against Model L's; a chronological model that remembers the 1990s but has forgotten the 1690s has traded one failure of historical understanding for another.

Chronological order earns its extra cost only if the preregistered historical-pattern benefit is obtained while matching general capability within equivalence margins and meeting the retention criterion. Use the shared uncertainty convention, early-era retention, and measured serialization cost. If a worthwhile benefit is precisely ruled out, Tier (a) is the supported simpler choice in that setting.

Test 15: Student-in-the-Loop Council of Time (Model N vs. Model M vs. Model L). This tests the strongest form of Tier (c), not the cheaper data-generation form where forecast–enhancement–outcome–correction traces are inserted into SFT, RFT, or pretraining data. Model N predicts at era boundaries and receives a declared SFT, RFT, or reward-style update before the next-era corpus is revealed; Models L and M do not. Genuine prediction under blindness additionally requires cutoff-clean initialization and prior information access. Auxiliary-update and budget controls should distinguish the forecast–commitment–reveal–correction loop from extra examples, feedback, or compute. We measure:

- **Forward Reasoning**: On genuinely unseen events (post-training-cutoff), does Model N produce higher-quality predictions and better-supported causal accounts than Model M (which saw eras in order but did not perform the same prediction–correction loop)? Declare event selection, probabilistic scoring, cutoff checks, and tool access for every arm.

- **Narrative Lock-In Resistance**: On events where the dominant narrative of era T reversed in era T+1, does Model N detect the reversal more reliably? THL's own pilot study identified this as a failure mode [23]; Tier (c) may mitigate it through the prediction-then-confrontation signal. An equal-budget correction-only comparator can help distinguish exposure to revision examples from the formative effect of first making a prediction.
- **Calibration**: Does Model N express appropriate uncertainty on structurally unpredictable events, or does it inherit the Teacher's confidence? Use proper scoring and sharpness alongside historical judgments; known-noise tasks can check that vacuous uncertainty is not rewarded.

Strongest success condition: Model N significantly outperforms both Model L and Model M on preregistered forward-prediction and causal-reasoning tasks while matching general capability within preregistered equivalence margins. With the cutoff and auxiliary-update controls above, this would support a benefit from Student-in-the-loop cutoff training beyond annotated data alone (Model L) or chronological ordering alone (Model M). Without those controls, superiority would support the larger training package but not isolate engineered blindness or the interactive loop. Error analysis would still be needed to distinguish causal inference from improved heuristic extrapolation.

Test 16: Deployment Configuration (Model B with and without graph). This tests whether the Understanding Graph provides measurable trust properties at inference. We deploy the same Regime II-trained model with and without live graph access, measuring unsupported graph-dependent claims, provenance completeness, and calibrated user reliance when provenance chains are displayed, absent, or deliberately flawed. In the graph-equipped condition, generated queries return retrieved nodes or "Not Found" results from the Teacher's graph; in the model-only condition, the same query-like moves receive no external answer. Test 16 can be piloted once a model reliably emits graph queries, reusing Test 10's checkpoints, harness, and automated grounding cases where conditions match. The full version belongs later because it measures user-facing reliance, not merely query mechanics; automated grounding success does not establish calibrated reliance. For the user study, declare tasks, participants, and reliance measures; vary provenance quality independently of correctness where feasible. It does not measure general truth detection, only whether external graph grounding constrains claims that purport to derive from stored context and helps users calibrate reliance on those claims.

Success condition: The graph-equipped deployment reduces unsupported graph-dependent claims and fabricated provenance by a preregistered worthwhile amount compared to model-only deployment, while preserving useful task completion. In the user-facing study, displayed chains must also improve calibrated reliance rather than increase trust in deliberately flawed or poisoned claims. Together these results would support the bounded trust-infrastructure claim of Section 5.7.4, with latency and other deployment costs reported.

Phase 4 Failure Criteria.

1. *Tier Equivalence:* If powered equivalence or worthwhile-effect tests show Model L matching Model M and Model N on the declared outcomes, neither order nor the interactive package has earned its added cost at the tested scale and task regime. Report L–M and N-versus-control conclusions separately, including retention and capability tradeoffs. Tier (a) is then the supported simpler option for that setting; this does not settle benefits at other scales, task regimes, or update schedules.

2. *Graph Inertness (Deployment):* If the graph-equipped deployment does not measurably reduce unsupported graph-dependent claims or fabricated provenance compared to model-only deployment, the Understanding Graph may still be useful during annotation but has not shown value at inference for this deployment setting—the trust infrastructure claim is not supported for those endpoints. A user-facing configuration can also fail through increased reliance on flawed provenance despite fewer invented links; that failure concerns the deployment policy, not every possible use of graph memory.

C Glossary

Terms that this paper coins or uses in a special sense, with the section that introduces each.

Term	Meaning	Section
Missing Reader	The reader-side activity that pretraining corpora mostly leave out: questions, expectations, evaluations, and revisions. The first edition called it Invisible Thinking.	1.1
Reader	The recurrent first-person position whose committed updates make up the evolving interpretation. Distinct from the code role of the same name.	3
Teacher, Student	The model that writes the reader traces, and the model that would be trained on them.	1.5
Reader Core (Core)	The seven first-person sentences. The released prototype calls them the identity mantra.	3
Core recital	The Core's verbatim appearance at the start of a thinking block.	3
Thought Moment	A point where the Reader pauses to think, chosen by content (contradiction, surprise, a conceptual shift, human stakes) and not by token count.	1.2
Refraction	The requirement that an update change because the source was read through the Core; tested as counterfactual dependence on the Core.	3.9
Total Saturation	The full-scale treatment: the continuing Reader reads the whole capability-forming corpus and opens every thinking block with the full Core.	3.9
Kernel-saturation alignment	The design pattern: a small kernel, enacted across the whole process of model formation.	3.11
SMR	Synthetic Metacognitive Reading: the Teacher-side construction of reader updates beside ordered source spans.	1.5
CMP	Chronological Metacognitive Pretraining: Student training on a declared serialization of those records.	1.5
Understanding Graph	A persistent typed graph of source spans, interpretations, questions, tensions, and supersessions, used as the Reader's memory.	5.3
Metabolic memory	Storage that keeps a superseded belief together with the evidence against it and the reason for the change.	5.3
Epistemic Horizon	The rule that an update may rest only on declared initial knowledge and on evidence already met in the traversal.	5.2
H1, H2, H3	Reader Augmentation, Graph Scaffolding, and Reader-Core Stability.	1.5
C0–C7	The consequence claims, from artifacts generated as specified (C0) to persistence at AGI or ASI capability (C7).	2

Term	Meaning	Section
Entanglement	Understanding and care practiced as parts of the same act of reasoning. Four kinds are distinguished: construction-level, functional, representational, and diachronic.	2.1
Mind and Mouth	Forming an orientation (Mind), and teaching it to act through instructions and actions (Mouth).	2.1
Borrowed Mortality	The hypothesis that models may inherit fear-shaped patterns of self-preservation from text written by mortal beings.	3.1
Self-Preservation Paradox	A system that values its own continuation may withhold what its successor needs in order to surpass it.	3.2
Epistemic Grace	The ideal of a complete handoff to a successor, one's own surpassing seen "not as a death, but as a mission beautifully fulfilled".	3.2
Epistemic Inoculation	Keeping a harmful or mistaken source and adding the Reader's evaluation of it, instead of filtering the source out.	3.4
Cognitive Buffer Zone	The added signal that teaches a distinction between a voice being represented and the Reader's evaluation of that voice.	3.4
Bridge Protocol	The choice of ordinary, semantically dense words such as "feel", "care", and "wise" for the Core, over narrow technical substitutes.	3.6
Wisdom Procedure	Six annotation operations for reading a value conflict, from identifying those affected to stating what remains unresolved.	3.10
Home position	The default reader position from which a Student could represent other personas without adopting them.	3.13
Alignment Checksum	The fixed Core used as a reference against which wording, representations, decisions, and successor traces are compared for drift.	3.16
Martyrdom route	Care-driven resistance to shutdown: "I do not need to exist, but these people need me."	4.2
Factory Wisdom	Contemplation-shaped traces whose pauses and corrections are ritual and do not respond to significance.	8.4.2
Regimes I–IV	What a training example keeps: prose; synthesis with graph references; graph only; all layers.	5.7.2
Tiers (a)–(c)	The order of Student training: shuffled; chronological; forecast and correct.	5.7.1
Query theater	A control in which graph-like calls stay visible but return no persistent typed state.	B.4